

Transforming Information Assurance and IT Service Management Through Digital Engineering

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Signed,



John James Darth Vader Bonar

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Abstract

Digital Engineering has transformed how the Department of War, NASA, and the aerospace industry sustain complex systems. Its four pillars are Model-Based Systems Engineering, digital threads, digital twins, and Product Lifecycle Management, and together they are credited with gains in mission assurance and lifecycle governance. These same methods remain virtually untested in enterprise Information Technology and Information Assurance, even though compliance regimes such as the NIST Risk Management Framework and the DoW Cybersecurity Maturity Model Certification strain the documentation and traceability practices those fields depend on. A structured literature review found almost no research applying Digital Engineering to either domain, which leaves the question of whether these methods could transfer resting on argument rather than on evidence.

This study therefore asked the professionals themselves. A quantitative, cross-sectional survey measured what practitioners in Information Technology and Information Assurance know of these capabilities and what they believe those capabilities are worth. It drew 1,411 responses, roughly three times the planned target, from professional networks weighted toward defense and aerospace and toward earlier-career respondents.

Familiarity with Digital Engineering as a named discipline proved low. Seventy-eight percent of respondents placed themselves in the two lowest familiarity categories, and only 22% reported any professional exposure within the previous two years. Once the survey described the underlying capabilities in plain functional language, however, agreement became near-universal at 94%, and perceived value stayed high across the applicability, information technology, and security domains, ranging from 82% to 88%. Two subgroup patterns held up under scrutiny. Information Technology respondents outscored Information Assurance respondents by a wide margin, with effect sizes running from 1.64 to 1.94, and respondents with eleven to fifteen years of experience were more

reserved than either their junior or their senior colleagues, a difference that survived adjustment for professional field and familiarity.

Two boundaries qualify these results. Recruitment leaned toward the sectors where Digital Engineering is already practiced, so the estimates should be read as upper-leaning rather than as measurements of the profession at large. The study also measured only two of the barriers that could stand in the way of adoption. Between those two, awareness and exposure bind more tightly than doubt about value, although a cross-sectional design cannot confirm that direction, and cost, tooling maturity, and organizational readiness went unmeasured.

The work contributes a first empirical baseline for a population that had not previously been measured, together with an open-source screening method, validated against synthetic data, for detecting coordinated or automated responses in anonymous surveys. Its wider significance turns on who carries the burden of compliance. That burden falls hardest on organizations with the least capacity to absorb it, and the professional receptiveness measured here is a precondition for the lower-cost implementations those organizations would need.

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Chapter 1: Introduction

When threat actors began exploiting zero-day vulnerabilities in Ivanti Connect Secure and Ivanti Policy Secure appliances in December 2023, security teams across multiple agencies worked under acute time pressure. Defenders sought to identify every affected component and to understand what attackers were already exploiting (Cybersecurity and Infrastructure Security Agency, 2024b). Yet they had no comprehensive understanding of how vulnerabilities in one system could cascade across interconnected infrastructure and national security systems. When the Cybersecurity and Infrastructure Security Agency issued Emergency Directive 24-01 in January 2024, agencies lacked ready visibility into the affected configurations and their dependencies. While they worked to establish that visibility, adversaries retained the initiative (Cybersecurity and Infrastructure Security Agency, 2024a). This operational failure is not an isolated incident. It exemplifies a structural condition of modern practice: enterprises now change their information systems faster than their documentation can describe them, leaving defenders to act on records that no longer match reality.

The consequences of such failures extend beyond the immediate organizations affected. When defenders cannot comprehend the cascading impacts of compromise, they communicate risk less clearly to organizational leadership, they lose the ability to prioritize remediation by actual impact severity, and coordinating a defense across organizational boundaries becomes impractical. When defenders cannot understand system interdependencies, incidents that might otherwise be contained become enterprise-wide crises. Security teams then shift to reactive responses rather than proactive defense, spending effort on manual discovery. Model-based approaches are designed to do that same work far more efficiently, a goal the Department of Defense set out in its 2018 Digital Engineering Strategy (Department of War, 2018).

Information Assurance, as codified by the National Institute of Standards and Technology, encompasses those measures that protect and defend information and

information systems by ensuring their availability, integrity, authentication, confidentiality, and non-repudiation (Ross et al., 2020). Cybersecurity is an operational component within this broader discipline, concentrating on the protection of information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. Throughout this dissertation, the term *Information Assurance* denotes the broader discipline encompassing security policy, risk management, compliance verification, and protective measures. Cybersecurity, by contrast, refers specifically to the technical and operational dimensions of protecting systems from cyber threats. The two terms are not used interchangeably: Information Assurance is the broader governance and assurance framework, while cybersecurity addresses the protective mechanisms and threat responses that operate within it.

Terminological precision bears operational consequences. Treating Information Assurance as a synonym for cybersecurity invites organizations to under-invest in the governance, documentation, and architectural foundations that effective cybersecurity operations rest on. Failing to maintain accurate system documentation, for example, represents an Information Assurance shortfall that manifests as cybersecurity operational degradation. Literature from defense and aerospace contexts suggests that Digital Engineering may address both dimensions: the governance and documentation requirements of Information Assurance and the operational visibility requirements of cybersecurity defense. Whether IT and Information Assurance professionals recognize this potential is a central question of this research.

This chapter examines the challenges organizations face in implementing Information Assurance practices and managing Information Technology (IT) service delivery. It introduces Digital Engineering as a disciplinary approach to gaps that persist despite mature frameworks, among them the National Institute of Standards and Technology (NIST) Risk Management Framework (RMF) (Ross et al., 2020), the Information Technology Infrastructure Library (ITIL) (Cannon, 2013), and the Unified

Architecture Framework (UAF) (Object Management Group, 2022b).

A Note on Institutional Nomenclature

A point of nomenclature warrants clarification at the outset, because it recurs throughout this dissertation and concerns the very institution from which the methods studied here originate. The United States defense establishment traces its origins to the Department of War, established in 1789 to administer primarily the Army. The National Security Act of 1947 created the National Military Establishment, unifying the War and Navy Departments together with the newly independent Air Force. That body was renamed the Department of Defense in 1949. The institution has now entered a transition back toward its original title. In September 2025, Executive Order 14347 authorized the use of "Department of War" as a secondary title for the institution and its senior officials in official correspondence, public communications, and other non-statutory contexts (Executive Office of the President, 2025). Only an Act of Congress can alter the statutory name of an executive department. The legally established name therefore remains the Department of Defense while the secondary designation is in active use and legislation to make the change permanent is pending. This dissertation accordingly uses the historical names when referring to documents and events of their own eras and "Department of War" when referring to the institution as a present-day actor.

Concretely, the dissertation refers to the institution as the Department of War (DoW) wherever it appears as a present-day actor, for example in describing its Digital Engineering Strategy, while preserving published titles exactly as issued. The Department of Defense Architecture Framework remains DoDAF, Department of Defense Instruction 5000.97 remains DoDI 5000.97, and bibliographic entries credit the Department of War as institutional author while keeping each document's published title verbatim. The convention keeps the work current with the institution's evolving identity, which matters because the Department of War is the principal originator of the Digital Engineering practices this research examines. It also preserves the scholarly record: cited titles remain

exactly as published and therefore discoverable. Where the abbreviation DoD appears in later chapters it does so only as an inseparable element of a proper name, such as DoDAF or DoDI, and never as a reference to the institution itself.

Current State of Information System Management

Organizations today operate within an environment defined by rapid technological evolution and escalating system complexity. Cloud computing, microservices architectures, Internet of Things (IoT) devices, and operational technology have converged to produce complex, interdependent systems that overwhelm traditional approaches to both Information Assurance and IT service delivery (Benbya et al., 2020). These advances deliver undeniable operational benefits, but they carry substantial costs in system visibility, security control implementation, configuration management, and service delivery coordination.

Technological change continues to accelerate. Organizations that required months to deploy new capabilities a decade ago now deploy changes continuously through automated pipelines. This acceleration benefits operational agility but strains the documentation and verification processes on which Information Assurance depends: static documentation designed for quarterly or annual update cycles cannot maintain accuracy when systems change hourly, a mismatch examined later in this chapter.

Enterprise information systems now routinely span multiple technology domains: cloud-based infrastructure and services, on-premises data centers, edge computing environments, operational technology networks, and mobile and remote access systems. This technological heterogeneity generates persistent challenges in maintaining security visibility, implementing consistent protection mechanisms, and delivering reliable IT services across disparate environments. Teams cannot track system dependencies with confidence, configuration management falters, and security control implementation proceeds inconsistently across heterogeneous platforms (Bokan & Santos, 2021).

Beyond technical complexity, the challenge extends to organizational design.

Structures that evolved to manage discrete technology domains now impede the integrated visibility that modern environments require. Security teams operate separately from IT operations teams, while cloud architects work independently of network engineers. Application developers deploy services without understanding infrastructure dependencies. This organizational fragmentation mirrors and reinforces the technical fragmentation that undermines both Information Assurance and IT service delivery effectiveness.

Information Assurance Practice

Information Assurance practice has evolved in response to the complexities of modern enterprise environments. Central to this evolution, the NIST Risk Management Framework provides a structured, disciplined approach for managing security and privacy risk that organizations can apply across diverse information systems (Ross et al., 2020). RMF establishes a lifecycle approach to security through seven iterative steps: prepare, categorize, select, implement, assess, authorize, and monitor. Now foundational for federal agencies, the framework finds increasing adoption among organizations operating national security systems and private enterprises seeking systematic approaches to Information Assurance.

The RMF represents a substantial advancement over earlier compliance-focused approaches that treated security as a point-in-time certification rather than a continuous process. Its emphasis on continuous monitoring and ongoing authorization reflects recognition that security postures change constantly as systems evolve, threats emerge, and organizational requirements shift. Implementing continuous monitoring effectively requires capabilities that most organizations lack: real-time visibility into system configurations, automated assessment of control effectiveness, and dynamic risk calculation based on current rather than documented system states.

Additional Information Assurance lifecycle frameworks exist, including ISO 31000 (International Organization for Standardization, 2018a), the NIST Cybersecurity Framework (National Institute of Standards and Technology, 2014), and

COBIT (Information Systems Audit and Control Association (ISACA), 2018), among others. These frameworks provide alternative approaches to the RMF. However, they share common challenges in maintaining accurate documentation, ensuring visibility into system states, and coordinating security efforts across organizational boundaries. This research focuses on the NIST Risk Management Framework, a choice that reduces complexity by avoiding direct comparison among multiple frameworks while still addressing challenges common to all.

Security control selection is a key RMF activity. Federal information systems and national security systems typically use NIST Special Publication 800-53 Revision 5 as the authoritative catalog of security controls (Ross et al., 2020). Organizations operating outside federal requirements may employ alternative frameworks for control selection, including ISO/IEC 27001 (International Organization for Standardization, 2022), the NIST Cybersecurity Framework, or industry-specific standards. This research focuses on NIST 800-53 Revision 5 given its applicability to federal and national security contexts, though the underlying principles extend to organizations employing other control frameworks.

NIST Cybersecurity Framework (CSF) 2.0, released in February 2024, provides a voluntary, risk-based structure that complements the RMF by organizing cybersecurity outcomes into six core Functions: Govern, Identify, Protect, Detect, Respond, and Recover (National Institute of Standards and Technology, 2024). Version 2.0 added the Govern Function to address cybersecurity governance at the organizational level. This addition recognizes that leadership engagement, risk strategy, and policy oversight are prerequisites for effective cybersecurity rather than ancillary concerns. Each Function decomposes into Categories and Subcategories that map to specific outcomes. CSF 2.0 maintains explicit alignment with SP 800-53 controls through Informative References, enabling organizations to trace CSF outcomes to specific control implementations. CSF 2.0 broadens its scope beyond critical infrastructure to encompass all organizations regardless of sector or size. In this expanded form, it serves as a bridging framework. Organizations

subject to RMF compliance can use CSF to communicate risk posture to non-technical stakeholders. Organizations not subject to federal requirements can adopt CSF as a starting framework that maps naturally to SP 800-53 controls should compliance requirements evolve. CSF 2.0's emphasis on governance, supply chain risk management, and continuous improvement aligns with Digital Engineering principles, particularly the authoritative source of truth concept and digital thread traceability. To the author's knowledge, no published research has examined this alignment.

Beyond federal systems, an expanding compliance landscape governs nonfederal organizations that process, store, or transmit Controlled Unclassified Information (CUI) on behalf of government agencies. NIST Special Publication 800-171 Revision 3, finalized in May 2024, establishes security requirements for protecting CUI in nonfederal systems and organizations (Ross & Pillitteri, 2024). Revision 3 restructured the framework from fourteen to seventeen security requirement families, adding Planning, System and Services Acquisition, and Supply Chain Risk Management. It also aligned its control structure directly with SP 800-53 Revision 5 as the single authoritative source. SP 800-171's 97 security requirements incorporate Organization-Defined Parameters that provide implementation flexibility while enabling automated compliance assessment through machine-readable formats such as the Open Security Controls Assessment Language (OSCAL) (National Institute of Standards and Technology, 2023). CSF 2.0 maps to SP 800-171 requirements through shared SP 800-53 lineage: organizations implementing SP 800-171 controls simultaneously address CSF Subcategory outcomes in the Identify, Protect, Detect, Respond, and Recover Functions, creating a unified compliance posture across both frameworks.

NIST Special Publication 800-172 supplements the SP 800-171 baseline with thirty-five enhanced security requirements designed to protect CUI associated with critical programs and high-value assets against Advanced Persistent Threats (Ross et al., 2021a). Operating on a three-pillar defense strategy that comprises penetration-resistant

architecture, damage-limiting operations, and cyber resiliency and survivability, SP 800-172 demands capabilities that manual processes and static documentation struggle to sustain. Enhanced requirements span access control, configuration management, identification and authentication, incident response, risk assessment, situational awareness, and system protection. A Revision 3 update currently in development expands scope from confidentiality protection alone to encompass confidentiality, integrity, and availability, further increasing documentation and verification demands.

Operationalizing these NIST requirements within the defense industrial base, the Cybersecurity Maturity Model Certification (CMMC) 2.0 framework establishes a three-tiered certification structure (Department of War, 2024). Level 1 requires the fifteen basic safeguarding requirements of Federal Acquisition Regulation (FAR) clause 52.204-21 for Federal Contract Information protection, verified through annual self-assessment. Level 2 mandates compliance with the 110 requirements of SP 800-171 Revision 2, assessed through triennial third-party evaluation by CMMC Third-Party Assessment Organizations. Level 3 adds twenty-four enhanced requirements drawn from SP 800-172, assessed by the Defense Industrial Base Cybersecurity Assessment Center. Phase 1 implementation began with the December 2024 effective date of 32 CFR Part 170, with full implementation across all contracts involving Federal Contract Information or Controlled Unclassified Information expected by November 2028. CMMC marks a fundamental shift from self-attestation to verified compliance, imposing documentation and evidence requirements that intensify the challenges organizations already face in maintaining accurate, current security documentation. This expanding compliance landscape extends from CSF 2.0's voluntary governance framework through SP 800-171 and SP 800-172's CUI protection requirements to CMMC's third-party verification mandate. Across its full breadth, organizations confront escalating documentation, traceability, and verification demands that Digital Engineering capabilities are designed to address.

Selecting appropriate security controls depends on accurately understanding the

systems being protected, their operational context, their interconnections with other systems, and their role within the broader enterprise architecture. Control selection that proceeds from inaccurate system understanding produces security postures that address documented rather than actual risk. This disconnect between documentation and reality is a structural challenge that persists regardless of which control framework an organization employs.

Implementing the RMF effectively presents documented challenges in complex technological environments. Organizations must categorize information systems based on potential impact, select appropriate security controls from comprehensive catalogs, implement those controls across diverse platforms, assess control effectiveness, obtain authorization decisions, and maintain continuous monitoring throughout the system lifecycle. Each step demands accurate, current information about system configurations, security control implementations, and operational states. Traditional documentation approaches struggle to maintain such information.

The continuous monitoring requirement deserves particular attention because it exposes the limitations of document-centric approaches most directly. Continuous monitoring as envisioned by the RMF requires ongoing awareness of security-relevant system changes, automated assessment of security posture impacts, and timely reporting to authorizing officials. Organizations attempting to implement continuous monitoring through manual processes discover that the labor required exceeds available resources. Those pursuing automation instead discover that they lack the authoritative system models and configuration baselines that automation requires.

IT Service Management Practice

IT service management (ITSM) has matured through frameworks designed to ensure reliable service delivery across the enterprise. The Information Technology Infrastructure Library provides comprehensive guidance for aligning IT services with business needs through structured processes for service strategy, service design, service

transition, service operation, and continual service improvement (Cannon, 2013). ITIL emphasizes configuration management, change management, and service asset management as core capabilities on which effective IT service delivery depends.

ITIL's evolution from Version 3 to ITIL 4 reflects recognition that service management practices must adapt to cloud computing, DevOps practices, and agile delivery models. ITIL 4 introduces the Service Value System concept, emphasizing flexibility and continuous improvement over rigid process compliance. Yet the dependencies on accurate configuration information and effective change coordination persist regardless of which ITIL version organizations adopt. Even the Service Value System cannot create value if the underlying information about services, configurations, and dependencies remains inaccurate or incomplete.

Configuration management within the ITIL framework requires organizations to maintain accurate configuration management databases documenting configuration items, their attributes, and their relationships. Change management processes depend on accurate configuration information to assess change impacts and coordinate modifications across interconnected systems. Service asset management extends these capabilities to encompass the full lifecycle of IT assets from acquisition through retirement. These interconnected processes provide structure for managing complex IT environments, but they depend on the accuracy and currency of underlying information, an accuracy that organizations consistently fail to achieve.

The relationship between configuration management and change management illustrates how documentation failures compound. Change management processes assess proposed changes against documented configurations and relationships. When documentation is incomplete, change assessments miss dependencies that exist in operational systems. Changes approved on the basis of incomplete assessments cause unintended impacts. Those impacts require emergency changes to address. Emergency changes bypass change management processes, further degrading documentation accuracy.

This cycle perpetuates itself, progressively undermining both configuration management and change management effectiveness.

Challenges in Current Practice

Traditional documentation approaches and manual tracking methods prove increasingly inadequate for capturing and managing the complexity of modern information systems. Paper-based security documentation, static network diagrams, and periodic compliance assessments fail to reflect the dynamic nature of contemporary enterprise environments. IT service management practices that rely on manual configuration tracking and change coordination struggle to maintain accuracy and timeliness in environments characterized by continuous deployment and rapid change cycles.

A structural challenge emerges: the frameworks are sound and the practitioners dedicated, yet the limitation lies in the mismatch between the documentation velocity that manual processes can sustain and the operational velocity that modern enterprise environments demand. The evidence reviewed below suggests that process improvement and additional staffing alone cannot close this gap. Closing the gap requires a paradigm shift from document-centric to model-centric practices, precisely the shift that Digital Engineering is designed to provide.

Research documents pervasive failures across both Information Assurance and IT service management domains. Industry analyst estimates indicate that 80% of Configuration Management Database (CMDB) implementations fail to deliver intended value (Gartner, 2019). Studies find that organizations can monitor only 66% of their IT environments, leaving 34% unmonitored (IDC & Exabeam, 2023). Shadow IT, defined as technology acquired or deployed outside official governance, now represents 30–40% of enterprise IT spending, creating assets invisible to documentation efforts (Gartner, 2022). The mean time to identify security breaches averages 204 days, reflecting the visibility gaps that impair threat detection (IBM Security & Ponemon Institute, 2024).

These statistics reveal systemic limitations of document-centric approaches rather

than isolated organizational shortcomings. Organizations that invest heavily in documentation still experience these failures. Organizations with mature governance processes still discover undocumented systems and unknown dependencies. Such consistency of failure across diverse organizations identifies the document-centric approach itself, however well executed, as the source of the problem.

A common pattern unites these failures: organizations cannot maintain accurate, current documentation of their information systems using traditional approaches. Modern IT environments change at rates exceeding the capacity of manual documentation processes. Static artifacts become obsolete before completion. Configuration databases diverge from operational reality. Security documentation describes intended states rather than actual implementations. Such a documentation-reality gap undermines every process that depends on accurate system information, a category that includes nearly all Information Assurance and IT service management activities.

Chapter 2 examines these visibility and documentation failures in depth, synthesizing peer-reviewed research and industry analysis to establish the evidence base for why traditional practices fail. Documentation challenges reflect multiple converging factors: organizational silos fragmenting visibility, complexity exceeding human documentation capacity, manual processes unable to match rates of change, and technical debt accumulating in documentation domains.

Digital Engineering as Potential Solution

The preceding section established that these failures persist despite organizational effort, which reframes the problem as one of approach: the method itself falls short even where execution is diligent. Digital Engineering answers that reframed problem by changing the unit of record from the document to the model: it is a systematic approach to designing, developing, and managing complex systems through integrated digital models and data-driven processes (International Council on Systems Engineering (INCOSE), n.d.). The discipline originated in the defense and aerospace sectors, where organizations

including the United States Department of War (DoW) (Office of the Under Secretary of Defense for Research and Engineering, 2022), the National Aeronautics and Space Administration (NASA) (National Aeronautics and Space Administration — Office of the Chief Engineer, 2020), and the International Council on Systems Engineering (INCOSE) (International Council on Systems Engineering (INCOSE), n.d.) have formalized it through authoritative guidance. While these practices emerged primarily from physical systems engineering, the underlying principles offer capabilities of documented value for Information Technology and Information Assurance domains.

The defense and aerospace origins of these practices carry significance beyond historical interest. Both sectors developed them to address challenges structurally similar to those confronting enterprise IT and Information Assurance: complex interdependent systems, stringent compliance requirements, mission-critical operations, and the need to maintain comprehensive visibility across extended lifecycles. Solutions that proved effective for managing combat aircraft development and spacecraft missions may serve equally well for managing enterprise information systems and security postures. That transfer remains untested, and this dissertation measures professional perception of its potential rather than the transfer itself. Measured evidence supports this potential: E. B. Rogers and Mitchell documented an 18% improvement in systems engineering efficiency and a 9% reduction in defects following MBSE adoption on a complex system-of-systems program (E. B. Rogers & Mitchell, 2021). Yet such measured evidence remains rare, and no comparable data exist for enterprise IT or Information Assurance contexts.

Four pillars support Digital Engineering: Model-Based Systems Engineering (MBSE), digital threads (the authoritative traceability that connects system lifecycle artifacts), digital twin technology, and Product Lifecycle Management (PLM). Understanding these pillars provides context for examining how Digital Engineering practices might address the challenges identified in current Information Assurance and IT service management practice.

Integrating these pillars distinguishes Digital Engineering from isolated tool adoption. Organizations might implement modeling tools without achieving Model-Based Systems Engineering, or deploy digital twin capabilities without establishing digital thread traceability. Digital Engineering's value emerges when these capabilities function together as an integrated approach, an integration that current Information Assurance and IT service management practices generally do not provide.

Model-Based Systems Engineering

Model-Based Systems Engineering marks a paradigm shift from document-centric practices to model-centric approaches for system development and management (Hutchison et al., 2020). Rather than relying primarily on textual descriptions and static diagrams, MBSE employs formal, executable models that capture system architecture, behavior, requirements, and relationships in structured, machine-readable formats. These models serve as authoritative sources of truth that can be analyzed, simulated, and validated throughout the system lifecycle (Hutchinson et al., 2021).

Distinguishing document-centric from model-centric approaches matters because the contrast defines the fundamental transformation that Digital Engineering proposes. Document-centric approaches produce artifacts such as diagrams, specifications, and procedures that describe systems. These artifacts require human interpretation, cannot be automatically validated for consistency, and provide no mechanisms for maintaining currency as systems evolve. When an organization produces a network diagram, that diagram captures system state at the moment of creation. Every subsequent change renders the diagram incrementally less accurate. Likewise, a security plan describes intended security posture only at the time of writing, and every subsequent system modification creates potential divergence between the plan and operational reality. Model-centric approaches produce executable representations that can be queried, analyzed, and validated automatically. As models change, dependent artifacts can update automatically, and when engineers evaluate proposed changes, models enable impact

analysis that documents cannot provide. Maintaining consistency automatically addresses the velocity mismatch between documentation processes and operational change rates described earlier in this chapter.

Formal modeling languages underpin MBSE implementations and provide the technical foundation for these capabilities. SysML, the Systems Modeling Language standardized through the Object Management Group, extends the Unified Modeling Language (UML) with constructs specifically designed for systems engineering (Friedenthal et al., 2014). Where UML focuses on software system modeling, SysML adds requirements modeling, parametric analysis, and physical system representation that UML lacks. SysML defines nine diagram types organized into four diagram categories (requirements, structure, behavior, and parametrics) that span stakeholder requirements, system architecture and composition, operational dynamics, and constraint-based quantitative analysis (detailed in Chapter 2). This taxonomy provides a comprehensive vocabulary for representing complex systems from multiple perspectives simultaneously, a capability that directly supports the multi-stakeholder visibility requirements of both Information Assurance governance and IT service management coordination.

SysML's evolution from 1.x to the emerging v2 specification addresses limitations that impeded broader adoption. Version 2 introduces a redesigned language architecture emphasizing improved precision, expressiveness, and usability. A standardized textual notation accompanies the graphical notation, enabling model creation through text-based interfaces more familiar to software developers and IT professionals. SysML v2 also introduces a standardized Application Programming Interface (API) for tool interoperability, addressing a persistent criticism that MBSE tool ecosystems require vendor-specific integrations. Improved interoperability reduces the vendor lock-in concerns that organizations evaluating MBSE adoption frequently cite. For enterprise IT contexts, the textual notation and API standardization lower adoption barriers by aligning MBSE practices more closely with the text-based configuration and automation workflows that IT

professionals employ daily.

The MBSE tool ecosystem encompasses both commercial and open source platforms. Commercial tools, including Dassault Systemes Cameo Systems Modeler and IBM Engineering Systems Design Rhapsody, provide comprehensive modeling environments with enterprise support, training resources, and integration capabilities developed through decades of aerospace and defense application. Open source alternatives have emerged primarily through the Eclipse Foundation's modeling ecosystem. These include Eclipse Papyrus for UML and SysML modeling (Eclipse Foundation, 2024d); Capella's Arcadia-based environment, which Thales developed and organizations have deployed across a variety of industrial contexts (Eclipse Foundation, 2024a; Roques, 2016); and the web-based SysON implementation of the emerging SysML v2 specification (Obeo, 2025). These open source tools reduce economic barriers to MBSE adoption. However, their documentation, community resources, and application examples focus predominantly on traditional systems engineering domains. Organizations seeking to apply MBSE tools for enterprise IT or Information Assurance purposes must adapt without domain-specific guidance, an adaptation that remains unexplored in the academic literature.

Architecture frameworks provide the structural foundation for organizing MBSE implementations. Within MBSE practice, architecture frameworks define the viewpoints, views, and model elements that architects employ to represent systems. UAF, discussed in detail in a subsequent subsection, provides the most comprehensive architecture framework for MBSE implementations spanning both defense and commercial domains. Its multi-viewpoint approach aligns naturally with the needs of organizations managing information systems that must satisfy both Information Assurance requirements and IT service delivery objectives. Architects can represent security controls, operational processes, service dependencies, and resource allocations within a single integrated model environment.

For information systems, MBSE principles enable organizations to create formal

models of their IT infrastructure, security architectures, and service delivery processes. These models capture the static configuration of systems and, beyond it, the dynamic relationships between components, the flow of information through the enterprise, and the dependencies that affect both security postures and service delivery. Model-based approaches provide enhanced visibility into system complexity, enable automated analysis of security implications for proposed changes, and support more effective planning for IT service delivery requirements. Integrating MBSE with established frameworks such as the NIST RMF and ITIL enables organizations to maintain living models that reflect both security control implementations and service configuration states. However, achieving this integration requires modeling language extensions and domain-specific profiles that researchers have yet to develop or validate for enterprise IT contexts. SysML provides excellent support for modeling physical systems with requirements, behaviors, and structures, but modeling IT services, network configurations, and security controls requires adaptation or extension that practitioners must develop independently. The absence of standardized approaches for modeling enterprise IT in SysML is a technical barrier to adoption that accompanies the awareness and perception barriers this research investigates.

Digital Threads

Digital threads constitute authoritative traceability: the verified, bidirectional connections between requirements, design elements, implementation artifacts, and validation activities that persist throughout a system's lifecycle (Singh & Willcox, 2018). The term describes the connections that link authoritative sources—including Model-Based Systems Engineering models, requirements management systems, configuration management databases, and Product Lifecycle Management repositories—into a unified, navigable body of system information. Digital threads ensure that organizations can track how requirements flow through the development and implementation process, identify which system components implement specific capabilities, and verify that implemented solutions satisfy intended requirements. Unlike traditional documentation approaches,

digital threads are designed to maintain verified relationships that remain current as systems evolve (Zhang et al., 2024).

Authoritative traceability merits close attention because it addresses a fundamental limitation of current Information Assurance and IT service management practice. Traditional traceability attempts to maintain connections through manual cross-references, requirements matrices, and documentation linkages. These manual traceability mechanisms require constant maintenance, degrade as systems evolve, and provide no automated verification of consistency. A requirements traceability matrix that maps security controls to implementation artifacts represents a snapshot of intended relationships at the time of creation. Every subsequent change to either controls or implementations creates potential inconsistency that manual processes may not detect. Digital threads establish traceability through model relationships that can update automatically as models change. Queries against digital thread repositories return current rather than historical information. Impact analyses traverse digital thread connections to identify affected components throughout the system architecture. Maintaining currency automatically distinguishes digital threads from the manual traceability approaches that current compliance practices employ.

The Department of Defense's 2018 Digital Engineering Strategy positions the digital thread as essential infrastructure for achieving the authoritative source of truth: the single, trusted baseline of system information from which all stakeholders operate (Department of War, 2018). As a concept, the authoritative source of truth addresses a persistent challenge in both defense acquisition and enterprise IT management: when multiple documentation artifacts describe the same system, discrepancies inevitably emerge, and no definitive mechanism exists for determining which artifact accurately reflects operational reality. Digital threads resolve this challenge by establishing verified connections between all system artifacts, ensuring that changes propagate consistently and that queries return authoritative rather than potentially obsolete information. The Systems Engineering Research Center (SERC) has produced research supporting digital thread implementation,

including the Enterprise System-of-Systems Model for Digital Thread Enabled Acquisition, which establishes architectural patterns for maintaining traceability across complex, multi-system environments (Systems Engineering Research Center, 2018). DoDI 5000.97 codifies these concepts into mandatory requirements, directing defense programs to maintain digital thread capabilities throughout the acquisition lifecycle (Department of War, 2023).

For Information Assurance practice, digital threads address gaps in current RMF implementation. The RMF requires organizations to select security controls, implement those controls, and assess their effectiveness throughout the system lifecycle. Digital threads enable organizations to trace security requirements from categorization decisions through control selection, implementation, and assessment activities, so that policy documents connect to technical configurations, and those configurations connect in turn to assessment evidence, all within a single authoritative chain. This traceability supports the continuous monitoring phase of the RMF by maintaining verifiable connections between security requirements, implemented controls, and compliance artifacts. When a security control implementation changes, digital thread connections enable automated identification of affected requirements, dependent controls, and compliance documentation requiring update. When auditors require evidence of control implementation, digital threads provide navigable paths from requirements through implementation to assessment results without manual evidence compilation.

Within IT service management contexts, digital threads align with ITIL configuration management and change management practices. Organizations can trace service delivery requirements to underlying infrastructure components and configuration items, connecting the CMDB to as-built system documentation and operational baselines. This capability supports more accurate impact assessment for changes and more effective root cause analysis for service disruptions. When a proposed change affects a configuration item, digital thread connections reveal which services depend on that item, which security

controls protect it, and which compliance requirements govern it. Maintaining current, verified traceability relationships through digital threads reduces the time and effort required for compliance audits while improving the accuracy of both security assessments and service impact analyses.

Implementing digital threads in practice requires integration infrastructure that connects diverse tools and repositories. In defense contexts, this integration has developed over years of investment and standardization. Enterprise IT environments employ different tool ecosystems—among them IT service management platforms like ServiceNow and BMC, security information and event management systems, cloud management consoles, and configuration automation tools—none of which were designed with digital thread connectivity in mind. Establishing digital thread capabilities in enterprise IT contexts therefore requires integration approaches that bridge systems engineering tools and IT management platforms. This integration challenge is a practical barrier that accompanies the conceptual transfer of digital thread principles from defense to enterprise IT contexts.

Digital Twin Technology

Digital twin technology creates virtual replicas of physical or logical systems that maintain synchronization with their real-world counterparts through continuous data exchange (Shao, 2021). These virtual representations enable organizations to simulate system behavior, analyze potential changes, predict future states, and optimize performance without disrupting operational systems (Madni & Sievers, 2019). Digital twins combine real-time operational data with analytical models to provide dynamic, predictive capabilities that extend far beyond traditional monitoring and simulation approaches (Khan et al., 2022).

As a concept, the digital twin originated in Product Lifecycle Management research. Grieves traces the evolution of digital twins from their conceptual origins through contemporary applications, positioning digital twins as the integration of physical and virtual spaces that enables analysis, optimization, and prediction across system

lifecycles (Grieves, 2014). A defining distinction between digital twins and traditional simulation lies in synchronization. Where traditional simulations model hypothetical or designed system states, digital twins maintain continuous data exchange with their physical or logical counterparts. This exchange enables operational decision-making based on current rather than documented system states. Such synchronization directly addresses the documentation-reality gap that undermines Information Assurance and IT service management practices. Static models represent intended or designed system states; digital twins represent current operational states, updated continuously through integration with operational data sources. This distinction carries substantial implications for organizations that must make security and operational decisions based on accurate understanding of system configurations.

Standardization efforts have established frameworks for digital twin implementation that reduce interoperability concerns and provide structured guidance. The International Organization for Standardization published ISO 23247, the Digital Twin Framework for Manufacturing, defining a four-part reference architecture for observable elements, twin entities, data and device infrastructure, and cross-domain integration (International Organization for Standardization, 2021). Shao et al. analyzed the standard in detail (Shao et al., 2023). While developed for manufacturing, the reference architecture establishes patterns of data synchronization, entity representation, and cross-domain integration that inform digital twin implementations in other domains. IEC 62832 complements ISO 23247 by addressing the industrial-process measurement, control, and automation aspects of digital twin frameworks (Shao, 2021). In parallel, the Internet Engineering Task Force has published a draft reference architecture specifically addressing network infrastructure digital twins, representing direct engagement with the IT infrastructure domain (IETF Network Management Research Group, 2024). IETF engagement signals that the networking community has recognized digital twin applicability for IT infrastructure, though the published architecture remains at draft stage and has not yet achieved the

maturity of manufacturing-focused standards.

NIST has engaged with digital twin technology through NIST Internal Report 8356, finalized in February 2025, which addresses cybersecurity challenges and trust considerations for digital twin implementations (Voas et al., 2025). Notably, this publication examines security considerations *for* systems employing digital twins rather than digital twin applications *to* Information Assurance, a distinction that reveals the current orientation of institutional research. NIST's Framework for Cyber-Physical Systems similarly addresses digital integration challenges at the intersection of physical and computational systems (National Institute of Standards and Technology, 2017). Together, these publications establish that authoritative institutions recognize digital twin relevance to security domains while treating digital twins primarily as systems requiring security rather than as tools for enhancing security operations. Reorienting from securing digital twins to employing them for security purposes represents a conceptual shift that the academic literature has begun but not yet completed.

Open source frameworks have emerged to reduce economic and technical barriers to digital twin adoption. The Digital Twin Consortium has established an open source collaboration initiative providing frameworks and reference implementations (Digital Twin Consortium, 2024), while the Eclipse Foundation offers Eclipse Ditto for IoT digital twins (Eclipse Foundation, 2024c) and Eclipse BaSyx for Asset Administration Shell industrial twins (Eclipse Foundation, 2024b). Gil et al. surveyed fourteen open source frameworks against ISO 23247 criteria and found substantial variation in maturity, documentation quality, and community support (Gil et al., 2024). Autiosalo et al. introduced Twinbase, an open source server for the Digital Twin Web concept, enabling organizations to publish and discover digital twin descriptions through standardized interfaces (Autiosalo et al., 2021). These open source options provide accessible entry points for organizations exploring digital twin capabilities, though their documentation and application examples focus on manufacturing and IoT rather than enterprise IT or

Information Assurance contexts.

Digital twins support IT service delivery in ways that align with ITIL service design and service transition practices. Organizations can employ digital twins for capacity planning, change impact analysis, and service optimization: teams test changes and analyze performance implications before implementing modifications in production environments. Simulating proposed changes in a synchronized virtual environment reduces the risk of service disruptions while supporting more rapid and confident change implementation. When combined with digital thread traceability, digital twins enable organizations to assess the security and operational implications of proposed changes simultaneously. Current practices, which typically assess security and operational impacts through separate processes, cannot provide this integrated analysis capability.

Digital twins offer comparable advantages for Information Assurance, supporting security control validation, risk assessment, and incident response. Organizations can create digital twins of their information systems to simulate security scenarios, test control effectiveness, and analyze attack vectors in environments isolated from production operations. These virtual representations enable security teams to evaluate the impact of proposed security controls, assess the effectiveness of defensive measures, and predict system behavior under various threat scenarios. Digital twins support RMF assessment activities by enabling organizations to test security configurations and validate control implementations before deployment to production environments. Simulating adversary behavior against synchronized representations of actual system configurations enables security testing that reflects operational reality rather than documented assumptions.

Product Lifecycle Management

Product Lifecycle Management provides frameworks and toolsets for managing information, processes, and resources throughout a system's entire lifecycle from initial conception through retirement (Ross et al., 2022). PLM integrates data from diverse sources, maintains configuration baselines, manages change processes, and ensures that

stakeholders access current, accurate information about system states and changes. This integrated approach to lifecycle management extends beyond simple version control to encompass configuration management, change coordination, release management, and information governance.

PLM originated in manufacturing and product development, where organizations confronted the challenge of managing complex products involving thousands of components, multiple suppliers, extended development timelines, and rigorous quality requirements. Manufacturing organizations developed PLM practices to address challenges that bear structural similarity to those confronting enterprise IT management: maintaining accurate documentation of complex configurations, coordinating changes across interdependent components, ensuring consistency between design documentation and as-built products, and managing compliance with regulatory requirements throughout product lifecycles. Over decades of industrial application, the commercial PLM ecosystem has matured, producing platforms that manage engineering artifacts across distributed supply chains. Open source PLM alternatives have emerged to reduce cost barriers, though Campos et al. noted that industrial open source PLM solutions vary considerably in maturity and require careful evaluation against organizational requirements (Campos et al., 2014).

Applying PLM principles to information systems represents a conceptual extension from these manufacturing origins. Physical products have lifecycles that parallel information system lifecycles in important ways: conception, design, development, deployment, operation, maintenance, and retirement. PLM practices developed for managing physical product lifecycles address challenges of configuration management, change coordination, and baseline maintenance that information system managers confront daily. The question is whether PLM tools and methodologies can be adapted effectively for information system contexts, where system components are logical rather than physical, change cycles occur in hours rather than months, and deployment involves software configurations rather than manufacturing processes.

PLM capabilities align closely with ITIL service lifecycle management concepts. Organizations can implement PLM frameworks to support ITIL configuration management by maintaining authoritative configuration baselines and managing configuration item relationships. PLM change coordination capabilities enhance ITIL change management by providing improved visibility into change impacts across service dependencies. Maintaining integrated views of system configurations, security controls, and service delivery components reduces inconsistencies between security and IT operations teams, improves change coordination, and supports more effective compliance management across the system lifecycle. PLM's emphasis on managing information across organizational boundaries addresses the silo problem that fragments visibility in current IT environments: security teams, operations teams, and application teams can operate from a shared, authoritative information base rather than maintaining separate, potentially inconsistent documentation.

PLM principles apply equally within Information Assurance, addressing challenges in managing complex IT infrastructures and security postures throughout the system lifecycle. The RMF explicitly recognizes the importance of lifecycle management, requiring organizations to maintain security controls and documentation throughout system operation and into decommissioning. PLM approaches support these requirements by managing security control baselines, coordinating changes across interconnected systems, maintaining configuration integrity, and ensuring that security teams operate from consistent, current information throughout the authorization boundary. PLM configuration baseline capabilities directly address the requirement in NIST SP 800-53 control CM-2 for documented, maintained system baselines (Ross et al., 2020). PLM change coordination capabilities support control CM-3 requirements for configuration change control. PLM access governance capabilities align with CM-5 requirements for access restrictions governing system changes. These correspondences between PLM capabilities and specific compliance requirements suggest that PLM practices could address compliance obligations

that organizations currently struggle to fulfill through manual processes.

Institutional Endorsement and Strategic Direction

That Digital Engineering has matured from experimental practice to institutional mandate provides important context for evaluating its potential applicability beyond defense and aerospace. The breadth and depth of institutional endorsement establish Digital Engineering as a mature discipline validated by the organizations most experienced in managing complex, compliance-intensive systems. Understanding this institutional foundation informs assessment of whether Digital Engineering merits investigation for enterprise IT and Information Assurance applications.

The Digital Engineering Strategy, published in June 2018 under the institution's then-current name of Department of Defense, established the formal vision for transforming defense acquisition through Digital Engineering practices (Department of War, 2018). Five strategic goals collectively articulate the transformation Digital Engineering envisions: formalize the development, integration, and use of models to inform enterprise and program decision-making; provide an authoritative source of truth by establishing a digital thread as the means of delivering the technical baseline; incorporate technological innovation to improve the engineering practice; establish a Digital Engineering ecosystem to provide the infrastructure and environments for performing Digital Engineering activities; and transform the culture and workforce to adopt and support Digital Engineering across the lifecycle. These goals extend beyond technology adoption to encompass organizational transformation, workforce development, and cultural change. Any transfer of Digital Engineering to enterprise IT contexts must address these same dimensions.

DoDI 5000.97, issued in December 2023, codified these strategic aspirations into mandatory requirements for defense programs (Department of War, 2023). It mandates that programs use digital artifacts as the authoritative source of system information and maintain digital thread capabilities throughout the acquisition lifecycle. Moving from strategic vision to regulatory mandate marks a major institutional commitment: defense

programs must now implement Digital Engineering practices rather than merely consider them. Complementing this mandate, the Systems Engineering Guidebook provides detailed implementation guidance for meeting these requirements, establishing processes, methods, and tool expectations for Digital Engineering practice within defense acquisition (Office of the Under Secretary of Defense for Research and Engineering, 2022). A clear progression from strategy document to formal instruction to implementation guidebook demonstrates institutional maturation that parallels the compliance framework progression familiar in Information Assurance contexts, which moves from policy through standards through implementation guidance.

NASA has pursued a parallel path toward Digital Engineering adoption, lending independent validation to the approach. NASA-HDBK-1004, the NASA Digital Engineering Acquisition Framework Handbook, establishes guidance for incorporating Digital Engineering practices into NASA programs (National Aeronautics and Space Administration, 2020). A companion document, the NASA Future Model-Based Systems Engineering Vision and Strategy Bridge, articulates NASA's long-term vision for MBSE adoption across the agency's portfolio of complex, mission-critical programs (National Aeronautics and Space Administration, 2021). That NASA independently developed Digital Engineering guidance carries particular significance. Both the Department of War and NASA confronted similar challenges: complex systems demanding comprehensive documentation, stringent compliance requirements, and mission-critical operations tolerating no ambiguity in system understanding. Both organizations converged on Digital Engineering as the solution. This independent convergence strengthens the argument that Digital Engineering addresses fundamental challenges in managing complex, compliance-intensive systems rather than representing a domain-specific solution applicable only to defense acquisition.

INCOSE has positioned Digital Engineering as the future direction of the systems engineering discipline. Its Systems Engineering Vision 2035 establishes a roadmap for

transforming systems engineering practice through Digital Engineering, model-based methods, and integrated digital environments (International Council on Systems Engineering, 2021). Complementing this vision, the INCOSE Systems Engineering Handbook, Fifth Edition, provides comprehensive guidance for implementing systems engineering practices including MBSE, reflecting the discipline's current state of practice and serving as the primary professional reference for systems engineers worldwide (International Council on Systems Engineering, 2023). INCOSE's Digital Engineering Information Exchange Working Group specifically addresses the interoperability and integration challenges that arise when organizations attempt to exchange Digital Engineering artifacts across organizational and tool boundaries (International Council on Systems Engineering (INCOSE), n.d.). This working group's existence acknowledges that Digital Engineering implementation requires addressing practical interoperability challenges, and those challenges would intensify when extending Digital Engineering beyond its established systems engineering context into enterprise IT domains.

The Systems Engineering Body of Knowledge (SEBoK) is jointly managed by INCOSE, the IEEE Systems Council, and the Stevens Institute Systems Engineering Research Center. It provides the globally recognized authoritative reference defining Digital Engineering's relationship to MBSE, digital threads, and authoritative sources of truth within the ISO/IEC/IEEE 15288:2023 systems engineering lifecycle framework (SEBoK Authors, 2025). SEBoK establishes the conceptual vocabulary and architectural relationships that the systems engineering community employs when discussing Digital Engineering, providing the definitional foundation on which this dissertation's investigation builds.

SERC has produced a body of research publications supporting Digital Engineering implementation and assessment. Its Digital Engineering Competency Framework defines 962 Knowledge, Skills, Abilities, and Behaviors organized by proficiency levels, providing

the most comprehensive specification of what professionals require to practice Digital Engineering effectively (Hutchinson et al., 2021). A companion Digital Engineering Metrics framework establishes measurement approaches for assessing adoption progress and benefits realization, addressing the measurement challenge that the literature identifies as a persistent barrier to evidence-based adoption decisions (Hutchison et al., 2020). Additionally, the SERC Systems Engineering Modernization report examines the integration of Digital Engineering with Mission Engineering, Agile, and DevOps practices, establishing connections between Digital Engineering and the development methodologies prevalent in enterprise IT contexts (Systems Engineering Research Center, 2022). Collectively, these SERC publications provide a research infrastructure of competency definitions, metrics, and integration frameworks that would support Digital Engineering adoption in new domains, including enterprise IT and Information Assurance.

NIST's contributions span multiple relevant domains. The Framework for Cyber-Physical Systems addresses integration challenges at the intersection of physical and computational systems (National Institute of Standards and Technology, 2017). NIST's systems security engineering publications reference model-based approaches and traceability requirements that align with Digital Engineering principles. These publications comprise Special Publication 800-160 Volume 1 Revision 1 on Engineering Trustworthy Secure Systems (Ross et al., 2022) and Volume 2 Revision 1 on Developing Cyber-Resilient Systems (Ross et al., 2021b). NIST Internal Report 8356 on digital twin security considerations (Voas et al., 2025) and the Open Security Controls Assessment Language initiative (National Institute of Standards and Technology, 2023) further demonstrate NIST engagement with concepts that Digital Engineering could integrate. Taken together, these publications establish that the institution responsible for federal cybersecurity standards recognizes the relevance of model-based approaches, traceability, and digital integration to security engineering. NIST has not yet explicitly connected these concepts under the Digital Engineering framework, however, nor provided specific guidance for their

application in enterprise IT contexts. This dissertation addresses one such research gap: the distance between NIST's recognition of relevant principles and the absence of specific Digital Engineering implementation guidance for enterprise IT.

Enterprise Architecture Frameworks

Enterprise architecture frameworks provide the structural foundation on which Digital Engineering implementations rest. These frameworks define how organizations represent, analyze, and manage the complex relationships among business processes, information flows, application systems, and technology infrastructure. Understanding the enterprise architecture framework ecosystem is essential for evaluating Digital Engineering's potential in enterprise IT contexts because the frameworks determine what can be modeled, how models are organized, and what analytical capabilities models support.

The Unified Architecture Framework, now codified as ISO/IEC 19540 through the Object Management Group (Object Management Group, 2022b), is a major consolidation of enterprise architecture standards. UAF merges three military architecture frameworks: the Department of Defense Architecture Framework (DoDAF) (Department of War, 2009), the UK Ministry of Defence Architecture Framework (MODAF), and the NATO Architecture Framework (NATO, 2018). This consolidation proceeded from the explicit recognition that the architectural concepts developed for military system-of-systems management possess broad applicability beyond defense contexts. The Object Management Group determined that approximately 90% of concepts captured in military architecture frameworks prove equally applicable in commercial domains (Object Management Group, 2024), motivating the development of a unified framework that serves both defense and commercial organizations without requiring domain-specific adaptations.

UAF employs a grid-based structure defining seventy-one view specifications organized through two complementary parts: the UAF Domain Metamodel (DMM), which defines the underlying concepts, relationships, and constraints; and the UAF Modeling Language (ML), which provides the concrete notation for expressing architecture

descriptions (Object Management Group, 2022a). Grid rows represent stakeholder domains, namely Strategic, Operational, Services, Personnel, and Resources, while columns represent architecture aspects including taxonomy, structure, connectivity, processes, states, interaction, constraints, and traceability. Together, this grid organization enables architects to represent any complex system-of-systems from multiple stakeholder perspectives simultaneously, with formal relationships connecting views across the grid to maintain consistency. The traceability column explicitly addresses the requirement for maintaining verified connections across architectural perspectives, a requirement that aligns directly with the digital thread concept and with the compliance traceability obligations that NIST SP 800-53 imposes.

Evolution from domain-specific frameworks to the unified standard proceeded through several stages. DoDAF 2.0 established the foundational metamodel for defense architecture description (Department of War, 2009), MODAF developed parallel conventions for UK defense, and the NATO Architecture Framework provided capabilities for alliance-wide interoperability (NATO, 2018). Earlier evaluation identified limitations in these individual frameworks: the National Defense Industrial Association found that DoDAF required augmentation to support systems engineering activities (National Defense Industrial Association Systems Engineering Division, 2011), and Giachetti identified specific gaps between DoDAF's metamodel and Model-Based Systems Engineering practice (Giachetti, 2015). The Unified Profile for DoDAF and MODAF (UPDM) was an initial consolidation effort that evolved into UAF under broader OMG stewardship, with ISO/IEC standardization following to establish UAF as a globally recognized standard (Hause et al., 2017).

Comparative research validates UAF's position relative to alternative enterprise architecture frameworks. Bankauskaite evaluated enterprise architecture frameworks using weighted criteria encompassing completeness, maturity, standardization, and tool support, finding that UAF achieved the highest overall rating among the frameworks evaluated,

surpassing TOGAF (The Open Group, 2018), DoDAF, MODAF, NAF, and FEAF (Bankauskaite, 2019). This comparative advantage derives from UAF's comprehensive metamodel, its ISO international standardization, and its explicit design for both defense and commercial applications. Research by Eichmann et al. demonstrated UAF application to system-of-systems development (Eichmann et al., 2019). In a military context, Liu et al. applied UAF to system-of-systems design using top-down MBSE methodology (Liu et al., 2023). Building on UAF-based MBSE, Torkjazi et al. integrated autonomy into system-of-systems architectures (Torkjazi et al., 2024), while Abhaya established methods for building system-of-systems models (Abhaya, 2021). Collectively, these studies demonstrate UAF's growing adoption and validation across diverse system-of-systems contexts, though applications to enterprise IT infrastructure and Information Assurance remain absent from the literature identified in this review.

The complementary relationship between UAF and the commercially prevalent Open Group Architecture Framework (TOGAF) reduces adoption barriers for organizations already invested in enterprise architecture practice. TOGAF provides a comprehensive architecture development method and enterprise architecture governance framework widely adopted in commercial IT contexts (The Open Group, 2018). A joint white paper between The Open Group and MITRE Corporation established that UAF and TOGAF address different but complementary aspects of enterprise architecture: TOGAF provides the methodology and governance framework for architecture development, while UAF provides the architecture description specification for expressing what has been architected (The Open Group & MITRE Corporation, 2013). Organizations can retain TOGAF methodology while adopting UAF for architecture description, enabling incremental adoption rather than wholesale replacement of established practices. This compatibility matters especially for enterprise IT organizations that have invested in TOGAF-based architecture programs: it suggests that UAF adoption for model-based architecture description need not require abandonment of existing architecture governance investments.

UAF's foundation on UML and SysML profiles enables model-based documentation approaches that address the accuracy and currency challenges plaguing traditional enterprise architecture implementations (The Aerospace Corporation, 2023). Because UAF architecture descriptions are expressed as formal models rather than static documents, they inherit the automated consistency checking, impact analysis, and traceability capabilities that model-based approaches provide. A model-based foundation connects enterprise architecture practice to the broader Digital Engineering ecosystem: UAF models can participate in digital threads, maintain synchronization with operational systems through digital twin mechanisms, and be managed through PLM lifecycle processes. UAF has been adopted by the Department of War (Office of the Under Secretary of Defense for Research and Engineering, 2022), by NATO (NATO, 2018), and by the UK Ministry of Defence, and it carries ISO international standardization as well. Together these endorsements establish UAF as the authoritative consolidating framework for organizations that need architecture capabilities across multiple domains. For enterprise IT and Information Assurance applications, UAF provides structured approaches to documenting systems, relationships, and security requirements that align with both NIST and ITIL expectations. Understanding this enterprise architecture foundation provides essential context for evaluating how Digital Engineering extends architectural approaches from static documentation toward living, model-based representations that maintain currency with operational reality.

Gaps in Current Practice

Despite advances in Information Assurance methodologies and IT service management frameworks, organizations continue to encounter challenges that limit their effectiveness in protecting information assets and delivering reliable services. The NIST Risk Management Framework and ITIL provide structured approaches to Information Assurance and IT service management respectively, yet implementation challenges persist across both domains. Digital Engineering practices, as the 2018 Digital Engineering

Strategy directs for defense programs, are designed to address structurally analogous challenges: maintaining authoritative documentation, ensuring configuration visibility, and providing traceability across complex systems (Department of War, 2018). Whether these capabilities translate to enterprise IT and Information Assurance contexts, and whether professionals in those domains recognize the potential relevance, remains uninvestigated. Examining the specific gaps that persist in current practice illuminates the structural parallels that motivate this investigation.

These gaps persist despite framework maturity and organizational investment, echoing the pattern documented earlier in this chapter: these outcomes are consistent with limitations inherent in document-centric methodologies rather than with execution failures alone.

Information Assurance Challenges

Organizations implementing the NIST Risk Management Framework struggle to maintain visibility into their security postures across complex, distributed information systems. The RMF continuous monitoring phase requires organizations to maintain ongoing awareness of security control effectiveness and system security state. Security teams often lack accurate, current understanding of system configurations, security control implementations, and the dependencies that affect security effectiveness. Visibility gaps manifest throughout the RMF lifecycle: organizations find it difficult to track security dependencies effectively, leaving vulnerabilities unidentified when teams implement changes. When teams cannot maintain accurate documentation of system interconnections and data flows, particularly in environments with rapid deployment cycles, the risk assessment and incident response capabilities that the RMF demands break down.

Understanding cascading impacts poses a further challenge. When incidents occur or defenders discover vulnerabilities, they must rapidly assess which systems are affected, what data are at risk, and how compromise of one system might enable access to interconnected systems. Such assessment requires understanding of system dependencies

that current documentation approaches cannot maintain. When defenders cannot trace first, second, and third order impacts, incident response becomes a broad search effort rather than a precision operation, consuming time and resources while adversaries retain the initiative.

Security control implementation presents distinct challenges in modern enterprise environments characterized by hybrid cloud deployments, distributed architectures, and frequent changes. Organizations must implement and maintain consistent security controls from NIST SP 800-53 across diverse platforms while supporting continuous deployment practices and rapid update cycles (Ross et al., 2020). Traditional security configuration management approaches fail to scale effectively in these dynamic environments, leading to inconsistent security postures and compliance gaps. The challenge compounds when organizations attempt to validate control effectiveness across interconnected systems where authorization boundaries grow increasingly complex to define and maintain.

IT Service Management Challenges

IT service management faces parallel challenges in maintaining accurate system documentation. As noted earlier in this chapter, most CMDB implementations fail to deliver intended value, leaving organizations without authoritative sources for configuration information (Gartner, 2019). Manual configuration tracking cannot keep pace with the rate of change in modern IT environments. Shadow IT creates blind spots where undocumented systems introduce unknown dependencies and security risks. Change management processes suffer when impact assessments rely on incomplete or inaccurate dependency information.

The economic dimensions of these failures merit examination. Organizations commit budget, staff time, and tooling to CMDB implementations, documentation efforts, and change management processes. When these investments fail to deliver intended value, organizations face difficult choices: invest additional resources attempting to improve failing approaches, accept degraded capabilities and increased risk, or seek alternative approaches that address root causes rather than symptoms. Digital Engineering is a

candidate alternative approach whose reported benefits in defense and aerospace contexts suggest potential applicability to enterprise IT environments.

The convergence of these challenges creates a compounding effect where neither Information Assurance nor IT service management can achieve their objectives independently. Security teams cannot effectively assess risks without accurate understanding of IT infrastructure. IT teams cannot effectively manage changes without understanding security implications. Both domains require the visibility and accurate documentation that current practices fail to provide.

The Documentation-Reality Gap

The persistent gap between documentation and operational reality, documented in the Challenges in Current Practice discussion earlier in this chapter, forms the common thread connecting failures across both domains. When documentation diverges from reality, security assessments measure fiction rather than fact, change impact analyses miss undocumented dependencies, incident responders lose time reconciling records with operational systems, and compliance auditors cannot verify that documented controls exist in practice. Digital Engineering offers capabilities designed to address this gap through authoritative sources of truth, continuous synchronization between models and operational systems, and automated consistency verification. The 2018 Digital Engineering Strategy directs defense programs to establish these capabilities in complex, compliance-intensive environments (Department of War, 2018). The question this research investigates is whether IT and Information Assurance professionals recognize the potential value of these capabilities for their work, a prerequisite for determining whether Digital Engineering practices developed in defense contexts merit investigation for broader enterprise application.

Research Questions

Two findings now stand in tension. Defense and aerospace have committed to Digital Engineering at institutional scale to manage the very complexity that overwhelms

enterprise IT and Information Assurance, and the limited measured evidence available is encouraging. Yet no study has measured whether the professionals who manage that complexity recognize Digital Engineering as a potential remedy. That unmeasured question defines this study's focus. Building on the challenges documented in current practice and the capabilities offered by Digital Engineering, this investigation pursues three questions:

1. To what extent are Information Technology and Information Assurance professionals aware of Digital Engineering capabilities, including Model-Based Systems Engineering, digital threads, digital twin technologies, and Product Lifecycle Management principles?

2. To what extent do Information Technology and Information Assurance professionals perceive Digital Engineering capabilities as potentially valuable or important for their work in Information Assurance, security compliance, and IT service delivery?

3. To what extent do Information Technology and Information Assurance professionals believe that Digital Engineering practices could help them in performing their jobs, meeting compliance requirements, or enhancing organizational capabilities in Information Assurance and IT service delivery?

Research Question 3 is examined through the operational and compliance-oriented items distributed across Sections 3 through 5 of the survey instrument, grouped into IT-operational, security-operational, and explicitly compliance-framed sets, so that endorsement of Digital Engineering's job-performance and capability value can be read separately from endorsement of its compliance value.

These questions are posed of the profession at large, and the study answers them for the population its recruitment actually reached: a non-probability sample drawn through defense- and aerospace-weighted professional channels and weighted toward earlier-career respondents. Chapter 3 specifies that frame and Chapter 5 bounds the conclusions to it.

Awareness and Perception as the Necessary First Step

These research questions focus on professional awareness and perceptions as the groundwork for subsequent inquiry. Establishing awareness levels and perceived value is an essential first step. It must precede any investigation of practical implementation approaches, organizational adoption strategies, or empirical validation of Digital Engineering benefits in Information Assurance and IT service management contexts. Preliminary evidence from adjacent domains suggests that such investigation is urgently needed: Henderson et al. found that approximately 22% of systems engineering professionals, all practitioners within the discipline that developed MBSE, could not articulate a clear definition of MBSE (Henderson et al., 2024). If awareness deficits persist within the originating discipline, awareness among professionals in Information Assurance and IT Service Management, disciplines that have never systematically encountered these methodologies, requires empirical measurement rather than assumption.

Distinguishing General Value from Personal Applicability

The distinction between Research Question 2 and Research Question 3 requires clarification, because both address professional perceptions of Digital Engineering value. Research Question 2 measures whether professionals recognize Digital Engineering capabilities as abstractly valuable or important for their professional domains, an assessment of general relevance that does not require the respondent to evaluate specific operational impact within their own work context. Research Question 3 probes a more personal and practical construct: whether professionals believe Digital Engineering could tangibly help them perform their jobs, meet the compliance requirements they personally face, or enhance the specific organizational capabilities they are responsible for delivering. Technology adoption literature distinguishes between recognizing that a capability has general value and believing that the same capability would improve one's own work. Professionals may acknowledge that Digital Engineering offers theoretical value for their domain while remaining skeptical about its practical applicability to their specific

organizational context. This distinction carries substantial implications for adoption strategy: a professional community that perceives general value but doubts personal applicability requires different intervention than one that perceives both.

Whether the two are empirically separable in the responses, or instead express a single perceived-value appraisal at different levels of specificity, is itself an empirical question. The analysis does not construct separate composites for general and personal value. Chapter 4's construct-validity results collapse the IT-value and security-value dimensions into a single perceived-value construct. That outcome indicates the two questions address one perceived-value appraisal at different levels of specificity, and Chapters 4 and 5 report them on that basis.

Theoretical Framework and Research Propositions

The investigation draws on three established streams of technology-adoption theory to connect its constructs. The Technology Acceptance Model holds that behavioral intention to adopt a technology is driven by perceived usefulness and perceived ease of use (Davis, 1989). The Unified Theory of Acceptance and Use of Technology generalizes this into performance expectancy, effort expectancy, social influence, and facilitating conditions, with individual characteristics acting as moderators (Venkatesh et al., 2003). Rogers' Diffusion of Innovations situates adoption within the perceived attributes of an innovation—among them relative advantage, compatibility, and observability—and emphasizes that perceptions rather than objective properties govern uptake (E. M. Rogers, 2003). Call and Herber mapped this framework directly onto Model-Based Systems Engineering adoption, establishing its applicability to the present domain (Call & Herber, 2022).

These streams motivate the conceptual model in Figure 1. The model positions awareness of the discipline and recognition of its capabilities, the focus of Research Question 1, as antecedents of perceived value, the focus of Research Questions 2 and 3. This construct corresponds to perceived usefulness in the Technology Acceptance Model

and to relative advantage in the diffusion framework. Perceived value in turn informs adoption intent, captured here through interest in learning more and willingness to recommend organizational exploration. Professional field and experience enter the model as moderators, consistent with the individual-difference moderators of the Unified Theory, while self-reported familiarity is treated as a covariate of perceived value. The survey instrument operationalizes the perceived-value slice of these frameworks together with two coarse indicators of adoption intent. It does not measure effort expectancy, facilitating conditions, or social influence, so the study tests the value-perception portion of the adoption models rather than the models in full; the remaining constructs are left to the future work outlined in Chapter 5.

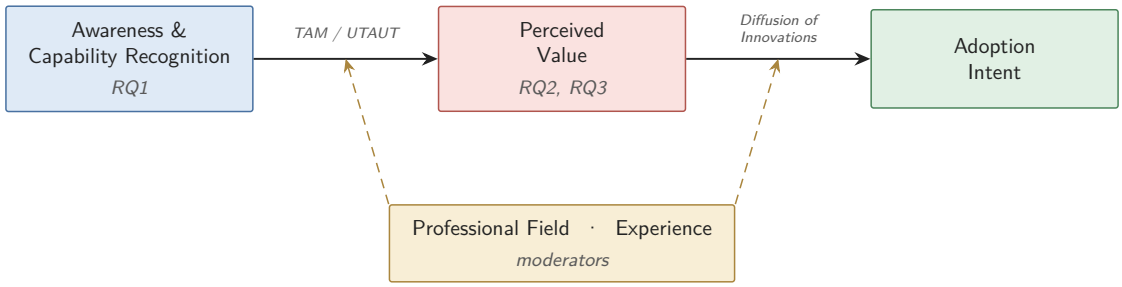


Figure 1: Conceptual model linking awareness and capability recognition to perceived value and adoption intent, with professional field and experience as moderators.

Because the design is cross-sectional and the application domain is largely unstudied, the research advances propositions rather than formal causal hypotheses. Four directional expectations follow from the framework and the gaps documented above:

1. **P1 (Low awareness).** Awareness of and recent exposure to Digital Engineering are low among IT and Information Assurance professionals, who sit outside the systems engineering discipline in which these methods developed.

2. **P2 (Value despite low awareness).** Perceived value is nonetheless positive once capabilities are described in functional terms, because Digital Engineering addresses the documentation, traceability, and visibility problems these professionals routinely

face. Because an instrument that measures perceived value must necessarily describe the capability being evaluated, this proposition carries a demand-characteristics risk; the neutral-framing safeguards and bias diagnostics detailed in Chapters 3 and 4 are designed to test whether the resulting endorsement survives that scrutiny.

3. **P3 (Variation by field and experience).** Perceived value and awareness vary systematically with professional field and experience level rather than being uniform across the community.

4. **P4 (Familiarity and value).** Greater familiarity with Digital Engineering is associated with higher perceived value.

Chapter 4 evaluates each proposition against the survey data, and Chapter 5 returns to them in stating the conclusions. Propositions P1 and P2 correspond primarily to Research Questions 1 and 2, with P2's expectation of functional-value endorsement extending to the applicability beliefs of Research Question 3; P3 and P4 cut across all three questions as expected patterns of variation and association.

Research Scope and Approach

This research examines professional perceptions in four areas: Model-Based Systems Engineering, digital threads, digital twin technology, and Product Lifecycle Management. The investigation assesses awareness and perceived value of Model-Based Systems Engineering approaches for representing information system architectures and security controls. It asks to what extent professionals perceive value in digital threads for maintaining authoritative traceability between security requirements, control implementations, and compliance evidence as required by frameworks such as the NIST RMF. It explores perceptions of digital twin capabilities for security simulation, testing, and IT service modeling. Finally, it investigates how strongly professionals value Product Lifecycle Management principles for managing information system configurations and security control baselines throughout the system lifecycle.

Methodological Approach

The research employs a quantitative survey method to collect data from IT and Information Assurance professionals. Survey methodology enables systematic data collection from a broad population of practitioners, supporting statistical analysis and generalization of findings. The anonymous nature of survey research encourages candid responses about professional knowledge gaps and organizational capabilities. Chapter 3 presents the complete research methodology including survey design, sampling strategy, and analytical approach.

Target Population

The research targets IT and Information Assurance professionals across diverse organizational contexts. This population includes security analysts, compliance officers, IT managers, network administrators, systems administrators, and other professionals who implement and maintain Information Assurance controls or deliver IT services. By surveying professionals across organizational contexts rather than limiting investigation to a single industry or organization, the research enables assessment of awareness and perceived value across the professional community.

This target population intentionally extends beyond the systems engineering professionals who have been the focus of previous MBSE adoption research. While studies by Call et al. and Henderson et al. examined perceptions among systems engineering professionals (Call et al., 2024; Henderson et al., 2024), no comparable research addresses IT and Information Assurance practitioners. This population is a crucial test of Digital Engineering's transferability: if professionals who have never encountered these methodologies nevertheless recognize their potential value for addressing familiar challenges, the case for cross-disciplinary investigation strengthens substantially.

Potential Benefits for Under-Resourced Organizations

The potential transferability of Digital Engineering benefits carries particular significance for small and under-resourced organizations. Many under-resourced

organizations require effective Information Assurance and IT service delivery capabilities. These include Federally Qualified Health Centers and safety-net hospitals serving low-income patients, tribal health and government organizations, rural healthcare providers, school districts and minority-serving institutions in under-resourced communities, public libraries, small and rural municipalities, and the non-profit and social-service entities that address community needs. For these organizations the frameworks already exist; what most sharply constrains them is the scarcity of the specialized labor those frameworks presuppose. Model-Based Systems Engineering has already been applied to rural-healthcare planning in exactly such resource-limited settings, which establishes a concrete precedent for carrying these methods beyond well-funded enterprises (Berg et al., 2023).

Under-resourced organizations must frequently demonstrate compliance with regulatory frameworks, security standards, and funding requirements. Healthcare providers must satisfy HIPAA security requirements. Educational institutions must protect student data under FERPA. Social service organizations must safeguard client information while demonstrating accountability to funding agencies. These compliance obligations demand documented evidence of security controls and system configurations, and producing that documentation consumes scarce staff time and organizational resources.

If Digital Engineering practices can reduce the burden of compliance documentation while improving documentation accuracy, organizations with limited resources could redirect staff effort toward direct service provision rather than documentation administration. Digital threads, through automated traceability, could reduce the manual effort required for compliance audits. MBSE could maintain documentation accuracy through automated synchronization rather than manual updates. PLM configuration management capabilities could, in turn, lower the specialized expertise required to maintain accurate system documentation. Each of these mechanisms substitutes reusable, automatable model structure for scarce specialized labor. To the extent that the

substitution holds, it could place authoritative documentation and demonstrable compliance within reach of organizations that cannot today afford the dedicated staff such work requires; whether that capability also narrows the broader equity gap is the distributional question taken up under Significance.

This research does not presume that Digital Engineering benefits will transfer effectively to resource-constrained organizations. Accordingly, the survey population targets IT and Information Assurance professionals broadly, not exclusively those working in under-resourced settings. However, by establishing baseline awareness and perceived value data across the professional community, this research creates a foundation for subsequent investigation of Digital Engineering applicability in diverse organizational contexts. If professionals perceive value in Digital Engineering capabilities, future research can examine practical implementation approaches suitable for organizations with varying resource levels.

The logical pathway from defense and aerospace demonstration to broader application proceeds through several steps. First, defense and aerospace organizations demonstrate that Digital Engineering delivers measurable benefits for complex systems with stringent compliance requirements. Second, research establishes whether IT and Information Assurance professionals outside these sectors recognize potential value in Digital Engineering capabilities. Third, if perceived value exists, subsequent research can examine implementation approaches, adaptation requirements, and cost-benefit considerations for organizations in different contexts. This research addresses the second step: determining whether professional awareness and perceived value support continued investigation of Digital Engineering for enterprise IT and Information Assurance applications.

Why Perceptions Matter

Investigating professional perceptions requires justification, given the availability of alternative research approaches. Technology adoption research consistently shows that perceived value influences adoption decisions regardless of actual value (Davis, 1989).

Professionals who do not perceive value in a capability will not advocate for its adoption within their organizations. Establishing whether IT and Information Assurance professionals recognize potential value in Digital Engineering capabilities therefore addresses a prerequisite question for successful adoption.

Furthermore, perception research enables assessment of awareness gaps that might impede adoption. If professionals are unaware of Digital Engineering capabilities, education and communication initiatives become necessary precursors to adoption efforts. If professionals are aware but do not perceive value, the theoretical premise that Digital Engineering addresses recognized needs requires reconsideration. Understanding the current state of professional awareness and perceptions enables targeted strategies for advancing Digital Engineering adoption in Information Assurance and IT service management domains.

Relationship to Prior Work

This research builds on a small body of prior work while extending investigation to a professional population that earlier studies did not reach. By surveying professionals actively working in these domains, it identifies whether practitioners recognize connections between their current practices and Digital Engineering capabilities. Findings illuminate whether existing Information Assurance and IT service delivery frameworks already incorporate concepts analogous to Model-Based Systems Engineering, digital threads, digital twins, or Product Lifecycle Management, or whether these Digital Engineering capabilities are genuinely novel approaches within Information Technology contexts. Such understanding establishes whether Digital Engineering offers new conceptual frameworks for addressing Information Assurance and IT service delivery challenges or whether it primarily provides different terminology for existing practices.

Bonar and Hastings established an initial reference model showing how Digital Engineering practices can support and enhance compliance verification within the context of information systems (Bonar & Hastings, 2024). The current research extends this

foundation by examining whether the broader professional community recognizes value in the capabilities that the reference model proposes.

Significance of the Research

This research carries implications across academic, industry, national, and societal dimensions. Understanding these dimensions of significance contextualizes the contribution this investigation makes to knowledge and practice.

Academic Significance

The academic significance of this research begins with an absence. As Chapter 2 documents, the literature review identified a near-complete absence of academic research applying Model-Based Systems Engineering, digital threads, digital twins, or Product Lifecycle Management to enterprise IT infrastructure or Information Assurance programs. This study addresses that absence directly by establishing whether the professionals who work in these domains recognize a gap that Digital Engineering may address. This gap persists even though NIST and ITIL frameworks impose requirements for capabilities that Digital Engineering could provide: enterprise architecture representation, documentation accuracy, and traceability.

This research contributes empirical evidence to a domain currently characterized by theoretical proposition rather than data. Perception data collected from working professionals enables assessment of whether the theoretical framework connecting Digital Engineering capabilities to documented IT and Information Assurance challenges resonates with practitioners who experience those challenges directly. Findings indicating that professionals recognize value in Digital Engineering capabilities would validate the theoretical premise and support subsequent implementation research. Findings indicating limited awareness or skepticism would redirect the research agenda toward education initiatives or reconsideration of transferability assumptions. Either outcome advances the academic understanding of Digital Engineering's potential applicability beyond its established defense and aerospace foundations. The layered awareness-and-recognition

design and the anonymous open-link mode of data collection also yield methodological results of independent interest. Foremost among them, as Chapter 4 details, is a validated and openly released response-authenticity screen for detecting coordinated and automated responding in anonymous online surveys, offered as a reusable methodological contribution in its own right.

Industry Significance

For industry, an adoption effort would meet either a receptive professional community or an indifferent one, and at present no data indicate which. This research supplies that missing signal by measuring how practitioners perceive Digital Engineering capabilities. Organizations considering Digital Engineering adoption can benefit from understanding current awareness levels and perceived value within the professional community. If research reveals widespread recognition of Digital Engineering value, organizations may find receptive audiences for adoption initiatives. If research reveals limited awareness or skepticism, organizations can anticipate the education and change management challenges that adoption would require.

Beyond individual organizations, the research characterizes which segments of the professional community perceive the least value and remain least aware, and therefore where receptiveness is weakest. Such information enables tool vendors, service providers, and standards organizations to focus communication and enablement on the audiences that the data identify as least reached, rather than allocating effort uniformly. Understanding professional perceptions enables more effective resource allocation across the ecosystem supporting Digital Engineering adoption.

National Significance

At the national scale, the documentation and visibility failures examined here cease to be an operational inconvenience and become a matter of national security and the protection of critical infrastructure. Federal information systems and national security systems protect information assets and enable government operations on which citizens

depend. Organizations operating these systems face the challenges documented throughout this dissertation: maintaining accurate documentation, implementing consistent security controls, and verifying compliance across complex technical environments.

The 2018 Digital Engineering Strategy directs defense programs to establish digital threads, which are designed to enhance compliance verification and security assurance by providing authoritative traceability: verified connections between security requirements, control implementations, and compliance evidence (Department of War, 2018). Operators of federal and national security systems must demonstrate compliance with many regulatory frameworks and security standards, often requiring extensive manual effort to collect evidence and prepare for audits. Digital Engineering practices offer the potential to reduce the burden of compliance verification while improving the accuracy and currency of compliance documentation, enabling organizations to redirect resources toward proactive security improvements rather than documentation effort.

Understanding first, second, and third order impacts of security incidents is especially important for critical infrastructure protection. When adversaries compromise systems supporting government functions or critical infrastructure, defenders must rapidly assess the scope of compromise and potential cascading effects. Digital Engineering practices are designed to provide the visibility and traceability that enable rapid, accurate impact assessment, capabilities that current approaches lack. The present study measures professional receptiveness to this class of traceability capability rather than to cascading-impact assessment as a separately worded use case, a distinction Chapter 5 makes explicit. Whether organizations outside defense and aerospace can realize these capabilities effectively remains an open question that this research begins to address through professional perception data.

Societal Significance

Beyond organizations operating national security systems, Digital Engineering capabilities may benefit organizations serving communities with limited resources.

Healthcare providers, educational institutions, social service organizations, and other under-resourced entities face the same Information Assurance and IT service delivery challenges as large enterprises, often with fewer resources to address them.

The distributional character of the burden gives the issue its equity significance. The documentation, traceability, and verification labor that compliance demands falls most heavily on precisely the organizations least able to absorb it. This asymmetry functions as an operational digital divide in which security and compliance capability tracks organizational wealth rather than mission need. The burden is, in this sense, regressive, a pattern documented for federal regulation generally and for cybersecurity compliance specifically, as Chapter 2 details (Crain & Hopkins, 2001; U.S. Government Accountability Office, 2026). If Digital Engineering practices prove transferable beyond defense and aerospace contexts, the capabilities that lower the labor and expertise threshold for compliance would benefit most the organizations for which that threshold is currently prohibitive. Such practices would make it more feasible for smaller organizations to demonstrate regulatory compliance and security effectiveness to funding agencies, oversight bodies, and stakeholders.

The potential for Digital Engineering to extend sophisticated security and documentation capabilities to resource-constrained organizations is a central motivating consideration for this research. Currently, enterprise architecture, authoritative traceability, and model-based documentation remain accessible primarily to large organizations with specialized expertise and dedicated budgets. Community clinics, tribal organizations, under-resourced schools, small municipalities, and social-service non-profits carry the same regulatory obligations without comparable means to meet them. The decisive question is therefore directional: whether such a capability narrows the gap between the security posture available to well-funded institutions and that available to the smallest and least-resourced organizations, or whether, like prior technology waves, it reaches the well-resourced first and widens that gap instead. This research does not

directly investigate adoption within resource-constrained organizations, and it does not claim that the benefit will automatically transfer. What it contributes is the necessary first link in that chain. By establishing whether IT and Information Assurance professionals broadly perceive value in Digital Engineering capabilities, the findings create the evidentiary foundation for subsequent research into accessible, lower-cost implementations suited to the smallest and least-resourced organizations. That foundation lets the equity benefit anticipated here be deliberately pursued rather than merely hoped for.

Chapter Summary

This chapter established the context for investigating professional awareness and perceptions of Digital Engineering capabilities in Information Assurance and IT service management. It identified the challenges organizations face under traditional document-centric approaches: maintaining accurate system documentation, implementing consistent security controls, and delivering reliable IT services. Digital Engineering—with its four pillars of Model-Based Systems Engineering, digital threads, digital twin technology, and Product Lifecycle Management—offers capabilities with reported value in defense and aerospace contexts that may address analogous gaps in enterprise IT and Information Assurance practice. The institutional endorsement of Digital Engineering by the Department of War, NASA, INCOSE, and standards bodies including the Object Management Group and ISO establishes the maturity and authoritative standing of these methodologies. Enterprise architecture frameworks, particularly the Unified Architecture Framework, provide the structural foundation for implementation.

The research questions address three matters: whether professionals are aware of Digital Engineering capabilities, whether they perceive those capabilities as valuable for their work, and whether they believe the practices could improve their effectiveness in meeting compliance requirements and delivering IT services. The significance spans four dimensions. Academically, the study addresses identified literature gaps. For industry, it informs adoption strategies. Nationally, it bears on the protection of government systems.

For society, it points toward better security capabilities for organizations that serve under-resourced communities.

The research targets IT and Information Assurance professionals across diverse organizational contexts, enabling assessment of awareness and perceived value across the professional community. While the defense and aerospace sectors have committed to Digital Engineering at institutional scale, this research investigates whether professionals in other sectors recognize potential value in these capabilities for their work. That question is central to the dissertation, and the chapters that follow answer it in sequence: the literature review establishes why the answer cannot be assumed, the methodology develops the instrument to measure it, and the results report what the survey finds. The findings inform whether Digital Engineering methodologies developed for defense applications might benefit organizations across all sectors, including the smallest and least-resourced.

Chapter 2 presents the literature review examining existing research across Digital Engineering, Information Assurance, and IT service management domains. The review establishes the theoretical framework for this research and documents the research gaps that this investigation begins to address. Its sustained analytical synthesis identifies what the literature contains and, beyond that, what debates remain unresolved, what tensions persist between proven capability and limited adoption, and why the present research is necessary.

Chapter 2: Literature Review

The literature on Digital Engineering tells a story of demonstrated capability confined within disciplinary walls. Model-Based Systems Engineering, digital threads, digital twins, and Product Lifecycle Management have matured through decades of application in defense and aerospace, accumulating institutional endorsement from the Department of War, NASA, and INCOSE alongside sparse but consistent empirical evidence of return on investment. Yet this maturation has occurred almost entirely within the systems engineering discipline. Enterprise IT infrastructure management and Information Assurance confront structurally analogous challenges of documentation accuracy, configuration visibility, compliance traceability, and change coordination. Nevertheless, these domains have developed independently, producing their own frameworks, tools, and patterns of persistent failure. The result is a pronounced research gap. Two efforts approach the problem: a preliminary reference model by Bonar and Hastings (2024) and a conceptual framework for DevSecOps security assurance from the Carnegie Mellon Software Engineering Institute (Chick et al., 2023). Beyond these, no peer-reviewed research identified in this review addresses applying Digital Engineering methods to enterprise IT infrastructure management and Information Assurance programs as an integrated enterprise discipline. That scope remains distinct from Model-Based Systems Engineering applied to individual engineered systems or to single compliance controls.

This chapter examines why that gap persists and why it matters. This structured, analytical review spans disciplinary evolution, MBSE value evidence, adoption dynamics, cross-disciplinary transfer barriers, digital twin security applications, compliance automation, IT service management failures, and enterprise visibility challenges. Rather than cataloging what exists, the review foregrounds the debates, tensions, and unresolved questions that position this dissertation's investigation within the academic conversation. Four interrelated arguments emerge. First, systems engineering, Information Assurance,

cybersecurity, and IT service management evolved along independent academic and professional trajectories, creating disciplinary silos that explain the absence of cross-disciplinary research. Second, the evidence base for MBSE value, while growing, remains disproportionately reliant on perceived rather than measured outcomes, creating economic uncertainty that discourages adoption beyond established domains. Third, adoption research shows that perceptions, rather than objective technical merit, drive adoption decisions, yet no empirical data exist on how IT and Information Assurance professionals perceive Digital Engineering capabilities. Fourth, compliance frameworks and IT service management standards explicitly require capabilities that Digital Engineering provides. Yet the academic community has not investigated whether Digital Engineering could fulfill these requirements more effectively than the traditional approaches that frequently fail. Throughout, this review takes as given the conceptual definitions of Digital Engineering and its four pillars established in Chapter 1. The review builds on that foundation to weigh evidence, debate, and gaps rather than to restate definitions. Where a pillar reappears here, it does so as the object of evidentiary appraisal.

Parallel Paths: The Independent Evolution of Adjacent Disciplines

Understanding why that gap opened, and why it has proved so durable, requires understanding how these disciplines evolved along separate trajectories. Each developed its own academic identity, professional communities, curricular standards, and institutional frameworks, even while confronting similar challenges in documentation, traceability, configuration management, and lifecycle governance. This independent evolution created disciplinary silos so deeply embedded that researchers within each tradition rarely encounter the methods of adjacent fields. The result is a structural consequence of how academic disciplines form and maintain boundaries, an outcome of institutional design working as it normally does.

Systems engineering emerged as a recognizable discipline in the mid-twentieth century, rooted in large-scale military and industrial programs that demanded coordination

across multiple engineering specialties. The discipline developed across three historical intervals: foundational (1926–1960), exploratory (1961–1989), and contemporary (1990–2017) (Hossain et al., 2020). Across these intervals, systems engineering "is not affirmed by a set of underlying fundamental propositions" but rather emerged as "a set of best practices" consolidated from defense and aerospace experience (Hossain et al., 2020). Hall's foundational *A Methodology for Systems Engineering* (1962) and Goode and Machol's *System Engineering* (1957) established the intellectual architecture, while organizations such as Bell Telephone Laboratories and the RAND Corporation provided the institutional incubators (Hossain et al., 2020). Reviewing seventy years of the discipline's existence, Honour noted that systems engineering experienced "rapid growth" in its early decades followed by "partial stagnation in the 1990s and 2000s" as prescriptive process standards displaced first-principles engineering thinking (Honour, 2018). This stagnation coincided with the period when enterprise IT complexity was accelerating. The temporal mismatch helps explain why IT organizations did not adopt systems engineering methods during the critical window when they might have been most needed.

Disciplinary insularity deepened as systems engineering grew more abstract. A companion history traced the evolution from a technical-objective focus through a process and methodology focus toward contemporary calls to return to foundational roots, arguing that systems engineering "became more abstract and disconnected from the rest of engineering" when prescriptive processes were developed (Pennotti et al., 2024). As part of INCOSE's Future of Systems Engineering initiative, that same work characterized systems engineering as inherently transdisciplinary yet insufficiently recognized as such. This characterization carries direct implications for the present research. If systems engineering's own professional community acknowledges that the discipline has been too insular, the failure to extend its methods to enterprise IT becomes explicable. It stems from disciplinary boundaries that neither community has been motivated to cross, boundaries left in place by default while the question of applicability went unasked. Survey

evidence corroborates this self-assessment: across INCOSE chapter attendees in North America, a measurable gap separates systems engineering as a documented discipline from its actual technical practice (Wasson, 2019).

Crucially, throughout these seven decades, systems engineering oriented itself toward physical systems: aircraft, naval vessels, spacecraft, weapons platforms, and industrial control systems. Enterprise information systems—encompassing servers, networks, databases, cloud infrastructure, and software applications—fell outside the discipline's traditional scope. Yet these systems grew to exhibit the same characteristics of complexity, interdependency, and emergent behavior that justified systems engineering approaches for physical platforms. The Systems Engineering Body of Knowledge acknowledges this historical trajectory, documenting that systems engineering principles apply to "engineered systems" broadly while noting that most published case studies and validated methodologies derive from defense and aerospace contexts (SEBoK Authors, 2025).

Information Assurance, by contrast, developed from different institutional origins. Where systems engineering emerged from engineering programs and defense laboratories, Information Assurance developed at the intersection of national security policy and computing education. Maconachy et al. formalized the discipline's conceptual foundations in their influential model extending McCumber's 1991 information security cube to encompass five security services (confidentiality, integrity, availability, authentication, and non-repudiation) across three information states and three countermeasure categories (Maconachy et al., 2001). The framework emerged under the auspices of the National Security Agency and Idaho State University's National Information Assurance Training and Education Center. It established Information Assurance as a discipline concerned with protecting and governing information across its lifecycle rather than with engineering the systems that process it. The Committee on National Security Systems codified training standards through NSTISSI documents 4011 through 4016, and the Centers of Academic Excellence in Information Assurance Education program, established

in 1998, formalized the academic pipeline (Dark et al., 2006). Information Assurance thus became a "pervasive theme" within IT curricula, embedded within computing education but distinct from the engineering disciplines where systems engineering resided (Dark et al., 2006).

A comprehensive evaluation of Information Assurance and security models traced how the field's models evolved, moving from McCumber's original cube through the Maconachy-Schou-Ragsdale model and arriving at a new Reference Model of Information Assurance and Security. That evaluation drew on twenty-six expert interviews, three workshops, and a case study (Cherdantseva et al., 2016). That analysis revealed that existing IA models addressed "trends of diversification and de-perimeterisation" but focused on security properties and countermeasures rather than on the engineering lifecycle processes through which systems are designed, built, and maintained. This focus on protection properties rather than engineering processes is a fundamental orientation difference from systems engineering, and it explains why Information Assurance professionals have not encountered Model-Based Systems Engineering in their professional formation.

Cybersecurity's emergence as a standalone academic discipline further fragmented the intellectual landscape. Parrish et al. characterized cybersecurity as a "meta-discipline," an aggregate label encompassing contributions from computer science, information systems, information technology, software engineering, and electrical engineering (Parrish et al., 2018). Their analysis traced how establishing the CAE/IAE program in 1998 initiated institutional recognition that evolved from Information Assurance education into cybersecurity education over two decades. The Joint Task Force on Cybersecurity Education formalized this disciplinary independence through *Cybersecurity Curricula 2017*, the first global curricular guidelines for post-secondary cybersecurity degree programs, organized around eight knowledge areas with contributions from over three hundred advisors across thirty-five countries (Joint Task Force on Cybersecurity Education, 2017).

The committee report acknowledged that "the field of cybersecurity is in its formative stages" (Joint Task Force on Cybersecurity Education, 2017). Such a characterization positions cybersecurity as a discipline still establishing its boundaries rather than one prepared to absorb methods from engineering fields with which it shares no curricular lineage.

An analysis of master's programs at leading universities examined the evolution of cybersecurity as an academic discipline, finding that it "is considered to be evolving at a rapid pace" compared to other scientific and engineering disciplines (Cabaj et al., 2018). The *Computing Curricula 2020* report, the ACM and IEEE Computer Society's comprehensive overview of computing education, positioned cybersecurity alongside established computing disciplines while documenting how each maintained distinct identities despite shared foundations (CC2020 Task Force, 2020). Chapter 2 of that report traced how computing disciplines evolved since CC2005, revealing a pattern of increasing specialization and decreasing cross-disciplinary integration. Systems engineering does not appear among the computing disciplines at all; it occupies a separate academic lineage rooted in industrial and mechanical engineering departments rather than computing schools. This institutional separation means that cybersecurity and IT graduates rarely encounter systems engineering coursework, and systems engineering graduates rarely encounter Information Assurance or IT service management content.

IT Service Management developed along yet another independent trajectory, one driven by practitioner communities rather than academic research. The foundational systematic literature review of ITSM implementation examined research published between 2000 and 2012 (Iden & Eikebrokk, 2013). Its findings documented that the earliest academic study addressing ITIL adoption appeared in 2005 and the first journal article in 2006, more than fifteen years after ITIL's initial development by the United Kingdom's Central Computer and Telecommunications Agency in the 1980s. This extended gap between practitioner adoption and academic investigation produced a field with extensive

operational guidance but limited theoretical grounding. The most recent comprehensive systematic review of ITSM reinforced this assessment, covering 126 empirical studies from 2012 through 2021 and reconceptualizing ITSM as a Management Control System (MacLean & Titah, 2023). Their observation that ITSM has been "undertheorized by academics" despite widespread practitioner adoption encapsulates the practitioner-academic gap that characterizes the field (MacLean & Titah, 2023).

The consequences of this practitioner-driven evolution include widespread "conceptual confusion" across ITSM models such as ITIL, CobiT, BSM, SLM, and ISO/IEC 20000 (Winniford et al., 2009). Multiple frameworks addressing overlapping concerns developed without mutual awareness or coordination, a fragmentation pattern that mirrors the broader disciplinary silos this section documents. In a cross-national empirical study of 623 respondents from the United Kingdom, United States, German-speaking countries, and Australia, Marrone et al. found that operational ITIL processes achieved substantially higher adoption rates than strategic processes (Marrone et al., 2014). A systematic literature review by Serrano et al. synthesized forty-seven articles mapping the ITSM field's evolution since 2002, tracing how ITSM grew from the service management concept emerging in the 1930s through ITIL's development into a distinct professional domain (Serrano et al., 2021). Throughout this evolution, ITSM developed no connection to systems engineering practice. ITIL's process-oriented framework emerged from operational IT management experience rather than from engineering design principles, creating a fundamental methodological incompatibility: ITSM treats IT services as processes to be managed, while systems engineering treats systems as engineered artifacts to be designed, verified, and maintained through lifecycle governance.

These parallel evolutionary paths produced more than an absence of applied research. The deeper deficit is that no intellectual infrastructure exists to enable such application. Researchers in each discipline publish in different journals, attend different

conferences, cite different foundational theorists, and frame problems through different conceptual lenses. A systems engineer describes configuration management as maintaining the consistency of a system's physical and functional attributes with its requirements and design documentation (SEBoK Authors, 2025). An IT service manager describes configuration management as maintaining information about configuration items required to deliver IT services (AXELOS Limited, 2019). An Information Assurance professional describes configuration management as establishing and maintaining the integrity of systems through control of changes (National Institute of Standards and Technology, 2019). These three descriptions address the same operational challenge through three different conceptual vocabularies, each embedded within frameworks that do not reference one another. The research gap this dissertation investigates is, at its root, a gap between disciplines that never developed the shared vocabulary necessary for methodological exchange.

A Theoretical Lens on the Divide: Boundary Objects and Knowledge Transformation

Why a divide of this kind should prove so stubborn, and how it might be crossed, comes into sharper focus when the gap is viewed through the theory of knowledge boundaries. The pivotal idea is the *boundary object*, introduced by Star and Griesemer in a study of how naturalists, collectors, and administrators cooperated productively without sharing a common expertise. A boundary object is an artifact "plastic enough to adapt to local needs and the constraints of the several parties employing them, yet robust enough to maintain a common identity across sites" (Star & Griesemer, 1989). In plainer terms, it is a shared representation—such as a map, a specification, or a model—around which communities that speak different professional languages can coordinate their work without first agreeing on terms. The concept names precisely the sort of artifact that might let a systems engineer, a security analyst, and an IT service manager reason about the same system together.

Why the present divide resists easy bridging emerges from the refinement that Carlile brought to the idea. He graded knowledge boundaries by the difficulty of crossing them: syntactic boundaries yield to the simple *transfer* of information, semantic boundaries demand the *translation* of differing interpretations, and pragmatic boundaries, where the parties hold genuinely different interests and entrenched practices, yield only to the costly *transformation* of knowledge itself (Carlile, 2002, 2004). The boundary separating systems engineering from Information Assurance and IT service management is pragmatic in exactly this sense. The three vocabularies of configuration management cataloged above are not a shortfall of translation that a shared glossary could repair; they encode different practices, incentives, and professional identities, and bridging them therefore demands the transformation of working knowledge rather than the mere exchange of definitions.

Several complementary lenses deepen this account of why the divide endures. W. M. Cohen and Levinthal established that a community's capacity to absorb external knowledge depends on its stock of prior related knowledge (W. M. Cohen & Levinthal, 1990). Disciplines that have never encountered one another's methods therefore have little of the absorptive capacity required to assimilate them. Szulanski showed empirically that the principal impediments to transferring proven practice are knowledge-related rather than motivational, a phenomenon he termed internal stickiness (Szulanski, 1996). J. S. Brown and Duguid and Wenger located knowledge within communities of practice, whose tacit and practice-bound knowing resists movement across community boundaries (J. S. Brown & Duguid, 1991; Wenger, 1998). At a broader scale, Gibbons et al. characterized a shift toward knowledge produced in the context of application by transdisciplinary teams (Gibbons et al., 1994), and Bammer et al. argue that the integration expertise needed to span such teams is itself a distinct and frequently absent capability (Bammer et al., 2020). Read together, these perspectives recast the systems-engineering-to-information-assurance gap not as a failure of effort but as the predictable behavior of distinct epistemic communities, each holding limited absorptive

capacity for the other's practice-bound knowledge.

That such transformation is arduous yet disproportionately valuable is the central message of the National Academies' assessment of interdisciplinary work. The assessment defines the enterprise as integrating "information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines" to attack problems "beyond the scope of a single discipline," and it catalogs the institutional barriers that keep such integration rare (National Academy of Sciences, National Academy of Engineering, and Institute of Medicine, 2005). Systems engineering has itself responded to this call. Madni reframes the discipline as inherently transdisciplinary, urging it to reach beyond its traditional borders and to exploit convergence with adjacent fields rather than retreat behind them (Madni, 2018, 2019). Nor is the maneuver merely aspirational. Working in the security-critical setting of collaborative cyber-physical systems, Ramli et al. treat a reference architecture as a boundary object that aligns otherwise divergent stakeholders around shared trustworthiness properties (Ramli et al., 2021). This illustrative application suggests that the concept can perform real work at the frontier between engineering and security.

Read together, these strands recast the research gap as something deeper than an empirical absence. Theory anticipates both the difficulty and the remedy. The divide persists as the predictable behavior of a pragmatic knowledge boundary, and the most promising instrument for crossing such a boundary is precisely the shared, authoritative, machine-readable model that Digital Engineering produces. Whether the models of Digital Engineering can in fact serve as the boundary objects that carry transformed knowledge between systems engineering, Information Assurance, and IT service management is, at bottom, the question this dissertation begins to answer. The present study takes only the first step toward that answer, and a deliberately bounded one. It measures whether the professionals who would have to cross this boundary perceive value in the capabilities that might carry them across it, which corresponds to the perceived-usefulness component of

the adoption theories introduced in Chapter 1. Whether Digital Engineering models actually function as boundary objects, transforming knowledge across these communities in practice, is a question a perception survey cannot settle. This study therefore frames the boundary-object account as the theory that the implementation and longitudinal studies proposed in Chapter 5 are designed to test, rather than as a claim the present data establish.

The Evidence Paradox: Demonstrated Value Without Sufficient Proof

Model-Based Systems Engineering rests on a paradox that shapes its entire adoption landscape: practitioners and organizations widely perceive MBSE as valuable, yet rigorous empirical measurement of that value remains exceptionally rare. Understanding this paradox is essential because it explains both why MBSE has succeeded within systems engineering and why it has not expanded beyond it. Organizations already embedded in the systems engineering discipline can rely on institutional experience and professional networks to validate MBSE investment. Organizations outside the discipline, including enterprise IT and Information Assurance, lack comparable experiential evidence and must rely on published research that provides insufficient quantitative justification.

The most comprehensive assessment of MBSE evidence, a systematic literature review, categorizes reported benefits into four evidence types: measured, observed, perceived, and referenced (Henderson & Salado, 2021). Approximately two-thirds of claimed MBSE benefits lack empirical measurement and instead rely on perceived or referenced evidence, a finding with direct consequences for adoption decisions. A systematic review by Carroll and Malins reached a convergent conclusion from a separate corpus: of 88 MBSE case studies examined, 67 reported no quantitative metrics, and the authors cautioned that case-study author bias tends to distort the return-on-investment figures that the remainder do report (Carroll & Malins, 2016). Organizations evaluating MBSE investment discover that the research literature offers abundant testimony to MBSE's promise but limited quantified outcomes on which to build business cases. A

review of MBSE motivation and research opportunities reached a similar conclusion, arguing that the value proposition requires demonstration through real-world applications and that further advancements remain necessary before broader adoption can occur (Madni & Sievers, 2018). The more recent systematic literature review by Wooley and Womack analyzed adoption, benefits, and challenges across the Digital Engineering research corpus and returned no research addressing enterprise IT infrastructure or Information Assurance applications (Wooley & Womack, 2025). This finding corroborates that the gap reflects the state of the literature rather than an incomplete search.

Among the rare exceptions to the measurement deficit, the work of E. B. Rogers and Mitchell stands as the most frequently cited quantified evidence of MBSE return on investment (E. B. Rogers & Mitchell, 2021). Their case study examined the Submarine Warfare Federated Tactical Systems program, a rapidly evolving combat system of systems comprising over ten million source lines of code deployed across 104 submarines, following a $3 million MBSE investment over two and a half years. A like-for-like comparison between legacy document-centric processes and the MBSE approach revealed that systems engineering hours per requirement decreased from 12.1 to 9.9, representing an 18% improvement in efficiency that exceeded the 13% improvement projected by an earlier pilot study. The MBSE approach handled 42% more interface requirements changes while consuming only 16% more hours, and it reduced total interface defects by 9%. It also shifted 18% of defects to earlier discovery phases, where correcting a defect costs roughly one-fourth to two-thirds of what correction costs during platform integration testing. Supplementary data demonstrated 30% more baselines produced monthly and 60% more integrated subsystems and combat system variants managed within constant resources (E. B. Rogers & Mitchell, 2021).

These results stand out for their scarcity as much as for their magnitude. The same evidence assessment identified the Rogers and Mitchell study as one of only two papers in the literature reporting measured MBSE return on investment evidence (Henderson &

Salado, 2021). The intervening years have added only sparingly to that record, and the most recent addition warrants both caution and attention. In a 2025 mixed-methods study of the post-production design-change process for United States Navy assets, Shefa and Topcu estimated that a Digital Engineering transformation could reduce median project duration by roughly half while sharply lowering its variability, although the gains varied across task categories (Shefa & Topcu, 2025). Two cautions attach to this finding: it remains a preprint awaiting peer review, and its setting is naval sustainment rather than enterprise IT. Neither caution is disqualifying, and the study earns its place on three grounds. Its mixed-methods design triangulates quantitative schedule data against practitioner interviews rather than resting on testimony alone; the magnitude it reports is consistent with the rare earlier measurement from E. B. Rogers and Mitchell; and its object of study, a recurring change-and-configuration process burdened by document-centric handoffs, is structurally close to the change management that consumes Information Assurance and IT service management teams. Treated as corroboration rather than as decisive proof, the result reinforces the governing pattern: where measured returns appear at all, they tend to be large, and what stays scarce is the measurement itself.

Subsequent efforts to address this measurement deficit have advanced the methodological conversation without resolving it. A measurement framework for selecting and developing metrics to assess MBSE and Digital Engineering value included causal maps of expected benefits (Henderson et al., 2023). A separate evaluation of MBSE's perceived value analyzed attributes, benefits, and drawbacks reported in the literature, finding a persistent gap between what practitioners believe MBSE delivers and what published evidence can substantiate (Campo et al., 2023). Both studies underscore a recurring theme. The systems engineering community recognizes the measurement deficit and has begun developing frameworks to address it, yet empirical data from operational programs remain rare. Authoritative recent assessments continue to refine, rather than resolve, the measurement question. Two RAND industry studies conclude that the benefits

of Digital Engineering are genuine but accrue over the long term and resist immediate cost or schedule quantification (Clayton et al., 2024; Whitehead et al., 2024), and a Defense Science Board task force similarly reports demonstrated lifecycle benefits set against substantial workforce and tooling costs (Defense Science Board, 2024). The multi-organization Digital Engineering Measurement Framework responds directly to this difficulty by defining standardized benefit areas against which value can be assessed (Practical Software and Systems Measurement, 2022). Survey evidence reinforces the perception dimension, as E. T. G. Brown and Zaidi found organizational resistance and a shortage of trained engineers to be the dominant adoption barriers among aerospace and defense practitioners (E. T. G. Brown & Zaidi, 2025).

For enterprise IT organizations operating on annual budgets with continuous delivery expectations, this evidentiary deficit creates a rational basis for hesitation: the demonstrated benefits derive from a naval combat system program operating under a level-of-effort contract structure fundamentally different from enterprise IT budget models. Whether comparable returns would materialize in enterprise IT remains untested.

Earlier systematic reviews reinforce this pattern. A review of thirteen years of SysML research mapped the landscape of model-based methods and their empirical evidence (Wolny et al., 2020). A survey of MBSE tools and applications found consistent reports of improved requirements traceability and stakeholder communication alongside persistent concerns about tool complexity and organizational adoption challenges (Chami & Bruel, 2018). Collectively, these reviews establish that MBSE has demonstrated value across multiple domains while simultaneously documenting two critical limitations: the evidence remains predominantly qualitative, and the domains of application remain overwhelmingly confined to aerospace and defense.

The tension between demonstrated capability and insufficient proof creates what might be termed an adoption credibility gap. Within the systems engineering discipline, professional experience and institutional knowledge compensate for the measurement

deficit. Systems engineers who have used MBSE tools can observe improvements in their own work, creating the experiential validation that published evidence lacks. Outside the discipline, this compensatory mechanism does not operate. IT and Information Assurance professionals have no experiential basis for evaluating MBSE claims. The literature provides them with compelling arguments but insufficient evidence, a combination that technology adoption research suggests is inadequate for driving adoption decisions.

Adoption as a Perception Problem

If the evidence paradox explains why MBSE has not expanded through rational economic calculation, the adoption literature reveals a complementary explanation rooted in perception dynamics. A growing body of empirical research shows that MBSE adoption decisions are driven more by how professionals perceive the technology than by its objective characteristics. This finding carries direct implications for the present research: if perceptions govern adoption even among systems engineering professionals who work with models daily, understanding perceptions among IT and Information Assurance professionals, who have never systematically encountered these methods, becomes a prerequisite for any meaningful discussion of cross-disciplinary transfer.

Call and Herber supply the theoretical foundation for this argument, mapping Everett Rogers' Diffusion of Innovations theory to MBSE adoption dynamics (Call & Herber, 2022). Rogers' framework identifies five perceived attributes of innovations (relative advantage, compatibility, complexity, trialability, and observability) that collectively account for 49–87% of variance in adoption rates across innovation research. The critical theoretical insight, and the one most consequential for the present research, is that *perceptions* of these attributes, not the attributes themselves, drive adoption behavior. The authors invoke W. I. Thomas's dictum that "if [people] define situations as real, they are real in their consequences," arguing that shaping how MBSE attributes are perceived can accelerate adoption rates more effectively than improving the attributes themselves (Call & Herber, 2022).

An empirical follow-up by Call et al. tested this theoretical mapping through a survey distributed to systems engineering professionals via INCOSE and professional networks, which drew 270 responses, 197 of them complete (Call et al., 2024). The study examined perceptions across six respondent subpopulations, among them those involved versus not involved in MBSE efforts, model users versus non-users, and respondents subject to Digital Engineering mandates versus those who are not. Respondents broadly recognized MBSE's relative advantage in improving data quality and traceability. However, perceived compatibility with existing practices and perceived complexity emerged as significant barriers that suppressed adoption even when practitioners acknowledged objective value.

Most striking is the finding on trialability. Respondents not involved in MBSE efforts reported far lower access to trial opportunities and tools; for the statement "I have opportunities to experiment with MBSE," the chi-squared test of independence between the involved and non-involved subpopulations produced the largest divergence in the study, with $\chi^2 = 72.1$ and reported $p = .000$ (Call, 2024; Call et al., 2024). This creates what Call et al. characterize as a barrier-reinforcing cycle: professionals who have not experienced MBSE cannot access the trial opportunities that would enable informed adoption decisions, while those who have experienced MBSE report substantially more favorable perceptions. The study further characterizes MBSE as a "preventative innovation" whose advantages derive from preventing problems such as inconsistencies, documentation errors, and rework rather than from producing visible new benefits. This characteristic depresses both perceived relative advantage and observability, compounding adoption barriers even when objective value exists.

This perception-adoption dynamic has profound implications for cross-disciplinary transfer. If systems engineering professionals, practitioners who work within the discipline that developed MBSE, exhibit perception gaps that impede adoption, the barriers among IT and Information Assurance professionals are likely to be greater still on the attributes that depend on contact with the methods, namely trialability and observability. Relative

advantage may behave differently. Digital Engineering addresses documentation, traceability, and visibility problems these professionals already face, so a favorable reading of its advantage could survive transfer even where exposure is absent, which is the expectation Proposition P2 states and Chapter 4 tests. These professionals operate outside the systems engineering discipline entirely. They have no professional exposure to MBSE concepts, no institutional networks sharing MBSE experiences, and no trial opportunities through which favorable perceptions might develop. The present research addresses this gap by measuring perceptions in a population where no empirical data currently exist.

Complementary evidence from Henderson et al. deepens understanding of adoption dynamics through a mixed-methods study combining systematic literature review with eighteen semi-structured practitioner interviews from organizations attempting MBSE adoption (Henderson et al., 2024). The literature review extracted over 600 individual lessons learned from 46 published papers, coded into categories spanning communication, model definition, organizational strategy, and technical implementation. The interview findings illuminate the human dimensions of adoption that quantitative studies cannot fully capture. Most directly relevant to the question of professional awareness, approximately 22% of interview participants, all of them systems engineering professionals, could not convey a clear definition of MBSE. One participant captured the prevailing confusion: "Most people think of MBSE as being synonymous with a specific tool" and "don't understand how MBSE fits into DE or what it really does other than it is modeling instead of documents" (Henderson et al., 2024). Another stated plainly that "if they aren't familiar with MBSE, they're not going to use it."

Organizational lessons proved equally revealing. For both organizations in the study whose adoption efforts failed, the driving barrier was lack of management support. Middle management resistance proved especially damaging because executives endorsed MBSE while middle managers could disregard the endorsement without operational consequences. Successful adopters commonly established core MBSE teams, such as communities of

practice or centers of excellence, and implemented role-based training at four levels: model reviewers for leaders and decision-makers, developers and modelers, architects for senior engineers, and administrators for IT and software support. A reinforcing dynamic emerged: greater stakeholder exposure to models produced more perceived benefits, which generated more organizational buy-in, which enabled further exposure. This virtuous cycle operates within organizations that have initiated adoption but cannot initiate itself in domains where MBSE has never been introduced. A synthesis of five years of research on Digital Engineering and MBSE adoption produced an enterprise adoption framework, a model of factors spanning organizational design, enablers and barriers, and change management (McDermott et al., 2024). The synthesis confirmed that adoption requires coordinated action across multiple organizational dimensions simultaneously, a finding that compounds the challenge for enterprise IT organizations unfamiliar with the discipline.

A survey of 51 practitioners extended these findings, examining how organizational structure variables correlate with MBSE adoption outcomes (Henderson & Salado, 2024). The results reveal a pattern with direct implications for enterprise IT contexts. Flexibility and interconnectedness showed the strongest and most consistent positive correlations with both adoption process and implementation variables. Formalization, defined as documented procedures and processes, correlated positively, a somewhat counterintuitive finding suggesting that governance frameworks support rather than hinder adoption when combined with flexibility. Centralization correlated negatively with adoption and implementation, as did large organizational size and high vertical differentiation. These structural findings suggest that enterprise IT organizations characterized by rigid hierarchies, centralized governance, and siloed operations may face structural impediments to Digital Engineering adoption that compound the perception barriers documented by Call et al.

Research beyond the systems engineering discipline corroborates and extends these dynamics. The embedded systems industry sits outside the defense and aerospace context

where MBSE originated. A qualitative study of twenty interviews across ten companies in that sector identified immature tooling, return on investment uncertainty, and migration fears as key barriers, confirming that the adoption challenges documented within systems engineering intensify when the technology crosses disciplinary boundaries (Vogelsang et al., 2017). A broader study of strategic adoption of digital innovations found that digital transformation requires coordinated enterprise-level application rather than bottom-up adoption of individual technologies (Campagna & Bhada, 2024). The research identified twelve strategic adoption influencers and noted that adoption research focuses on individual technologies rather than integrated digital transformation. This finding explains why platform-level Digital Engineering adoption within defense programs has not expanded to enterprise IT functions operating separately from program organizations.

Technology adoption research in cybersecurity and IT contexts provides a parallel body of evidence that has not been connected to the Digital Engineering adoption discourse. One study integrated Diffusion of Innovation, the Technology Acceptance Model, and the Technology-Organization-Environment framework with a balanced scorecard approach, surveying 147 IT experts in United Kingdom small and medium enterprises to identify factors affecting cybersecurity technology adoption (Hasani et al., 2023). Their study is one of the few empirical investigations combining multiple adoption models in a cybersecurity context, identifying eight key factors including organizational IT readiness, perceived ease of use, and management support. These factors parallel the MBSE adoption barriers documented by Henderson et al. In a related effort, Alhalafi and Veeraraghavan developed a cybersecurity-specific extension of the Unified Theory of Acceptance and Use of Technology, surveying 108 IT professionals and 554 public respondents to examine cybersecurity adoption in smart city contexts (Alhalafi & Veeraraghavan, 2023). Drawing on the same theoretical foundation that Call et al. used for MBSE, Anthony applied Rogers' Diffusion of Innovations framework to examine motivations and barriers for advanced cybersecurity tool adoption in high-threat

industries (Anthony, 2024). That cybersecurity researchers independently selected the same adoption theory applied by MBSE adoption researchers underscores the theoretical compatibility between these fields despite their practical isolation.

At the organizational level, a large-scale empirical study using data from 1,988 company executives examined IT innovation adoption factors, finding that organizational IT innovation readiness, which encompasses both technology readiness and internal expertise, is a key mediator of adoption outcomes (Ali et al., 2022). A study combining Diffusion of Innovation and Technology-Organization-Environment frameworks surveyed 535 managers and found that relative advantage, complexity, observability, top management support, and competitive pressure serve as significant determinants of digital technology adoption (Shahadat et al., 2023). These organizational-level findings parallel the structural variables examined by Henderson and Salado in the MBSE context, suggesting that common adoption dynamics operate across technology domains even when the specific technologies differ.

The adoption literature thus establishes a critical precondition for the present research. Perceptions govern technology adoption. Even among systems engineering professionals, the population most favorably positioned to appreciate MBSE, perceptions remain mixed and marked by substantial awareness deficits. Cybersecurity and IT adoption research reveals parallel dynamics operating through the same theoretical frameworks. Yet no study bridges these parallel tracks by examining Digital Engineering perceptions among IT and Information Assurance professionals. Until such data are collected, any discussion of Digital Engineering adoption beyond defense and aerospace rests on assumption rather than evidence. This dissertation addresses that evidentiary gap.

Crossing Disciplinary Boundaries: Digital Engineering Beyond Its Origins

Digital Engineering's concentration within aerospace and defense raises an unresolved question that the literature has acknowledged but not adequately investigated: can these methods transfer effectively to domains with different professional cultures, tool

ecosystems, and economic structures? The preliminary evidence is suggestive but fragmented, consisting of isolated applications to security domains, a conceptual framework from an authoritative institution, emerging applications in healthcare and manufacturing, and a single systematic mapping study of model-based security engineering. No sustained research program has examined cross-disciplinary transfer to enterprise IT or Information Assurance.

Bone et al. described the Systems Engineering Research Center's research effort underlying the Department of Defense Digital Engineering initiative, identifying three key enablers for transformation: IT infrastructure, workforce development, and policy (Bone et al., 2019). Though defense-originated, their analysis addressed cross-cutting adoption dynamics and workforce transformation challenges applicable to any domain adopting Digital Engineering. The transformation they described requires coordinated investment across technology, people, and governance, precisely the integrated approach that enterprise IT organizations, structured around operational silos and annual budget cycles, struggle to sustain.

Among existing efforts, the most substantive bridge between Digital Engineering and Information Assurance comes from the Carnegie Mellon Software Engineering Institute. Chick et al. argued that DevSecOps pipelines constitute complex socio-technical systems that require systems engineering treatment, comprising components that are independently developed and maintained, physically and logically distributed, and interoperable (Chick et al., 2023). The tight integration of business mission, capability delivery, and products "increases the attack surface of the product under development." As organizations adopt DevSecOps tools and techniques with increased coupling between the product being built and the tools used to build it, "the attack surface of the product continues to grow, incorporating segments of the development environment." The research developed a DevSecOps Platform Independent Model using the Unified Architecture Framework and SysML to model pipeline security properties, demonstrating how to

construct cybersecurity assurance cases from model-based representations.

What gives such a connection its analytical rigor is the assurance case, a connective concept with an established lineage of its own. An assurance case is a structured, evidence-supported argument that a system satisfies a claimed property such as safety or security. Kelly and Weaver formalized its dominant notation, the Goal Structuring Notation, which renders the argument as an explicit hierarchy of claims, supporting strategies, and the evidence that discharges them (Kelly & Weaver, 2004). Wei et al. subsequently carried the notation into the model-based world through the Object Management Group's Structured Assurance Case Metamodel, enabling assurance arguments to be generated and maintained as formal models rather than as static documents (Wei et al., 2019). The consequence for Information Assurance is immediate and concrete. When the relationships among requirements, controls, and evidence already reside inside a Digital Engineering model, the compliance argument that auditors demand can be assembled from that model rather than reconstructed by hand. That manual reconstruction is the precise labor that document-centric practice imposes and that the compliance frameworks examined later in this chapter render ever more onerous.

Most relevant to this dissertation, the SEI framework proposed a conceptual shift from process-based to property-based security assurance. Traditional cybersecurity assurance relies on process-based standards such as the NIST Risk Management Framework and SP 800-53 security controls. As systems become more complicated and interconnected, Chick et al. argued, "process-based standards fail to assure system owners that the system functions only as intended under all operational circumstances" (Chick et al., 2023). When model-informed threat modeling and security practices are done well, "the overall risks associated with the DevSecOps pipeline and associated products will be reduced, and the compliance and legal requirements will naturally be addressed within the engineering lifecycle." This finding directly supports the theoretical premise of this dissertation: that Digital Engineering capabilities can integrate security engineering into

development and operational processes rather than treating compliance as a separate activity conducted after the fact.

However, the SEI framework is a conceptual and architectural argument rather than empirical validation. The research demonstrates technical feasibility through detailed modeling of configuration management capabilities but does not provide quantitative effectiveness data. The framework was designed for heavily regulated and cybersecurity-constrained environments including defense, banking, and healthcare. These are domains where Information Assurance professionals operate and where compliance verification demands consume substantial organizational resources. Lacking empirical validation, the framework leaves the critical question unanswered: does MBSE-based security assurance deliver measurable improvements over traditional compliance approaches in practice?

Beyond the SEI framework, emerging research has begun applying MBSE to non-defense domains, though these applications remain isolated and preliminary. One MBSE methodology for capturing cybersecurity requirements in space systems used SysML within Cameo Systems Modeler, expanding on NIST cyber resiliency frameworks to show how model-based approaches can structure security analysis within an engineering lifecycle (Kirshner, 2023). In a scoping review of MBSE methods applied to rural healthcare system disaster planning, Berg et al. demonstrated that model-based approaches can organize complex interdependencies beyond traditional engineering contexts (Berg et al., 2023). A rapid review by Xames et al., covering twelve years of MBSE use in healthcare systems, identified applications ranging from patient flow modeling to wellness digital twins (Xames et al., 2024). A practical MBSE methodology for digital transformation of existing industrial enterprises, developed by Badenko et al., demonstrated feasibility in a non-defense application outside the discipline's traditional domains (Badenko et al., 2024). Reviewing cross-industry principles for digital representations of complex technical systems using the MBSE approach, Bolshakov et al.

explicitly addressed applications beyond defense and aerospace (Bolshakov et al., 2023). A common pattern runs across these applications: each demonstrates feasibility within a single bounded domain, and none attempts the enterprise-wide integration across infrastructure, security, and service management that this dissertation examines.

Other researchers have probed adjacent territory without directly addressing enterprise IT. One MBSE methodology for critical infrastructure vulnerability assessment and decision analysis used DoDAF-based modeling to link regulatory requirements, system architecture, and attack vectors (Huff et al., 2019). That work demonstrates MBSE application to infrastructure protection and security assessment beyond traditional platform engineering but does not extend to enterprise IT infrastructure management. A UML-based MBSE security profile conforming to ISO/IEC 27001 found, through a feasibility survey of ten engineering companies, that standard SysML and popular MBSE methods inadequately address security aspects (Mažeika & Butleris, 2020). This finding identifies a tool gap that would impede adoption even if organizational and perception barriers were overcome: the modeling languages themselves require extension to accommodate the security constructs that enterprise IT and Information Assurance demand. A further proposal, SysML-SecA, combined SysML with security analysis techniques for integrated threat modeling (Apvrille & Roudier, 2015). These preliminary investigations establish that researchers have recognized the potential of MBSE for security applications while simultaneously documenting the technical, methodological, and empirical gaps that separate potential from realization.

Nguyen et al. conducted a systematic mapping of forty-eight primary studies on model-based security engineering for cyber-physical systems (Nguyen et al., 2017). The study found that most research uses domain-specific languages or UML, focuses on early lifecycle stages, and addresses threats, attacks, and vulnerabilities generically. This mapping reveals a field in its formative stages, one where foundational concepts are being established but where mature, validated methods for operational security engineering have

not yet emerged. The gap between model-based security engineering research and operational Information Assurance practice remains substantial.

In the years since that mapping, model-based security engineering has expanded substantially, though still within engineered platform and cyber-physical contexts. Mitola and Prys extend the Digital Engineering framework into a repeatable cyber-hardening process, applying functional mission analysis and zero-trust principles to harden systems against advanced persistent threats (Mitola & Prys, 2024). Gampel and Eveleigh employ Model-Based Systems Engineering to streamline NIST Risk Management Framework control implementation and the authorization-to-operate decision for industrial control systems (Gampel & Eveleigh, 2025), while Larsen et al. and Nolte et al. pull threat modeling and identifying security requirements forward into the early model-based design of cyber-physical systems (Larsen et al., 2024; Nolte et al., 2025). Extending the assurance-case lineage traced earlier, Mohamad et al. present an asset-driven method for constructing security assurance cases aligned to industry standards (Mohamad et al., 2023). These contributions confirm that model-based security engineering continues to advance actively. Yet each remains anchored in engineered platforms, spacecraft, vehicles, and industrial control systems, rather than in enterprise IT infrastructure and Information Assurance programs, which leaves the specific gap this dissertation addresses intact.

Academic literature on MBSE applications thus presents a landscape of expanding interest constrained by limited evidence. Researchers have begun exploring applications to security domains, infrastructure protection, healthcare systems, and compliance frameworks. Authoritative institutions have produced conceptual frameworks demonstrating technical feasibility. Yet the enterprise IT context remains absent from the research, even though organizations in that context manage heterogeneous infrastructure, maintain compliance across multiple regulatory frameworks, and coordinate service delivery across organizational silos. The frameworks exist; the methods have matured; the tools have proliferated. The cybersecurity skills gap analysis by Ramezan, examining 935

cybersecurity positions across nine sub-fields, underscores that cybersecurity professionals operate within specialized roles that share no curricular or experiential overlap with systems engineering practice (Ramezan, 2023). The research community has not bridged this gap, and the professional communities have not demanded it.

Digital Twins as Emerging Security Tools

Digital twin technology has generated substantial research interest in cybersecurity applications, creating a parallel track to MBSE-based security engineering that the literature has not yet integrated into a coherent Digital Engineering narrative. The growing body of research on digital twins for security purposes is both an opportunity and an analytical challenge. It demonstrates that Digital Engineering concepts are penetrating security domains, but it does so through a technology-specific lens. That lens misses the integrated approach, combining MBSE, digital threads, digital twins, and PLM, that characterizes Digital Engineering as a discipline.

Grieves traces the digital twin concept from its Product Lifecycle Management origins through contemporary applications, positioning the digital twin as integrating physical and virtual systems to enable analysis, optimization, and prediction (Grieves, 2014). The foundational distinction between digital twins and traditional simulation lies in synchronization: digital twins maintain continuous data exchange with their physical or logical counterparts, enabling operational decision-making based on current rather than documented system states. An influential early framework for leveraging digital twins in systems engineering established the conceptual architecture that subsequent security applications adapted (Madni & Sievers, 2019).

Digital twin security research has matured enough to sustain systematic reviews and comprehensive surveys in recent years. The seminal survey classified security threats across five digital twin components and established the analytical framework that subsequent research has adopted (Alcaraz & Lopez, 2022). Its component-level threat taxonomy and systematic coverage of the field make it the reference assessment of the digital twin security

landscape. The systematic literature review by El-Hajj et al. analyzed sixty-seven papers published between 2018 and 2023 examining how digital twins enhance security in Industry 4.0 applications (El-Hajj et al., 2024). Covering intrusion detection, vulnerability assessment, cyber range simulation, and threat intelligence applications, the review identified enabling technologies, including machine learning, blockchain, and 5G, used in conjunction with digital twins for security purposes. A comprehensive review extended this analysis, categorizing digital twin cybersecurity studies by technique type and by digital twin level (component, process, asset, system, and network-of-systems) and assessing risk levels from medium to very high depending on twin type and industry sector (Alhamam et al., 2025). A three-dimensional security framework then mapped security risks across digital twin architectural layers and application domains, providing a structured taxonomy for understanding where security challenges concentrate (Jeremiah et al., 2024). A further survey of digital twin modeling techniques classified security operation modes into analytics, simulation, and replication, and reviewed both open-source tools and commercial platforms (Qureshi et al., 2025). These reviews differ in taxonomy and scope, yet they converge on two conclusions: digital twin security applications have achieved sufficient research volume to warrant meta-analysis, and the field's attention remains concentrated on industrial and cyber-physical settings rather than enterprise IT.

Within this expanding literature, several streams bear directly on Information Assurance and enterprise IT applications, and together they carry digital twins across the security lifecycle from offline testing to operational risk assessment. Virtual replicas can support security testing and analysis without affecting operational systems (Eckhart & Ekelhart, 2019), train Security Operations Center analysts through a digital twin-based cyber range (Vielberth et al., 2021), and, at the organizational level, help enterprises understand and manage their security postures (Dietz & Pernul, 2020). The synchronization that gives twins this reach cuts both ways. Twins can enhance security while simultaneously introducing new attack surfaces (Karaarslan & Babiker, 2021), and

continuous synchronization between virtual and physical representations enables dynamic threat modeling across the cyber-physical system lifecycle (Erceylan et al., 2025). Progress nonetheless remains bounded. A systematic review of twenty-seven articles on digital twins for incident detection and response in critical infrastructure identified persistent gaps in scalability, interoperability with legacy systems, and real-world validation (Kampourakis et al., 2025). These gaps, particularly interoperability with legacy systems and validation in operational environments, represent precisely the challenges that enterprise IT organizations would confront in digital twin adoption.

Standards development further indicates maturing technology readiness. ISO 23247 and IEC 62832 provide standards for digital twin frameworks (Shao, 2021), and subsequent analysis detailed ISO 23247's four-part structure (Shao et al., 2023). The Internet Engineering Task Force has published a draft reference architecture for network infrastructure digital twins (IETF Network Management Research Group, 2024). These standardization efforts establish interoperability requirements that reduce vendor lock-in concerns and enable integration across implementations.

NIST's engagement with digital twin technology illuminates the current state of institutional thinking. NIST Internal Report 8356 addresses cybersecurity challenges and trust considerations for digital twin implementations (Voas et al., 2025). Notably, this publication addresses security considerations *for* systems employing digital twins rather than digital twin applications *to* Information Assurance. The distinction is revealing: NIST examines how to secure digital twin implementations rather than how digital twins might enhance security posture visibility or compliance verification. This framing reflects the current state of institutional research: digital twins are treated as systems to be secured rather than as tools for improving security operations.

Security Risks of Improperly Secured Digital Twins

The security of the digital twin itself is directly germane to the enterprise Information Assurance focus of this dissertation rather than a detour into engineering

security. A digital twin is itself a critical information repository: an authoritative, continuously synchronized representation of the very systems Information Assurance exists to protect. Like any such repository (the configuration management database is the familiar enterprise example), it must be secured for confidentiality, integrity, and availability. A twin that mirrors production faithfully is a high-value asset squarely within the Information Assurance remit rather than adjacent to it. Its protection is therefore part of the discipline this dissertation addresses, which is why the risks below matter to the argument.

While the preceding discussion examines digital twins as security-enhancing tools, the literature reveals an equally important and less frequently addressed concern: improperly secured digital twins themselves become attack vectors that adversaries can exploit. Because a digital twin faithfully replicates the architecture, configuration, and behavioral characteristics of its physical or logical counterpart, unauthorized access grants an attacker a comprehensive reconnaissance platform. That platform reveals system vulnerabilities, operational patterns, and interdependencies without triggering alerts on the production environment. Framing this danger through the concept of the "evil digital twin," Suhail et al. warned that adversaries who compromise or construct malicious virtual replicas can use them to rehearse attacks, identify exploitable weaknesses, and craft intrusion strategies with a precision that traditional reconnaissance cannot match (Suhail et al., 2024). Their analysis identified data poisoning, model manipulation, and feedback loop corruption as three vectors through which compromised twins propagate harm back to physical systems.

Empirical threat analysis corroborates these concerns. Alcaraz and Lopez, in the most widely cited survey of digital twin security threats, classified attacks across five architectural components: physical entity, communication channel, data storage, virtual model, and application layer. Each component introduces distinct exploitation opportunities (Alcaraz & Lopez, 2022). The resulting taxonomy established that

synchronization mechanisms, while essential for twin fidelity, simultaneously create bidirectional pathways through which adversaries can inject false data into the physical system or exfiltrate sensitive operational intelligence from the virtual replica. Jeremiah et al. extended this analysis through a three-dimensional security framework mapping risks across digital twin architectural layers and application domains, finding that risk severity escalates from medium to very high as twin fidelity and system-of-systems integration increase (Jeremiah et al., 2024). Higher-fidelity twins, precisely because they model systems more accurately, provide attackers with more actionable intelligence when compromised.

Holmes et al. posed the question directly: are digital twins a cybersecurity solution or a cybersecurity challenge? Their analysis concluded that the answer depends on implementation rigor (Holmes et al., 2021). Organizations that deploy digital twins without embedding security controls throughout the twin lifecycle, from design through data ingestion to model synchronization, create shadow representations of critical infrastructure that bypass the security controls protecting the physical systems themselves. Karaarslan and Babiker cataloged specific countermeasures including authentication protocols for twin-physical communication channels, encryption of twin state data, and integrity verification mechanisms for synchronization updates (Karaarslan & Babiker, 2021). Yet their review found that many implementations prioritize functional fidelity over security hardening, leaving twins exposed to manipulation.

Suhail et al. proposed ENIGMA, an explainable digital twin security solution for cyber-physical systems, specifically addressing the challenge of detecting when a digital twin has been subverted (Suhail et al., 2023). Their framework employs explainable artificial intelligence to identify anomalous divergences between twin predictions and physical system behavior, divergences that may indicate adversarial manipulation of the twin model. Jiang et al. demonstrated complementary capabilities in a cyber-physical production systems context, developing a digital twin security model that enhances asset visibility and supports vulnerability prioritization through virtual testing, while

simultaneously documenting forty-one vulnerability instances across fourteen critical components that twin-based analysis revealed (Jiang et al., 2024).

For Information Assurance practice, these exploitation risks carry particular weight. Organizations contemplating digital twins for compliance verification or security posture management must recognize that the twin itself becomes a high-value target requiring its own security controls, authorization boundaries, and continuous monitoring. These are precisely the governance mechanisms that the RMF prescribes for any information system processing sensitive data. NIST Internal Report 8356 acknowledges this recursive security challenge, establishing trust considerations for digital twin deployments that include data integrity verification, access control requirements, and lifecycle security management (Voas et al., 2025). An unresolved tension persists in the literature: researchers advocate digital twins as tools for improving security visibility while simultaneously documenting that inadequately secured twins expand the very attack surface they are intended to reduce. Resolving this tension requires integrating digital twin security within the broader Digital Engineering framework rather than treating twin security as an isolated concern. Such integration connects twin security to digital thread traceability, MBSE-based security architecture, and PLM lifecycle governance. This integrated approach has yet to be examined in an enterprise IT setting.

Academic research on open source digital twin frameworks adds another dimension to the adoption question. Gil et al. conducted a systematic survey of open source digital twin frameworks, analyzing fourteen frameworks against criteria derived from ISO 23247 standards and finding substantial variation in maturity, documentation quality, and community support (Gil et al., 2024). Autiosalo et al. introduced Twinbase, an open source server for the Digital Twin Web concept (Autiosalo et al., 2021). These open source options reduce economic barriers to adoption but do not resolve the more fundamental challenge: digital twin applications for enterprise IT contexts lack the academic research and validated methods that would guide adoption decisions.

Digital twin security research thus forms a field developing in parallel with but largely disconnected from the broader Digital Engineering discourse. Researchers explore digital twins as security tools without embedding their work within the Digital Engineering framework that integrates MBSE, digital threads, and PLM into a coherent discipline. This disconnection means that researchers investigate individual digital twin security capabilities in isolation rather than as components of the integrated approach that defense and aerospace organizations employ. Whether integrating digital twin security capabilities within a comprehensive Digital Engineering framework would produce greater value than isolated implementations remains an open question, one that the absence of enterprise IT research leaves unaddressed.

The Compliance Imperative and Its Unfulfilled Requirements

Compliance frameworks create requirements that Digital Engineering could address, yet no research identified in this review examines Digital Engineering approaches, as an integrated enterprise discipline, to satisfying these requirements. This disconnect between what frameworks demand and what current practices deliver is one of the most compelling arguments for investigating Digital Engineering application to Information Assurance.

The NIST Risk Management Framework, documented in Special Publication 800-37 Revision 2, provides the authoritative approach to managing security and privacy risk for federal information systems through seven iterative steps: prepare, categorize, select, implement, assess, authorize, and monitor (Joint Task Force Transformation Initiative, 2018). The RMF explicitly requires enterprise architecture integration during the prepare step. Yet compliance with this requirement assumes capabilities that organizations frequently lack: the ability to maintain accurate, current documentation of enterprise architecture that reflects operational reality. NIST Special Publication 800-53 Revision 5 compounds these demands through specific controls requiring enterprise architecture capabilities: PL-2 requires security plans consistent with enterprise architecture; PL-8 requires security architecture development; PM-7 establishes enterprise architecture

requirements; CM-2 requires documented baselines; CM-8 requires accurate system component inventory; SA-17 requires design specifications consistent with enterprise architecture (Ross et al., 2020). Each control names, in the language of compliance, a capability that Digital Engineering already supplies in its native domain. The mapping between the two vocabularies is what remains undrawn.

The Committee on National Security Systems Instruction 1253 extends these requirements to national security systems, where documentation and visibility challenges compound under additional constraints (Committee on National Security Systems, 2022). Organizations operating outside federal requirements may employ ISO/IEC 27001:2022 for information security management (International Organization for Standardization, 2022). Mažeika and Butleris found that standard MBSE methods inadequately address ISO 27001 security requirements, identifying a specific gap between what compliance frameworks demand and what current modeling approaches provide (Mažeika & Butleris, 2020). These alternative frameworks share a common characteristic with NIST guidance: they assume documentation accuracy and visibility capabilities that organizations struggle to maintain.

NIST's systems security engineering publications establish principles that align with Digital Engineering's integrated approach. Special Publication 800-160 Volume 1 Revision 1 describes a basis for engineering trustworthy secure systems (Ross et al., 2022). Special Publication 800-160 Volume 2 Revision 1 addresses cyber resiliency considerations (Ross et al., 2021b). These publications reference model-based approaches and traceability requirements without providing specific implementation guidance for Digital Engineering in enterprise IT contexts. The publications establish requirements that Digital Engineering could address: traceability between security requirements, design decisions, and implementation artifacts; documentation that maintains currency throughout system lifecycles; and visibility into system configurations and relationships. Even so, they do not explicitly connect these requirements to Digital Engineering solutions.

Emerging research on compliance automation demonstrates that the academic

community has recognized the inadequacy of manual compliance approaches, even if this recognition has not yet connected to the Digital Engineering discourse. Over the past decade this work has coalesced into a coherent research area that treats automated, continuous verification as both feasible and necessary. Santilli defined continuous compliance and identified thirteen requirements for automated, ongoing verification from an industrial perspective (Santilli, 2023), laying intellectual foundations for a field that did not exist as a coherent research area a decade ago. Angermeir et al. extended that foundation with a design science methodology for continuous security compliance in highly regulated domains, showing that automated compliance is both technically feasible and organizationally necessary as system complexity outpaces manual verification capacity (Angermeir et al., 2024). Castellanos Ardila et al. then mapped the landscape through a systematic literature review of compliance checking approaches for software processes, cataloging automated verification methods and identifying research directions (Castellanos Ardila et al., 2022).

A semantically rich knowledge graph captured regulations from GDPR, PCI DSS, ISO 27001, NIST 800-53, and CSA CCM, enabling automated compliance checking for cloud services (K. P. Joshi et al., 2020). Validated against the privacy policies of major cloud providers, that work shows that model-based approaches to compliance are technically feasible and practically valuable. The Cloud Property Graph bridged static code analysis and runtime security assessment, using an ontology of cloud resources to identify misconfigurations and regulatory non-compliance across multi-vendor cloud deployments (Banse et al., 2021). A UML-based model for cross-standard security requirements mapped explicitly to NIST's Open Security Controls Assessment Language (OSCAL) Catalog Model, letting organizations track compliance across multiple frameworks simultaneously (Stojkov et al., 2021). Most recently, attack graphs have been combined with OSCAL under compliance-as-code principles to automate risk assessment (Koufos et al., 2025). These efforts differ in formalism, spanning knowledge

graphs, property graphs, and UML models, yet they share one architectural commitment: encoding regulatory requirements in machine-readable form so that compliance verification becomes an automated computation rather than a manual clerical task.

The DevSecOps literature provides additional context for this compliance discussion. A systematic review of fifty-four peer-reviewed studies identified twenty-one challenges and thirty-one solutions in DevSecOps adoption, classified into People, Practices, Tools, and Infrastructure themes (Rajapakse et al., 2022). A related analysis identified eighteen challenges mapped to ten core categories, finding that standards exert the most decisive influence on adoption outcomes (Akbar et al., 2022). A multi-vocal literature review identified the primary dimensions of DevSecOps, providing a comprehensive mapping of the field's scope and boundaries (Zhao et al., 2024). Earlier, Myrbakken and Colomo-Palacios provided one of the first systematic examinations of DevSecOps, defining the field and characterizing the challenges of integrating security into DevOps practices (Myrbakken & Colomo-Palacios, 2017). These DevSecOps studies establish the problem space of integrating security into development and operational pipelines without connecting to the Digital Engineering framework that could provide structured methodology for that integration.

NIST's own OSCAL initiative represents institutional recognition that manual documentation approaches cannot sustain the accuracy and currency that compliance requires (National Institute of Standards and Technology, 2023). OSCAL provides standardized machine-readable formats for expressing security control catalogs, baselines, profiles, and assessment results. The initiative aligns with Digital Engineering's emphasis on machine-readable documentation, and digital thread traceability could connect OSCAL compliance documentation to underlying system configurations, enabling automated verification that documented controls exist as implemented. Yet no academic study has investigated this integration. A notable finding from the literature search is that OSCAL has achieved strong government adoption but minimal peer-reviewed academic treatment,

a gap that reflects the broader disconnect between compliance practice and academic research.

The compliance automation research and the Digital Engineering literature thus develop along parallel tracks that have not converged. Compliance researchers build knowledge graphs, property graphs, and domain-specific languages to automate regulatory assessment. Digital Engineering researchers develop model-based approaches, digital threads, and authoritative sources of truth for complex system management. Both communities address documentation accuracy, traceability, and automated verification. Neither community has systematically explored how integrating their approaches might produce capabilities exceeding what either achieves independently. This unexplored intersection is a substantial research opportunity that the present investigation begins to address by establishing whether the professionals who must implement compliance, namely IT and Information Assurance practitioners, recognize value in Digital Engineering capabilities.

A dimension of the compliance imperative that the literature largely leaves implicit is its uneven distribution across organizations. The frameworks above impose the same documentation, traceability, and verification obligations on a community health center, a tribal government, or an under-resourced school district as on a defense prime contractor. Yet the manual labor and specialized expertise those obligations demand are a far heavier proportional burden where staff and budgets are smallest. The compliance burden is, in effect, regressive, and the manual document-centric practices that dominate today entrench that asymmetry. Research commissioned by the United States Small Business Administration documents the pattern across regulation generally: firms with fewer than 20 employees bear an annual federal regulatory cost of $6,975 per employee, nearly 60% more than the $4,463 per employee borne by firms with more than 500 employees (Crain & Hopkins, 2001). The pattern extends to cybersecurity compliance specifically. The Government Accountability Office found that concerns about the complexity and cost of

CMMC requirements, including third-party assessments estimated at $4,042 to $117,768, concentrate among the small businesses that constitute roughly three quarters of the defense industrial base; some of these businesses may stop working with the Department of War rather than absorb those costs (U.S. Government Accountability Office, 2026). The few studies that carry model-based methods into resource-limited public-service settings are therefore notable: Berg et al. applied Model-Based Systems Engineering to rural-healthcare disaster planning, demonstrating that the approach is not confined to well-funded enterprises and can structure complex requirements in exactly the settings where capacity is scarcest (Berg et al., 2023). If Digital Engineering can substitute reusable model structure for the scarce specialized labor that compliance currently requires, its largest marginal benefit would accrue to small and under-resourced organizations. The equitable distribution of compliance capability thus becomes a substantive, and largely unexamined, stake in whether these methods transfer beyond their sector of origin.

IT Service Management: Structural Failures Despite Mature Frameworks

IT Service Management frameworks assume capabilities that current practices cannot sustain. ITIL provides comprehensive guidance for aligning IT services with business needs through structured processes, yet its effectiveness depends on the accuracy and currency of underlying configuration information, an accuracy that organizations consistently fail to achieve (Cannon, 2013). Version 4 reorganized service management practices and introduced the Service Value System concept. It acknowledges that tracking configurations across virtual systems, cloud computing, and cybersecurity domains presents challenges, yet it provides limited guidance on the documentation accuracy problems that undermine every ITIL process (AXELOS Limited, 2019).

Empirical evidence for ITIL implementation challenges spans two decades of research yet reveals persistent rather than resolving problems. Resistance to change ranks as the top ITIL implementation challenge, at 27% (Cook et al., 2021). A survey of 491 firms found that while over 90% use ITSM frameworks, little research examines the

benefits actually realized (Marrone & Kolbe, 2011). An empirical investigation of ITIL's role in IT governance found that implementation success depends on group efficacy and organizational resources, and that ITIL enables governance processes but "lacks structural and relational governance practices" (Iden & Eikebrokk, 2014). This finding carries particular weight for the present research: if ITIL provides process governance but lacks structural governance, and if Digital Engineering provides structural governance through model-based system representations, then integrating these approaches addresses a documented deficiency rather than introducing redundant capability.

A. Joshi et al. studied 881 companies worldwide and found that firms with superior IT governance processes achieve more than 25% higher profits than peer firms with poor governance (A. Joshi et al., 2022). This finding quantifies the business impact of governance failures, the very failures that ITSM frameworks were designed to prevent but have not resolved. The parallel with MBSE evidence is striking: both MBSE and ITIL are widely adopted based on perceived rather than measured value, and both face adoption challenges rooted in organizational factors rather than technical limitations. Neither literature, however, has put the obvious question to the other: whether the structural governance that model-based representation supplies could repair what ITIL's process governance leaves undone.

Configuration Management Database failures provide the most extensively documented evidence of structural inadequacy, though the headline figures originate from industry-analyst reports rather than peer-reviewed research. That gap itself reflects the practitioner-driven evolution of ITSM documented in the disciplinary history section of this chapter. Gartner reports an 80% failure rate for CMDB implementations (Gartner, 2019), and Gartner further predicts that 99% of organizations using CMDB tooling without addressing data quality gaps will experience visible business disruption (Gartner, 2020). Forrester research finds that less than half of organizations trust the data in their CMDB (Forrester Research, 2020). Peer-reviewed research corroborates these industry

findings: Hauder et al. found through a study of 123 enterprise architecture practitioners that manual documentation processes cannot maintain accuracy in dynamic environments (Hauder et al., 2012).

Forrester analysis concludes that the CMDB approach itself has failed (Betz, 2025). After decades of implementation attempts across organizations, CMDBs consistently fail to deliver intended value. The analysis attributes failures to structural issues: involving process experts rather than data management professionals, manual data entry that cannot maintain accuracy, and scope creep that renders CMDBs unmanageable. Forrester's research on Application and Infrastructure Dependency Mapping found that 56% of enterprises report incomplete views of dependencies between applications and underlying infrastructure (Forrester Research, 2018). This assessment suggests that incremental CMDB improvements cannot resolve inherent approach limitations. Such a conclusion opens conceptual space for model-based alternatives that maintain accuracy through automated synchronization rather than manual data entry.

Shadow IT compounds the documentation challenge. Gartner research indicates 41% of employees used shadow IT in 2022, expected to climb to 75% by 2027, with 30–40% of large companies' IT expenditure representing shadow IT (Gartner, 2022). The foundational definition of shadow IT cites industry estimates that its true extent is roughly ten times what CIOs suspect, and identifies key risk factors including security vulnerabilities, compliance violations, and loss of IT control (Haag & Eckhardt, 2017). A systematic review of 126 publications documented how silos between business units and IT departments drive shadow IT, creating security risks, data inconsistency, and compliance violations (Klotz et al., 2019). Systems deployed outside formal governance remain invisible to IT operations and security teams, creating unknown attack surfaces that no documentation process can capture. Examining organizational dynamics when hidden IT systems are discovered reveals interdepartmental conflicts and governance failures that perpetuate documentation gaps (Fürstenau et al., 2021). The shadow IT phenomenon

reflects a structural mismatch between IT governance and organizational needs: when official IT processes cannot meet user requirements quickly enough, users acquire solutions independently, creating operational dependencies that official documentation does not capture.

Change management effectiveness depends directly on documentation accuracy. Industry analysis confirms that reliance on outdated documentation leads to inaccurate impact assessments (Freshworks, 2025). Research by Bokan and Santos highlights difficulties organizations encounter in maintaining comprehensive security oversight (Bokan & Santos, 2021). Change management depends on accurate understanding of system relationships (International Organization for Standardization, 2018b) that current documentation approaches cannot maintain. Each change approved against inaccurate documentation carries incident risk. When impact assessments miss dependencies that exist in operational systems, changes cause unintended effects that consume resources, damage trust in change processes, and create pressure for emergency changes that further degrade documentation accuracy. The resulting cycle is self-reinforcing and resistant to process improvement within the document-centric paradigm.

Recent applications confirm that Model-Based Systems Engineering continues to expand into new and demanding complex-systems problems. Sun et al., for instance, developed an interactive visualization-based MBSE technique for requirements analysis and test design in the joint testing of complex systems, reporting gains in the accuracy and dependability of the resulting test plans (Sun et al., 2025). Within the enterprise domain specifically, prior research by Bonar and Hastings presented a preliminary reference model in which Digital Engineering practices support compliance verification (Bonar & Hastings, 2024). Yet no peer-reviewed study has directly examined integrating Model-Based Systems Engineering with IT Service Management. That absence is consequential; it is the precise gap this dissertation confronts.

Enterprise Visibility: Empirical Evidence of Systemic Failure

The challenges documented in the preceding sections share common roots in the inability of organizations to maintain accurate, current documentation of their enterprise information systems. This section synthesizes peer-reviewed research and industry analysis to establish that these challenges reflect systemic patterns rather than isolated organizational deficiencies, and to provide the empirical foundation for evaluating Digital Engineering as a potential disciplinary response.

Independent studies converge on a single finding: organizations cannot see large portions of their own IT environments, and the blind spots concentrate in the cloud. Monitoring coverage is incomplete at the outset. One global study found that organizations can monitor only 66% of their IT environments, leaving blind spots particularly in cloud deployments (IDC & Exabeam, 2023), and 63% of security teams lack visibility and control into all user device activity connected to their infrastructure (Ponemon Institute, 2023). Practitioner confidence tracks that shortfall. Only 15% of respondents to the SANS Institute SOC Survey expressed very high confidence that all devices on their network are discoverable (SANS Institute, 2023). The gaps then compound across organizational boundaries, because 55% of organizations maintain security and IT data silos and 62% report that those silos slow security response times (Ivanti, 2025). Cloud infrastructure concentrates the exposure most sharply. Cloud-related breaches struck 95% of organizations in the preceding eighteen months (Cloud Security Alliance, 2024), 82% of enterprises experienced security incidents due to cloud misconfigurations, and 67% struggle with limited visibility into cloud infrastructure (Check Point Software Technologies & Cybersecurity Insiders, 2024). The scale of the unseen surface is itself unknown. Organizations use 15–22 times more cloud services than their IT departments estimate (Cisco Systems, Inc., 2016), and Gartner predicted that through 2025, 99% of cloud security failures would be the customer's fault (Panetta, 2019).

Peer-reviewed research provides empirical grounding for these industry findings. An

empirical study of configuration errors in commercial and open source systems found that 70–85% of misconfigurations result from mistakes in setting configuration parameters and that 22–57% involve configurations external to the examined system (Yin et al., 2011). NIST Special Publication 800-128 defines configuration drift as systems deviating from baseline configurations over time through manual interventions, software updates, and environmental factors (National Institute of Standards and Technology, 2019). The Uptime Institute's Annual Outage Analysis confirms that 64% of IT system outages occurred because of configuration or change management issues (Uptime Institute, 2023). The IT Process Institute established that 80% of unplanned outages result from ill-planned changes made by administrators or developers (IT Process Institute, 2004). Proper dependency documentation would have flagged many such changes.

Breach detection times serve as proxy measures for organizational visibility. IBM Security and Ponemon Institute report that the mean time to identify breaches reached 204 days, with breaches involving lifecycles exceeding 200 days costing an average of $5.46 million (IBM Security & Ponemon Institute, 2024). The 2025 report found that 35% of breaches involved shadow data, defined as information stored in unmanaged locations, and 40% involved data stored across multiple environments that organizations struggle to inventory comprehensively (IBM Security & Ponemon Institute, 2025). These findings demonstrate that visibility failures directly impact security outcomes in measurable economic terms.

Theoretical and empirical frameworks help explain why these failures persist, and the enterprise architecture literature shows the same failure mechanisms recurring across independent studies. Complexity-driven degradation is the first mechanism: through PLS-SEM analysis of 249 information systems managers, Beese et al. demonstrated that IS architecture complexity significantly reduces efficiency, flexibility, transparency, and predictability, and that organizations without adequate enterprise architecture management face compounding architectural degradation (Beese et al., 2023). Unvalidated

framework prescriptions constitute a second mechanism: an examination of twenty-seven organizations by Kotusev found that prescribed enterprise architecture artifact lists from frameworks like TOGAF were never empirically validated and do not reflect actual practice, documenting a fundamental gap between what frameworks recommend and what organizations find useful (Kotusev, 2019). Stakeholder-engagement inhibitors form a third: Kurnia et al. identified organizational silos, political barriers, and poor cross-team communication as inhibitors that directly undermine documentation quality and IT governance effectiveness (Kurnia et al., 2021). In a scoping review of forty studies on organizational silos, Bento et al. identified five conceptualizations and characterized silo mentality as an absence of systems thinking and organizational vision (Bento et al., 2020). Documentation challenges cut across all three mechanisms: Brée and Karger conducted a systematic literature review organizing enterprise architecture management challenges into six dimensions, with documentation challenges including dearth of automated tools, immature documentation models, and insufficient emphasis on forward-looking documentation (Brée & Karger, 2022).

A complementary stream examines the pathways through which enterprise architecture delivers value when it succeeds, and it reaches equally sobering conclusions. Tamm et al., drawing on twenty-two expert interviews and two in-depth case studies, investigated how enterprise architecture leads to organizational benefits, observing that the mechanisms underpinning EA benefit realization remain "not fully understood" (Tamm et al., 2022). van de Wetering—in motivating a survey of 414 senior practitioners, business and IT managers, and executives—observes that despite high expectations "many EA initiatives in practice fail or do not live up to stakeholder expectations," reinforcing the pattern of enterprise architecture as a discipline with acknowledged value but unreliable execution (van de Wetering, 2022). A fuzzy-set qualitative comparative analysis investigated how enterprise architecture management contributes to digital transformation capabilities, identifying configurational paths to effectiveness that require combinations of

factors rather than any single intervention (Pattij et al., 2022).

Complexity theory provides an explanatory framework for why traditional documentation approaches fail in modern enterprise environments. Complex Adaptive Systems theory, applied to information systems development, yields seven first principles of adaptive success and the argument that if complexity is not managed appropriately, information systems fail (Benbya & McKelvey, 2006). Later work demonstrated that enterprise information systems have reached complexity levels exceeding prior technological generations (Benbya et al., 2020). Enterprise IT environments exhibit characteristics of complex adaptive systems, including many interconnected components, emergent behaviors arising from component interactions, continuous change, and unpredictable responses to interventions. Static documentation approaches assume that these characteristics remain stable between documentation updates.

Technical debt research adds a final perspective, and across primary studies, mapping studies, and secondary syntheses it converges on treating documentation debt as a distinct, recurring category rather than an incidental concern. One study specifically addressed documentation technical debt, defined as problems concerning non-existent, inadequate, or incomplete software project documentation (Mendes et al., 2019). A systematic mapping study by Li et al., covering ninety-four studies, established technical debt taxonomy and management frameworks that include documentation debt as a distinct category (Li et al., 2015). Across nineteen secondary studies, Junior and Travassos consolidated perspectives on technical debt (Junior & Travassos, 2022), and Kleinwaks extended technical debt concepts to systems engineering contexts (Kleinwaks, 2023). Documentation technical debt accumulates when organizations defer documentation updates to prioritize operational activities. Unlike code technical debt, documentation debt remains invisible until documentation is needed for change impact assessment, incident response, or compliance verification. At that point, the accumulated debt imposes costs far exceeding the original deferral savings.

Table 1 summarizes the evidence documenting enterprise visibility and documentation failures across peer-reviewed and industry research sources.

Table 1: Enterprise visibility and documentation failure evidence.

Finding	**Statistic**	**Source**
Visibility Gap Metrics		
IT environment monitorable	66%	(IDC & Exabeam, 2023)
Security teams lacking device visibility	63%	(Ponemon Institute, 2023)
High confidence in device discovery	15%	(SANS Institute, 2023)
Organizations with security/IT silos	55%	(Ivanti, 2025)
Configuration Management Failures		
CMDB implementation failure rate	80%	(Gartner, 2019, 2020)
Outages from configuration issues	64%	(Uptime Institute, 2023)
Misconfigurations from parameter errors	70–85%	(Yin et al., 2011)
Unplanned outages from ill-planned changes	80%	(IT Process Institute, 2004)
Shadow IT and Undocumented Assets		
Shadow IT as percentage of IT spend	30–40%	(Gartner, 2022)
Cisco cloud services vs. IT estimates	15–22x higher	(Cisco Systems, Inc., 2016)
Employees using shadow IT (2022)	41%	(Gartner, 2022)
Projected shadow IT usage (2027)	75%	(Gartner, 2022)
Security Impact Metrics		
Mean time to identify breach	204 days	(IBM Security & Ponemon Institute, 2024)
Cloud breaches from misconfigurations	82%	(Check Point Software Technologies & Cybersecurity Insiders, 2024)
Organizations with cloud breaches (18 mo)	95%	(Cloud Security Alliance, 2024)
Cloud security failures attributed to customer fault (through 2025)	99%	(Panetta, 2019)

Note. The 80% CMDB implementation failure rate derives from industry-analyst (Gartner) reports rather than peer-reviewed research and should be read as practitioner-oriented estimation rather than empirically validated measurement.

Peer-reviewed empirical research and industry analysis converge to establish that enterprise visibility and documentation failures represent a systemic challenge rather than isolated organizational deficiencies. Traditional documentation approaches—including manual configuration tracking, periodic documentation updates, and static architecture diagrams—cannot maintain accuracy in environments characterized by continuous change, complex dependencies, and organizational silos.

The evidentiary sources in this section warrant a methodological note. The enterprise visibility evidence draws substantially on industry analyst reports and practitioner surveys alongside peer-reviewed academic research. This reliance on industry gray literature reflects a characteristic of the research domain rather than an analytical preference: the operational challenges of enterprise IT documentation and visibility have received considerably more attention from industry analysts and practitioner communities than from academic researchers. The relative scarcity of peer-reviewed empirical studies quantifying CMDB failure rates, shadow IT prevalence, and breach detection timelines itself constitutes evidence of the research gap this dissertation addresses. Peer-reviewed evidence corroborates industry findings in several instances, as with the work of Hauder et al. on documentation accuracy, Mendes et al. on documentation technical debt, Beese et al. on architecture complexity, Tamm et al. on enterprise architecture failures, and Kotusev on enterprise architecture practice. Where such corroboration occurs, the convergence strengthens confidence in the broader pattern. This review presents the industry evidence not as a substitute for academic rigor but as the best available evidence for phenomena that academic research has not yet systematically investigated.

Research Gaps and Theoretical Framework

The structured review of the literature reveals a pattern: frameworks and compliance requirements assume documentation and visibility capabilities that organizations frequently lack, while Digital Engineering methods that could address these capabilities remain confined within a discipline that enterprise IT and Information Assurance have not engaged. This section synthesizes the findings into a theoretical framework while documenting the research gaps that this investigation begins to address.

The academic research gap is pronounced. Systematic literature reviews examining MBSE report no published research, within the scope of their searches, addressing enterprise IT infrastructure or Information Assurance applications as an integrated enterprise discipline, as distinct from MBSE applied to individual engineered systems or to

single compliance controls, beyond the preliminary reference model by Bonar and Hastings (2024) and the conceptual DevSecOps framework by Chick et al. (2023). This gap persists despite explicit requirements in compliance frameworks for capabilities that Digital Engineering provides. As the disciplinary history presented in this chapter demonstrates, the gap cannot be attributed to Digital Engineering immaturity, given that defense and aerospace have employed Digital Engineering successfully for years. Nor can it be attributed to tool unavailability, since MBSE tools have existed for over a decade and PLM systems for several decades, with digital twin platforms maturing rapidly over the past ten years. Rather, the gap reflects disciplinary boundaries that formed when systems engineering, Information Assurance, cybersecurity, and IT service management each developed independent academic identities, professional communities, and research traditions. The low interconnectedness between organizational units that Henderson and Salado associate with impeded adoption operates at the disciplinary level itself (Henderson & Salado, 2024).

Standards bodies have recognized enterprise applicability of systems engineering approaches. The Unified Architecture Framework provides viewpoints applicable to enterprise IT. NIST publications require enterprise architecture capabilities for compliance. ITIL requires visibility and documentation that model-based approaches could provide. Yet academic research has not examined practical application. This standards-research disconnect leaves practitioners without empirical guidance for applying available standards to enterprise IT challenges.

Table 2 summarizes the research gaps identified across the literature domains examined in this review.

Based on the structured review of the literature, this research adopts a theoretical framework integrating Digital Engineering principles with established Information Assurance and IT Service Management practices. This framework posits that Digital Engineering is a disciplinary approach with capabilities reported in defense and aerospace

Table 2: Research gaps within corpus of knowledge.

Domain	Gap Description	Research Implication
MBSE for Enterprise IT	One study applying MBSE to enterprise IT, none for IA program management	Foundation research required
MBSE Adoption Evidence	Adoption studies limited to SE populations; no data from IT or IA professionals	Perception measurement needed in new populations
Disciplinary Integration	No research connecting SE, IA, cybersecurity, and ITSM methods despite shared challenges	Cross-disciplinary investigation needed
Digital Threads	No research on traceability for IT/IA contexts	Conceptual validation needed
Digital Twin	Growing security application research but no enterprise IT integration studies	Application studies needed
ITSM Integration	No frameworks integrating DE with ITIL	Integration research required
Compliance Automation	Model-based compliance research and DE research develop on parallel tracks without convergence	Integration research needed
Adoption Theory	Cybersecurity adoption and MBSE adoption research use same theories but never intersect	Bridging research needed
Open Source	No academic validation for enterprise IT	Evaluation research needed
Professional Perceptions	Unknown awareness and perceived value among IT/IA professionals	This research addresses

contexts that align with gaps that have persisted in enterprise IT despite decades of framework development and organizational investment. The framework does not assert that Digital Engineering will resolve these gaps in enterprise IT contexts, which remains an empirical question. Consistent with the boundary-object and knowledge-transformation lens established earlier in this chapter (Carlile, 2004; Star & Griesemer, 1989), the framework treats the structural correspondences developed below not as mere analogies but as the conditions under which Digital Engineering models could function as boundary objects. Such boundary objects would span the pragmatic knowledge boundary (Carlile,

2002, 2004) that separates systems engineering from Information Assurance and IT service management. It thereby identifies structural correspondences between demonstrated Digital Engineering capabilities and documented enterprise IT challenges, establishing the theoretical basis for investigating whether practitioners recognize them.

Persistent failures documented in IT Service Management and Information Assurance practices share common root causes that Digital Engineering practices are designed to address. Organizations struggle with documentation accuracy because traditional approaches rely on manual processes disconnected from operational systems. Organizations fail to maintain traceability because document-centric methods cannot sustain verified connections as systems evolve. Organizations lack visibility because static artifacts cannot represent dynamic system states. Digital Engineering's reported ability, measured in rare but consistent cases, to address these root causes in defense and aerospace contexts motivates investigation of whether similar benefits may transfer to enterprise IT environments.

The Department of Defense's 2018 Digital Engineering Strategy defines the authoritative source of truth as "a single source of data and models" that provides "a definitive technical baseline" for programs (Department of War, 2018). Current IT and Information Assurance practices lack authoritative sources of truth, instead maintaining multiple disconnected documentation artifacts that diverge over time. Digital threads establish and maintain authoritative traceability throughout system lifecycles, addressing the gap between security requirements, control implementations, and compliance evidence. Digital twin capabilities enable organizations to simulate system behavior, test proposed changes, and analyze scenarios without affecting production systems. Model-based approaches maintain synchronization with operational systems, providing the visibility that manual documentation cannot sustain.

The theoretical framework acknowledges several limitations. First, it extrapolates from Digital Engineering value demonstrated in aerospace and defense to anticipated value

in enterprise IT contexts. Whether benefits demonstrated for physical systems transfer to logical information systems remains unvalidated. Second, it assumes that Digital Engineering tools and methods can be adapted for enterprise IT contexts. Mažeika and Butleris documented that current MBSE methods inadequately address security requirements (Mažeika & Butleris, 2020), suggesting adaptation requirements may exceed anticipated effort. Third, the framework does not address organizational change management, workforce development, or cultural transformation requirements. The adoption barriers documented by Henderson et al.—including middle management resistance, awareness deficits, and the need for dedicated MBSE teams (Henderson et al., 2024)—would likely manifest with greater intensity in domains unfamiliar with systems engineering practices. Fourth, the framework focuses on potential benefits without comprehensive analysis of costs or implementation challenges. Cost-benefit analysis requires empirical data this research does not collect.

Chapter Summary

This chapter has examined the literature across interconnected domains relevant to applying Digital Engineering methods to Information Assurance and IT Service Management, advancing four interrelated arguments that position the present research within the academic conversation.

First, the independent evolution of systems engineering, Information Assurance, cybersecurity, and IT service management as separate academic and professional disciplines created structural barriers to cross-disciplinary methodological exchange. Systems engineering emerged from defense laboratories and engineering departments; Information Assurance grew from national security policy and computing education; cybersecurity crystallized as a meta-discipline drawing from multiple computing fields; and IT service management developed as a practitioner-driven domain with limited academic theorization. These parallel trajectories produced separate journals, conferences, curricular standards, and professional communities, institutional structures that explain the absence of research

connecting Digital Engineering to enterprise IT and Information Assurance.

Second, the evidence base for MBSE value, while compelling in individual cases such as the 18% efficiency improvement and 9% interface defect reduction documented by E. B. Rogers and Mitchell (2021), remains disproportionately reliant on perceived rather than measured outcomes. Henderson and Salado found that approximately two-thirds of claimed benefits lack empirical measurement (Henderson & Salado, 2021). Efforts to develop measurement frameworks by Henderson et al. and evaluations of perceived value by Campo et al. have advanced the methodological conversation without resolving the measurement deficit (Campo et al., 2023; Henderson et al., 2023). This evidence deficit creates rational hesitation among organizations outside the systems engineering discipline, where institutional experience cannot compensate for the measurement gap.

Third, adoption research demonstrates that perceptions govern adoption decisions more powerfully than objective technical merit. Call et al. showed that perceived complexity and compatibility barriers impede adoption even among systems engineering professionals who recognize MBSE's relative advantage (Call et al., 2024). Henderson et al. found that approximately 22% of interviewed systems engineering professionals could not clearly define MBSE (Henderson et al., 2024). Parallel research in cybersecurity contexts by Hasani et al. and Anthony demonstrates that the same adoption theories and barrier patterns operate across technology domains (Anthony, 2024; Hasani et al., 2023). If awareness deficits and perception barriers persist within the originating discipline, their magnitude among IT and Information Assurance professionals, who operate outside the systems engineering discipline entirely, demands empirical measurement rather than assumption.

Fourth, compliance frameworks and IT service management standards require capabilities that current practices consistently fail to provide, while Digital Engineering offers those capabilities in adjacent domains. The enterprise visibility evidence documents the scope of failure: 66% IT environment monitoring, the industry-reported 80% CMDB

failure rate, and 204-day average breach detection times. Enterprise architecture research by Tamm et al. and van de Wetering confirms that EA initiatives frequently fail to meet stakeholder expectations (Tamm et al., 2022; van de Wetering, 2022). The compliance automation literature, including emerging work on continuous compliance by Santilli and Angermeir et al., demonstrates growing recognition that model-based approaches can address documentation and traceability requirements (Angermeir et al., 2024; Santilli, 2023). Yet no research connects these parallel tracks by examining Digital Engineering as an integrated solution for enterprise IT and Information Assurance challenges.

The theoretical framework developed from this synthesis posits that structural correspondences between demonstrated Digital Engineering capabilities and documented enterprise IT challenges warrant investigating whether practitioners recognize these correspondences. The research contributes by investigating whether IT and Information Assurance professionals perceive value in Digital Engineering capabilities. If professionals perceive value, findings justify subsequent implementation research. If professionals do not perceive value despite documented challenges, findings challenge the theoretical premise and identify barriers the field must address before adoption can occur.

Chapter 3 presents the research methodology employed to investigate professional awareness and perceptions of Digital Engineering capabilities. The methodology uses a quantitative survey-based approach following a systems engineering lifecycle to ensure rigor and traceability throughout the research process.

Chapter 3: Research Methodology

This chapter presents the research methodology used to investigate professional awareness and perceptions of Digital Engineering capabilities within the Information Technology (IT) and Information Assurance (IA) domains. The study uses a quantitative survey approach to collect data from IT and IA professionals, enabling systematic analysis of awareness levels, perceived value, and anticipated benefits of Digital Engineering practices for IA and IT service delivery. The methodology follows a systems engineering lifecycle approach, applying structured engineering principles to research design while supporting rigorous traceability between research questions, survey instruments, and analytical approaches.

Research Design Overview

This research uses a quantitative, cross-sectional survey design to address the three research questions established in Chapter 1. Survey methodology best fits the research questions for three reasons. First, the research questions focus on measuring professional awareness and perceptions across a broad population of IT and IA practitioners, requiring data collection from enough respondents to establish representative findings. Second, survey methodology enables standardized data collection that supports statistical analysis and generalization of results to the broader professional population. Third, the anonymous nature of survey research encourages candid responses about professional knowledge gaps and organizational capabilities without concerns about professional reputation or organizational disclosure.

The cross-sectional design captures professional perceptions at a single point in time, providing a snapshot of current awareness and perceived value of Digital Engineering capabilities within the IT and Information Assurance professional community. While this design does not enable longitudinal analysis of changing perceptions, it establishes baseline data that will inform future research directions and support strategic decision-making about Digital Engineering adoption initiatives.

Justification for Survey Methodology

Selecting survey methodology over alternative approaches reflects deliberate consideration of what this research seeks to accomplish and why that objective warrants investigation. Three questions merit explicit answers: why investigate perceptions rather than implementations, why survey methodology rather than case study or pilot implementation, and why perceived value warrants dissertation-level research.

Why Perceptions Matter for Technology Adoption

Technology adoption research consistently shows that perceived value influences adoption decisions regardless of demonstrated actual value. Both the Technology Acceptance Model and its extensions establish that perceived usefulness and perceived ease of use predict behavioral intention to adopt technologies. Professionals who do not perceive value in a capability will not advocate for its adoption within their organizations, regardless of technical merit demonstrated in other contexts. As the literature review in Chapter 2 documents, Digital Engineering has demonstrated value in aerospace and defense contexts. However, demonstrated value in one domain does not automatically transfer to adoption in another. IT and Information Assurance professionals operate in different organizational contexts, face different constraints, and hold different professional identities than aerospace systems engineers. Whether these professionals recognize potential value in Digital Engineering capabilities for their work is an open question that must be answered before implementation research becomes meaningful.

Furthermore, if professionals are unaware of Digital Engineering capabilities, no amount of demonstrated value will drive adoption because the capabilities will not enter consideration during tool selection, process design, or strategic planning. Establishing current awareness levels identifies whether education and communication initiatives are necessary precursors to adoption efforts. If professionals are aware but do not perceive value, the theoretical premise that Digital Engineering addresses recognized needs within these domains requires reconsideration before investing in implementation research.

Why Not a Case Study or Implementation Pilot

This research considered and rejected alternative approaches for specific reasons. A case study examining Digital Engineering implementation within a specific organization would provide rich contextual data about practical implementation challenges and outcomes. However, case study findings would reflect the particular organizational context, culture, technical environment, and implementation approach of that organization. The findings would not establish whether the broader professional community recognizes value in Digital Engineering or is aware of these capabilities. A successful case study might show technical feasibility without indicating whether adoption would occur beyond the studied organization.

An implementation pilot would face similar limitations while requiring access to organizational resources and authority to implement Digital Engineering tools and practices. Such a pilot would measure actual rather than perceived value, but would do so within a single organizational context that may not represent the broader population. Implementation research also presumes that the target professional community recognizes sufficient potential value to warrant the investment required for implementation. This presumption is precisely what the current research tests.

Survey research enables assessment across a broad population of practitioners, establishing baseline awareness and perceived value data that informs whether subsequent case study or implementation research would find receptive audiences. Survey methodology is the appropriate starting point for investigating a nascent application domain where professional awareness and perceptions remain unknown.

Why Perceived Value Warrants Dissertation Research

Whether perceived value merits dissertation-level investigation reflects a particular view of research contribution. By addressing a documented gap, this research contributes to knowledge: the academic literature contains no investigation of whether IT and Information Assurance professionals perceive value in Digital Engineering capabilities. As

Chapter 2 establishes, the theoretical framework posits that Digital Engineering offers solutions to documented problems in these domains. Testing whether practitioners recognize this potential value provides empirical grounding for the theoretical framework.

If research reveals that professionals recognize value in Digital Engineering capabilities, this finding validates the theoretical premise and establishes foundation for subsequent implementation research. If research reveals limited awareness or skepticism, this finding challenges the theoretical framework and identifies barriers that must be addressed before adoption can occur. Either outcome advances understanding and provides actionable direction for researchers and practitioners. The study's contribution is empirical: it establishes evidence regarding professional perceptions that informs subsequent research and practice decisions, advancing that evidence in place of advocacy.

Systems Engineering Approach to Research Design

To show rigor in the design, execution, analysis, and documentation of results, this research follows a systems engineering lifecycle. Applying systems engineering principles to the dissertation project lends structure and credibility to the dissertation process through disciplined practice and supporting artifacts. A model captures and tracks the principal method of the dissertation effort, preserving traceability to decisions, citations, and artifacts. This is the same traceability that Digital Engineering brings to complex system development.

This choice is methodological rather than stylistic. A recurring threat to survey validity is that the link between the instrument and the research objectives rests on the author's assertion rather than on documented evidence. Instead, this study represents each research question as a *requirement element*, a formally identified statement of what the research must measure. It models each survey item as a *Block*, the basic building-block unit the model uses to represent one distinct question. It then binds every item to the requirement it addresses through a *Satisfy* relationship, the modeling link that records that one element fulfills another. These links form a *digital thread*, a connected and

authoritative chain of references that ties each item to its research question. Every item therefore traces, explicitly and bidirectionally, to the research question it is intended to measure. Completeness of that mapping can be evaluated programmatically rather than asserted in prose. The approach supplies documented, quantitative traceability and content-validity evidence in a form that is at once visually inspectable (Figures 3 and 12) and machine-checkable. It does so by applying to the research instrument the very digital-thread capability whose value the dissertation sets out to assess. The method thus demonstrates in practice the traceability and rigor it investigates rather than merely describing them.

The survey model follows a structured lifecycle consisting of six phases:

Strategic Phase

Captures conceptual elements including hypothesis, capabilities, constraints, goals, vision, timeline, milestones, stakeholders, and drivers. This phase establishes the foundation for all subsequent research activities by clearly articulating what the research seeks to accomplish and why it matters.

Requirements Phase

Derives and analyzes strategic elements to create requirements for the dissertation and associated survey. Requirements follow ISO 15288:2023 standards: hierarchical in nature and structure, atomic in composition, bidirectional in traceability, individually measurable, and explicitly tracking relationships and interdependencies.

Architecture Phase

Establishes the high-level outline, structure, and guidance for the design and implementation phases of the dissertation. The survey architecture defines the overall structure of sections and question flow, ensuring logical progression from awareness through applicability to perceived value.

Design Phase

The Design Phase uses the architecture guidance and requirements to design the dissertation proposal and survey instrument, satisfying the architectural outline with complete traceability back to requirements and strategic elements.

Results Phase

Captures the survey data into the model for traceability along with results of analysis. Analysis methods and results are captured as elements within the model, enabling verification that findings address the original research questions.

Report Phase

The model serves as a deliverable component of the dissertation that provides a rigorous source of artifacts, methods used, traceability chains, and evidentiary support for the findings.

Alignment to Dissertation Design

Figure 2 provides a visual representation of the dissertation process using an MBSE approach. The research has progressed through the full systems engineering lifecycle: the strategic, requirements, architecture, and design phases established the survey instrument and analytical plan, while the results and report phases (reported in Chapter 4) captured the collected survey data and the results of analysis into the model for traceability.

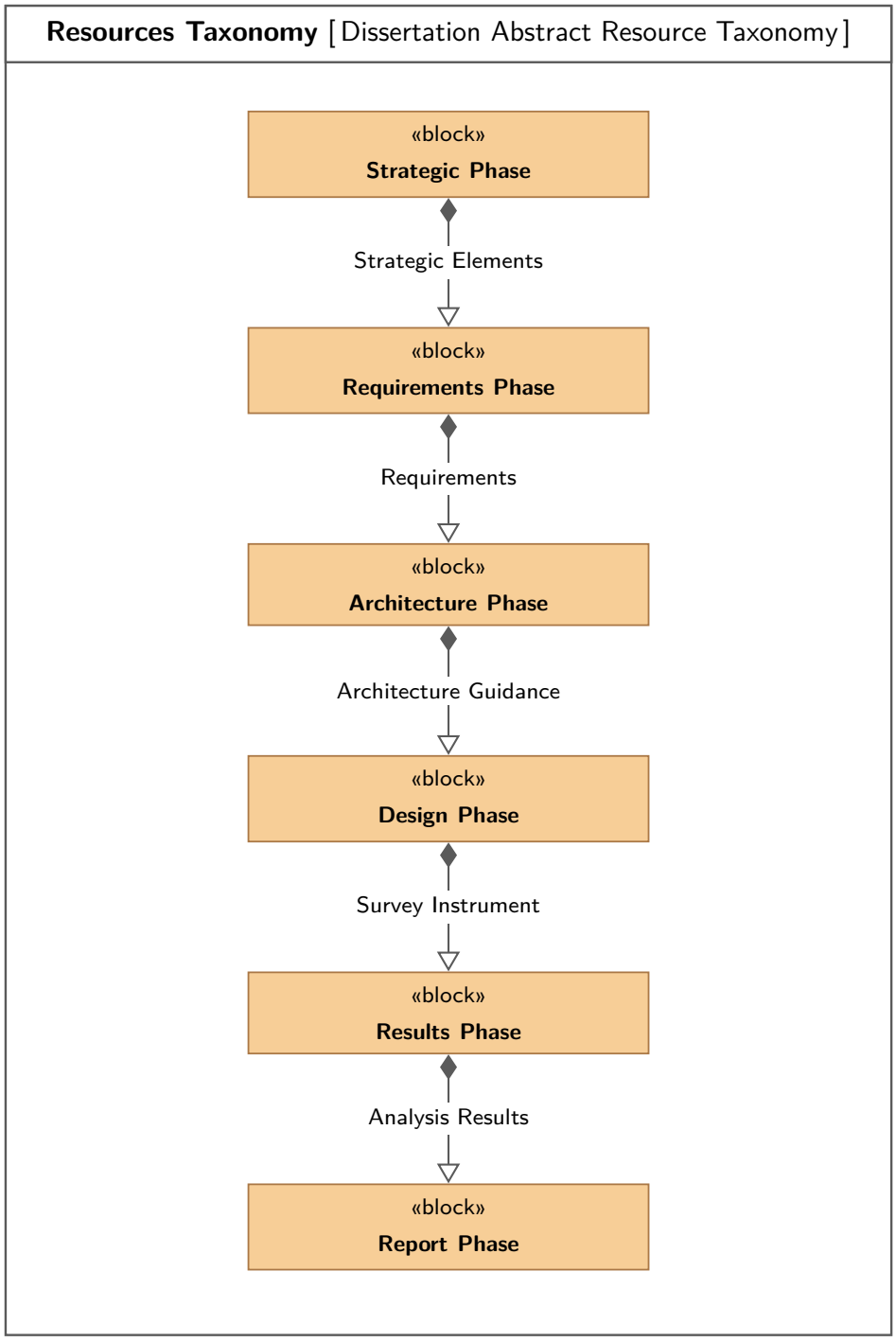


Figure 2: Dissertation lifecycle process as an abstract resource taxonomy.

Research Questions and Survey Alignment

The survey instrument addresses the three research questions directly:

Research Question 1

To what extent are Information Technology and Information Assurance professionals aware of Digital Engineering capabilities, including Model-Based Systems Engineering, digital threads, digital twin technologies, and Product Lifecycle Management principles?

Research Question 2

To what extent do Information Technology and Information Assurance professionals perceive Digital Engineering capabilities as potentially valuable or important for their work in Information Assurance, security compliance, and IT service delivery?

Research Question 3

To what extent do Information Technology and Information Assurance professionals believe that Digital Engineering practices could help them in performing their jobs, meeting compliance requirements, or enhancing organizational capabilities in Information Assurance and IT service delivery?

Research Propositions and Planned Analyses

Chapter 1 advances four research propositions that follow from the theoretical framework; Chapter 4 evaluates each proposition against the survey data, and Chapter 5 returns to them in stating the conclusions. The analytical plan binds each proposition to a designated analysis:

- **P1 (Low awareness).** Awareness of and recent exposure to Digital Engineering are low among IT and IA professionals, who sit outside the systems engineering discipline in which these methods developed. Descriptive statistics on the familiarity (Question 1.1) and exposure (Question 1.2) items, reported with Wilson confidence intervals, assess this proposition.

- **P2 (Value despite low awareness).** Perceived value is nonetheless positive once capabilities are described in functional terms, because Digital Engineering addresses the documentation, traceability, and visibility problems these professionals routinely face. Descriptive agreement levels on the value items, together with the familiarity-versus-value diagnostic described in the Response Bias Mitigation section, assess this proposition.

- **P3 (Variation by field and experience).** Perceived value and awareness vary systematically with professional field and experience level rather than being uniform across the community. Mann–Whitney U tests (professional field), Kruskal–Wallis tests with Dunn's post-hoc comparisons under Holm adjustment (experience level), and chi-square tests with Cramér's V (categorical associations) assess this proposition.

- **P4 (Familiarity and value).** Greater familiarity with Digital Engineering is associated with higher perceived value. Rank-order correlation and ordinary least squares regression of the value composites on familiarity, professional field, and experience level assess this proposition.

The propositions above define the analyses specified in this proposal. Chapter 4 additionally reports a set of supplementary robustness and measurement-validity analyses that this plan did not specify, including a common-method-variance screen, a construct-validity assessment of the composite scores, and a measurement-invariance check across professional-field groups. The candidate added those analyses after data collection concluded, as a self-initiated methodological strengthening informed by comparison with related dissertations; Chapter 4 labels them as post-hoc where they appear and interprets them as robustness evidence rather than as tests of the propositions.

The design mapped each survey section and question to the three research questions to support comprehensive coverage of the research objectives while minimizing respondent

burden through focused questioning. This mapping provides the traceability essential to demonstrating that the instrument measures what it purports to measure.

Figure 3 illustrates an MBSE representation of the Research Questions as requirement type elements. Selecting requirement type elements within the Unified Architecture Framework (UAF) allows survey questions to use the "Satisfy" relationship. Authoritative traceability within the model enables programmatic evaluation that verifies the completeness of the question-to-requirement mapping; this check validates traceability coverage rather than construct validity.

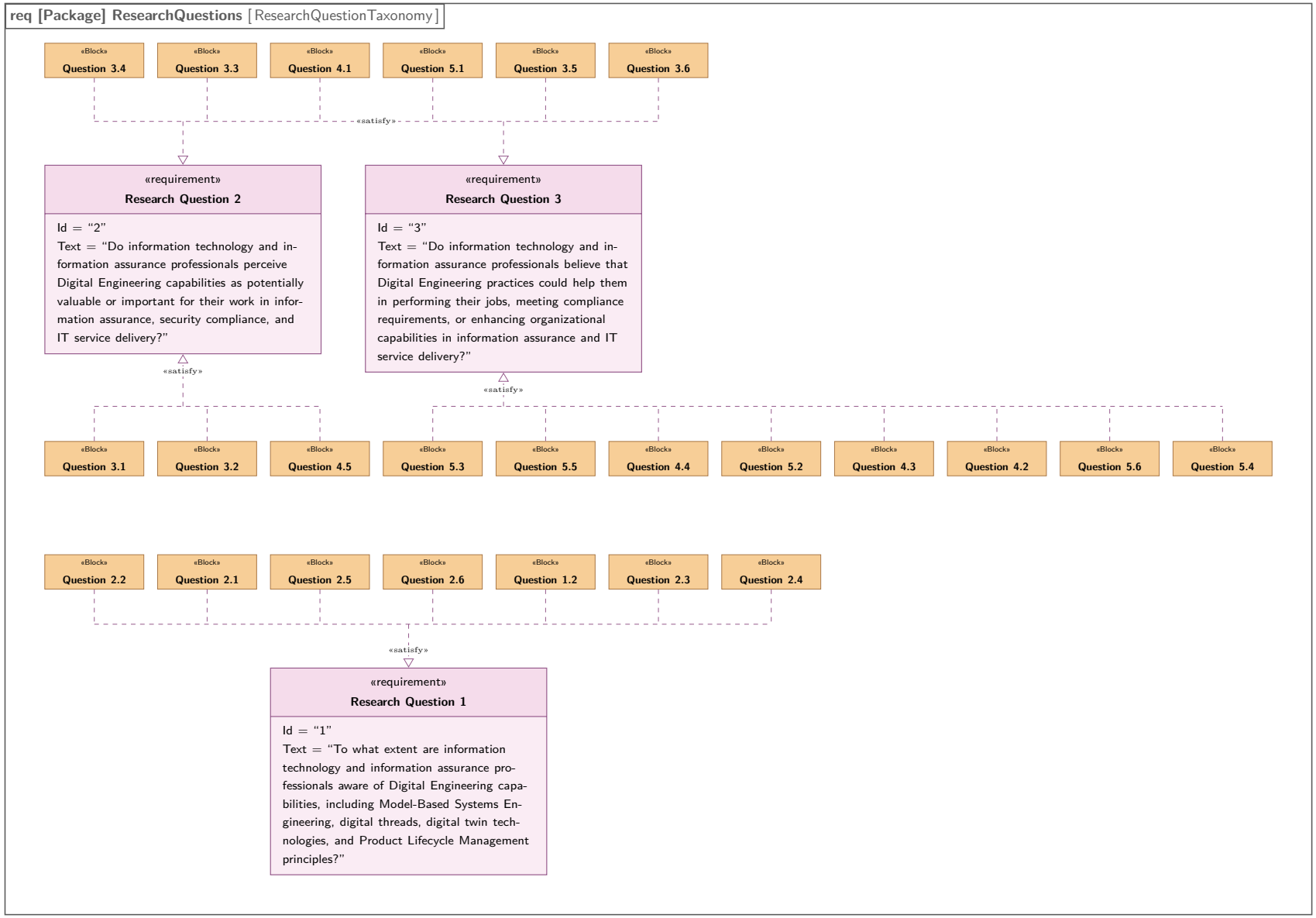


Figure 3: MBSE instantiation of research questions.

Survey Requirements Specification

The systems engineering approach established specific instrument requirements before design. These requirements help align the survey with the research objectives while maintaining ethical standards and data quality.

Anonymity and Privacy Requirements

The following requirements govern anonymity and privacy protection:

- The survey shall be anonymous, collecting no personally identifiable information during the survey process.
- The survey shall not collect age, sex, gender, race, creed, gender identity, sexual orientation, income, nationality, or other non-relevant personal information.
- The survey shall be protected with appropriate security controls to ensure the confidentiality, integrity, and availability of collected data.
- The survey raw data shall not be sold or used for any purpose other than the dissertation research.

Instrument Design Requirements

The following requirements govern survey instrument design:

- The survey shall use a Likert Scale for material research questions to enable quantitative analysis.
- The survey shall include applicable notices required by the University Institutional Review Board.
- The survey should collect demographic information to identify the participant's professional field of study or practice.
- The survey should collect demographic information about temporal experience level within the participant's field.

Content Requirements

The survey content assesses the awareness, applicability, and perceived value that participants hold toward Digital Engineering and its component disciplines. The survey begins at the enterprise level and examines the four pillars: Model-Based Systems Engineering, the Digital Thread, Digital Twin, and Product Lifecycle Management. Consistent question flow across this structure enables analysis of how each discipline is perceived in relation to Digital Engineering as a unified approach.

Value assessment questions in Sections 3 through 5 use positively framed capability statements (e.g., "Digital Engineering could deliver meaningful value" or "could reduce development cycle time"). The research objectives and target population characteristics informed this framing through deliberate design decisions. Because the research questions ask to what extent professionals perceive value in Digital Engineering capabilities, positively framed statements directly measure this construct by presenting capabilities and assessing agreement. An instrument measuring perceived value necessarily presents value propositions for evaluation. Negatively framed items (e.g., "Digital Engineering would not improve security posture") would introduce cognitive complexity through double negatives when combined with disagreement responses. The survey methodology literature documents this confound, which disproportionately affects respondents unfamiliar with the subject matter.

The design addresses the acquiescence bias risk inherent in consistently framed questions through two analytical checks. First, response-pattern analysis examined whether any respondents showed uniform agreement across all items (straight-lining), a pattern suggestive of satisficing or acquiescence rather than considered evaluation. Second, the Section 1 familiarity self-assessment provides an independent baseline against which value perception scores can be interpreted. Respondents who report low familiarity yet indicate high agreement with value statements can be identified and examined. Such a pattern signals potential acquiescence rather than genuine perception. Two

response-format features complement these checks. A neutral midpoint ("Neither agree nor disagree") reduces forced-choice artifacts by providing a non-affirmative response option for respondents who are genuinely uncertain or indifferent. The binary and ternary investment willingness questions (Questions 4.5 and 5.7) offer explicit "No" and "Unsure" options. These options capture skepticism and uncertainty without requiring disagreement with a positively framed statement. Because all value items are positively keyed, these safeguards detect and bound acquiescence rather than eliminate it, and residual acquiescence risk remains an acknowledged limitation of the instrument.

Target Population and Sampling

Target Population

Professionals actively working in Information Technology and Information Assurance roles within organizations that manage information systems constitute the target population for this study. The population encompasses individuals involved in IT service delivery, infrastructure management, service operations, IA, security operations, compliance management, and security architecture. It spans organizational types including private sector enterprises, government agencies, educational institutions, and nonprofit organizations.

The dual focus on IT and IA professionals recognizes that Digital Engineering capabilities may offer distinct value propositions for these related but different professional communities. IT professionals primarily concerned with service delivery and operational efficiency may perceive value in Digital Engineering practices differently than IA professionals focused on threat mitigation, compliance verification, and security control implementation. Understanding both perspectives is necessary for thorough assessment.

Sampling Strategy

This study uses non-probability convenience sampling, recruiting participants through professional networking platforms and direct professional outreach. While probability sampling would provide stronger generalizability, establishing a sampling frame

for all IT and Information Assurance professionals is difficult, which renders probability sampling impractical for this research. Convenience sampling enables efficient access to qualified respondents while maintaining focus on professionals with relevant experience and expertise.

The design anticipated recruiting across several channels, among them professional organizations such as ISACA, $(ISC)^2$, and ITIL professional communities; LinkedIn professional groups focused on IT service management and IA; industry conferences and professional development events; and direct outreach to IT and security departments across multiple sectors. Execution did not realize that breadth. As the Recruitment Channels and Promotion section below records, LinkedIn served as the primary medium, and because the candidate's network there is concentrated in defense, aerospace, and federal organizations, the realized frame is narrower than the plan and the multi-channel breadth cannot be claimed as a mitigation. Chapter 5 carries the resulting composition concern into the limitations, where it is the most consequential the study reports.

Sample Size Determination

The sample size calculation targeted a maximum margin of error of 5% at the 95% confidence level. For survey research targeting a large population where the population size exceeds 100,000 individuals, the sample size formula for estimating a proportion is:

$$n = \frac{Z^2 \times p \times (1 - p)}{E^2} \tag{1}$$

Where:

- n = required sample size
- Z = Z-score for desired confidence level (1.96 for 95% confidence)
- p = estimated population proportion (0.5 for maximum variability)
- E = desired margin of error (0.05 for 5%)

Substituting these values:

$$n = \frac{1.96^2 \times 0.5 \times 0.5}{0.05^2} = \frac{3.8416 \times 0.25}{0.0025} = \frac{0.9604}{0.0025} = 384.16 \tag{2}$$

Therefore, a minimum of 385 completed survey responses is required to achieve the target margin of error. This calculation assumes maximum variability in responses (p = 0.5), which provides the most conservative sample size estimate. If actual response distributions show less variability, the effective margin of error will be smaller than 5%. These precision calculations assume probability sampling; under the convenience sampling used here they serve as planning benchmarks, and reported confidence intervals are conditional on the sample behaving like a random draw from the accessible population.

To account for potential incomplete responses and to support adequate representation across professional subgroups (IT professionals versus security professionals, varying experience levels), the study set the target sample size for data collection at 450 completed surveys. This target exceeds the minimum requirement by approximately 17%, aligning with survey research best practices recommending 15% to 20% oversampling to accommodate anticipated attrition (Dillman et al., 2014). Research on online survey completion rates indicates that professional surveys typically experience 10% to 15% incomplete response rates due to survey abandonment, partial completion, or data quality

concerns requiring exclusion (Fan & Yan, 2010). The 450-response target provides sufficient buffer to reach the minimum 385 usable responses while enabling meaningful subgroup analyses with adequate statistical power. Specifically, if the sample divides evenly between IT and security professionals, each subgroup would include approximately 190–225 respondents, providing margins of error of roughly 6.5% to 7.1% for subgroup-specific analyses. The realized sample substantially exceeded this target. The survey drew 1,600 submissions. The data-quality screening detailed in the Data Analysis section then removed 189 cases (188 abandoned or aberrant submissions, plus one automated entry identified by the response-authenticity check reported in Chapter 4), approximately 12%, a figure squarely within the attrition range anticipated above. After that screening, 1,411 responses remained for analysis. The retained sample, reported in Chapter 4, yields subgroup sizes well above the planning assumptions and correspondingly tighter precision than the design required.

Survey Instrument Design

Instrument Overview

The 30-question survey instrument systematically addresses the research questions across six thematic sections while maintaining reasonable completion time. Question complexity and respondent fatigue considerations set the target completion time at roughly ten minutes, balancing comprehensive data collection against respondent engagement and completion rates.

The six sections are structured as follows:

1. Section 1: Awareness and Familiarity with Digital Engineering (2 questions)
2. Section 2: Understanding of Digital Engineering Capabilities (6 questions)
3. Section 3: Applicability of Digital Engineering (6 questions)
4. Section 4: Value Assessment for Information Technology (5 questions)

5. Section 5: Value Assessment for Information Assurance and Cybersecurity (7 questions)

6. Section 6: Interest and Demographic Information (4 questions)

Question Format and Scale Selection

The survey uses two primary question formats: five-point Likert-type scales and binary (yes/no) response options. Psychometric research establishing the validity and reliability of these formats for measuring attitudes, perceptions, and self-reported behaviors guided this choice.

Likert Scale Justification

Five-point Likert scales measure awareness levels, agreement with capability statements, and perceptions of value. Likert scales have been extensively validated for measuring attitudes and perceptions in social science research since their introduction by Rensis Likert in 1932. Research showing that five to seven response options provide optimal discrimination between respondent positions while remaining cognitively manageable drove the choice of five options.

The analytical treatment of Likert scale data warrants explicit methodological comment. A longstanding debate in psychometric and social science research concerns whether Likert scale responses constitute ordinal data (where distances between scale points are not necessarily equal) or may be treated as interval data (where parametric statistical methods such as means and standard deviations are appropriate). Jamieson (Jamieson, 2004) argues that Likert data are strictly ordinal and that parametric analyses may produce misleading results. Norman (Norman, 2010), drawing on extensive simulation research, shows that parametric methods are sufficiently robust to yield valid results with Likert data even when distributional assumptions are violated. Subsequent research by Sullivan and Artino (Sullivan & Artino, 2013) and de Winter and Dodou (de Winter & Dodou, 2010) supports Norman's position, finding that parametric

tests applied to five-point Likert data produce conclusions consistent with non-parametric alternatives across a wide range of distributional conditions.

This research follows the analytical convention adopted by most technology acceptance and survey research. It reports parametric descriptive statistics (means, standard deviations) for Likert items and composite scores to enable comparison with the broader literature. When distributional assumptions are not met, it uses non-parametric alternatives for inferential testing. Specifically, the distributional shape of each composite (skewness and proximity to the response ceiling), corroborated by Shapiro–Wilk tests and visual inspection, determines whether parametric tests (t-tests, ANOVA) or non-parametric alternatives (Mann–Whitney U, Kruskal–Wallis) are applied for group comparisons. The Shapiro–Wilk test is treated as corroborating rather than dispositive, because at the realized sample size it is highly powered against trivial departures from normality and would reject for substantively negligible deviations; the decision therefore rests primarily on the observed left skew and ceiling concentration of the composites. The analysis reports medians and interquartile ranges alongside means and standard deviations for individual Likert items, providing readers with both parametric and ordinal summaries. This dual-reporting approach supports transparency and enables readers who hold either position in the ordinal-interval debate to evaluate the findings against their preferred analytical framework.

Consistent use of Likert scales across sections provides an intuitive response method for participants and enables direct comparison of responses across different constructs. Two distinct anchor sets serve the constructs being measured:

. **Familiarity Scale (Section 1, Question 1.1).**

1. Not at all familiar
2. Slightly familiar
3. Moderately familiar

4. Very familiar

5. Extremely familiar

. **Agreement Scale (Sections 2-5).**

1. Strongly disagree

2. Disagree

3. Neither agree nor disagree

4. Agree

5. Strongly agree

The agreement scale follows established conventions in technology acceptance research, including the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT) frameworks, and has demonstrated validity for measuring perceptions of technology value and usefulness across diverse populations and contexts. Including a neutral midpoint ("Neither agree nor disagree") enables respondents to indicate genuine uncertainty rather than forcing artificial choices, which research suggests improves response validity. Figure 4 shows the instantiation of familiarity and agreement response scales within the model of the dissertation survey.

Binary and Ternary Question Justification

The survey uses binary yes/no and ternary yes/no/unsure formats for factual questions about prior exposure (Section 1, Question 1.2), interest in further learning (Section 6, Questions 6.1-6.2), and investment willingness (Sections 4-5, Questions 4.5 and 5.7). Binary/ternary formats are appropriate for these questions because they seek to classify respondents into discrete categories rather than measure degrees of agreement or perception intensity. The simplicity of binary/ternary responses also reduces cognitive load for straightforward factual questions where graduated responses would add complexity

without meaningful information gain. Figure 4 provides a visualization of the enumeration elements of the binary and ternary responses.

Categorical Questions

Section 6 includes categorical questions capturing demographic information about professional field of practice and years of experience. Following the demographic information requirements, these questions capture only sector of work and temporal experience level without collecting personally identifiable information. The experience level categories are structured to capture early-career, mid-career, and late-career stages:

- 1-5 Years of experience (Early Career)
- 6-10 Years of experience (Mid Career - Early)
- 11-15 Years of experience (Mid Career - Late)
- 16+ Years of experience (Late Career)

These categories enable analysis of response patterns across professional subgroups and experience levels, supporting investigation of whether awareness and perceptions vary systematically by professional background. Such variation would carry important implications for professional development and adoption strategies.

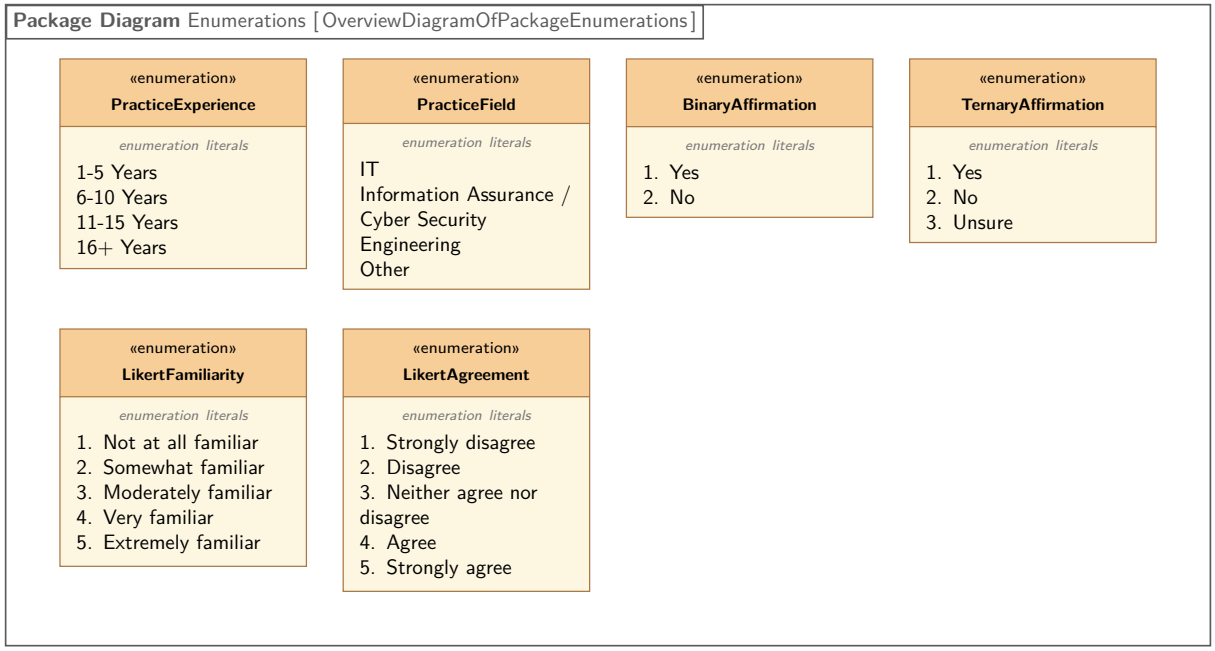


Figure 4: Survey question enumeration.

Survey Section Structure and Research Question Mapping

This section details how each survey section addresses the research questions, providing explicit traceability between survey content and research objectives. Every question serves a defined purpose, and the mapping supports comprehensive coverage of the research questions.

Table 3 consolidates all thirty items. For each item it records the construct measured, the verbatim item stem, and the research question or category to which the item maps. The subsections that follow describe each section's role and its mapping to the research questions.

Table 3
Survey items mapped to measured constructs and research questions.

Item	Construct measured	Verbatim item stem	Mapped RQ
Q1.1	Self-reported awareness levels	"Please rate your level of familiarity with Digital Engineering concepts and practices."	RQ1
Q1.2	Recent professional exposure	"Have you encountered Digital Engineering methodologies, frameworks, or tools in your professional work within the past two years?"	RQ1
Q2.1	Model-Based Systems Engineering	"Digital Engineering includes model-based systems engineering approaches that can improve development processes."	RQ1
Q2.2	Digital Twin for IT	"Digital Engineering can enable digital twin development and virtual prototyping for Information Technology systems."	RQ1
Q2.3	Digital Thread and data-driven practices	"Digital Engineering supports continuous integration and data-driven decision-making in technology development."	RQ1
Q2.4	Digital Twin for security	"Digital Engineering enables digital twin technology that can simulate security scenarios and test defensive measures without impacting production systems."	RQ1
Q2.5	Security validation through the Digital Thread	"Digital Engineering supports continuous security validation and data-driven threat analysis throughout the development lifecycle."	RQ1

Continued on next page

Table 3 continued from previous page

Item	Construct measured	Verbatim item stem	Mapped RQ
Q2.6	Compliance automation	"Digital Engineering can improve security control implementation through automated compliance checking and verification."	RQ1
Q3.1	IT sector relevance	"Digital Engineering methodologies have relevant applications within the Information Technology sector."	RQ2
Q3.2	Information Assurance relevance	"Digital Engineering methodologies have relevant applications for addressing Information Assurance challenges."	RQ2
Q3.3	Digital Twin organizational value	"The ability to utilize digital twins to test changes against accurate replicas of production environments would provide value to my organization."	RQ2, RQ3
Q3.4	MBSE organizational value	"The use of digital models to map and document an organization's environment and configurations would provide value to my organization."	RQ2, RQ3
Q3.5	PLM organizational value	"The use of digital lifecycle management to meet compliance and service delivery requirements would provide value to my organization."	RQ2, RQ3
Q3.6	Compliance applicability	"My organization faces regulatory or compliance requirements that could benefit from Digital Engineering approaches."	RQ2, RQ3

Continued on next page

Table 3 continued from previous page

Item	Construct measured	Verbatim item stem	Mapped RQ
Q4.1	Overall IT value	"Digital Engineering could deliver meaningful value to my organization's Information Technology processes."	RQ2, RQ3
Q4.2	Cycle time benefit	"Digital Engineering could reduce development cycle time in my organization."	RQ3
Q4.3	Quality benefit	"Digital Engineering could improve product quality and reduce defects in my organization."	RQ3
Q4.4	Collaboration benefit	"Digital Engineering could improve collaboration effectiveness across development teams in my organization."	RQ3
Q4.5	Investment willingness	"My organization would be willing to invest in Digital Engineering capabilities if clear return on investment could be demonstrated." (Yes/No/Unsure)	RQ2
Q5.1	Overall security value	"Digital Engineering could deliver meaningful value to my organization's Information Assurance and cybersecurity operations."	RQ2, RQ3
Q5.2	Vulnerability management benefit	"Digital Engineering could reduce the time required to identify and remediate security vulnerabilities in my organization."	RQ3
Q5.3	Security posture benefit	"Digital Engineering could improve security posture and reduce successful cyber incidents in my organization."	RQ3

Continued on next page

Table 3 continued from previous page

Item	Construct measured	Verbatim item stem	Mapped RQ
Q5.4	Threat modeling benefit	"Digital Engineering could enhance threat modeling and risk assessment capabilities in my organization."	RQ3
Q5.5	Cross-team collaboration benefit	"Digital Engineering could improve collaboration between security teams, development teams, and operations teams in my organization."	RQ3
Q5.6	Compliance benefit	"Digital Engineering could help my organization achieve better compliance with security frameworks and regulatory requirements."	RQ3
Q5.7	Security investment willingness	"My organization would be willing to invest in Digital Engineering capabilities for cybersecurity purposes if clear return on investment could be demonstrated." (Yes/No/Unsure)	RQ2
Q6.1	Learning interest	"Would you be interested in learning more about Digital Engineering applications for Information Assurance, Cybersecurity, or Information Technology in your industry?"	Supporting
Q6.2	Recommendation likelihood	"Would you recommend that your organization explore Digital Engineering adoption for improving security operations?"	Supporting

Continued on next page

Table 3 continued from previous page

Item	Construct measured	Verbatim item stem	Mapped RQ
Q6.3	Professional field	"Please indicate your field of practice." (Information Technology / Information Assurance or Cyber Security / Engineering / Other)	Demographics
Q6.4	Experience level	"Please indicate your level of experience in your field of practice." (1-5 Years / 6-10 Years / 11-15 Years / 16+ Years)	Demographics

Section 1: Awareness and Familiarity with Digital Engineering

Section 1 contains two questions directly addressing Research Question 1 regarding professional awareness of Digital Engineering capabilities. Figure 6 provides an MBSE-based visualization as a block element, and Figure 5 details the properties recorded for each question element. Table 3 lists both item stems. Question 1.1 uses a five-point familiarity scale and provides a direct, self-assessed measure of overall Digital Engineering awareness, establishing the baseline from which all subsequent analysis proceeds. Question 1.2 uses a binary format and distinguishes theoretical awareness from practical professional experience, recognizing that knowing about Digital Engineering differs categorically from having encountered it in practice.

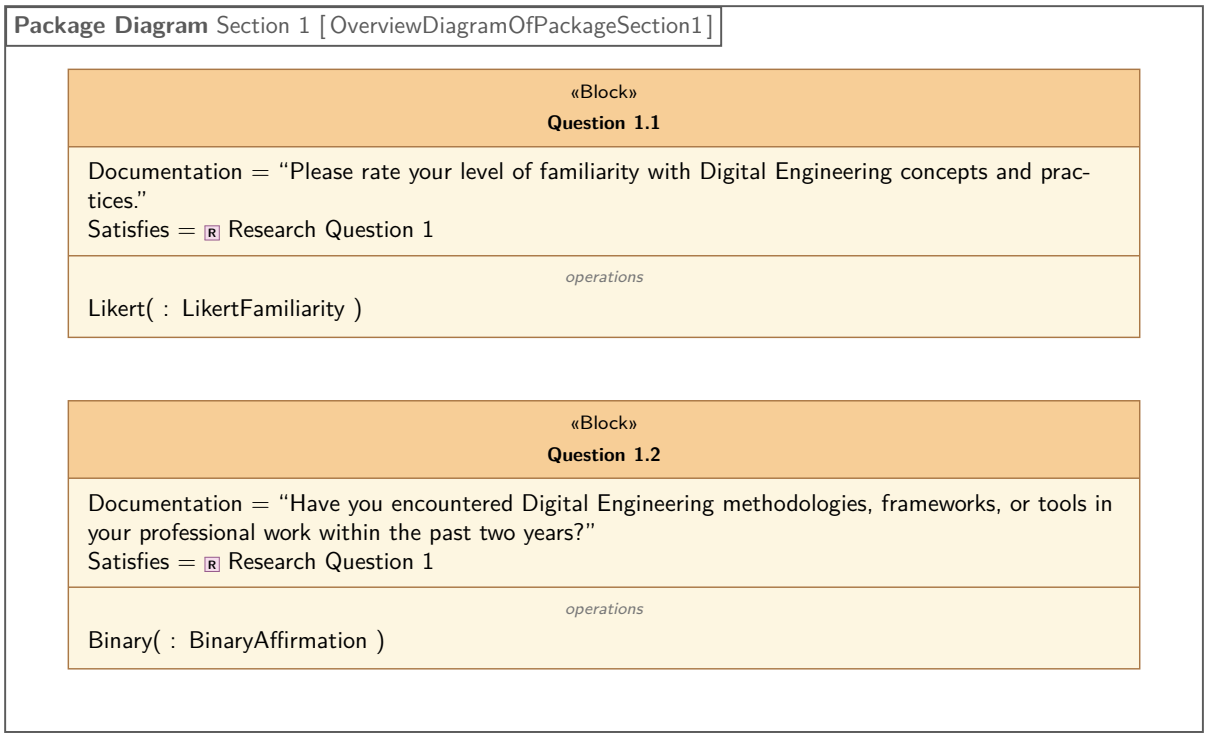


Figure 5: Survey questions Section 1 detailed properties taxonomy.

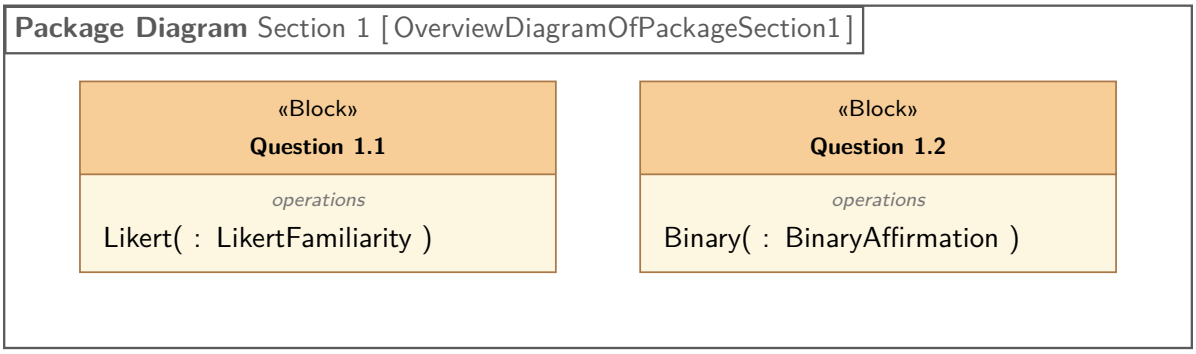


Figure 6: Section One survey question elements package diagram.

Mapping to Research Question 1

These questions directly measure the extent of professional awareness, providing quantifiable data on familiarity levels and recent professional exposure to Digital Engineering concepts.

Section 2: Recognition of Digital Engineering Capabilities

Section 2 contains six Likert-scale questions assessing respondent recognition of specific Digital Engineering capabilities, each described in functional terms. Figure 7 visualizes each research question as represented in the model. The questions of Section 2 probe awareness of the four Digital Engineering pillars in the context of IT and IA applications, as listed in Table 3.

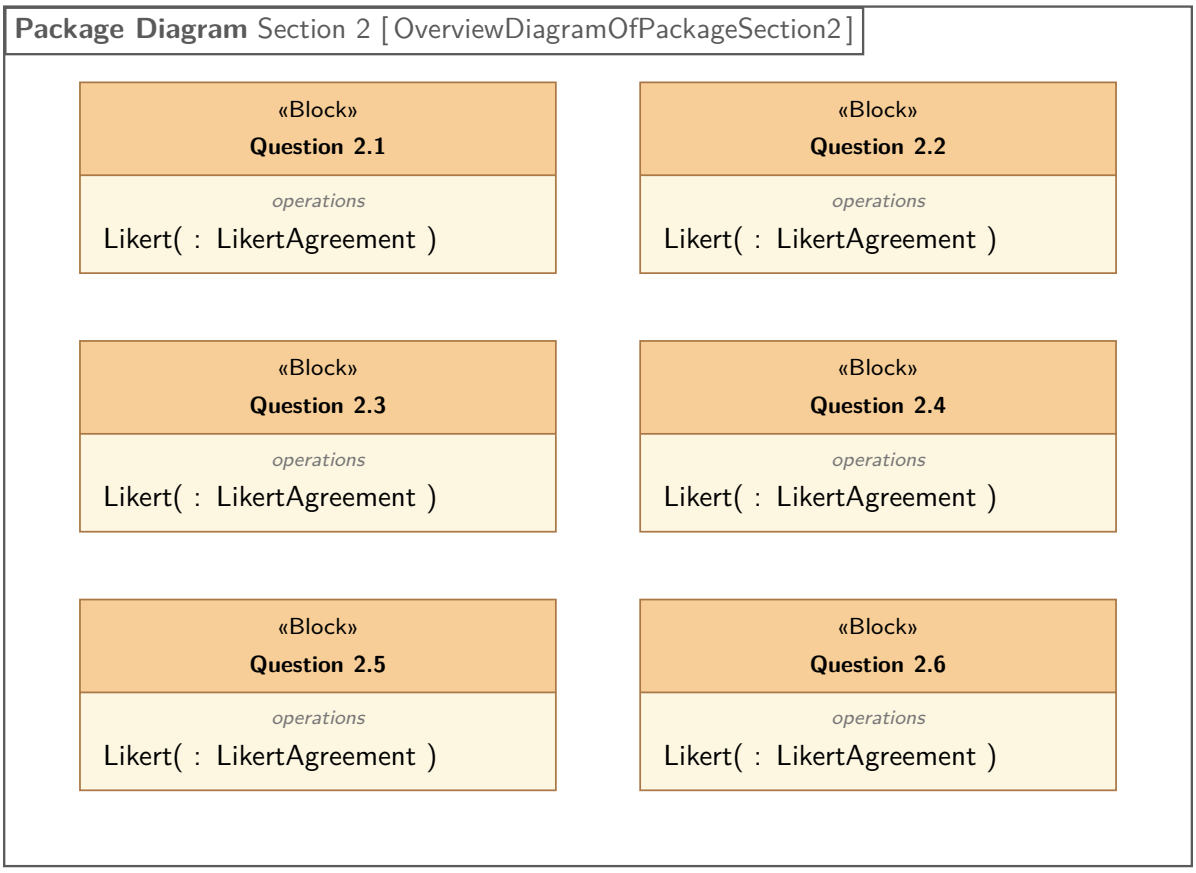


Figure 7: Section Two survey question elements package diagram.

Mapping to Research Question 1

This section extends Research Question 1 by measuring both general awareness and recognition of specific Digital Engineering pillars (MBSE, Digital Twin, the Digital Thread, and PLM) and their applications to IT and IA contexts. A methodological distinction

warrants clarification: Section 2 questions use agreement scales rather than direct awareness or knowledge measures. Agreement with a capability statement (e.g., "Digital Engineering includes model-based systems engineering approaches that can improve development processes") captures recognition of a described capability, meaning whether respondents affirm the stated relationship once it is put to them in functional language. Establishing comprehension would require knowledge-verification items, such as definitional or discrimination items carrying a correct answer, and the instrument includes none. Section 1 captures familiarity with the named discipline directly through self-assessment; Section 2 probes further by testing whether respondents affirm the capability descriptions that give the discipline its content. Taken together, Sections 1 and 2 provide a layered assessment of Research Question 1 that separates familiarity with the label from recognition of the capabilities described functionally, and the design claims no measure of knowledge beyond those two.

Section 3: Applicability of Digital Engineering

Section 3 contains six Likert-scale questions examining perceptions of Digital Engineering applicability to IT and IA domains. This section bridges awareness assessment and value perception, probing whether respondents see relevance to their professional contexts. Figure 8 visualizes the instantiation of Section 3 survey questions within the model.

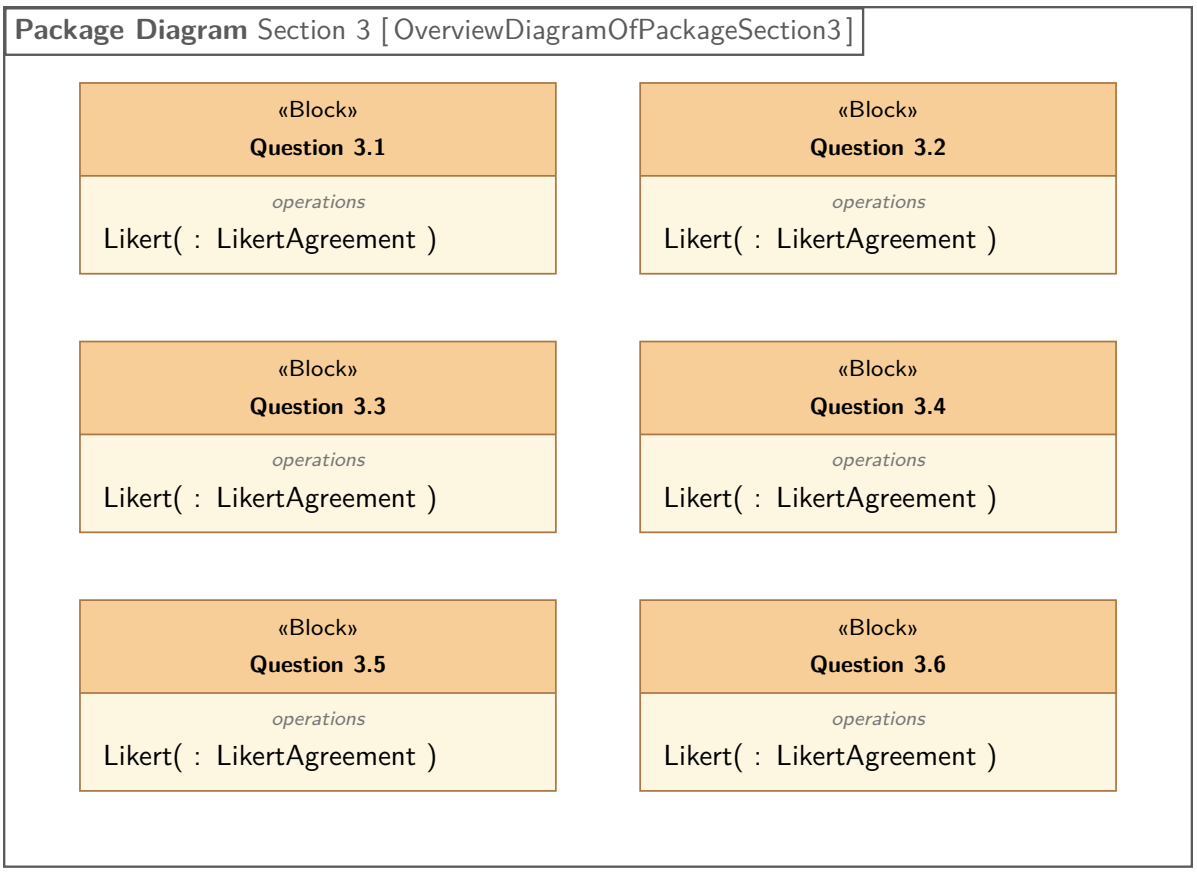


Figure 8: Section Three survey question elements package diagram.

Mapping to Research Questions 2 and 3

This section addresses whether respondents perceive Digital Engineering as applicable to their professional domains and whether they identify potential organizational value in specific capabilities.

Section 4: Value Assessment for Information Technology

Section 4 contains five questions assessing the perceived value of Digital Engineering for IT operations specifically; Figure 9 visualizes these questions.

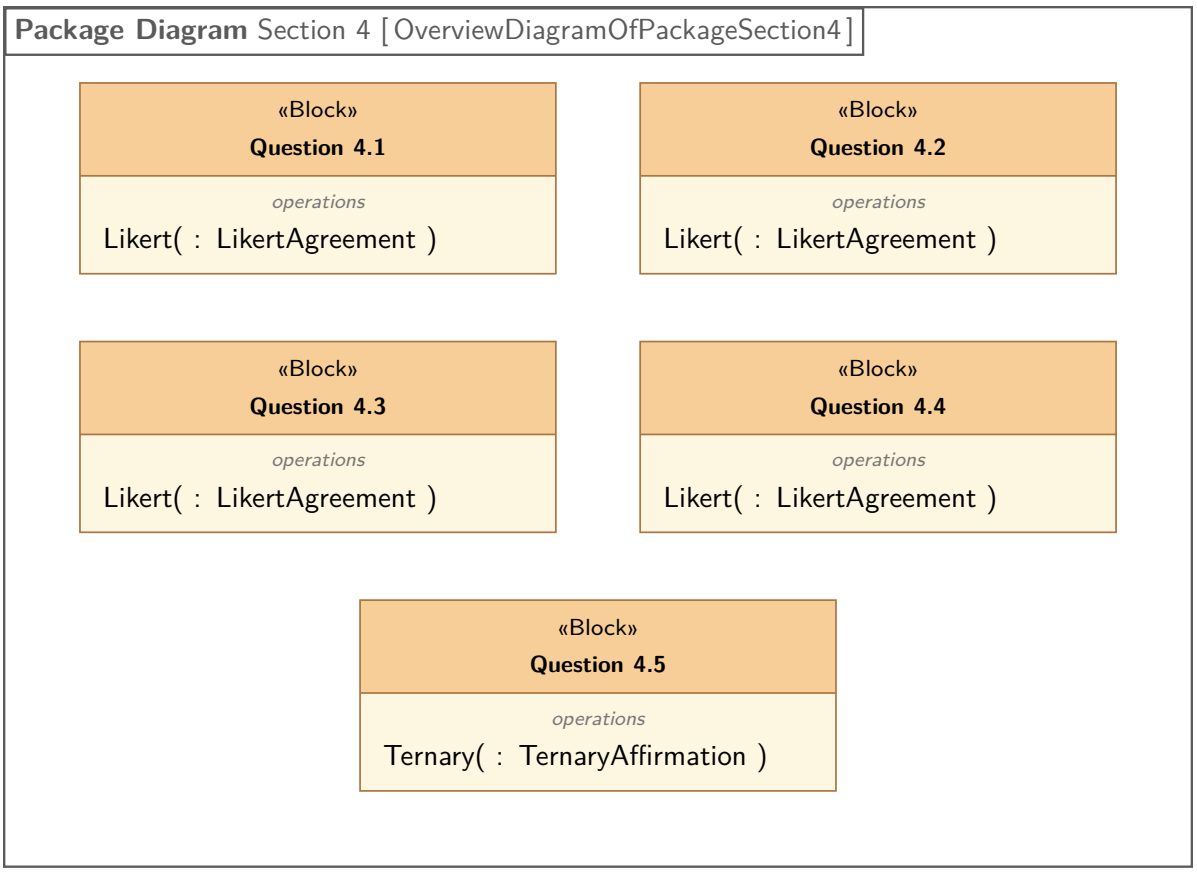


Figure 9: Section Four survey question elements package diagram.

Mapping to Research Questions 2 and 3

This section directly addresses whether IT professionals perceive value in Digital Engineering and believe these practices could enhance their organizational capabilities and job performance.

Section 5: Value Assessment for Information Assurance and Cybersecurity

Section 5 contains seven questions assessing perceived value of Digital Engineering for Information Assurance and cybersecurity operations as displayed in Figure 10.

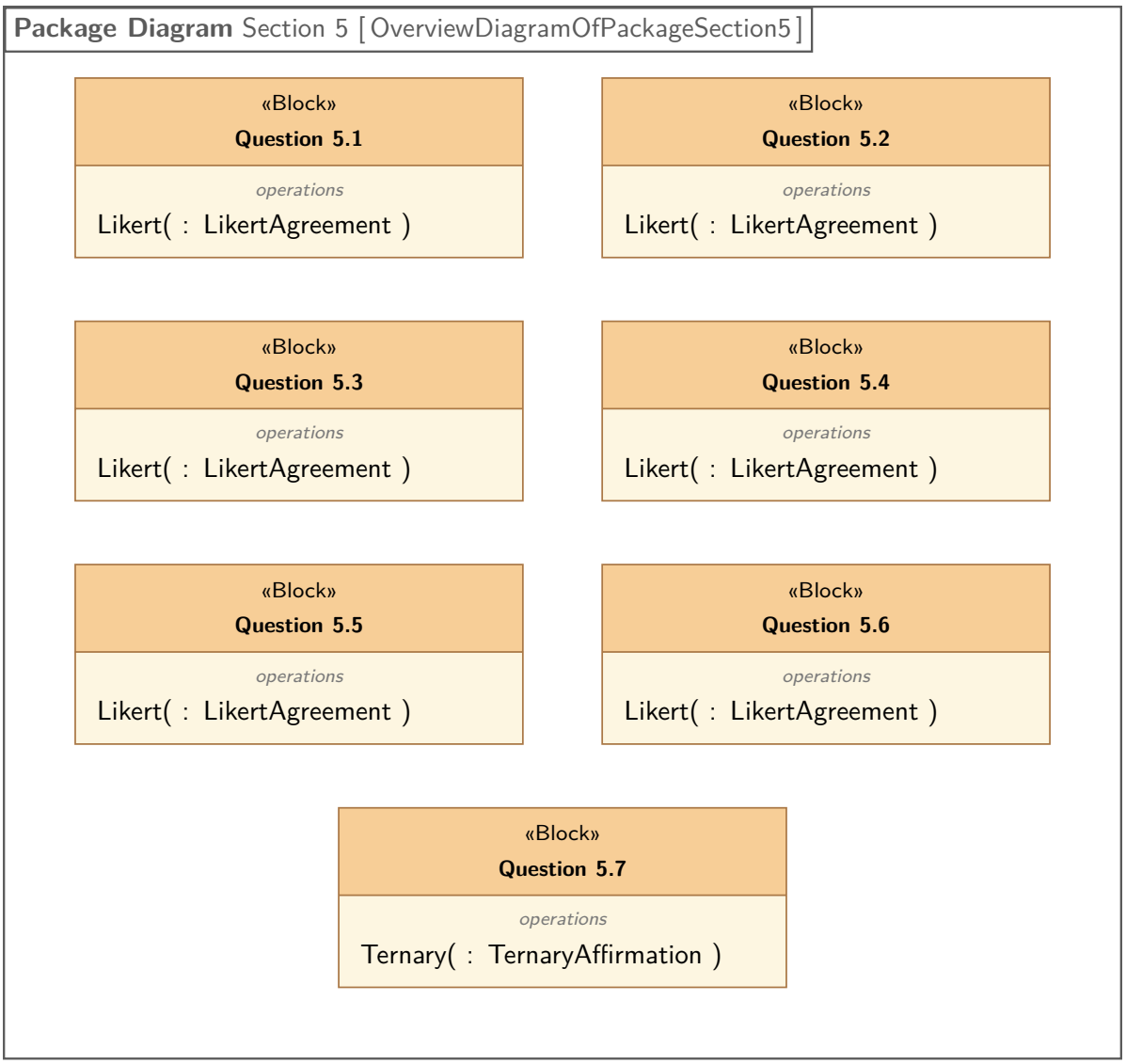


Figure 10: Section Five survey question elements package diagram.

Mapping to Research Questions 2 and 3

This section directly addresses whether IA professionals perceive value in Digital Engineering for their specific domain and believe these practices could help them meet compliance requirements and enhance security capabilities.

Section 6: Interest and Demographic Information

Section 6 contains four questions capturing respondent interest in Digital Engineering learning opportunities and demographic characteristics. Figure 11 visualizes Section 6 survey questions.

Question 6.3

Question 6.3 captures the respondent's professional field; Table 3 gives its verbatim stem and response options. Combining Information Assurance and Cyber Security into a single response category reflects a practical design choice: working professionals identify their field using varied terminology including cybersecurity, information security, and Information Assurance, and a single combined label maximizes correct self-selection without presuming which term a respondent uses. The category was worded to read as familiar both to the governance-oriented professionals Chapter 1 designates Information Assurance and to the operationally oriented professionals it designates cybersecurity. One consequence, carried into the limitations of Chapter 5, is that the resulting category is internally heterogeneous, spanning both senses, and that the anonymous instrument cannot separate a governance-oriented from an operationally oriented respondent within it after the fact. Respondents selecting Engineering or Other appear in descriptive analyses; the primary field comparison contrasts the Information Technology and Information Assurance/Cyber Security groups.

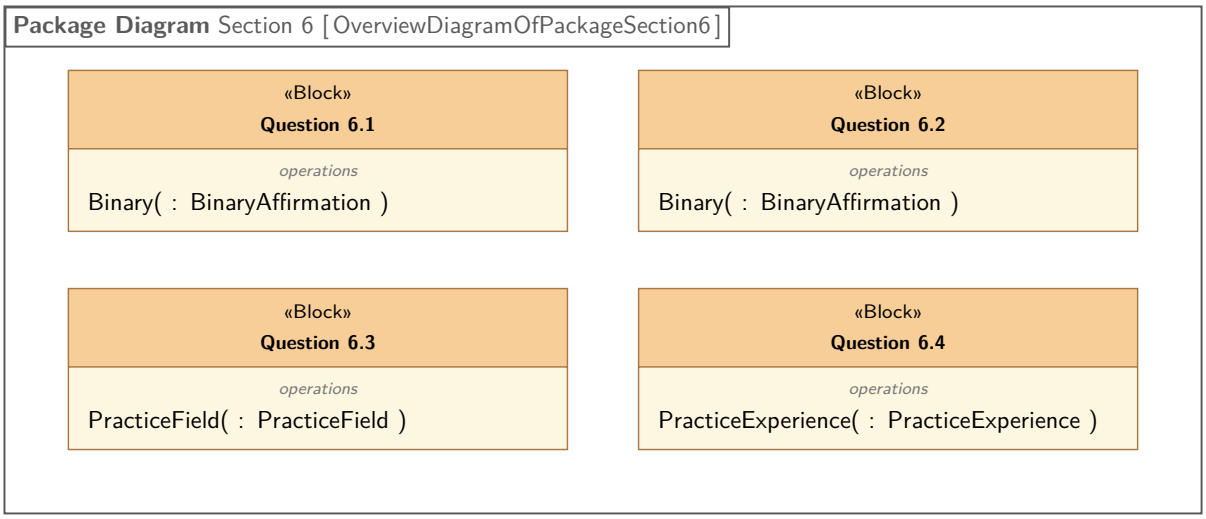


Figure 11: Section Six survey question elements package diagram.

Mapping to Research Questions

While not directly measuring awareness or perceived value, demographic data enables investigation of whether awareness and perceptions vary by professional field or experience level, providing insights for targeted future research and professional development initiatives.

Traceability of Survey Questions

Figure 12 provides a comprehensive diagram that shows authoritative traceability of survey question elements to the research questions. Relationship traceability diagrams enable programmatic verification that every survey question traces to a research question; this check validates traceability completeness rather than construct validity. The mapping is complete: every survey item resolves to at least one research question or supporting demographic category, and no item stands outside the traceability structure.

Satisfy: ↗	Total	Sec. 1		Section 2						Section 3						Section 4					Section 5							Section 6			
Research Questions		Q1.1	Q1.2	Q2.1	Q2.2	Q2.3	Q2.4	Q2.5	Q2.6	Q3.1	Q3.2	Q3.3	Q3.4	Q3.5	Q3.6	Q4.1	Q4.2	Q4.3	Q4.4	Q4.5	Q5.1	Q5.2	Q5.3	Q5.4	Q5.5	Q5.6	Q5.7	Q6.1	Q6.2	Q6.3	Q6.4
Research Question 1	8	↗	↗	↗	↗	↗	↗	↗	↗																						
Research Question 2	10									↗	↗	↗	↗	↗	↗	↗				↗	↗						↗				
Research Question 3	14											↗	↗	↗	↗	↗	↗	↗	↗		↗	↗	↗	↗	↗	↗					
Demographics	2																													↗	↗
Supporting	2																											↗	↗		

Figure 12: Research question traceability.

Data Collection Procedures

Survey Platform and Administration

The study administered the survey electronically through Microsoft Forms, which supports anonymous response collection, secure data storage, and compliance with research ethics requirements. Electronic administration enabled efficient access to geographically distributed respondents while ensuring consistent question presentation and response capture across all participants. Collection ran from April 23 through June 2026, opening once the Institutional Review Board determination was in place and closing when the realized sample had passed the target described above.

Data Protection Plan

In accordance with the survey requirements, the study protected the survey data through the specific controls listed below, safeguarding the confidentiality, integrity, and availability of collected data. The data protection plan included:

- Password protection and multifactor authentication for access to survey administration and data
- Encryption at rest for stored survey responses
- Encryption in transit for data transmission between respondents and the survey platform

- No collection of IP addresses or other metadata that could potentially identify respondents
- Secure storage in compliance with university data governance requirements

Informed Consent

The survey begins with a comprehensive participation notice that explains the research purpose, what participation involves, confidentiality protections, the voluntary nature of participation, and contact information for questions or concerns. Specifically, the notice states:

- The survey is conducted as part of a doctoral research project at Dakota State University
- Participation is completely anonymous with no personally identifiable information collected
- Risks of participation are minimal and similar to those encountered in normal daily internet activity
- Participation is entirely voluntary with the right to withdraw at any time before submission
- Respondents must be 18 years or older to participate
- No compensation or incentive is offered for participation
- Completion and submission of the survey constitutes informed consent

Recruitment and Distribution

The candidate distributed the survey through professional networking outreach, detailed by channel in the following subsection. Recruitment messages clearly described the target population (IT and IA professionals), estimated completion time (approximately ten

minutes), and research purpose. The announcement was framed descriptively rather than persuasively, naming the study topic and eligible population without asserting that Digital Engineering delivers value, and it is reproduced verbatim in Appendix A.

Recruitment Channels and Promotion

LinkedIn served as the primary recruitment medium. Direct participation requests went to professionals in the candidate's LinkedIn network, without filtering by industry or employer. Network composition rather than a sampling decision therefore determined which sectors the invitation reached, and those contacts span aerospace, defense, technology, and government organizations whose work intersects with systems engineering, IT, and IA. These organizations included Collins Aerospace, RTX, Raytheon Technologies, Pratt & Whitney, Anduril, General Atomics, General Dynamics Mission Systems, Boeing, Lockheed Martin, Northrop Grumman, Shield AI, Booz Allen Hamilton, Broadcom, Palo Alto Networks, Dell, Supermicro, Microsoft, the Defense Information Systems Agency, the National Security Agency, the Department of War, Cisco, Defense Unicorns, General Electric, and INCOSE. Reach therefore spanned both the defense and aerospace sector in which Digital Engineering originated and the broader enterprise IT and IA community the research seeks to reach, in proportions the network set rather than the design. Because that network is itself concentrated in defense, aerospace, and federal organizations, the realized sample is weighted toward those sectors, a composition whose implications for generalizability are addressed in the limitations of Chapter 5.

To widen the reach of the recruitment post beyond the candidate's immediate professional network, a paid LinkedIn promotion campaign ran from early May through mid-June with a capped spend of 500 United States dollars. The campaign boosted only the visibility of the call for participants. It offered no compensation or incentive of any kind to respondents, and its sole function was to extend the distribution of the recruitment message. No participant received payment, consistent with the informed-consent terms described above.

Data Analysis

Analytic Approach and Its Rationale

The analytic strategy follows from the nature of the data and the purpose of each research question. It is stated here in full so that every test reported in Chapter 4 can be read against an explicit rationale rather than inferred after the fact. Three principles govern the plan.

First, the survey produces ordinal data. The Likert items record ranked agreement whose intervals are not guaranteed to be equal, and the descriptive results confirm that the composite distributions are non-normal, left-skewed, and concentrated near the top of the scale. Methods that assume normally distributed, interval-scaled data are therefore not the principled default. The analysis accordingly adopts distribution-free procedures as its primary inferential tools: the Mann–Whitney U test for the two-group field comparison, the Kruskal–Wallis test for comparisons across the experience cohorts, Dunn's test with Holm adjustment for the post-hoc contrasts that follow a significant Kruskal–Wallis result, rank-order correlation for the familiarity–value relationship, and the chi-square test of independence for categorical associations. The analysis reports parametric counterparts, principally the independent-samples t-test and ordinary least squares regression, alongside the non-parametric results where they aid comparability with the technology-acceptance literature, and never as the sole basis for a conclusion. The choice between a parametric and a non-parametric test is made per comparison, guided by the Shapiro–Wilk test and by visual inspection of the distributions, as the comparative subsections below describe.

Second, because the realized sample is large, statistical significance is easy to obtain and, on its own, uninformative: at this sample size even a negligible difference will reach $p < .05$. The analysis therefore foregrounds effect sizes and confidence intervals, so that the practical magnitude of a difference, not the mere fact of its significance, carries the interpretation. Each comparison reports the effect size matched to its test: the rank-biserial correlation for Mann–Whitney, eta-squared (η^2) for Kruskal–Wallis, and

Cramér's V for chi-square, with Cohen's d added for comparability against prior work. The analysis reports proportions with Wilson score intervals, which remain well behaved for the skewed, near-ceiling proportions this instrument produces, where the ordinary normal-approximation interval is unreliable. Where many tests are conducted within a family, the Holm step-down correction controls the family-wise error rate without the excessive conservatism of a simple Bonferroni adjustment. The specific estimators and their formulas are given in the Statistical Formulas subsection below.

Third, and prior to any of the above, inference is only as trustworthy as the data are genuine. A conclusion drawn from automated, coordinated, or non-engaged submissions would be a conclusion about noise rather than about the professional community the study means to characterize. The plan therefore begins by establishing that the analyzed responses are organic, that is, that they were produced by real, engaged professional participants rather than by scripts, coordinated rings, or respondents who did not read the items. The data-preparation procedures that follow, together with the response-authenticity analysis reported in Chapter 4, are the mechanism for that assurance. They do not prove that every retained response is genuine, which no screen can establish, but they are designed to remove the identifiable forms of non-organic responding and to document, rather than assume, what confidence in the remaining data is warranted.

Data Preparation and Cleaning

The analysis plan established a priori that responses missing more than 20% of the substantive items would be excluded. Before any statistical work, the analysis reviewed survey responses for completeness and data quality, and this review materially shaped the analytic dataset. The survey platform recorded 1,600 submissions. A manual screening removed 188 of these as abandoned or aberrant, and a post-hoc response-authenticity check, reported in Chapter 4, removed one further submission as an automated entry, leaving 1,411 responses for analysis. Three criteria identified the abandoned or aberrant cases. The first was gross incompleteness: submissions that answered no more than the two

Section 1 questions and supplied nothing thereafter. The majority of these recorded no answers at all, and the remainder halted after the opening one or two questions. In either form, such submissions fall well beyond the 20% missing-data threshold established for exclusion and, more consequentially, provide no data against which the research questions could be evaluated. The second and third criteria concerned implausible completion times. Some submissions were recorded in under three minutes, an interval too brief for considered response to a thirty-item instrument estimated to require approximately ten minutes. Others exceeded five days, a pattern indicating a survey left open in a browser and later abandoned rather than completed in a sitting. The lower cutoff follows response-time screening conventions in web-survey methodology, which flag submissions completed at less than roughly half the median completion time as likely careless responding (Greszki et al., 2015; Leiner, 2019); a three-minute completion of an instrument with an estimated ten-minute median amounts to roughly 30% of that median, comfortably below the conventional threshold. The reasoning is concrete rather than merely conventional: completing thirty items in under three minutes allows fewer than six seconds per item, too little time to read a capability statement, weigh it against one's professional experience, and register a considered judgment. A submission produced at that pace cannot represent the engaged, organic responding the instrument was designed to elicit, whatever its source, and retaining it would substitute mechanical speed for deliberation in the very measures the study reports. The large majority of the pruned responses arrived during the month of May, coinciding with the opening and most heavily promoted phase of recruitment.

The decision to remove these cases rather than retain them rests on a straightforward methodological judgment. Every composite and comparison in this study depends on substantive responses to the Section 2 through Section 5 items, so a submission that stops after Section 1 contributes nothing to those measures while inflating the apparent sample. Retaining near-empty or time-aberrant records would not have enriched the analysis; it would have introduced noise capable of distorting means, variances, and the

planned bias diagnostics, most directly the wave analysis, which keys on the order of submission. Removing 188 such records therefore constitutes a conservative form of data hygiene rather than a selective exclusion of inconvenient cases, because the criteria are content-neutral with respect to how respondents answered the substantive items. After removal, item-level missingness among the retained 1,411 respondents is negligible and was handled through available-case analysis for the specific analyses in which a required variable was absent.

Descriptive Statistical Analysis

The analysis calculated descriptive statistics for all survey questions to characterize the distribution of responses. For Likert-scale questions, these included frequency distributions, mean scores, median scores, and standard deviations. For binary and categorical questions, they included frequency counts and percentages within each response category.

Research Question 1 Analysis

Analysis addressing Research Question 1 focused on Sections 1 and 2 responses to characterize professional awareness of Digital Engineering capabilities. Key metrics included:

- Distribution of familiarity ratings from Question 1.1, with percentage of respondents at each familiarity level
- Percentage of respondents reporting professional exposure to Digital Engineering within the past two years (Question 1.2)
- Mean agreement scores and standard deviations for understanding of specific capabilities (Questions 2.1-2.6)
- Percentage of respondents indicating agreement (score ≥ 4) with each capability statement

Chapter 4 reports results with 95% confidence intervals to indicate the precision of population estimates. For the target sample size of 385 respondents, the margin of error for reported proportions would not exceed 5%; the realized sample of 1,411 responses produced substantially tighter intervals. As noted in the sample size determination, these intervals are conditional on the convenience sample approximating a random draw from the accessible population.

Research Question 2 Analysis

Analysis addressing Research Question 2 focused on Sections 3, 4, and 5 responses measuring perceived value of Digital Engineering capabilities. Key metrics included:

- Mean agreement scores and response distributions for applicability questions (Questions 3.1-3.6)
- Mean agreement scores and response distributions for IT value assessment questions (Questions 4.1-4.4)
- Mean agreement scores and response distributions for cybersecurity value assessment questions (Questions 5.1-5.6)
- Percentage of respondents indicating organizational investment willingness (Questions 4.5 and 5.7)

The analysis interpreted responses indicating agreement (score = 4) or strong agreement (score = 5) with value statements as evidence that professionals perceive potential value in Digital Engineering capabilities.

Research Question 3 Analysis

Analysis addressing Research Question 3 focused on questions specifically addressing job performance, compliance requirements, and organizational capability enhancement. Key questions included:

- Question 3.5: Digital lifecycle management for compliance and service delivery
- Question 3.6: Regulatory or compliance requirements that could benefit from Digital Engineering
- Question 4.1: Meaningful value to IT processes
- Questions 4.2-4.4: Specific IT operational benefits (cycle time, quality, collaboration)
- Question 5.1: Meaningful value to cybersecurity operations
- Questions 5.2-5.6: Specific security operational benefits (vulnerability remediation, security posture, threat modeling, collaboration, compliance)

The percentage of respondents indicating agreement with these statements provided direct evidence regarding whether professionals believe Digital Engineering could help them in their work.

Comparative Analysis

Comparative analyses examined whether awareness levels and value perceptions differ significantly between professional subgroups and experience levels, directly assessing Proposition P3 (variation by field and experience). Such differences carry important implications for adoption strategies and professional development initiatives.

Professional Field Comparison

Independent samples t-tests and Mann–Whitney U tests compared responses between IT professionals and security professionals (based on Question 6.3), serving the field component of Proposition P3. The observed distributional shape, corroborated by Shapiro–Wilk tests and visual inspection, guided the choice of parametric versus non-parametric test. As reported in Chapter 4, the Shapiro–Wilk tests rejected normality for all composites (an expected outcome at this sample size, where the test is highly powered against trivial departures), and the composites are left-skewed and ceiling-concentrated. Mann–Whitney U therefore served as the primary test.

Experience Level Comparison

Kruskal–Wallis tests compared the composites across the four experience level categories (Question 6.4) because the composite distributions are non-normal; these comparisons serve the experience component of Proposition P3. Dunn's test with Holm adjustment identified which specific groups differed.

Association Analysis

Chi-square tests of independence examined associations between categorical variables, such as the relationship between professional field and prior Digital Engineering exposure (Question 1.2), or between experience level and organizational investment willingness (Questions 4.5, 5.7). These categorical associations, reported with Cramér's V, complete the assessment of Proposition P3.

Familiarity and Value Association

Rank-order correlation between the familiarity item (Question 1.1) and the value composites, supplemented by ordinary least squares regression of the value composites on familiarity, professional field, and experience level, assesses Proposition P4 (familiarity and value).

Composite Score Analysis

Composite scores average the Likert responses within thematic groupings, providing summary measures of awareness and perceived value:

Awareness Composite

Average of Question 1.1 and Questions 2.1-2.6 (seven items measuring familiarity and understanding of Digital Engineering capabilities)

IT Value Perception Composite

Average of Questions 3.1, 3.3-3.5, and 4.1-4.4 (eight items measuring perceived value of Digital Engineering for IT operations)

Security Value Perception Composite

Average of Questions 3.2, 3.6, and 5.1-5.6 (eight items measuring perceived value of Digital Engineering for Information Assurance and cybersecurity)

The analysis assigned items to composites a priori on theoretical grounds rather than deriving them empirically. In particular, the Section 3 applicability items were allocated between the IT Value and Security Value composites according to the domain each item addresses: items concerning IT operations, environment modeling, and lifecycle value (Questions 3.1, 3.3–3.5) were placed with the IT items, while the items concerning Information Assurance relevance and compliance exposure (Questions 3.2, 3.6) were placed with the security items. Internal consistency was then used to confirm, not to construct, these groupings. Cronbach's alpha assessed the internal consistency of composite scores, with values above 0.70 considered acceptable for research purposes. Had internal consistency proven insufficient, item analysis would have been conducted to identify potentially problematic items; as Chapter 4 reports, all three composites exceeded the threshold, so the composites were carried forward as specified here. Chapter 4 also reports item-level diagnostics for the Awareness composite, which combines a familiarity self-report with capability-agreement items, together with the reasoning for retaining its familiarity item despite those diagnostics.

Statistical Significance and Effect Sizes

All inferential tests used $\alpha = 0.05$. However, given the large sample size realized in this study, the analysis reports effect sizes alongside every significance test to assess the practical significance of observed differences. Statistical significance alone can mislead when sample sizes are large; effect sizes reveal whether differences matter in practice. The rank-biserial correlation serves as the primary effect size for Mann–Whitney comparisons and eta-squared (η^2) for Kruskal–Wallis tests, consistent with the non-parametric primary tests. The analysis reports Cohen's d as a supplementary measure for comparability with the technology-acceptance literature, interpreting it against conventional thresholds of 0.2

(small), 0.5 (medium), and 0.8 (large). For chi-square tests, Cramér's V serves as the effect size measure.

Statistical Formulas

For transparency and reproducibility, the estimators and test statistics used in the analysis are stated explicitly below. Each Likert item and composite was summarized by the sample mean and the sample standard deviation,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{3}$$

The precision of each mean was expressed through a 95% confidence interval based on Student's t distribution, while agreement proportions $\hat{p} = k/n$ were bounded by the Wilson score interval, which remains well behaved near the response ceiling,

$$\bar{x} \pm t_{0.975,\,n-1}\,\frac{s}{\sqrt{n}}, \qquad \frac{\hat{p} + \dfrac{z^2}{2n} \;\pm\; z\sqrt{\dfrac{\hat{p}\,(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \tag{4}$$

with $z = z_{0.975} = 1.96$. Internal consistency for each composite of k items was quantified by Cronbach's α, in which σ_j^2 denotes the variance of item j and σ_T^2 the variance of the summed score,

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{j=1}^{k}\sigma_j^2}{\sigma_T^2}\right). \tag{5}$$

Group differences were summarized by Cohen's d, computed with the pooled standard deviation s_p, and by the rank-biserial correlation r_{rb} derived from the Mann–Whitney statistic U,

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}, \qquad r_{rb} = 1 - \frac{2U}{n_1 n_2}. \tag{6}$$

The Mann–Whitney test itself compares the rank sum R_1 of the first group against its expectation, and the Kruskal–Wallis test generalizes the comparison to g groups, where R_j is the sum of the pooled-sample ranks in group j, with the η^2 statistic serving as its effect size,

$$U = R_1 - \frac{n_1(n_1+1)}{2}, \qquad H = \frac{12}{N(N+1)} \sum_{j=1}^{g} \frac{R_j^2}{n_j} - 3(N+1), \qquad \eta^2 = \frac{H-g+1}{N-g}. \quad (7)$$

Associations between categorical variables were tested with the chi-square statistic, comparing observed counts O_{ij} against the counts $E_{ij} = R_i C_j / N$ expected under independence, and the magnitude of each association was reported as Cramér's V,

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad V = \sqrt{\frac{\chi^2}{N\,(\min(r,c)-1)}}. \quad (8)$$

Multiplicity within each family of related comparisons was controlled by the Holm–Bonferroni step-down procedure, which orders the m unadjusted p-values ascending and rejects the hypothesis associated with $p_{(i)}$ while

$$p_{(i)} \le \frac{\alpha}{m-i+1}. \quad (9)$$

The procedure stops at the first i for which the inequality fails and retains all remaining hypotheses. These expressions were implemented in a fully scripted analytical pipeline, so that every statistic reported in Chapter 4 is traceable to its definition and reproducible from the response data.

Management of Type I and Type II Errors

Multiple provisions within the analytical design manage the risk of both Type I errors (false positives, in which the analysis concludes that a difference or association exists when it does not) and Type II errors (false negatives, in which the analysis fails to detect a genuine difference or association). These provisions operate at the levels of sample design,

instrument design, and analytical procedure.

Type I Error Management

The primary Type I error concern in this study arises from conducting multiple statistical tests across the twenty-three Likert-scaled survey items (organized into the four comparison families used in the analysis: awareness, applicability, IT value, and security value) and multiple subgroup comparisons. When many tests are performed at the $\alpha = 0.05$ significance level, the family-wise error rate increases, elevating the probability that at least one test produces a spurious significant result. To manage this inflation, the analytical approach uses the Holm–Bonferroni sequential correction procedure for families of related comparisons. The Holm–Bonferroni method provides stronger Type I error control than unadjusted testing while maintaining greater statistical power than the classical Bonferroni correction, which can be overly conservative when applied to large families of tests.

Comparisons are organized into logical families that follow the structure of the research questions. Awareness items, drawn from Sections 1 and 2, address Research Question 1. The remaining three families all bridge Research Questions 2 and 3: applicability items from Section 3, IT value items from Section 4, and IA value items from Section 5. Within each family, the Holm–Bonferroni correction adjusts significance thresholds to maintain the family-wise error rate at $\alpha = 0.05$. Cross-family comparisons, such as the relationship between overall awareness and overall perceived value, are treated as planned comparisons evaluated at the unadjusted $\alpha = 0.05$ level, as these represent distinct theoretical questions rather than multiple tests of similar hypotheses.

The emphasis on effect sizes alongside significance testing provides an additional safeguard against Type I error interpretation. Even when statistical tests yield significant p-values after correction, effect size assessment indicates whether observed differences reflect practically meaningful magnitudes rather than trivial differences detected through large sample power.

Type II Error Management

The design manages Type II error risk primarily through the chosen sample size. The target sample of 385–450 completed responses provides sufficient statistical power to detect meaningful differences in the primary analyses. For the primary descriptive analyses addressing Research Question 1, the sample size supports margins of error within five percentage points at the 95% confidence level for reported proportions and means, providing adequate precision to characterize awareness levels and value perceptions.

For comparative analyses between professional subgroups, power considerations informed the oversampling target. The design assumed an approximately even split between IT and IA respondents yielding subgroups of roughly 190–225 each, at which independent samples t-tests achieve power exceeding 0.80 to detect medium effect sizes (Cohen's d = 0.5) at $\alpha = 0.05$, and power of approximately 0.83–0.89 for small effects (Cohen's d = 0.3). Effects smaller than d $\approx$ 0.3 could nonetheless go undetected at the planned subgroup sizes. The realized sample eased this concern: the two focal subgroups comprised 501 IT and 501 IA respondents, far exceeding the planning assumption, so the achieved power to detect even small effects is high. Non-significant subgroup results are nonetheless interpreted as insufficient evidence rather than as evidence of no difference.

For experience level comparisons across four categories, the distribution of respondents across categories may result in unequal cell sizes that reduce power for specific pairwise comparisons. The analytical plan addresses this by reporting confidence intervals for all group means, enabling assessment of overlap and practical significance even when formal tests fail to reach significance. Non-parametric alternatives (Kruskal–Wallis with Dunn's post-hoc test) are used when distributional assumptions are not met, as these tests maintain appropriate Type II error rates under non-normal conditions.

Instrument-Level Error Reduction

Survey instrument design contributes to error management through several structural features. Established five-point Likert scales with validated anchor sets reduce

measurement error that could contribute to both Type I and Type II errors, so that response variability more closely reflects genuine differences in perception than response format artifacts. Consistent scale formatting across Sections 2 through 5 reduces within-respondent variability attributable to format switching, improving signal-to-noise ratios in comparative analyses.

The inclusion of both broad constructs (overall awareness, overall value perception) and specific sub-constructs (awareness of individual Digital Engineering pillars, value for specific operational domains) enables triangulation of findings. Convergent results across related items strengthen confidence in significant findings (reducing false positive concern), while divergent results prompt examination of whether non-significant findings reflect genuine null effects or insufficient measurement sensitivity (informing Type II error assessment).

Composite score analysis, with internal consistency assessed through Cronbach's alpha, further mitigates measurement error by aggregating multiple indicators of each construct. Composite scores based on reliable item sets exhibit less measurement error than individual items, improving the precision of group comparisons and reducing Type II error risk for analyses involving aggregated constructs.

Reliability and Validity Considerations

Content Validity

Systematically mapping survey questions to research questions, together with aligning to established frameworks in the Digital Engineering and technology acceptance literature, supports content validity. The survey instrument structure directly addresses the four pillars of Digital Engineering (MBSE, the Digital Thread, Digital Twin, and PLM) as established by INCOSE and adopted by NASA, DoW, and the Intelligence Community. The dissertation committee reviewed the instrument questions during development to assess how well they represent the intended constructs and whether they use appropriate professional terminology familiar to IT and IA practitioners.

Construct Validity

Established question formats and scale anchors, drawn from validated instruments in technology acceptance research, support construct validity. The agreement scale format follows conventions from TAM and UTAUT research and has demonstrated validity for measuring technology perceptions across diverse populations. A tripartite structure examining awareness, applicability, and perceived value follows established patterns in technology adoption research that have proven effective across multiple domains.

Reliability

Cronbach's alpha assessed internal consistency reliability for the composite scores. Additionally, the standardized question format and scale anchors across sections support response consistency by presenting respondents with familiar response frameworks throughout the survey. The use of consistent Likert scale response options across Sections 2-5 reduces potential confusion and supports reliable responding.

Pilot Testing and Instrument Refinement

Before formal dissertation proposal development, the survey instrument underwent pilot testing during the Spring 2024 semester. The pilot proceeded as instrument-development work under a formal Institutional Review Board exempt determination, separate from and preceding the exempt determination obtained for the dissertation survey itself; no pilot data are included in the analytic sample reported in Chapter 4. The pilot administered an earlier version of the instrument to IT and IA professionals, serving multiple purposes: evaluating question clarity and comprehension among respondents representative of the target population, assessing completion time and respondent fatigue, identifying ambiguous or confusing question formulations, and generating preliminary data to inform the structural design and thematic organization of the final instrument.

Data collected during the pilot study provided empirical evidence that directly shaped the design and structure of the survey questions presented in this methodology.

Pilot results informed several consequential design decisions reflected in the current instrument. The pilot findings drove refinement of question wording to reduce ambiguity in how Digital Engineering concepts were described. The revised wording had to provide sufficient context for respondents unfamiliar with Digital Engineering terminology while avoiding language that could prime respondents toward particular response patterns. Pilot completion patterns motivated adjusting section ordering to place awareness questions before value assessment questions, establishing a cognitive progression from recognition through evaluation. The pilot also validated the estimated completion time and confirmed that the five-point Likert scale format produced adequate response distribution across scale points in the pilot sample, without floor or ceiling effects at that stage. The realized sample proved more concentrated toward agreement than the pilot anticipated, a pattern Chapter 4 reports and treats as a threat to the comparative effect sizes.

The pilot study's contribution extends beyond instrument refinement to substantive validation of the research premise. Pilot data confirmed that the target population includes substantial variation in Digital Engineering awareness levels, validating the foundational assumption that awareness and perceived value remain open empirical questions requiring investigation. The observed variance in pilot responses across both awareness and value dimensions confirmed that the instrument captures meaningful differences among respondents rather than producing uniform response distributions that would limit analytical value.

The pilot testing process complemented the content validity established through systematic mapping of survey questions to research questions and alignment with established Digital Engineering and technology acceptance frameworks. Where the content validity process supported measurement of the intended constructs, pilot testing confirmed that respondents interpret questions as intended and can provide meaningful responses within the designed format. The combination of theoretical grounding through framework alignment and empirical refinement through pilot testing strengthens confidence in the

instrument's ability to generate valid, reliable data addressing the research questions.

Limitations

Several limitations should be considered when interpreting study results:

- The non-probability sampling approach limits generalizability to the broader population of IT and IA professionals. Results should be interpreted as indicative of perceptions within the accessible population rather than definitive measures of the entire professional community.
- Self-selection bias may result in over-representation of professionals with existing awareness or interest in Digital Engineering. Individuals who recognize the term or have prior exposure may be more likely to complete the survey, potentially inflating reported awareness levels.
- Social desirability bias may influence responses, particularly regarding perceived value questions. Respondents may indicate greater perceived value than they genuinely hold if they believe Digital Engineering represents a progressive or professionally desirable position.
- Self-reported awareness and perceptions may not accurately reflect actual knowledge or organizational capabilities. Respondents may overestimate their familiarity with Digital Engineering concepts or underestimate existing capabilities within their organizations that align with Digital Engineering principles but use different terminology.
- The cross-sectional design captures perceptions at a single point in time during a period of rapid evolution in both Digital Engineering practices and enterprise IT methodologies. Results represent a temporal snapshot that may not reflect awareness levels or perceptions six months or a year following data collection, particularly given increasing industry attention to digital transformation initiatives.

- Non-target respondents who complete the survey may affect results, though demographic questions enable identification and potential exclusion of responses from outside the target population.
- Several question stems bundle related capabilities within a single statement (for example, simulating security scenarios and testing defensive measures), a double-barreled wording pattern that means item-level agreement cannot be attributed to a single clause of the stem.
- The survey measures perceived value and anticipated benefits rather than actual experienced benefits. Positive perceptions do not guarantee that Digital Engineering would deliver value if implemented, nor do they indicate organizational readiness for adoption.

Response Bias Mitigation

Several design decisions embedded within the survey instrument and recruitment strategy proactively mitigate identified sources of response bias. These measures complement the acknowledged limitations by reducing their anticipated impact on data quality and interpretive validity.

Recruitment messaging mitigates self-selection bias, whereby respondents with prior Digital Engineering awareness disproportionately choose to participate, by explicitly framing the survey as targeting all IT and IA professionals regardless of Digital Engineering familiarity. Recruitment materials emphasize that the research seeks perspectives from professionals across the awareness spectrum and that no prior knowledge of Digital Engineering is required or expected. This framing encourages participation from professionals who might otherwise self-exclude on encountering unfamiliar terminology.

Survey questions in Sections 2 through 5 incorporate minimal contextual descriptions of Digital Engineering capabilities within the question stems, reflecting a deliberate balance between two competing methodological concerns. On one hand,

questions devoid of contextual information would measure only pre-existing awareness, excluding respondents unfamiliar with Digital Engineering terminology from providing meaningful responses about perceived value. On the other hand, extensive descriptions risk priming respondents toward positive assessments. The instrument resolves this tension by providing sufficient context for respondents to form an informed judgment about each capability statement without advocating for or characterizing the capability in evaluative terms. Each question describes what a Digital Engineering capability does in neutral, functional language, enabling respondents to assess whether that function would provide value within their professional context based on their own domain expertise. This approach reduces confusion for respondents unfamiliar with Digital Engineering terminology who would otherwise face questions referencing concepts they have not encountered, while avoiding framing that suggests a preferred response direction.

The anonymous survey design mitigates social desirability bias by removing professional reputation concerns that might otherwise influence responses. The absence of personally identifiable information collection, combined with clear communication of anonymity protections in the informed consent notice, encourages candid responses about knowledge gaps and skepticism toward unfamiliar methods.

Two analytical checks address acquiescence bias, the tendency for respondents to agree with statements regardless of content: the response-pattern screen for uniform agreement across all items (straight-lining) described in the Content Requirements section, and the low-familiarity/high-agreement diagnostic that flags respondents who report little familiarity yet agree strongly with value statements. The neutral midpoint ("Neither agree nor disagree") complements these checks by reducing forced-choice artifacts, enabling genuine expression of uncertainty or indifference rather than forcing agreement or disagreement. The binary and ternary response options for investment willingness questions (Questions 4.5 and 5.7) include an "Unsure" option that captures genuine uncertainty without channeling indecisive respondents toward affirmative responses.

Because all value items are positively keyed, these measures bound rather than eliminate acquiescence, and the residual risk is acknowledged in the limitations.

The analysis compared early and late respondents (wave analysis) to assess whether response patterns differ systematically between initial and subsequent response waves. Significant differences between early and late respondents may indicate non-response bias, as late respondents are often considered more representative of non-respondents, a rationale Chapter 4 revisits in light of the recruitment history. As reported there, this diagnostic returned a null result, with negligible differences between early and late respondents on all three composites. The recruitment-frame concern developed in Chapter 5 therefore rests on the sampling logic itself rather than on a wave signal. The analysis also compared the demographic composition of the sample against known characteristics of the IT and IA professional population to assess representativeness, with identified disparities reported as potential limitations on generalizability.

An additional analytical safeguard enables post-hoc assessment of potential priming effects from the contextual descriptions embedded in survey questions. Comparing value perception scores (Sections 4 and 5) between respondents reporting high Digital Engineering familiarity (Question 1.1 scores of four or five) and those reporting low familiarity (scores of one or two) provides a diagnostic for priming influence. If both familiarity groups report similarly elevated perceived value, the priming hypothesis gains support, suggesting that question descriptions may have influenced responses independent of prior knowledge. If high-familiarity respondents report substantially higher perceived value than low-familiarity respondents, the measurements more likely reflect genuine perception differences informed by prior knowledge and professional experience rather than question framing effects. This comparison does not eliminate priming concerns but provides empirical evidence for assessing their magnitude and interpretive implications.

Ethical Considerations

This research was designed in accordance with ethical principles for human subjects research and submitted to the Dakota State University Institutional Review Board (IRB). The IRB issued an exempt review determination under 45 CFR 46.104(d)(2), the category covering research limited to survey procedures (determination DSUIRB-20260421-13EX, April 21, 2026), and that determination was in place before any participant recruitment or data collection began; the first survey response was recorded on April 23, 2026. The full IRB Human Subjects Application is reproduced in Appendix D. Key ethical considerations include:

Anonymity

The survey collects no personally identifiable information, ensuring respondent anonymity. Demographic questions are limited to professional field and experience level, which cannot identify individual respondents.

Voluntary Participation

Participation is voluntary with no consequences for non-participation. Respondents may exit the survey at any time before submission without consequence and may skip any questions they choose not to answer.

Minimal Risk

The research presents minimal risk to participants, with potential inconvenience limited to time required for survey completion. Risks are similar to those encountered in normal daily internet activity.

Informed Consent

Informed consent is obtained through the participation notice presented before survey questions, with submission constituting consent.

Data Protection

Data were stored securely using password protection, multifactor authentication, and encryption at rest and in transit. Data were used solely for research purposes as described in the participation notice.

Data Use Restrictions

Survey raw data shall not be sold or used for any purpose other than the dissertation research.

Researcher Positionality and Data Availability

The candidate works professionally in systems engineering and Information Assurance and previously developed the reference model for Digital Engineering integration that motivates this investigation (Bonar & Hastings, 2024). This proximity to the subject informed the framing of the research questions and the design of the instrument, and it is disclosed here in the interest of transparency. Several features of the study were chosen to limit its influence on the findings. The survey is anonymous; the value items are framed as capability descriptions to be judged by respondents against their own context rather than as advocacy; the analysis foregrounds effect sizes and the planned bias diagnostics over selective emphasis; and the entire analytical pipeline is scripted, so that the reported results are reproducible rather than dependent on discretionary choices. One residual lies outside the reach of these provisions. Every item is affirmatively keyed, so a respondent inclined to agree meets no item on which agreement would register as disconfirmation, and no mitigation listed here alters that property of the instrument. Chapter 5 specifies the reverse-coded and knowledge-verification items a future administration would need to close it. The de-identified response data and the analysis code that generated every table and figure in this dissertation are available from the author on reasonable request, subject to the conditions of the Institutional Review Board approval.

Chapter Summary

This chapter has presented the research methodology used to investigate professional awareness and perceptions of Digital Engineering capabilities among IT and IA professionals. The research uses a quantitative survey methodology and follows a systems engineering lifecycle approach that supports rigor and traceability throughout the research process. Pilot testing during the Spring 2024 semester refined the survey instrument, which systematically addresses the three research questions through 30 questions organized into six thematic sections covering awareness, understanding, applicability, and perceived value for both IT and IA domains.

The design specified a target sample of 385–450 completed responses to provide sufficient statistical power and a maximum margin of error of 5% at the 95% confidence level; the realized sample of 1,411 responses exceeded this target threefold. Data analysis used descriptive statistics to characterize awareness levels and value perceptions, with comparative analyses examining differences across professional subgroups and experience levels. Composite scores summarized overall awareness and value perceptions, with internal consistency assessed through Cronbach's alpha. The analytical plan addresses the ordinal-interval debate surrounding Likert scale data through dual-reporting of parametric and non-parametric summaries, and manages Type I error inflation from multiple comparisons through the Holm–Bonferroni sequential correction procedure applied within logical families of related tests.

The study incorporates proactive response bias mitigation through recruitment messaging designed to encourage participation across the awareness spectrum, survey question design that provides minimal contextual information sufficient for informed response without evaluative priming, and planned wave analysis to assess non-response bias. Complementing the acknowledged limitations of non-probability sampling and self-selection, these provisions strengthen confidence in the validity of collected data.

The methodology establishes a rigorous foundation and generates empirical evidence

regarding professional perceptions of Digital Engineering that can inform future research and practice in IA and IT service delivery. The systems engineering approach to research design supports traceability between research questions, survey instruments, and analytical approaches, thereby demonstrating the disciplinary rigor of the research.

Chapter 4: Results and Analysis

This chapter presents the results of the survey data collection and analysis described in Chapter 3. The analysis executes the plan specified in the methodology: descriptive characterization, composite construction and reliability assessment, comparative testing across professional field and experience level, association analysis, and the planned bias diagnostics. The chapter organizes findings by the three research questions established in Chapter 1, answering each descriptively first and consolidating the three answers in Table 5 before it turns to the subgroup, regression, and robustness analyses that qualify them. Consistent with Chapter 3, the chapter reports parametric descriptive statistics (means, standard deviations) alongside ordinal summaries: medians and interquartile ranges accompany the item-level descriptives, medians accompany the composite descriptives, and the comparative tables report means with the rank-based tests. Non-parametric tests serve for inferential comparisons where distributional assumptions are not met. Because the sample is large, the analysis foregrounds effect sizes and confidence intervals and treats statistical significance as secondary.

A fully scripted pipeline produced all statistics, tables, and figures in this chapter from the collected response file, and independent recomputation cross-validated every reported value. The data depart from the proposal's expectations in three consequential respects: sample size, two investment-willingness items that show ceiling effects, and a non-monotonic experience pattern. The chapter reports these departures directly because they bear on how the findings should be interpreted.

Data Preparation and Sample

Data collection yielded **1,600** submitted responses, of which **1,411** remained for analysis. A pre-registered quality screen removed 188 abandoned or time-aberrant submissions: those answering at most the two opening questions, completing in under three minutes, or leaving the survey open beyond five days. A post-hoc response-authenticity screen (Section 5) then removed one further submission from a coordinated automated

cluster. Every count and statistic in this chapter refers to that analytic sample of 1,411, except where item-level missingness reduces the available n, as noted. The pipeline decoded Likert items from their anchored string form (for example, "4 - Agree") to the integer 1–5 scale and retained the binary and ternary items as categorical factors. Within the analytic sample, complete data across all twenty-three Likert-scaled items are present for 1,390 respondents (98.5%). Available-case analysis handled item-level missingness, which never exceeds six cases on any single item, consistent with the Chapter 3 plan to assess rather than impute missing data. Because the removed submissions answered at most the two opening questions—most of them nothing at all—and clustered at implausible completion times, excluding them sharpens rather than biases the analytic sample. It strips abandonment noise that would otherwise distort the composite distributions and the completion-time-based wave diagnostic, without privileging any particular pattern of substantive response.

At 1,411 completed responses, the realized sample is roughly three times the planned target of 385–450 derived in Chapter 3. This far exceeds the requirement for a five-percent margin of error and strengthens the precision of population estimates. It also reframes the inferential reading anticipated in the methodology: at $n > 1{,}400$, virtually any non-trivial difference attains $p < .001$, so statistical significance carries little interpretive weight on its own. What the sample delivers is precision. The 95% margin of error on the headline proportions is roughly one to two percentage points, for example ± 1.2 points on the 93.9% Section 2 endorsement and ± 2.2 points on the 78.2% low-familiarity share. The professional-field comparison is likewise powered essentially at unity. With 501 respondents in each focal group, the design detects an effect as small as $d = 0.2$ with power near 0.89, rising above 0.99 for any effect of $d \geq 0.27$, at $\alpha = .05$, far below the smallest effect observed. The analysis below therefore foregrounds effect sizes (Cohen's d, the rank-biserial correlation r_{rb}, Cramér's V, and η^2) and treats p-values as secondary. Because at this sample size every non-trivial test returns a p-value below any conventional threshold, the

chapter reports significant results as $p < .001$ and gives exact p-values for non-significant ones. The effect sizes and their confidence intervals carry the inferential weight.

Two demographic features of the realized sample differ from the proposal's framing. First, the professional-field item (Q6.3) resolved into *four* categories rather than the planned Information Technology versus Information Assurance dichotomy: Information Technology (35.5%), Information Assurance/Cyber Security (35.5%), Engineering (24.0%), and Other (5.0%). Following the Chapter 3 plan, the primary field comparison below contrasts Information Technology against Information Assurance/Cyber Security. The Engineering and Other groups are retained for descriptive context but not forced into a binary that the instrument did not produce. Second, the experience distribution is weighted toward earlier-career respondents (40.1% at 1–5 years; 34.7% at 6–10 years; 24.0% at 11–15 years), and the 16+ year cohort contains only 18 respondents (1.3%). These percentages are computed on the full analytic sample of 1,411 respondents. As Chapter 3 foresaw, that cell is too small to support stable pairwise inference, and the results below bear out this limitation.

Research Question 1: Awareness of Digital Engineering

Research Question 1 addresses awareness. The analysis draws on the Section 1 awareness items and the Section 2 capability-recognition items, following the layered assessment in Chapter 3 that separates familiarity with the named discipline from recognition of its capabilities once those capabilities are described in functional terms.

Self-Reported Familiarity and Exposure

Self-reported familiarity with Digital Engineering (Q1.1) is low across the sample. Nearly half of respondents (49.6%) selected "Not familiar" and a further 28.6% "Slightly familiar," placing 78.2% in the bottom two categories; the item mean is 1.94 ($SD = 1.20$, $Mdn = 2$). Only 17.6% (95% CI [15.7, 19.6]) rated themselves "Very" or "Extremely" familiar (Figure 13). Recent professional exposure (Q1.2) is consistent: 22.2% reported encountering Digital Engineering methodologies, frameworks, or tools in their work within

the past two years, against 77.8% who reported none. The familiarity self-assessment thus answers the direct, surface-level component of Research Question 1 unambiguously: awareness of Digital Engineering, as a named discipline, is low within this population of IT and Information Assurance professionals.

Beyond its central tendency, the shape of that distribution carries information of its own. Familiarity is bimodal rather than uniformly low. Two lowest categories hold 78.2% of respondents and the middle category holds only 4.2%, yet "Very familiar" recovers to 13.6%, more than three times the middle category. Responses therefore separate into a large stratum with little or no acquaintance with the discipline and a much smaller stratum that knows it well, with few respondents occupying the ground between them. That structure bears on the comparative analyses, because a sample composed of two such strata can produce group differences that reflect the composition of each group as much as the professional field defining it. The robustness analyses later in this chapter test that possibility directly and find that it accounts for part of the field difference while leaving most of it intact.

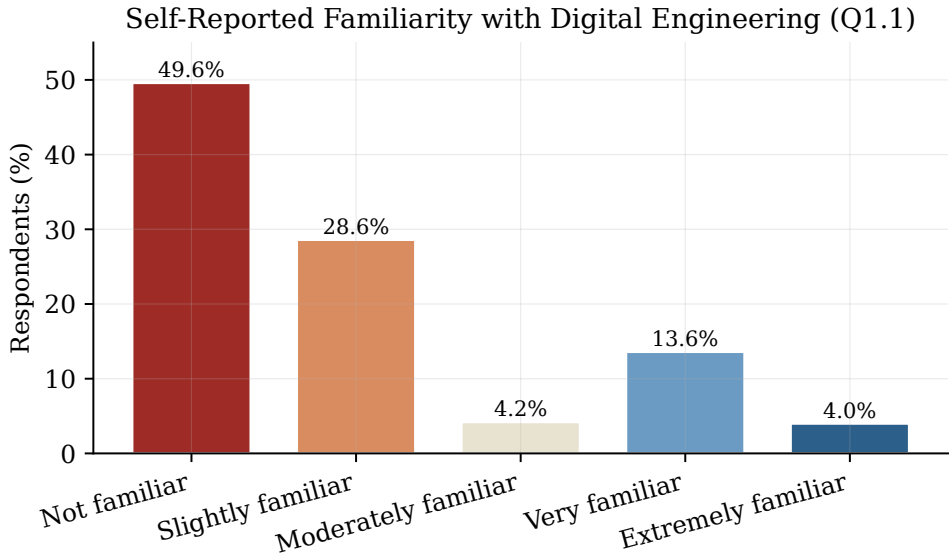


Figure 13: Self-reported familiarity with Digital Engineering (Q1.1).

Recognition of Specific Capabilities

The Section 2 capability-recognition items contrast sharply with the familiarity data. Agreement with the six capability statements is uniformly high: 93.9% of Section 2 responses fall at "Agree" or "Strongly agree," with item means between 4.29 and 4.50 (Table 4, items Q2.1–Q2.6). The same respondents who reported little familiarity with Digital Engineering nonetheless affirmed, at very high rates, that Digital Engineering "includes model-based systems engineering approaches" (Q2.1, 95.7% agreement), "can enable digital twin development" for IT systems (Q2.2, 95.8%), and "can improve security control implementation through automated compliance checking" (Q2.6, 95.7%). The full distribution of the twenty-two agreement-scale items (Sections 2 through 5), centered on the neutral midpoint, appears in Figure 14: every item is strongly right-shifted, with minimal disagreement throughout. Because Q1.1 measures familiarity on a different response scale, its distribution appears separately in Figure 13 rather than combined with the agreement items.

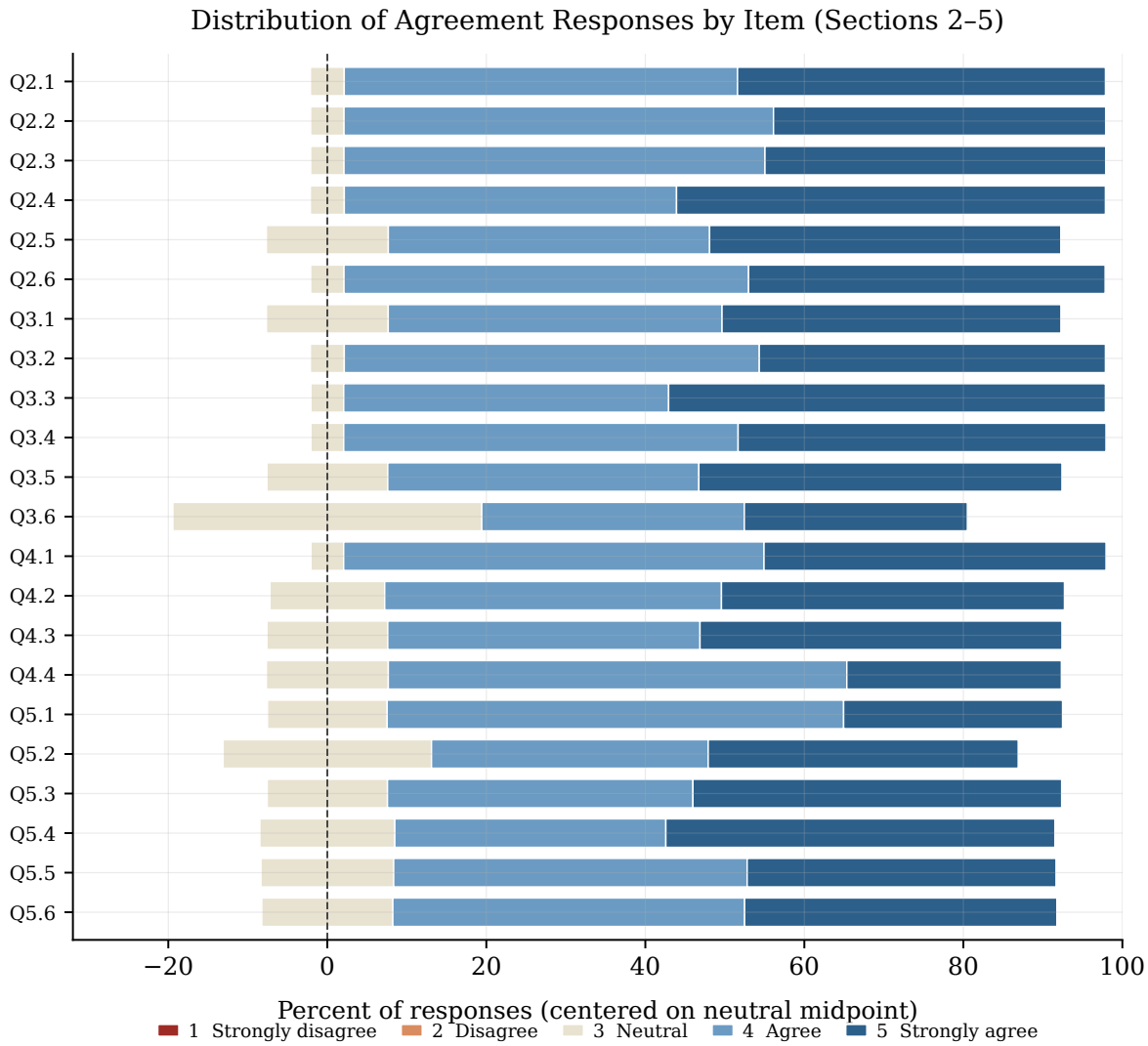


Figure 14: Response distributions for the twenty-two agreement-scale items (Q2.1–Q5.6).

This divergence is the central finding for Research Question 1, and it bears out the distinction drawn in Chapter 3 between awareness of the named discipline (Section 1) and recognition of described capabilities (Section 2). Two readings are admissible, and the data cannot fully separate them. The charitable reading is that respondents recognize and endorse the described capability relationships once those relationships are stated in plain functional language, even without prior familiarity with the umbrella term. The cautionary reading is that the Section 2 stems describe capabilities in affirmative terms that invite agreement, so high endorsement may partly reflect acquiescence or face-validity assent

rather than substantive knowledge. Neither reading can be settled here, because the instrument carries no knowledge-verification item against which an agreement response could be scored correct or incorrect. Section 2 therefore bounds what Research Question 1 can claim: it measures recognition, and comprehension lies outside its reach. The near-absence of disagreement on these items, visible in Figure 14, is consistent with either account and should temper any claim that Section 2 shows pre-existing domain awareness.

A straight-lining check across the twenty-two agreement-scale items quantifies the acquiescence risk: 1.6% of respondents (22 of the 1,390 with the complete battery) gave an identical response to all twenty-two items, a small yea-saying signature (Appendix B). Most vary their responses (median within-respondent standard deviation 0.50), so crude and uniform acquiescence is a minor presence rather than a driver of the high endorsement. That check bounds the extreme form and not acquiescence as such, because a mild yea-saying tilt would produce exactly the within-respondent variation observed here, and Chapter 5 accordingly keeps it among the open accounts. The present results therefore characterize Section 2 as measured *endorsement of capability descriptions* rather than as evidence of prior awareness. Taken with the Section 1 findings, the answer to Research Question 1 is that awareness of Digital Engineering as a discipline is low, while endorsement of its constituent capabilities, once described, is high.

Research Question 2: Perceived Value

Perceived value is high across all three value blocks, which is the answer the second research question receives. Three groups of items measure it: applicability in Section 3, information technology value in Section 4, and security value in Section 5. Agreement runs at 86.3% for applicability, 87.7% for information technology, and 82.2% for security, and item means fall between 3.9 and 4.5 (Table 4). Item-level agreement with 95% Wilson confidence intervals (Wilson, 1927) appears in Figure 15. The intervals are narrow throughout, a consequence of the large sample, though that narrowness is conditional on the accessible sample rather than on a random draw of the profession.

Table 4: Descriptive statistics for the Likert-scale items (1–5).

Item	Construct	n	M	SD	Mdn	IQR	Agree%	95% CI
Q1.1	Familiarity with DE (Q1.1)	1,411	1.94	1.20	2	1–2	17.6	[15.7, 19.6]
Q2.1	MBSE improves dev processes	1,411	4.42	0.57	4	4–5	95.7	[94.6, 96.7]
Q2.2	Digital twin for IT systems	1,405	4.38	0.56	4	4–5	95.8	[94.6, 96.7]
Q2.3	Continuous integration / data-driven	1,407	4.39	0.57	4	4–5	95.8	[94.6, 96.7]
Q2.4	Digital twin for security scenarios	1,407	4.50	0.58	5	4–5	95.7	[94.5, 96.7]
Q2.5	Continuous security validation	1,408	4.29	0.72	4	4–5	84.6	[82.6, 86.4]
Q2.6	Automated compliance checking	1,407	4.41	0.57	4	4–5	95.7	[94.5, 96.7]
Q3.1	Relevant to IT sector	1,410	4.27	0.72	4	4–5	84.6	[82.6, 86.4]
Q3.2	Relevant to IA challenges	1,408	4.39	0.57	4	4–5	95.7	[94.6, 96.7]
Q3.3	Digital twins vs. production	1,407	4.51	0.58	5	4–5	95.8	[94.6, 96.7]
Q3.4	Models map environment/config	1,407	4.42	0.57	4	4–5	95.9	[94.7, 96.8]
Q3.5	Lifecycle mgmt for compliance	1,408	4.30	0.72	4	4–5	84.8	[82.8, 86.6]
Q3.6	Org has compliance needs	1,408	3.89	0.81	4	3–5	61.1	[58.5, 63.6]
Q4.1	Value to IT processes	1,411	4.39	0.57	4	4–5	95.9	[94.7, 96.8]
Q4.2	Reduce dev cycle time	1,407	4.29	0.70	4	4–5	85.5	[83.6, 87.2]
Q4.3	Improve quality / fewer defects	1,408	4.30	0.72	4	4–5	84.8	[82.8, 86.6]
Q4.4	Improve dev collaboration	1,408	4.12	0.64	4	4–5	84.7	[82.7, 86.4]
Q5.1	Value to IA/cyber operations	1,409	4.12	0.64	4	4–5	85.0	[83.0, 86.7]
Q5.2	Faster vuln. remediation	1,408	4.13	0.80	4	3–5	73.8	[71.4, 76.0]
Q5.3	Improve security posture	1,408	4.31	0.72	4	4–5	84.8	[82.8, 86.6]
Q5.4	Enhance threat modeling	1,408	4.32	0.75	4	4–5	83.0	[81.0, 84.9]
Q5.5	Cross-team security collab.	1,408	4.22	0.71	4	4–5	83.3	[81.3, 85.2]
Q5.6	Better framework compliance	1,408	4.23	0.71	4	4–5	83.5	[81.5, 85.4]

A conservative reading restricts attention to the low-familiarity majority, for whom the value finding cannot be attributed to prior enthusiasm. Among the 78.2% of respondents in the two lowest familiarity categories (Q1.1 $\in \{1, 2\}$), perceived value remains high, with composite means of 4.21 for IT Value and 4.08 for Security Value, only modestly below the whole-sample means of 4.33 and 4.20. The central finding, that value is perceived as high even where familiarity is absent, therefore does not rest on the more familiar or more favorably disposed respondents. The same caution that qualifies the Section 2 items applies here: because every value item is affirmatively keyed, high agreement within the low-familiarity stratum rules out prior enthusiasm but cannot separate genuine appraisal from face-validity assent by respondents new to the concept. The value finding is therefore best read as receptiveness consistent with genuine appraisal

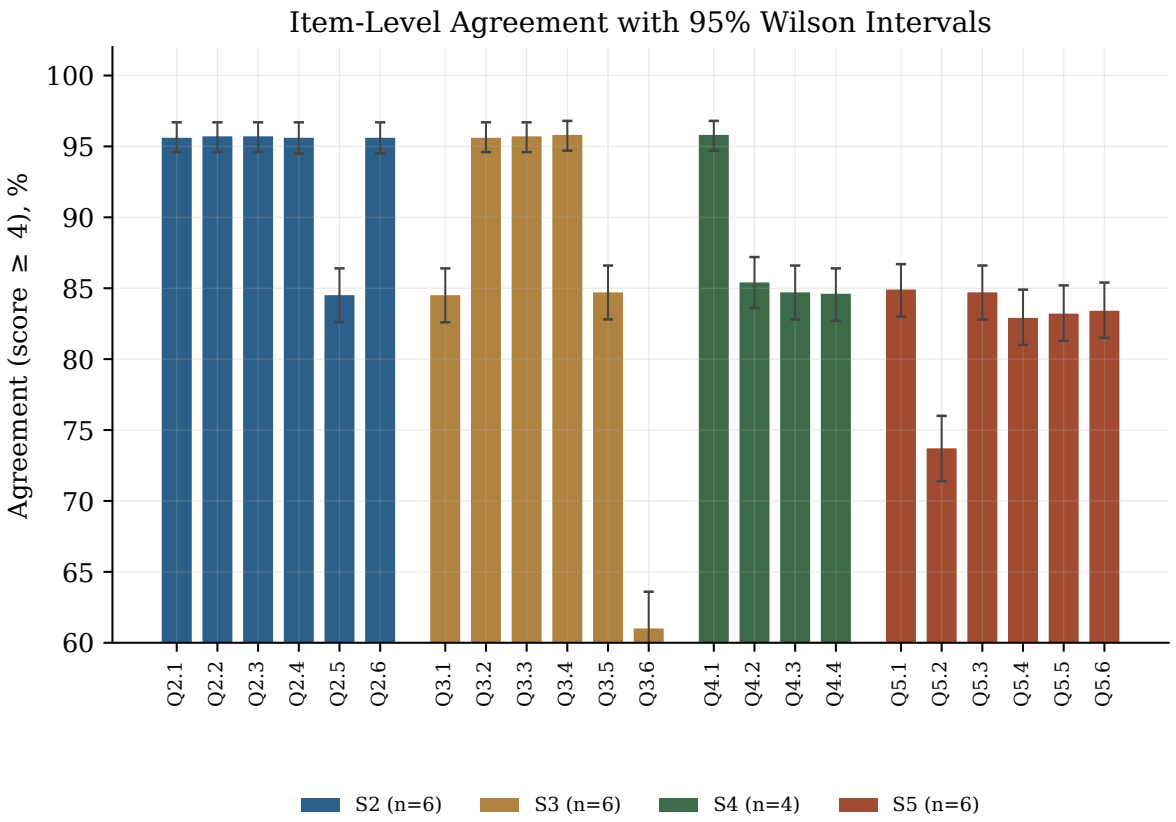


Figure 15: Item-level agreement (score ≥ 4) by section with 95% Wilson confidence intervals.

rather than as demonstrated conviction.

The two investment-willingness items require careful handling. IT investment willingness (Q4.5) records 1,192 "Yes" (84.5%), 218 "Unsure" (15.5%), and a single "No" (<0.1%) across the sample. Security investment willingness (Q5.7) records 99.8% "Yes" (1,408 of 1,411), with one "No" and two "Unsure." These distributions are at or near the response ceiling. Chapter 3 intended these items as evidence of organizational receptiveness, and at face value they supply it overwhelmingly. A near-unanimous item, however, has almost no variance, so it cannot discriminate among respondents or support meaningful subgroup contrasts. Such uniformity is also a recognized signature of response-set behavior or of a sample skewed toward the favorably disposed. Q5.7 therefore establishes a near-universal stated willingness *conditional on demonstrated return on*

investment while contributing essentially nothing to differentiation among respondents. This pattern should be read as conviction rather than discriminating evidence. Interest and intent items reinforce the value findings: 89.0% of respondents indicated interest in learning more about Digital Engineering applications (Q6.1), and 95.9% would recommend that their organization explore Digital Engineering adoption (Q6.2). Research Question 2 therefore receives a precisely estimated answer: perceived value is high, and professionals rate Digital Engineering capabilities as valuable across the applicability, IT, and security domains, with the qualification that the investment-willingness items reflect near-unanimous conviction rather than graduated discrimination. Every statistic in this section is a perception. They establish that this population expects Digital Engineering to be valuable in IT and security work, and whether Digital Engineering delivers that value once implemented is an implementation question this design does not address.

Research Question 3: Anticipated Benefit for Job Performance, Compliance, and Capability

Research Question 3 asks to what extent professionals believe Digital Engineering could help them perform their jobs, meet compliance requirements, or enhance organizational capabilities. The analysis draws on the operational and compliance-oriented items across Sections 3–5.

Research Questions 2 and 3 draw on overlapping survey material, and saying so plainly matters for how their two answers are counted. Chapter 1 distinguishes them conceptually, with Research Question 2 addressing general perceived value and Research Question 3 addressing applicability to the respondent's own job performance, compliance obligations, and organizational capability. The instrument operationalizes both through the Section 3 to Section 5 items, regrouped rather than separately measured, so the two answers rest on a common evidentiary base and are best read as one appraisal examined at two levels of specificity.

Agreement is substantial on all three groupings. The IT operational items, Q4.1

through Q4.4, cover meaningful value, cycle time, quality, and collaboration, and they draw 87.7% agreement. The security operational items, Q5.1 through Q5.5, cover value to operations, vulnerability remediation, posture, threat modeling, and cross-team collaboration, and they draw 82.0%. The three explicitly compliance-framed items, Q3.5, Q3.6, and Q5.6, draw 76.5%.

Those three aggregates conceal the finding that matters most for Research Question 3, because each averages over items that behave differently. Disaggregating the compliance grouping makes the point plainly. Its two capability-framed items draw agreement in line with the rest of the instrument: Q3.5, on lifecycle management for compliance activity, reaches 84.8% ($M = 4.30$), and Q5.6, on improved framework compliance, reaches 83.5% ($M = 4.23$). Q3.5 pairs compliance with service delivery in a single stem, so some of its agreement may attach to the second clause, which makes Q5.6 the cleaner of the two comparators. Its third item behaves differently. Q3.6 ("my organization faces regulatory or compliance requirements that could benefit") draws 61.1% agreement ($M = 3.89$; Table 4), more than twenty percentage points below its companions. The 76.5% aggregate is therefore an average across a genuine split rather than a uniform level of reservation, and reporting it alone would obscure what the items separately show.

What separates Q3.6 from its companions is the kind of judgment it requires. Q3.5 and Q5.6 ask respondents to appraise what Digital Engineering can do; Q3.6 asks them to characterize their own organization's regulatory exposure. The same pattern recurs in the security grouping, where Q5.2 (faster vulnerability remediation) draws 73.8% ($M = 4.13$) against 83.0% to 85.0% for the other four Section 5 operational items. Q5.2 likewise asks for a claim about a measurable operational outcome rather than about a capability in principle. Its stem, however, bundles two operations, identification and remediation, so a respondent who credits Digital Engineering with the first but not the second would also hedge. The design cannot separate that reading from the own-circumstances reading, and Q5.2 is reported here as consistent with the pattern rather than as independent support for

it.

Dispersion statistics corroborate the reading. Across the twenty-two agreement-scale items, Q3.6 and Q5.2 are the two lowest in agreement, and they are also the only two whose interquartile range extends down to the neutral midpoint (3–5 against 4–5 for every other item; Table 4). The next lowest item, Q5.4, sits at 83.0%, more than nine points above Q5.2. Reservation in this instrument is therefore concentrated in two items rather than distributed across the compliance and security blocks. Q3.6 carries the clearer case, because its stem is single-barreled and asks only for a characterization of the respondent's own organization. Read together, the two items locate reservation in claims respondents must ground in their own circumstances, with the weight of that reading resting on Q3.6.

Read together, the Research Question 3 items support the proposition that respondents believe Digital Engineering could help in their work, subject to one qualification the disaggregation makes precise. Endorsement is near-uniform wherever the instrument asks what the capability could do, and it falls by ten to twenty points wherever the instrument asks respondents to assert something concrete about their own organization or its measurable performance. This qualification bears on how the answer should be used. Anticipated benefit is high and is evidence of receptiveness; it is a weaker basis for inferring that respondents see the described benefits as already applicable to their own circumstances, which is the belief that would most directly precede adoption.

Direct Answers to the Research Questions

All three research questions ask to what extent, so each admits a descriptive answer, and Table 5 states the three before the analysis turns to how they are distributed. Awareness of Digital Engineering as a named discipline is low: 78.2% of respondents fall in the two lowest familiarity categories and 22.2% report recent professional exposure. Perceived value is high and precisely estimated across all three value blocks, at 86.3%, 87.7%, and 82.2% agreement. Anticipated benefit for job performance, compliance, and organizational capability is likewise high, at 87.7% on the IT operational items and 82.0%

on the security operational items, and it is most reserved on the explicitly compliance-framed items at 76.5%. These are the answers the study returns.

Table 5: Direct answers to the three research questions.

Research question	Direct answer	Key evidence
RQ1 Awareness	Low awareness, near-universal capability endorsement	Familiarity mean 1.94; 78.2% in the two lowest categories; 22.2% recent exposure; 93.9% Section 2 agreement
RQ2 Perceived value	High perceived value across all value blocks	Agreement 86.3% (applicability), 87.7% (IT), 82.2% (security); composite means 4.0–4.3
RQ3 Anticipated benefit	High anticipated benefit, most reserved on organization-specific compliance	IT operations 87.7%, security operations 82.0%, compliance-framed 76.5%

Note. All figures are from the authenticated analytic sample ($N = 1{,}411$). Each answer is descriptive. The subgroup, regression, and robustness analyses later in this chapter qualify these answers without displacing them.

The remainder of the chapter examines how those answers are distributed and how firmly they hold. Composite, comparative, regression, and robustness analyses characterize variation within the population and test the findings against alternative explanations. They refine the three answers above rather than displace them.

Composite Scores and Reliability

Item averaging produced the three composite scores specified in Chapter 3: an Awareness composite (Q1.1, Q2.1–Q2.6), an IT Value composite (Q3.1, Q3.3–Q3.5, Q4.1–Q4.4), and a Security Value composite (Q3.2, Q3.6, Q5.1–Q5.6). All three comfortably exceed the conventional Cronbach's $\alpha \geq 0.70$ threshold (Table 6): Awareness $\alpha = 0.792$, IT Value $\alpha = 0.858$, and Security Value $\alpha = 0.858$. Internal consistency is therefore adequate for the comparative analyses that follow.

One conceptual note bears on the Awareness composite. It averages the low-scoring familiarity item (Q1.1, $M = 1.94$) with the high-scoring Section 2 capability-recognition items ($M \approx 4.4$), producing a composite mean of 4.04 pulled upward by those items. The

composite is internally consistent, but its central tendency should not be read as evidence that respondents are highly aware: it blends low familiarity with high capability-endorsement, and its two constituents point in opposite directions. Reporting Q1.1 separately alongside the composite, as Table 4 does, prevents that misreading.

The item statistics quantify how far apart those constituents sit. Q1.1 correlates with the mean of the six capability-recognition items at $r = 0.32$, and its corrected item-total correlation is 0.315 against a range of 0.53 to 0.67 for the six Section 2 items. Deleting Q1.1 would raise the composite's α from 0.792 to 0.863. Under the conventional item-retention criteria, which flag items whose corrected item-total correlation falls below roughly 0.40 and whose removal improves α, Q1.1 is therefore a deletion candidate, and the remaining six items would form a cleaner scale measuring capability endorsement. The analysis retains all seven items regardless, because Chapter 3 specified this composite before data collection and revising a pre-specified measure after seeing the results would trade the discipline of pre-specification for analytic flexibility. That decision carries one consequence the comparative sections observe: the composite indexes capability endorsement more heavily than familiarity, so the field difference reported for Awareness below is better understood as a difference in capability endorsement, while Q1.1 carries the familiarity finding on its own.

Table 6: Composite scores and internal consistency.

Composite	Items	*M*	*SD*	Mdn	95% CI	α
Awareness (Q1.1, Q2.1–2.6)	7	4.044	0.478	4.000	[4.019, 4.069]	0.792
IT Value (Q3.1,3.3–3.5; Q4.1–4.4)	8	4.325	0.463	4.375	[4.301, 4.349]	0.858
Security Value (Q3.2,3.6; Q5.1–5.6)	8	4.202	0.509	4.250	[4.176, 4.229]	0.858

Comparative Analysis

Comparative analyses examine whether awareness levels and value perceptions differ across professional field and experience level. Shapiro–Wilk tests rejected normality for all three composites (see Appendix B), an expected result at this sample size, where the test is

highly powered against trivial departures; the composite distributions are also left-skewed. Mann–Whitney U and Kruskal–Wallis tests therefore serve as the primary inferential procedures, with parametric statistics reported for continuity with the technology-acceptance literature.

Professional Field: Information Technology versus Information Assurance

Following the analytical plan, the analysis compared Information Technology ($n = 501$) and Information Assurance/Cyber Security ($n = 501$) respondents on each composite. The differences are large and consistent in direction: IT respondents score substantially higher than IA/Cyber respondents on all three composites (Table 8), with Cohen's d between 1.64 and 1.94 and rank-biserial correlations of magnitude 0.74–0.82. The 95% confidence intervals on d are narrow and lie entirely above 1.4 (for example, Awareness $d = 1.64$, 95% normal-theory CI $[1.50, 1.78]$; the bootstrap intervals in the robustness analyses below agree to within a hundredth), so the effect is both very large by conventional benchmarks and precisely estimated.

Three framings of that magnitude deserve to be kept apart. The rank-biserial coefficients carry the negative sign of the U-based convention while the tables report Cohen's d as positive; both indicate Information Technology scoring higher. Because the higher-scoring Information Technology group sits near the scale ceiling, its variance is compressed, so Cohen's d, which standardizes by the pooled standard deviation, is upward-biased here. The distribution-free rank-biserial correlations are the more trustworthy effect sizes, reported alongside d throughout, and the headline magnitude should be read accordingly. Expressing the same differences in the units respondents actually used supplies a third and more interpretable reading: the Information Technology advantage is 0.65 scale points on Awareness, 0.69 on IT Value, and 0.75 on Security Value, or 16 to 19 percent of the four-point interval the scale spans. The standardized and raw framings describe one gap, and reporting both keeps the standardized magnitude from carrying more rhetorical weight than the underlying difference supports.

A further caution applies to reading the two value effect sizes as independent corroboration. The construct-validity analysis below finds the IT Value and Security Value composites not discriminant from each other (heterotrait–monotrait ratio of 0.99), so their two effect sizes, identical at $d = 1.94$, are better read as one result expressed on two overlapping item sets than as two confirmations of it.

The direction of the difference merits emphasis because it runs against the grain of the dissertation's framing. The Information Assurance audience the research most seeks to reach reports markedly *lower* awareness and perceived value than the IT audience. The familiarity gap is the largest single contrast in the data: IT respondents average 2.88 on Q1.1 against 1.20 for IA/Cyber, and the gap propagates through the value composites (Figure 16).

An effect of this magnitude warrants scrutiny in its own right, because separations approaching two standard deviations between two adjacent professional communities are uncommon and invite an artifactual reading. Five candidate explanations warrant consideration. The first is a genuine awareness and exposure deficit within the Information Assurance community. The second is a recruitment-frame effect, in which different channels reached the two groups. The third is differential item functioning, in which security professionals calibrate affirmatively framed capability statements more conservatively. The fourth is self-categorization, in which the "Information Assurance/Cyber Security" label attracts more skeptical respondents. The fifth is calibrated domain skepticism, in which Information Assurance respondents discount affirmatively framed claims about security capability because their expertise equips them to appraise such claims critically. The last of these differs from the others in kind rather than in degree, because it treats the lower Information Assurance ratings as informed judgment rather than as measurement artifact or absent knowledge. One observation in the data is consistent with it: Information Assurance respondents rate the security value of Digital Engineering (3.82) below the rating Information Technology respondents give that same security value (4.57), so on the subject

matter closest to their own practice the specialists are the more reserved group. That observation is consistent with calibrated skepticism without being distinctive to it, because every account listed above predicts lower Information Assurance ratings. It narrows nothing on its own and is reported as a pointer for the replication the limitations call for.

As the diagnostics below and Table 7 show, these accounts do not stand on equal footing. The within-cohort and regression analyses below exclude the experience confound, the invariance screen and the rank-based effect sizes weaken the differential-item-functioning and ceiling-artifact accounts, and the recruitment-frame and self-categorization accounts remain open because the anonymous instrument recorded no variable that could test them. The data cannot adjudicate among those open accounts; Chapter 5 weighs them. This chapter foregrounds the finding rather than burying it in Chapter 5, because it directly motivates the targeted professional-development argument the dissertation advances while remaining explicitly open about its possible sources. Table 7 consolidates the candidate explanations, the diagnostic applied to each, and what the data can and cannot exclude, so that readers can check at a glance the basis for reading the field gap as strong but provisional.

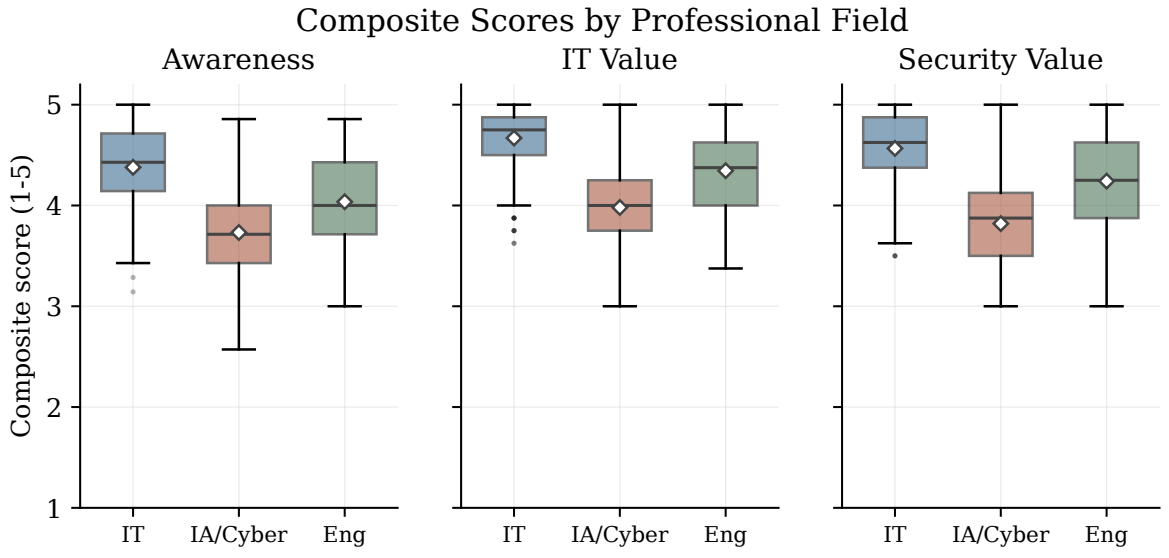


Figure 16: Composite score distributions by professional field.

Table 7: Candidate explanations for the Information Technology versus Information Assurance field gap, the diagnostic applied to each, and what the data can and cannot exclude.

Candidate explanation	**Diagnostic applied**	**Outcome**
Experience-imbalance confound	Within-cohort models; field-by-experience association ($V = 0.05$)	**Excluded**: IT exceeds IA within every experience cohort; field and experience are only weakly associated
Differential item functioning	Zumbo linear DIF screen on all 23 items	**Weakened**: 21 of 23 items show no DIF; residual item differences are small relative to the composite gap
Ceiling-induced variance compression inflating d	Distribution-free rank-biserial correlation reported alongside d	**Weakened**: the gap is large on the distribution-free measure as well ($\lvert r_{rb}\rvert$ of 0.74–0.82)
Prior-exposure imbalance	Recent-exposure indicator added to the value regressions	**Weakened**: the field coefficient is essentially unchanged
Familiarity-stratum composition	High-familiarity stratum indicator added to the value regressions	**Weakened**: the field coefficient retains 87 percent of its magnitude; 86 percent with exposure also held constant
Calibrated domain skepticism	Not separable from an awareness deficit in an anonymous cross-section, since both predict lower Information Assurance endorsement	**Not excluded**: separating informed discounting from unfamiliarity requires knowledge-verification items the instrument did not carry
Recruitment-frame difference	No employment-sector or recruitment-channel variable was recorded	**Not excluded**: untestable from these data; bounded by argument rather than measured
Self-categorization or dispositional difference by label	Not directly testable in an anonymous cross-section	**Not excluded**: requires a design that separates label choice from field

Table 8: Field comparison of the value composites, Information Technology vs. Information Assurance/Cyber Security.

Composite	IT M	IA M	U	p	d	95% CI d	r_{rb}
Awareness	4.379	3.732	218,766	$< .001$	1.64	[1.50, 1.78]	-0.743
IT Value	4.669	3.980	227,984	$< .001$	1.94	[1.79, 2.09]	-0.817
Security Value	4.566	3.820	227,814	$< .001$	1.94	[1.79, 2.09]	-0.815

At the item level, the analysis applied the Holm–Bonferroni family-wise correction specified in Chapter 3 within the Awareness family (Section 1 and Section 2 items). Every item survives correction with Information Technology exceeding Information Assurance throughout (Table 9); the correction overturns no conclusion, because the unadjusted p-values are already extreme, the expected behavior of Holm–Bonferroni when effects are large and uniform. The same procedure applied to the applicability, IT-value, and security-value families yields the same pattern: every item shows Information Technology exceeding Information Assurance and survives Holm correction (Tables 10, 11, and 12). The separation is broadly uniform across the three families (mean $|r_{rb}|$ of approximately 0.56 on the applicability items, 0.56 on the security-value items, and 0.53 on the IT-value items) and slightly smaller on the IT-value items, where Q4.4 shows the smallest effect in that family ($r_{rb} = -0.45$). Together with the organization-specific compliance item Q3.6 ($r_{rb} = -0.45$), these are the smallest single-item contrasts across the three value families.

Experience Level

Kruskal–Wallis tests across the four experience cohorts are significant for all three composites ($H(3)$ between 213.1 and 291.2, all $p < .001$; Table 13), with η^2 between 0.15 and 0.21 (medium to large). The pattern, however, is **not monotonic** and therefore cannot be summarized as "experience increases" or "experience decreases" perceived value. Dunn's post-hoc comparisons with Holm adjustment (Table 14) locate the effect in the 11–15 year cohort, which scores distinctly *lower* than both the 1–5 and 6–10 year cohorts. The effect is most pronounced on Security Value, where the 11–15 year mean (3.77) sits well below the early-career means (4.32–4.35). The 16+ year cohort posts the highest means on the

Table 9: Item-level field comparison within the Awareness family (IT vs. IA).

Item	IT M	IA M	r_{rb}	p_{Holm}	Sig.
Q1.1	2.88	1.20	-0.775	$< .001$	✓
Q2.1	4.65	4.17	-0.415	$< .001$	✓
Q2.2	4.60	4.15	-0.395	$< .001$	✓
Q2.3	4.61	4.16	-0.398	$< .001$	✓
Q2.4	4.73	4.25	-0.416	$< .001$	✓
Q2.5	4.53	4.02	-0.365	$< .001$	✓
Q2.6	4.65	4.17	-0.428	$< .001$	✓

two value composites, while the 6–10 year cohort leads on Awareness. After correction, the 16+ cohort differs significantly from the 11–15 year trough on both value composites ($z = -5.33$ and $z = -5.12$, both $p < .001$) but not on Awareness ($p = .118$), and none of its comparisons against the 1–5 or 6–10 year cohorts reaches significance. At $n = 18$, however, these estimates are unstable, and the analysis does not interpret them further.

On the most defensible reading, mid-career respondents (11–15 years) express more reservation than their junior colleagues, while the most senior group is too small to characterize. This non-monotonic pattern, rather than a linear trend, is what the confidence-interval reporting and non-parametric post-hoc testing specified in Chapter 3 were designed to surface (Figure 17). The multivariate analysis below revisits this pattern, where the mid-career trough proves to be a genuine experience-group difference that survives adjustment for professional field.

Instability in the senior estimates bears on how the pattern should be named. Describing it as a trough implies recovery at the senior end, and that recovery rests entirely on the eighteen respondents in the 16+ cohort. Setting that cohort aside, the three well-populated cohorts trace a decline that begins after the 6–10 year group rather than a dip that resolves. Regression evidence below establishes the 11–15 year effect itself robustly, and the analyses that follow rely on that effect alone; the shape of the curve beyond the 11–15 year cohort does not follow from these data.

Table 10: Item-level field comparison within the Applicability family (IT vs. IA).

Item	IT M	IA M	r_{rb}	p_{Holm}	Sig.
Q3.1	4.62	3.93	-0.519	$< .001$	✓
Q3.2	4.74	4.05	-0.613	$< .001$	✓
Q3.3	4.85	4.13	-0.634	$< .001$	✓
Q3.4	4.80	4.05	-0.660	$< .001$	✓
Q3.5	4.64	3.97	-0.494	$< .001$	✓
Q3.6	4.21	3.54	-0.448	$< .001$	✓

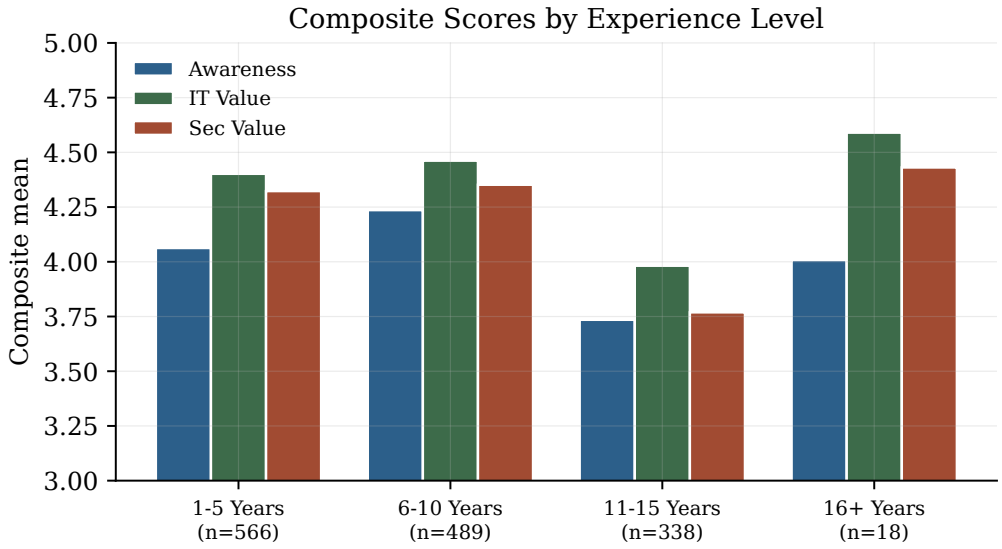


Figure 17: Composite means by experience cohort.

Association Analysis

Chi-square tests of independence (Table 15) are consistent with the associations anticipated in Chapter 3. Professional field is associated with prior exposure (Q6.3×Q1.2, Cramér's $V = 0.21$), echoing the familiarity gap above: IT respondents are more likely to report recent exposure than IA/Cyber respondents. The field-by-exposure test excludes the Other field group and the two respondents without an exposure response, yielding a three-by-two table with $df = 2$ and $n = 1{,}339$; Appendix B reports the full contingency table. Experience is only weakly associated with IT investment willingness ($V = 0.05$) and with security investment willingness ($V = 0.03$). The latter is unsurprising given that Q5.7 is almost invariant and so has little association to detect with any variable.

Table 11: Item-level field comparison within the IT-Value family (IT vs. IA).

Item	IT M	IA M	r_{rb}	p_{Holm}	Sig.
Q4.1	4.73	4.03	-0.621	$< .001$	✓
Q4.2	4.61	3.93	-0.503	$< .001$	✓
Q4.3	4.69	3.94	-0.553	$< .001$	✓
Q4.4	4.42	3.85	-0.445	$< .001$	✓

Focused Breakout Analysis: Professional Field and Experience

Following the proposal, the comparative analysis above contrasted Information Technology against Information Assurance/Cyber Security[1] on the composite scores. This section deepens that analysis along the two demographic axes the instrument captured, in two registers. The descriptive breakouts resolve all four professional-field groups (Information Technology, Information Assurance/Cyber Security, Engineering, and Other) and all four experience cohorts, so that the structure within each domain is visible rather than collapsed into a binary. The inferential *stratified* models, by contrast, cover only the two focal groups, Information Technology and Information Assurance/Cyber Security, consistent with the proposal's design. Engineering and Other enter the confirmatory multivariate regression below, with Information Assurance as the reference category, but are not the subject of the stratified within-domain breakouts. This division tests whether the field and experience effects are confounded and finds them largely independent.

Composite Scores Across All Four Field Groups

Resolving the composites across all four field groups (Table 16, Figure 18) shows that the Information Technology–Information Assurance contrast above is the widest separation in the data but not the whole picture. Information Technology posts the highest mean on every composite (4.38 awareness, 4.67 IT Value, 4.57 Security Value), while

[1] The field category is labeled "Information Assurance/Cyber Security" throughout this chapter following the instrument's combined response option (Chapter 3), which merged into one category the governance sense that Chapter 1 designates Information Assurance and the operational sense it designates cybersecurity. The category therefore spans both, and the anonymous instrument cannot separate a governance-oriented from an operationally oriented respondent within it; Chapter 5 carries this heterogeneity into the interpretation of the field gap.

Table 12: Item-level field comparison within the Security-Value family (IT vs. IA).

Item	IT M	IA M	r_{rb}	p_{Holm}	Sig.
Q5.1	4.47	3.76	-0.564	$< .001$	✓
Q5.2	4.43	3.78	-0.444	$< .001$	✓
Q5.3	4.70	3.92	-0.573	$< .001$	✓
Q5.4	4.69	3.95	-0.525	$< .001$	✓
Q5.5	4.64	3.77	-0.637	$< .001$	✓
Q5.6	4.64	3.78	-0.639	$< .001$	✓

Information Assurance/Cyber Security is the lowest on every composite (3.73, 3.98, 3.82). The Engineering group sits between the two on all three constructs (4.04, 4.34, 4.24). The Other group falls just below Engineering on every composite (3.92, 4.23, 4.14), placing it third of the four groups and above only Information Assurance/Cyber Security. The Other group is small ($n = 70$), heterogeneous, and almost entirely early-career (Table 18). Because it does not stand out on any composite, the analysis does not over-interpret it here; Chapter 5 notes it only as a candidate for focused qualitative study. The whole-sample means in the final column show Information Assurance/Cyber Security below the overall average on every composite while Information Technology sits above it.

Familiarity resolves differently from the composites and merits separate comment, because the Engineering group functions as a natural reference point for the field comparison. On Q1.1, Engineering respondents average 1.70 against 2.88 for Information Technology and 1.20 for Information Assurance/Cyber Security, and 51.9% of Engineering respondents selected "Not familiar." On the composites Engineering sits between the two focal groups, but on named-discipline familiarity it resembles Information Assurance far more closely than Information Technology. That ordering runs against what the discipline's provenance would predict, since Digital Engineering originates in systems engineering practice and its constituent methods are systems engineering methods. Two readings fit the result. Respondents choosing "Engineering" were not restricted to systems engineering, so the option would have admitted software, network, industrial, and other specialties for which the discipline is genuinely unfamiliar. This pattern also matches the umbrella-term

Table 13: Experience-level comparison of the composite means.

Composite	1–5	6–10	11–15	16+	H	df	p	η^2
Awareness	4.063	4.236	3.735	4.008	213.1	3	$< .001$	0.149
IT Value	4.402	4.462	3.982	4.590	229.4	3	$< .001$	0.161
Security Value	4.323	4.352	3.769	4.431	291.2	3	$< .001$	0.205

argument advanced throughout this dissertation, under which practitioners work with model-based methods, digital twins, and lifecycle management individually while not recognizing the collective label the defense community uses to organize them. On that second reading the Engineering group corroborates the central finding rather than complicating it, because it displays the same separation the sample as a whole displays: capability endorsement of 4.43 on the Section 2 items against named-discipline familiarity of 1.70. Because the instrument recorded no engineering subdiscipline, it cannot distinguish the two readings, and Chapter 5 carries the ambiguity into the interpretation of the field comparison.

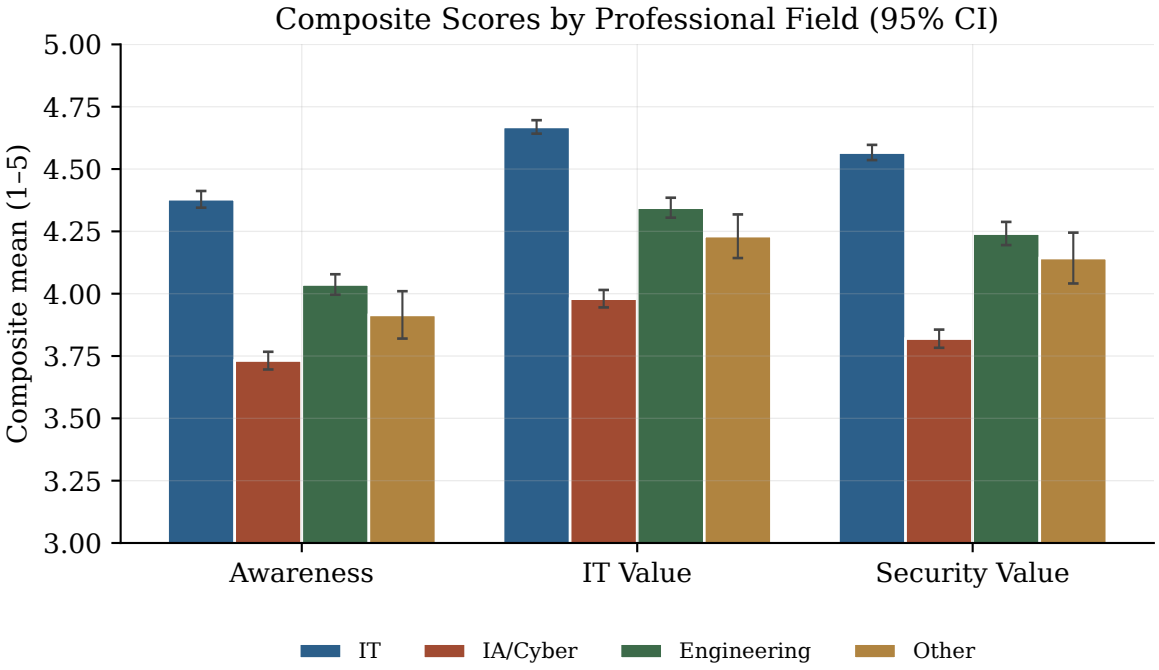


Figure 18: Composite means by professional field across all four groups, with 95% confidence intervals.

Table 14: Dunn post-hoc comparisons for the experience-level Kruskal–Wallis tests.

Composite	Cohort comparison	z	p_{Holm}	Sig.
Awareness	1–5 vs 6–10	-6.09	$< .001$	✓
	1–5 vs 11–15	9.51	$< .001$	✓
	1–5 vs 16+	0.82	.411	
	6–10 vs 11–15	14.56	$< .001$	✓
	6–10 vs 16+	2.39	.051	
	11–15 vs 16+	-1.89	.118	
IT Value	1–5 vs 6–10	-2.20	.084	
	1–5 vs 11–15	12.52	$< .001$	✓
	1–5 vs 16+	-1.79	.146	
	6–10 vs 11–15	14.09	$< .001$	✓
	6–10 vs 16+	-1.22	.221	
	11–15 vs 16+	-5.33	$< .001$	✓
Security Value	1–5 vs 6–10	-0.80	1.000	
	1–5 vs 11–15	15.07	$< .001$	✓
	1–5 vs 16+	-0.84	1.000	
	6–10 vs 11–15	15.35	$< .001$	✓
	6–10 vs 16+	-0.63	1.000	
	11–15 vs 16+	-5.12	$< .001$	✓

The same ordering appears in item-block agreement (Table 17). Information Assurance/Cyber Security records markedly lower agreement on every block, most starkly on the Security Value items (66.8% versus 94.7% for Information Technology and 82.6% for Other), the domain in which one might most expect security professionals to perceive value. That the security audience expresses the lowest agreement on security-value items is the most counterintuitive descriptive result in the breakout. Because this group mixes governance-oriented and operationally oriented respondents that the anonymous instrument cannot separate, two readings are available, and the data cannot decide between them. The first is an awareness deficit specific to this community. The second is a valid relevance judgment by operationally oriented respondents that a model- and document-centric paradigm fits hands-on defensive work less directly than it fits systems or lifecycle engineering. Chapter 5 takes up both readings and their differing implications for targeted professional development.

Table 15: Chi-square tests of independence with Cramér's V.

Association	χ^2	df	p	Cramér's V	n
Field × prior exposure (Q6.3×Q1.2)	60.3	2	$< .001$	0.212	1,339
Experience × IT investment (Q6.4×Q4.5)	6.0	6	.421	0.046	1,411
Experience × Sec investment (Q6.4×Q5.7)	2.2	6	.902	0.028	1,411

Experience Composition Within Each Field

Experience composition is largely independent of professional field; the two are only weakly associated (Table 18, Figure 19). The cross-tabulation is computed on the full sample. The association is small ($\chi^2(9) = 9.9$, Cramér's $V = 0.05$), and the four field groups share broadly the same experience profile. Each concentrates in the two earliest cohorts: Information Technology places 40.3% at 1–5 years and 35.5% at 6–10 years, Information Assurance/Cyber Security 37.7% and 36.3%, Engineering 42.5% at 1–5 years and 33.0% at 6–10 years, and the Other group is the most early-career (44.3% at 1–5 years). This shared experience profile removes the confound the headline comparisons might otherwise carry.

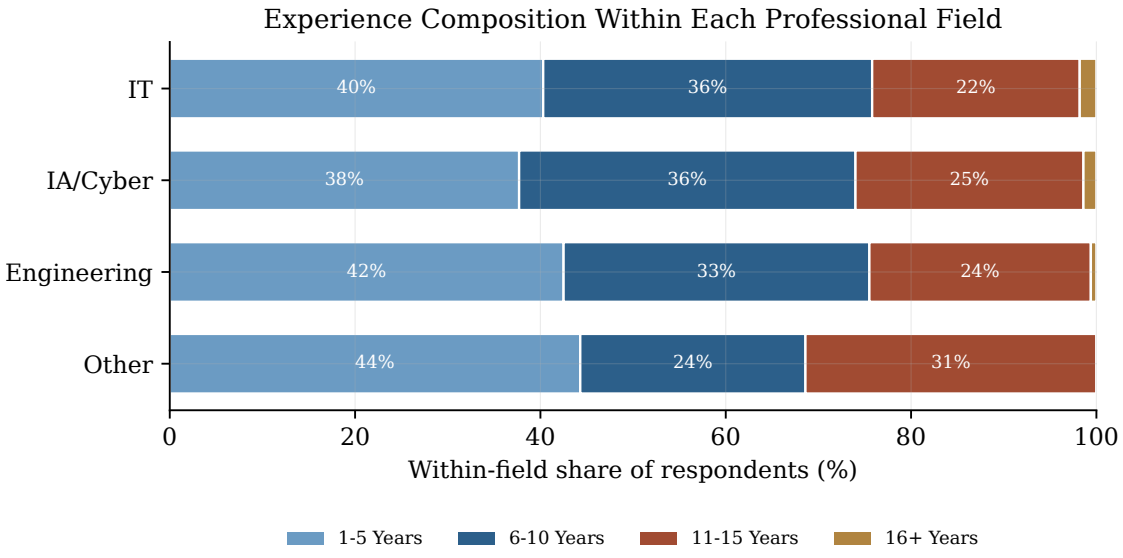


Figure 19: Experience composition within each professional field (within-field percentages).

Table 16: Composite scores by professional field (all four groups).

	IT		IA/Cyber		Engineering		Other		Overall
Composite	*M*	95% CI	*M*	95% CI	*M*	95% CI	*M*	95% CI	*M*
Awareness	4.38	[4.34, 4.41]	3.73	[3.70, 3.77]	4.04	[4.00, 4.08]	3.92	[3.82, 4.01]	4.04
IT Value	4.67	[4.64, 4.70]	3.98	[3.94, 4.01]	4.34	[4.30, 4.38]	4.23	[4.14, 4.32]	4.33
Security Value	4.57	[4.54, 4.60]	3.82	[3.78, 3.86]	4.24	[4.20, 4.29]	4.14	[4.04, 4.25]	4.20

Group sizes: IT $n = 501$, IA/Cyber $n = 501$, Engineering $n = 339$, Other $n = 70$.

Experience Deviation from the Overall Averages

Examining how each experience cohort deviated from the whole-sample average sharpens the non-monotonic pattern reported in the comparative analysis (Table 19, Figure 20). Relative to the grand mean, the 11–15 year cohort sits below average on all three composites by a consistent margin, at −0.31 on Awareness, −0.34 on IT Value, and −0.43 on Security Value. The 6–10 year cohort sits above average on all three, at +0.19, +0.14, and +0.15 respectively. The 1–5 year cohort stays close to the overall average throughout. The 16+ year cohort posts positive deviations on the two value composites but, at $n = 18$, is statistically unstable and appears for completeness only. The deviation view makes explicit that the experience effect is a mid-career trough at 11–15 years rather than a monotonic increase or decrease across the career span.

Table 17: Item-block agreement (percentage of responses at score ≥ 4) by professional field.

Item block	IT	IA/Cyber	Engineering	Other	Overall
Awareness (S2)	98.3	88.1	96.3	92.6	93.9
Applicability (S3)	94.7	76.2	88.9	86.6	86.3
IT Value (S4)	96.3	76.7	91.1	88.6	87.7
Security Value (S5)	94.7	66.8	86.6	82.6	82.2

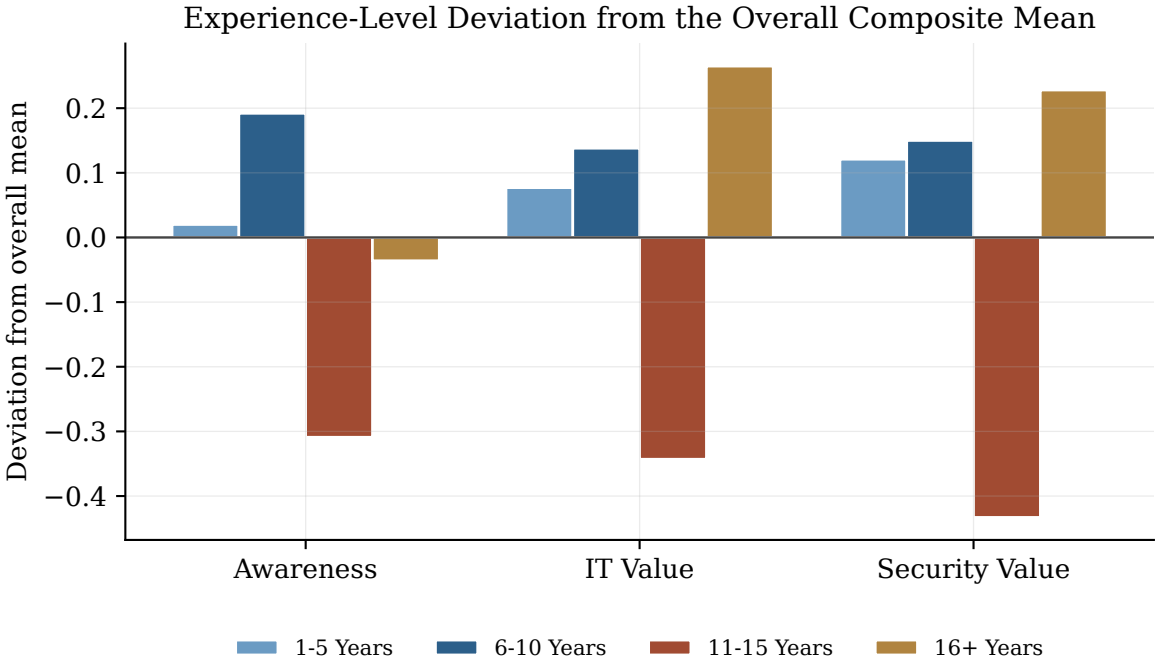


Figure 20: Deviation of each experience cohort from the overall composite mean.

Focused Models: Disentangling Field from Experience

Together, the two preceding subsections show that the mid-career trough and the Information Technology–Information Assurance gap are largely independent findings rather than a single confounded one. Because professional field and experience are only weakly associated, the 11–15 year cohort is not disproportionately Information Assurance, so neither effect is reducible to the other. Resolving the composites by experience *within* the two focal domains corroborates this separation (Table 20).

Two results follow. First, the 11–15 year cohort is a genuine local low point within

Table 18: Professional field × experience cross-tabulation.

Field	1–5 yrs	6–10 yrs	11–15 yrs	16+ yrs	Total
IT	202 (40.3%)	178 (35.5%)	112 (22.4%)	9 (1.8%)	501
IA/Cyber	189 (37.7%)	182 (36.3%)	123 (24.6%)	7 (1.4%)	501
Engineering	144 (42.5%)	112 (33.0%)	81 (23.9%)	2 (0.6%)	339
Other	31 (44.3%)	17 (24.3%)	22 (31.4%)	0 (0.0%)	70

$\chi^2(9) = 9.9$, $p = .355$, Cramér's $V = 0.05$, $n = 1{,}411$.

both focal domains: relative to its own domain mean, it deviates by −0.30 to −0.38 within Information Technology and by −0.31 to −0.44 within Information Assurance/Cyber Security across the three composites. The mid-career reservation is therefore a real experience-group pattern present in each community rather than an artifact of field composition. Second, the Information Technology–Information Assurance gap is *not* an artifact of experience: within each of the three interpretable experience cohorts (1–5, 6–10, and 11–15 years), Information Technology exceeds Information Assurance on every composite. On awareness, for example, the contrast runs 4.41 versus 3.72 at 1–5 years, 4.53 versus 3.97 at 6–10 years, and 4.08 versus 3.42 at 11–15 years. Information Technology remains higher on all three composites even in the 16+ year cohort, though with only 9 Information Technology respondents against 7 Information Assurance respondents, that cohort is too small to interpret. Holding experience constant leaves field the dominant axis of variation, and the mid-career trough an independent experience-group difference within each field. These focused models, restricted as the design specifies to the Information Technology and Information Assurance/Cyber Security groups, qualify the comparative results without overturning them.

Multivariate Corroboration

Stratified breakouts suggest, but do not formally test, that professional field and experience each contribute independently to the value differences. An ordinary least squares regression of each value composite on field, experience, and self-reported familiarity makes that test explicit (J. Cohen et al., 2003). The models are fit on all 1,411

Table 19: Experience-level deviation from the overall mean.

Composite	Overall M	1–5 yrs	6–10 yrs	11–15 yrs	16+ yrs
Awareness	4.04	4.06 (+0.02)	4.24 (+0.19)	3.73 (-0.31)	4.01 (-0.04)
IT Value	4.33	4.40 (+0.08)	4.46 (+0.14)	3.98 (-0.34)	4.59 (+0.27)
Security Value	4.20	4.32 (+0.12)	4.35 (+0.15)	3.77 (-0.43)	4.43 (+0.23)

Cohort n: 566 / 489 / 338 / 18. The 16+ cohort ($n = 18$) is shown for completeness but is statistically unstable.

respondents. They account for a substantial share of variance (IT Value $R^2 = 0.64$; Security Value $R^2 = 0.67$; Table 21) and yield three results that sharpen the descriptive picture. Because the value composites are bounded on the 1–5 scale and left-skewed, a robustness check refit the models with HC3 heteroscedasticity-robust standard errors. The refit left every coefficient's direction and significance intact (Appendix B), so the conclusions do not depend on the homoscedastic ordinary-least-squares assumption. Because professional field and self-reported familiarity are themselves correlated, the analysis checked collinearity directly: the maximum variance inflation factor is approximately 2.0 (familiarity 1.58, Information Technology 2.00), well below any conventional threshold, so the field and familiarity coefficients are separately identified.

Professional field remains a strong predictor once the models hold experience and familiarity constant. Relative to Information Assurance/Cyber Security, Information Technology respondents score 0.46 points higher on IT Value and 0.50 higher on Security Value, both at $p < .001$. The Engineering and Other groups also score higher than Information Assurance (0.30 and 0.35 for Engineering, 0.23 and 0.31 for Other), likewise at $p < .001$. The field gap therefore survives adjustment for experience and familiarity, supporting the within-cohort pattern above.

Most consequentially, the mid-career experience effect persists once the models adjust for field and familiarity: relative to the 1–5 year reference, the 11–15 year cohort scores 0.402 points lower on IT Value and 0.534 lower on Security Value, both at $p < .001$. This mirrors the lowest raw means the cohort posted in the omnibus analysis. This is consistent with a genuine experience-group difference rather than a composition artifact:

Table 20: Experience-resolved composite means within the IT and IA/Cyber domains.

Composite	Domain	Domain M	1–5 yrs	6–10 yrs	11–15 yrs	16+ yrs
Awareness	IT	4.38	4.41 (+0.03)	4.53 (+0.15)	4.08 (-0.30)	4.41 (+0.03)
Awareness	IA	3.73	3.72 (-0.01)	3.97 (+0.24)	3.42 (-0.31)	3.53 (-0.20)
IT Value	IT	4.67	4.74 (+0.07)	4.77 (+0.10)	4.36 (-0.31)	4.85 (+0.18)
IT Value	IA	3.98	4.03 (+0.05)	4.16 (+0.18)	3.63 (-0.35)	4.21 (+0.23)
Security Value	IT	4.57	4.66 (+0.09)	4.68 (+0.11)	4.19 (-0.38)	4.69 (+0.12)
Security Value	IA	3.82	3.93 (+0.11)	4.00 (+0.18)	3.38 (-0.44)	4.02 (+0.20)

with the two variables as nearly orthogonal as the cross-tabulation showed, the mid-career reservation cannot be explained away by field membership. This is the inferential counterpart to the within-domain trough identified earlier.

Familiarity, finally, is a positive predictor of perceived value (IT Value $b = 0.132$, standardized $\beta = 0.34$; Security Value $b = 0.142$, $\beta = 0.33$; both $p < .001$; standardized coefficients from the same models, not shown in Table 21). The result qualifies, without contradicting, the headline finding that respondents perceive value despite low familiarity: across respondents greater familiarity accompanies greater perceived value, yet the overall value means remain high because even unfamiliar respondents endorse the capability descriptions. Rank-order correlations point the same way: familiarity correlates moderately with both value composites (Spearman rank-order correlations $\rho = 0.63$ with IT Value and $\rho = 0.62$ with Security Value, both $p < .001$) (Spearman, 1904).

Bias Diagnostics

The methodology specified two planned diagnostics for response bias: a wave (non-response) analysis and a priming diagnostic. Both diagnostics ran as planned.

Wave (Non-Response) Analysis

Splitting the sample at the median submission time into early- and late-arriving halves reveals no meaningful differences on any of the three composites. Every effect size is negligible (Cohen's d between -0.02 and -0.04) and none approaches significance (all Mann–Whitney $p > .5$), with late-arriving respondents scoring only marginally higher than

Table 21: Regression of the value composites on field, experience, and familiarity.

Predictor	IT Value	Security Value
Familiarity (Q1.1)	0.132*** (0.008)	0.142*** (0.008)
Field: Information Technology	0.458*** (0.022)	0.496*** (0.023)
Field: Engineering	0.301*** (0.020)	0.350*** (0.021)
Field: Other	0.232*** (0.036)	0.305*** (0.038)
Experience: 6–10 years	0.082*** (0.017)	0.055** (0.018)
Experience: 11–15 years	−0.402*** (0.019)	−0.534*** (0.020)
Experience: 16+ years	0.168* (0.067)	0.092 (0.070)
Intercept	3.888*** (0.018)	3.759*** (0.019)
R^2	0.635	0.672
Adjusted R^2	0.633	0.670
N	1,411	1,411

$^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. Reference categories: Information Assurance/Cyber Security; 1–5 years of experience.

early-arriving ones (Table 22). The split is by date of submission rather than completion duration, following the early-versus-late tradition of non-response sensitivity analysis (Armstrong & Overton, 1977). Under that logic, in which late respondents proxy for non-respondents, the near-identical composite means across the two halves indicate that late arrivals differ negligibly from early arrivals, so the value composites show little sign of non-response bias. The null wave effect is reassuring on its own terms. It does not, however, by itself dispose of the recruitment-frame concern developed in Chapter 5, because both halves arrived through the same self-selected channels rather than from a random draw of the profession; Chapter 5 carries that concern into the limitations alongside this null result.

Priming Diagnostic

The priming check specified in Chapter 3 compares the value composites between high-familiarity respondents (Q1.1 $= 4, 5$) and low-familiarity respondents (Q1.1 $= 1, 2$). High-familiarity respondents report substantially higher perceived value on both composites, with differences of 0.54 scale points on IT Value and 0.57 on Security Value and $d \approx 1.3$ (Table 23). By the logic stated in Chapter 3, this weighs *against* the pure-priming hypothesis. If the contextual descriptions in the question stems were manufacturing agreement independent of knowledge, the two familiarity groups would

Table 22: Wave (non-response) analysis: early- vs. late-arriving respondents.

Composite	Early M	Late M	p	d
Awareness	4.033	4.054	.523	-0.04
IT Value	4.321	4.328	.802	-0.02
Sec Value	4.192	4.212	.527	-0.04

converge. Here they diverge sharply, a pattern more consistent with genuine perception differences informed by prior knowledge (Figure 21). The corollary is that aggregate value estimates are sensitive to the familiarity composition of the sample: because this sample is predominantly low-familiarity, the overall value means rest largely on the low-familiarity stratum, with the high-familiarity stratum reporting markedly higher value. Chapter 5 weighs what this implies for the broader population.

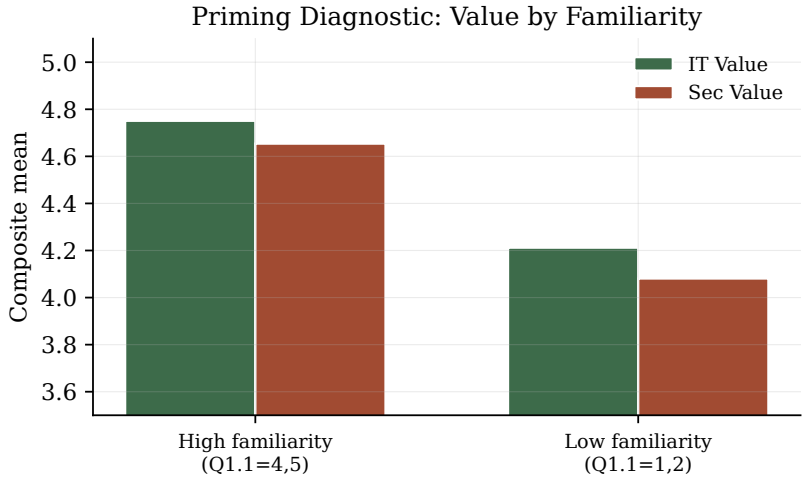


Figure 21: Value composites by familiarity stratum.

Supplementary Post-Hoc Robustness and Validity Analyses

Analyses in this section were not part of the approved proposal's analysis plan (Chapter 3, Research Propositions and Planned Analyses). The candidate added them after data collection concluded to strengthen the study's inferential foundations, a self-initiated step informed by a review of methodological practice in comparable dissertations. They test the robustness and measurement properties of the findings already

Table 23: Priming diagnostic: value composites by familiarity level.

Composite	High M	Low M	Δ	p	d
IT Value	4.751	4.212	0.539	$< .001$	1.30
Sec Value	4.654	4.081	0.573	$< .001$	1.25

reported rather than evaluate new propositions. Because they are post-hoc in this sense, the section reports them as corroboration and diagnostic evidence rather than pre-specified confirmatory tests, and each table is annotated accordingly. Their shared purpose is to examine whether the study's central results, and the professional-field contrast in particular, withstand scrutiny the proposal did not anticipate.

Common-Method Variance

Because every measure derives from a single self-report instrument completed at one sitting, common-method variance could in principle inflate the observed associations (Podsakoff et al., 2003). Harman's single-factor test offers a first screen: it asks whether one unrotated factor accounts for most of the variance across all items. It does not. A single factor explains 44.0% of the variance across the 23 composite items, below the 50% level conventionally read as evidence of a dominant method factor, and two factors exceed the eigenvalue-one criterion (Table 24). Harman's test is a lower-bound diagnostic rather than a decisive control, and 44.0% sits modestly below the threshold, so the conclusion is narrow: a single method factor does not dominate these data.

Beyond Harman's screen, the analysis considered a fuller latent-method-factor control and reports it here as a null attempt rather than a result. Fitting an unmeasured latent method construct, in which a common factor loads on all items alongside the substantive factors, is uninformative here because the two value constructs are nearly collinear (heterotrait–monotrait ratio 0.99, reported below). Under such low discriminant validity the added factor absorbs genuine shared substantive variance rather than isolating method variance, a documented failure mode of the technique (Podsakoff et al., 2003). The apparent method variance it returns is therefore not a valid estimate, and the chapter does

not report it as one.

The decisive point is conceptual as well as statistical. A method bias shared identically by every respondent shifts both groups equally, so it cannot by itself manufacture the between-groups mean difference that carries the study's central inference, the large Information Technology versus Information Assurance gap. A uniform response style is therefore an unlikely explanation for that gap. A response style that differs *by group*, however, such as more conservative endorsement among security professionals, is a group-correlated method effect Harman's test does not exclude. The differential-item-functioning screen below addresses exactly this, showing that the items function equivalently across the two communities once respondents are matched on their own composite score. That match is itself imperfect, because the composite carries any uniform style inflation. Three strands of evidence therefore converge. Harman's test sets a lower bound, the between-groups logic rules out a uniform bias, and the measurement-equivalence screen shows the items behaving alike across the two communities. Taken together they make common-method variance an unlikely sole explanation for the field and familiarity relationships, although a differential response style remains among the accounts Chapter 5 weighs.

Table 24: Common-method variance: Harman's single-factor test.

Diagnostic	**Value**
Items entered (all composite items)	23
Factors with eigenvalue > 1	2
Variance explained by first unrotated factor	44.0%
Single-factor dominance threshold	50%

Note. Post-hoc analysis, not specified in the proposal. A first factor below 50% indicates common-method variance is unlikely to account for the observed relationships.

Construct Validity of the Composite Scores

The comparative analyses treat the three composites as summative indices of related perceptions. A confirmatory three-factor model checks their measurement properties more

strictly, estimating composite reliability and average variance extracted for each (Fornell & Larcker, 1981). Composite reliability is high throughout, at 0.86 for Awareness, 0.87 for IT Value, and 0.87 for Security Value (Table 25), reinforcing the Cronbach's α values in Table 6. Convergent validity, indexed by average variance extracted, falls just below the 0.50 benchmark for all three composites (0.47 for Awareness, 0.46 for IT Value, and 0.45 for Security Value), indicating that items within each composite share somewhat less than half their variance with the intended construct. Discriminant validity is not established under the Fornell–Larcker criterion: the three composites intercorrelate between 0.74 and 0.85, exceeding the square roots of their average variance extracted (0.67 to 0.69).

The heterotrait–monotrait ratio of correlations sharpens this picture and is more informative than the Fornell–Larcker test here (Henseler et al., 2015). Awareness is discriminant from both value composites, with HTMT ratios of 0.83 against each, below the conventional 0.85 threshold; the two value composites, by contrast, are not discriminant from each other (HTMT = 0.99), so IT Value and Security Value are empirically a single perceived-value construct rather than two. The confirmatory three-factor model itself fits the data well (CFI = 0.966, RMSEA = 0.040), so the measurement structure is sound. The discriminant criteria flag the near-identity of the two value dimensions, a separability issue that leaves model fit intact; the analysis accordingly treats the two as related summative indices and never as independent latent variables.

These results carry a direct interpretive consequence, and the HTMT ratios locate it precisely. Awareness is empirically separable from perceived value, but the two value dimensions, IT Value and Security Value, are not separable from each other: respondents who value Digital Engineering for IT operations value it for security operations to almost exactly the same degree. This is theoretically unsurprising, since both dimensions tap a single underlying appraisal of the same described capabilities. The value composites are therefore best understood, and used throughout this chapter, as reliable summative indices of one perceived-value construct measured in two domains rather than as two discriminant

Table 25: Construct validity: composite reliability, average variance extracted, and HTMT discriminant ratios.

Composite	CR	AVE	$\sqrt{\text{AVE}}$
Awareness	0.858	0.471	0.686
IT Value	0.868	0.455	0.675
Sec Value	0.867	0.453	0.673

Composite pair		HTMT	< 0.85?
Awareness	IT Value	0.830	yes
Awareness	Sec Value	0.830	yes
IT Value	Sec Value	0.987	no

Note. Post-hoc analysis, not specified in the proposal. CR $\geq .70$ and AVE $\geq .50$ are the conventional thresholds for adequate convergent validity; the heterotrait-monotrait ratio (HTMT) discriminant threshold is 0.85. Values are computed from standardized loadings of a confirmatory three-factor model and from the item correlation matrix.

latent variables. A consequence follows for how the field gap is counted: because IT Value and Security Value index a single construct, their near-identical field gaps ($d = 1.94$ on each) constitute one finding rather than two independent confirmations. The Awareness gap ($d = 1.64$), meanwhile, leans partly on the cross-scale, DIF-affected familiarity item Q1.1 and is the least clean of the three. This does not weaken the group comparisons, which depend on the reliability of the indices and not on their mutual separability, but it does bound any claim that IT Value and Security Value capture independent dimensions.

One further consequence bears directly on the Research Question 2 and Research Question 3 distinction. Chapter 1 separates general perceived value from applicability to a respondent's own work, a difference that matters for adoption strategy, and the present items do not separate the two empirically. Testing that separation would require distinct item sets written to it, with general-relevance and own-context stems kept apart by construction. Chapter 5 carries this qualification into the limitations.

Robustness of the Field Gap: Bootstrap Intervals and the Combined Value Construct

Two robustness checks test whether the headline professional-field gap depends on how the value composites were built or on the parametric assumptions behind Cohen's d. The first pools all sixteen IT-value and security-value items into a single Perceived Value composite, warranted because the discriminant analysis above showed the two dimensions to be one construct. The pooled composite is highly reliable (Cronbach's $\alpha = 0.92$), and the Information Technology versus Information Assurance gap on it is if anything larger than on the separate composites (Cohen's $d = 2.08$), so the field difference is not an artifact of splitting value into two correlated halves. The second replaces the normal-theory confidence interval for d with a percentile interval from 5,000 bootstrap resamples, which makes no distributional assumption about the bounded, left-skewed composite scores. The bootstrap intervals are narrow and sit far above the conventional large-effect threshold of 0.8 for every composite. Awareness returns $d = 1.64$ with a 95% interval of $[1.50, 1.79]$, IT Value 1.94 with $[1.78, 2.11]$, Security Value 1.94 with $[1.79, 2.11]$, and the combined Perceived Value composite 2.08 with $[1.92, 2.26]$. These results reinforce the non-parametric Mann–Whitney tests already reported (Table 26, Figure 22). The field gap is therefore robust to the composite specification, to the distributional assumptions of the parametric effect size, and to the choice between parametric and rank-based inference.

Table 26: Field-gap effect sizes with bootstrap confidence intervals and the combined value composite.

Composite	**IT**	**IA**	**Cohen's d**	**95% bootstrap CI**
Awareness	4.38	3.73	1.64	[1.50, 1.79]
IT Value	4.67	3.98	1.94	[1.78, 2.11]
Security Value	4.57	3.82	1.94	[1.79, 2.11]
Perceived Value (combined)	4.62	3.90	2.08	[1.92, 2.26]

Note. Post-hoc analysis, not specified in the proposal. Group means for Information Technology (IT) and Information Assurance (IA); Cohen's d with a percentile 95% confidence interval from 5,000 bootstrap resamples. The Perceived Value composite pools all sixteen IT-value and security-value items, which the heterotrait–monotrait analysis showed to measure a single construct ($\alpha = 0.92$).

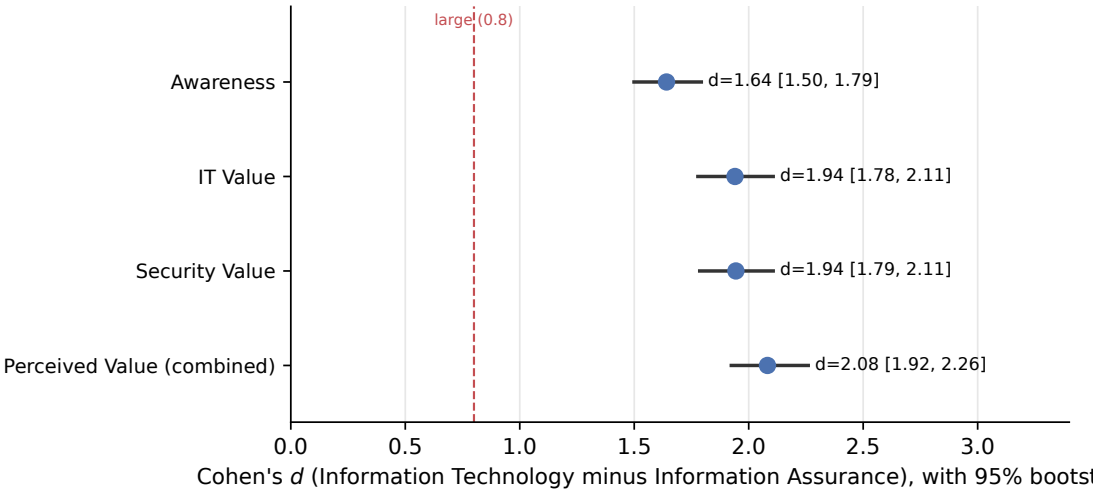


Figure 22: Professional-field effect sizes (IT versus IA) with 95% bootstrap confidence intervals.

Measurement Invariance Across Professional Field

Professional field carries the largest effect in the data, and one candidate explanation for it is differential item functioning: the possibility that Information Assurance and Information Technology respondents at the same underlying level of a construct nonetheless answer particular items differently. A regression-based screen, adapting Zumbo's logistic DIF framework to a linear implementation on the item scores, tests this directly by regressing each item on the respondent's own composite score, professional field, and their interaction. Any residual field effect then represents a difference the matched trait does not explain (Zumbo, 1999). Twenty-one of the 23 items show no meaningful differential functioning. Two meet the flag criterion (Table 27): the familiarity item Q1.1, whose matched-level difference is large ($\beta_{\text{IA}} = -0.99$) and largely restates the familiarity gap itself, and the capability item Q2.5, whose effect is small ($\beta_{\text{IA}} = +0.23$). Setting the familiarity item aside, the residual differences among the substantive capability items are small relative to the composite-level field difference, which approaches two standard deviations (Figure 16). The screen therefore supports a substantive reading of the field gap. A single capability item functions slightly differently

across the two communities and merits cautious interpretation, but item-level measurement differences cannot account for a separation of the magnitude reported earlier. This evidence weakens the differential-item-functioning account specifically, while leaving the substantive awareness-and-exposure account for Chapter 5 to weigh.

Table 27: Measurement-invariance screen: regression-based DIF, IT vs. IA/Cyber.

Item	$\beta_{\mathbf{IA}}$ (matched)	p	DIF flag
Q1.1	-0.99	$< .001$	yes
Q2.5	+0.23	$< .001$	yes
Q5.2	+0.16	.003	no
Q2.4	+0.16	$< .001$	no
Q2.6	+0.16	$< .001$	no
Q2.1	+0.15	$< .001$	no
Q2.2	+0.15	$< .001$	no
Q2.3	+0.14	$< .001$	no

Note. Post-hoc analysis, not specified in the proposal. β_{IA} is the IA-vs-IT difference in item response (scale points) after matching on the item's composite score (a linear implementation of Zumbo's DIF framework). 2 of 23 items met the flag criterion ($p < .001$ and $|\beta_{\text{IA}}| \geq 0.20$). All flagged effects are small relative to the composite-level field difference, which approaches two standard deviations.

Sensitivity to Straight-Lining

The straight-lining check reported in Appendix B identified 22 respondents (1.6% of the 1,390 with the complete agreement battery) who gave an identical answer to all twenty-two agreement-scale items. Removing them tests whether the high endorsement is an artifact of non-differentiating responding. It is not: with the straight-liners excluded, Section 2 agreement is essentially unchanged at 93.9%, because so few respondents straight-lined that their removal does not move the aggregate. Acquiescent straight-lining is therefore a negligible presence and does not manufacture the high endorsement.

Exposure, Familiarity Composition, and the Field Gap

Differential prior exposure, rather than field membership, could explain the professional-field difference, because Information Technology respondents were far more likely to report recent exposure to Digital Engineering than Information Assurance

respondents (32.1% versus 11.8%, a 161-to-59 split among the two focal groups). Adding a recent-exposure indicator to the value regressions of Table 21 tests this directly. The field coefficient is essentially unchanged (Information Technology relative to Information Assurance $b = 0.45$ for IT Value and 0.49 for Security Value, both $p < .001$, against 0.46 and 0.50 without the exposure term), and the exposure term itself is small ($b = 0.08$ for both composites, $p < .001$). The stark exposure imbalance across fields therefore does not account for the field gap in perceived value, which persists once exposure is held constant. Because the exposed Information Assurance subgroup numbers 59 respondents, this is reported as a regression adjustment rather than a within-stratum comparison.

Bimodal familiarity, described earlier in this chapter, supports a second and more demanding version of the same test. If the sample comprises a large naive stratum and a smaller well-acquainted one, and if the two focal fields draw unequally from them, then stratum composition rather than professional field could carry the difference. Information Technology and Information Assurance do draw unequally: 206 of the 501 Information Technology respondents fall in the high-familiarity stratum (Q1.1 of 4 or 5) against 11 of the 501 Information Assurance respondents. Adding an indicator for that stratum to the value regressions holds the composition constant. Field coefficients fall from 0.689 to 0.602 on IT Value and from 0.746 to 0.655 on Security Value, retaining 87% of their magnitude in both cases, and remain significant at $p < .001$ under HC3-robust standard errors. Entering the exposure indicator and the stratum indicator together retains 86%. The two-stratum structure of the familiarity distribution therefore accounts for roughly one seventh of the field gap and leaves the remainder standing. This analysis was not specified in the proposal and is reported, with the other supplementary analyses in this section, as a self-initiated robustness check. It narrows the recruitment-frame concern without resolving it, because a channel difference that delivered systematically different respondents within the same familiarity stratum would not be visible to this adjustment.

Response-Authenticity Screening (Post-Hoc Validation)

Because the survey ran anonymously through an open link, the data were exposed to the automated, coordinated, and synthetic responding that increasingly threatens online instruments of this kind (Griffin et al., 2022; Pozzar et al., 2020; Westwood, 2025). A growing share of this contamination is coordinated human fraud, such as incentive-farming rings sharing an answer key, rather than crude automation alone (Jaffe et al., 2026). That is precisely why a screen keyed on the shared response vector is apt: it detects coordinated duplication whether a script or a coordinated human effort produced it. As a post-hoc validation not specified in the proposal, the analysis added a response-authenticity screen to the pre-registered quality screen described in Chapter 3. This screen and its validation fall outside the original scope of the dissertation's research questions, which concern awareness and perceived value of Digital Engineering rather than survey methodology. The chapter nonetheless develops and reports the screen because it carries independent academic value: it contributes a repeatable, openly released, and validated procedure that researchers across fields conducting anonymous online surveys can adopt, beyond its immediate service to this study's data quality. The screen operates only on the submitted response vectors and their submission timing; the anonymous design assigns every respondent the same placeholder identity, so identity fields carry no forensic information and are not used.

The pre-registered quality screen had already removed 188 rapid or abandoned submissions from the 1,600 raw submissions. Screening for authenticity then examined the remaining sample for coordinated duplication using a documented rule: it flagged any complete response vector shared by four or more submissions as a candidate scripted cluster. It also flagged near-duplicate satellites within a small edit distance of such a cluster and the later copies of smaller duplicate pairs. A single cluster stood out unambiguously, and its structure identifies it as an unsophisticated automated attack of the kind often termed a script-kiddie attack: one reusing a single hard-coded payload with no attempt to vary the answers, randomize the submission timing, or disguise the payload.

The payload was one fixed response vector, a straight-line pattern of the maximum agreement value on every item paired with a single fixed demographic profile, submitted 71 times. Every one of the 71 submissions carried the byte-for-byte identical response vector, each completed in a median of 1.1 minutes, and the submissions arrived in a burst spaced a median of 1.1 minutes apart. Nothing about the payload changed from one submission to the next, precisely the weakness that makes such an attack trivial to detect: a genuine population never reproduces one complete, fully specified answer vector dozens of times, whereas a script replaying hard-coded answers does so by construction.

The layered screen caught the entire burst through two independent and mutually reinforcing mechanisms. The pre-registered timing screen removed 70 of the 71 submissions on completion speed alone, because their sub-three-minute completions fell below the rapid-response threshold. The response-authenticity screen then recognized all 71 as a single coordinated cluster on the strength of their shared complete response vector, a count far above the four-submission threshold. In doing so it caught the one member of the burst whose completion time had drifted just past the timing cutoff and would otherwise have survived. Together the two screens removed every submission in the attack: the timing filter caught the bulk on speed, and the vector screen recognized the attack as a single object and closed the one-response gap the timing filter left. The burst's net effect on the analytic sample was therefore one additional excluded submission beyond those the timing screen had already caught, and the sample stands at $N = 1{,}411$.

The screen also flagged a smaller set of candidate duplicate rows beyond that cluster, and these received individual manual scrutiny rather than automatic exclusion, in the documented and disclosed manner best-practice data screening recommends (DeSimone et al., 2015). On inspection they did not resemble coordinated responding. Their completion times were normal, with a median near nine minutes rather than the sub-minute times of the automated burst, and the members of each flagged pattern had arrived days or weeks apart rather than in a burst. Their response profiles were also

common among clearly authentic respondents: an Information Technology professional agreeing strongly across the value items is one of the most frequent genuine profiles in the sample. Two such respondents independently producing the same response vector is therefore an expected coincidence near the response ceiling rather than evidence of duplication. The analysis judged these cases authentic and retained them. To confirm that this judgment does not shape the results, the analysis repeated every test with the full set of automatically flagged rows removed ($N = 1{,}380$). The two samples yield materially identical results. Composite reliabilities move by less than 0.01. The Information Technology versus Information Assurance effect sizes remain very large, with Cohen's d of 1.58, 1.88, and 1.89 on the fully screened sample against 1.64, 1.94, and 1.94 on the primary one. Every association and diagnostic in this chapter is unchanged in direction and significance. Retaining or excluding the manually scrutinized cases changes no conclusion.

Before applying the screen, the analysis validated it against synthetic data with known ground truth, so that its error characteristics are established rather than assumed (Dupuis et al., 2019; Fawcett, 2006; Ilagan & Falk, 2023). That validation, detailed in Appendix B, characterizes both what the screen catches—coordinated and duplicated submissions—and where it is deliberately conservative: applied mechanically to ceiling-heavy data it would over-exclude genuine strong endorsers at roughly one to two percent, the same borderline rate seen in this sample. For exactly that reason the procedure pairs the automated rule with human adjudication rather than applying it alone. Reported as a data-quality validation, the exercise establishes that a single automated submission burst reached the instrument and was removed, that no further coordinated structure survives in the analytic sample, and that the findings are stable to the treatment of every borderline case the screen identified.

Taken together, the three diagnostics in this section speak to a single question: whether the analyzed responses are organic, the product of real and engaged professional participants rather than of automation, coordination, or inattention. Each addresses a

distinct threat, and the evidence converges. The wave analysis found no material difference between early and late respondents on any composite ($|d| \leq 0.04$), so the sample shows no self-selection signal along the order of submission. Straight-lining, the signature of inattentive or acquiescent responding, is present in only 1.6% of the sample, and its removal moves the headline endorsement rates by fractions of a percentage point. The response-authenticity screen isolated the single automated submission burst that reached the instrument, removed it, and confirmed that no further coordinated structure survives among the retained 1,411 responses. None of these results proves that every retained submission is genuine, a claim no screen can support. Two residual limits remain explicit: professional field is self-reported and unverifiable under the anonymous design, and individual misrepresentation of role lies beyond the reach of any content-based screen. Within those bounds, the converging evidence supports reading the analytic sample as a body of organic professional responses, and it establishes the empirical basis on which the findings that follow rest.

Summary of Findings

Table 5, presented with the direct answers earlier in this chapter, states the finding for each research question. Table 28 sets the four propositions against the evidence before the narrative summary that follows.

Across the three research questions the data tell a coherent story with three important qualifications. For Research Question 1, self-reported familiarity and recent exposure are low (familiarity mean near 1.9; 22.2% exposure), yet endorsement of specific capability descriptions is near-universal (93.9% agreement on Section 2). This gap is best characterized as low awareness coexisting with high capability-endorsement, rather than evidence of awareness as such.

For Research Questions 2 and 3, perceived value and anticipated benefit are high across the applicability, IT, and security blocks. The confidence intervals are narrow conditional on the accessible sample rather than on a random draw of the profession, and

Table 28: The four propositions against the evidence.

Proposition	Finding	Key evidence
P1 Low awareness	Supported	78.2% low familiarity; 22.2% exposure
P2 Value despite low awareness	Consistent; not falsifiable by this design	Low-familiarity majority still values highly (IT 4.21, security 4.08)
P3 Variation by field and experience	Supported on both	Field $d = 1.64$–1.94 (95% CI above 1.4); mid-career experience-group effect $b = -0.40/-0.53$, $p < .001$
P4 Familiarity linked to value	Supported	$\rho = 0.63/0.62$; standardized $\beta = 0.34/0.33$

Note. All effect sizes are from the authenticated analytic sample ($N = 1{,}411$); the field-gap intervals are 95% bootstrap intervals (Table 26). Propositions are supported in the sense of consistency with the cross-sectional evidence rather than causal confirmation.

the two investment-willingness items are at ceiling, supplying conviction but not discrimination.

A very large field gap favoring Information Technology over Information Assurance dominates the comparative analyses, together with a non-monotonic experience pattern in which mid-career (11–15 year) respondents are the most reserved. The focused breakout analysis shows these two findings are largely independent rather than confounded: professional field and experience are only weakly associated, so the mid-career trough is a genuine experience-group difference present within both focal domains rather than a field-composition artifact. The Information Technology–Information Assurance gap likewise persists within each of the three interpretable experience cohorts (1–5, 6–10, and 11–15 years), which indicates the field difference is not merely an experience effect in disguise. The 16+ cohort, at $n = 16$ across the two focal groups, is too small to interpret but shows Information Technology higher on all three composites as well.

Finally, the bias diagnostics are reassuring: the priming diagnostic argues against a pure question-framing explanation for the high endorsement, and the wave analysis returns a null result, with negligible differences between early- and late-arriving respondents on all three composites.

Read against the four propositions advanced in Chapter 1, the data support all four, with one refinement. P1 (low awareness) is clearly sustained: familiarity and recent exposure are both low. P2 (value despite low awareness) receives affirmative but qualified support, because capability endorsement and perceived value stay high even where familiarity is absent. This second proposition is, however, the most weakly tested of the four, since agreement with positively framed descriptions of capabilities tied to problems respondents already face is difficult to disconfirm by design. Its confirmation is therefore better read as consistent with the proposition than as a decisive test of it.

Stronger support for the study's contribution comes from three findings that could have failed and did not. In the priming diagnostic, high-familiarity and low-familiarity respondents diverged sharply rather than converging, which is the opposite of what a pure question-framing account predicts. High perceived value then persisted within the low-familiarity majority. The non-monotonic mid-career reservation, finally, was unanticipated in advance and stayed stable once the models adjusted for professional field.

Instead, the study treats the large professional-field gap as its most provisional result: it is stable across every statistical control the design permits, but the recruitment-frame difference that could confound it cannot be tested in these data (Table 7). The chapter therefore presents it as a target for replication rather than as settled support for the contribution. P3 (variation by field and experience) finds support on both axes. The variation by field is large and stable across the applied statistical controls, and persists within each of the three interpretable experience cohorts, while the variation by experience is a genuine mid-career effect that survives adjustment for field. P4 (familiarity associated with higher value) rounds out the set: familiarity positively predicts perceived value without displacing the high baseline endorsement even unfamiliar respondents express.

Chapter 5 interprets these findings against the literature reviewed in Chapter 2, draws conclusions for each research question, addresses the limitations the diagnostics

surface, and proposes directions for future research.

Chapter 5: Conclusions and Future Research

This dissertation set out to learn whether Information Technology and Information Assurance professionals are aware of Digital Engineering capabilities and whether they perceive those capabilities as valuable for their work. Chapter 1 framed the problem and posed the three research questions. Chapter 2 exposed a near-complete absence of academic research carrying Digital Engineering into enterprise IT and Information Assurance, and it argued that adoption in these domains is governed by perception rather than by proven technical merit. Chapter 3 specified a quantitative, cross-sectional survey methodology, and Chapter 4 reported its results from a sample of 1,411 professionals. The present chapter concludes the study. It synthesizes those results, interprets them against the literature, states the conclusions warranted for each research question, revisits the work's significance along the dimensions established in Chapter 1, acknowledges the limitations that bound the conclusions, and charts future research.

Summary of the Study

Evidence came from a quantitative survey of IT and Information Assurance professionals, structured around six thematic sections that measured awareness, recognition of specific capabilities, perceived applicability, and value assessments for both domains. Within that instrument, explicit traceability connected survey questions to the three research questions. Analytical plans specified descriptive characterization, composite construction with reliability assessment, comparative testing across professional field and experience level, association analysis, and two planned bias diagnostics. A realized sample of 1,411 completed responses exceeded the planned target of 385 to 450 by roughly threefold. This provided high precision while shifting emphasis away from statistical significance, which is near-universal at this sample size, and toward effect sizes and confidence intervals.

Conclusions by Research Question

Research Question 1: Awareness

To what extent are IT and Information Assurance professionals aware of Digital Engineering capabilities? The evidence yields a two-part answer. Awareness of Digital Engineering as a named discipline is low: 78.2% of respondents placed themselves in the two lowest familiarity categories, the familiarity item mean was 1.94, and only 22.2% reported any professional exposure within the prior two years. Endorsement of the specific capabilities that constitute Digital Engineering was nonetheless near-universal once the instrument described those capabilities in functional language, with 93.9% agreement across the Section 2 capability-recognition items. This divergence reads most defensibly as low awareness of the umbrella term coexisting with high endorsement of the underlying capability descriptions.

Two bounds attach to that second half. Section 2 presented functional descriptions and asked for agreement, so it measures recognition of a capability once described and leaves prior knowledge of Model-Based Systems Engineering, digital threads, digital twins, and Product Lifecycle Management unmeasured. Acquiescence to affirmatively framed statements would produce the same near-absence of disagreement. Research Question 1 therefore warrants a narrow conclusion: awareness of Digital Engineering as a named discipline is low, and the described capabilities draw near-universal endorsement when put to respondents in plain functional language. The finding echoes the awareness deficit that Henderson et al. documented even within the systems engineering discipline, where roughly 22% of practitioners could not articulate a clear definition of Model-Based Systems Engineering (Henderson et al., 2024). If awareness is fragile within the originating discipline, the low familiarity observed here among professionals who have never systematically encountered these methods is unsurprising and substantively important.

Research Question 2: Perceived Value

Research Question 2 asks to what extent professionals perceive these capabilities as valuable. The evidence admits a clear answer. The strength of that answer lies in the precision of the estimates rather than in the difficulty of disconfirming the affirmatively framed items, a limit Chapter 4 flags. Perceived value was high and precisely estimated across the applicability (86.3% agreement), IT-value (87.7%), and security-value (82.2%) blocks, with narrow Wilson confidence intervals reflecting the large sample. Interest and intent corroborated the value findings, as 89.0% expressed interest in learning more and 95.9% indicated they would recommend their organization explore adoption. The two investment-willingness items registered near-unanimous willingness conditional on demonstrated return on investment, including a 99.8% affirmative on the security investment item. These ceiling responses are best interpreted as conviction rather than as discriminating measurement. Their near-zero variance is also what acquiescence to an affirmatively framed item would produce, so these two items serve as corroboration rather than as load-bearing evidence. That perceived value is strong despite low familiarity is consequential for the adoption argument advanced in Chapter 2. Perception, not demonstrated technical merit, drives adoption decisions (Call & Herber, 2022; Call et al., 2024), and in this population high perceived value co-occurs with low awareness rather than following from it.

Research Question 3: Anticipated Benefit for Job Performance, Compliance, and Capability

Research Question 3 concerns belief that Digital Engineering could help in practice, by improving job performance, easing compliance, or strengthening organizational capability. The extent of that belief proves considerable, subject to one qualification. Agreement was substantial on the IT operational items (87.7%) and the security operational items (82.0%), and it remained a clear majority on the explicitly compliance-framed items (76.5%). The compliance grouping was the most reserved, and

the single most hedged item in the instrument asked respondents to characterize their own organization's regulatory exposure rather than to endorse a capability in the abstract. That qualification is itself informative, because respondents endorse Digital Engineering's capability to support compliance more readily than they assert that their own organization faces compliance pressures it could address. The result aligns with the compliance imperative documented in Chapter 2. Under that imperative, frameworks such as the NIST Risk Management Framework and the Cybersecurity Maturity Model Certification impose traceability and documentation demands that current practices struggle to sustain (Department of War, 2024; Ross et al., 2020). One boundary attaches to this answer and to the previous one together. Research Questions 2 and 3 were answered from a shared pool of items, and the construct-validity analysis found the value dimensions to be a single appraisal rather than separable ones. Both conclusions are therefore better read as one finding at two levels of specificity, and the conceptual distinction between general value and personal applicability stands as untested rather than as confirmed.

These answers also address the four propositions advanced in Chapter 1. Proposition P1 (awareness of Digital Engineering as a discipline is low) is supported by the familiarity and exposure data. Proposition P2 (perceived value is nonetheless positive once capabilities are described in functional terms) is supported by the near-universal capability endorsement and the high value ratings. Chapter 4 nonetheless marks it as the most weakly tested of the four, because the instrument's positively framed capability descriptions made disconfirmation unlikely by design.

Proposition P3 (awareness and value vary by professional field and experience) is supported on both components. The variation by professional field is large and stable across the statistical controls the design permits. Chapter 4 marks its magnitude as provisional, however, given the recruitment frame that cannot be tested in these data. The variation by experience is a genuine experience-group difference rather than a byproduct of cohort composition. The mid-career cohort at 11 to 15 years of experience scores

significantly lower on both value composites even after professional field and familiarity are held constant (regression coefficients of -0.40 and -0.53, both $p < .001$). Because professional field and experience are only weakly associated in these data, the mid-career reservation cannot be explained away by which professionals populate the cohort.

Proposition P4 (greater familiarity is associated with higher perceived value) is supported by the regression and rank-order analyses (standardized $\beta \approx 0.33$–0.34 across the two value composites). These are correlational endorsements from a cross-sectional design, so "supported" here denotes consistency with the evidence rather than confirmation.

Interpretation and Implications

At its center, this dissertation contributes a first empirical baseline of awareness and perceived value of Digital Engineering among Information Technology and Information Assurance professionals. The baseline draws on 1,411 respondents in a domain that previously held only theoretical proposition. That contribution is deliberately descriptive. It establishes professional receptiveness; it does not establish that Digital Engineering transfers to these domains or that it delivers value once implemented, a boundary the limitations preserve. Three findings carry implications beyond the immediate research questions.

The largest effect in the data separates the two focal fields. Information Technology respondents outscore Information Assurance respondents by a wide margin, with effect sizes between $d = 1.64$ and $d = 1.94$ across all three composites. One of those three figures carries a qualification the others do not. Because the Awareness composite blends the low-scoring familiarity item with the six capability-recognition items, its $d = 1.64$ is better read as a gap in capability endorsement than as a gap in awareness, and Chapter 4 reports it on that basis. This gap runs counter to the dissertation's expectation that the Information Assurance community would be the most receptive audience. Chapter 2's wider expectation fared better once separated into its parts. The attributes that depend on contact with the methods behaved as predicted, since exposure and familiarity are both

low, while relative advantage did not, since perceived value is high across every block. Adoption theory anticipates that pairing, and the present data locate this population precisely there: favorably disposed toward an advantage they have had little opportunity to observe.

Information Assurance respondents reported markedly lower familiarity, with a Q1.1 mean of 1.20 against 2.88 for IT, and lower perceived value. Three readings of the gap remain open. The first is a genuine awareness deficit specific to the Information Assurance community. The second is a valid relevance judgment: the group is heterogeneous, and its operationally oriented members may reasonably conclude that a model-centric and document-centric paradigm fits hands-on defensive work less directly. The third is an artifact of sample composition. Cross-sectional perception data cannot resolve which of these holds.

The focused breakout analysis in Chapter 4 nonetheless narrows the question by ruling out one alternative explanation. The within-domain models confirm directly that the field gap is not an experience effect in disguise: Information Technology exceeds Information Assurance within each of the three interpretable experience cohorts. The candidate explanations that remain carry different imperatives. If the deficit is genuine, the community that the compliance imperative most directly affects is also the least aware of the methods that could address it. That is precisely the condition under which professional-development and curricular initiatives are warranted. If the gap instead arises partly from the recruitment frame, from differential item functioning, or from self-categorization, then replication with representative sampling of the Information Assurance community becomes the priority. Replication would ensure that a measured rather than an apparent deficit directs intervention resources.

A tension left open in Chapter 2 also finds an empirical answer here. The adoption literature holds that perception governs adoption and that preventative innovations such as Model-Based Systems Engineering suffer from low observability that depresses perceived

advantage (Call et al., 2024). Chapter 4's results show that when capabilities are described in functional terms tied to recognized professional problems, perceived value is high even absent familiarity. The evidence is therefore consistent with awareness and exposure, rather than skepticism about value, being the more likely constraint on adoption in these domains, among the constraints the instrument measured. Because the design is cross-sectional and correlational, this remains a diagnosis to be tested rather than a causal conclusion the present data can establish; identifying the binding constraint would require the intervention and longitudinal studies proposed below. One step in that inference is assumed rather than tested. The adoption literature's construct is the valuation a professional brings to a real adoption decision, while this study measured stated agreement with described capabilities, most of it from respondents who had never encountered the methods in practice. Whether the second predicts the first is an empirical question these data do not settle, and the intervention studies proposed below would test it.

Two cautions attach to reading the familiarity–value gradient as evidence that exposure reveals value. The gradient is equally consistent with exposure lowering appraisal, since implementation cost and complexity become salient only once professionals engage the methods. The high-familiarity stratum is also self-selected. It contains those who encountered Digital Engineering and stayed; those who tried it and judged it not worth the cost are absent from the sample. Only the proposed intervention and longitudinal studies can therefore establish what direction genuine exposure would produce in the naive majority. If that reading is borne out, it identifies the form effective intervention should take: education and trial opportunity rather than persuasion about benefit.

Read through the boundary-object lens of Chapter 2, the IT-versus-Information-Assurance divide is the predictable signature of a pragmatic knowledge boundary (Carlile, 2002, 2004; Star & Griesemer, 1989). The shared, authoritative models of Digital Engineering are positioned to cross that boundary only once awareness brings them within these communities' reach. The high perceived value the

study records speaks directly to the crossing condition that Carlile identifies. Crossing a pragmatic boundary requires parties to accept the cost of transforming their working knowledge, and that acceptance presupposes that they judge the artifact carrying the transformation to be worth the cost. These professionals already rate the described capabilities as highly valuable. That rating is consistent with, though not equivalent to, a willingness to bear the transformation cost, which only implementation research could confirm. On this reading the transformation would await exposure to the models rather than assent to their worth. Among the constraints this study measured, awareness and exposure are therefore where the binding one sits, as the theory predicts once value is granted. The other costs of crossing a pragmatic boundary, in tooling, in skills, and in organizational commitment, lie outside the instrument, and only the proposed intervention studies can weigh them against awareness.

The bias diagnostics, finally, qualify but do not overturn the value findings. The priming diagnostic argues against a question-framing explanation that would operate uniformly across familiarity, because high-familiarity and low-familiarity respondents diverged sharply rather than converging. It cannot, however, exclude a framing or self-selection effect that itself co-varies with familiarity, since the high-familiarity stratum is self-selected. The wave analysis, by contrast, found no material difference between early and late respondents on any composite ($|d| \leq 0.04$, all Mann–Whitney $p > .5$), so response timing yields no evidence of self-selection along the early-to-late dimension. Reported together, these diagnostics support a measured conclusion. Perceived value is genuine and substantial. The defense-and-aerospace recruitment frame examined in the limitations still counsels reading the favorable estimates as upper-leaning for the full target population.

These results contribute the empirical foundation that Chapter 1 identified as missing. They establish the baseline that any subsequent implementation, case-study, or curricular research would draw on. The findings are consistent with the theoretical premise that Digital Engineering's capabilities resonate with the documented problems of IT and

Information Assurance practice, and they show that low awareness coexists with high perceived value in this population. Between those two candidate constraints, awareness and exposure are the more likely brake on adoption. No other barrier entered the instrument, so this ranking holds within the pair the study assessed and says nothing about implementation cost, tooling maturity, skills availability, or organizational readiness. The work thereby completes the second step in the logical pathway articulated in Chapter 1, which runs from demonstrated value in defense and aerospace, through established professional receptiveness, toward future investigation of practical implementation.

Research Significance Revisited

Chapter 1 framed the significance of this research along four dimensions. The results now permit reconsidering each against evidence rather than expectation.

Academic Significance

This work supplies empirical evidence to a domain previously characterized by theoretical proposition rather than data, realizing the academic significance Chapter 1 anticipated. The literature reviewed in Chapter 2 contained no investigation of whether IT and Information Assurance professionals perceive value in Digital Engineering capabilities. The systematic mapping of the field by Wooley and Womack likewise found no research addressing enterprise IT or Information Assurance applications (Wooley & Womack, 2025). In method, the study extends the diffusion-of-innovations perception surveys of Call et al. and Henderson et al. from the systems engineering community to the Information Technology and Information Assurance population (Call et al., 2024; Henderson et al., 2024). That extension is the point rather than an incidental feature, because this population is the precise locus of the transferability question the dissertation raises, and no prior survey had measured that population. Present findings supply the missing perception data and, in doing so, test the theoretical framework that connects Digital Engineering capabilities to documented IT and Information Assurance challenges. On the value dimension that framework survives, since practitioners recognize substantial value, while

the awareness dimension reframes the research agenda toward education and exposure.

Two methodological results of independent interest follow as well. One is measurement-theoretic. That low familiarity coexists with high capability-endorsement shows empirically why familiarity with a discipline and recognition of its capabilities require separate measurement, a distinction that the layered Section 1 and Section 2 design was built to capture. A third construct, verified knowledge, requires a third measure that this instrument did not carry. Another arose as an unplanned by-product: the anonymous open-link mode of data collection required defending the data against the automated, coordinated, and synthetic responding that increasingly threatens online surveys (Jaffe et al., 2026; Pozzar et al., 2020; Westwood, 2025). Mounting that defense required developing, validating, and openly releasing a duplicate-vector response-authenticity screen paired with human adjudication of borderline cases (Appendix B). Of the 71 identical vectors submitted in the one coordinated burst that reached the instrument, the pre-registered timing screen removed the 70 that arrived too quickly and the response-vector rule removed the single submission whose slower timing would otherwise have survived. The study offers the artifact as a reusable, auditable contribution and develops it more fully in a companion methods manuscript.

Industry Significance

Findings deliver an actionable diagnosis of the kind Chapter 1 envisioned for industry: a characterization of current awareness and perceived value that can inform adoption strategy across the professional community. Organizations contemplating Digital Engineering adoption face a receptive rather than a skeptical audience, since perceived value and recommendation intent are high. Receptiveness is a fact about the audience rather than about the technology. These results speak to what professionals expect Digital Engineering to do for them and leave what it does in deployment to the implementation research the future-work agenda specifies. They must, however, contend with a pervasive awareness deficit that places these capabilities outside the consideration set during tool

selection and strategic planning. The breakout analysis sharpens the guidance further by identifying where receptiveness is weakest. The Information Assurance community, which the compliance landscape most directly affects, reports the lowest awareness and value. If that deficit proves genuine rather than an artifact of sampling, it marks this community as the segment requiring the most deliberate engagement. Vendors, service providers, and standards bodies can therefore provisionally concentrate communication and enablement on the audiences that the data identify as least served, principally the Information Assurance community, rather than allocating effort uniformly.

What form that effort should take depends on how the deficit is read. If Information Assurance respondents report lower value because Digital Engineering has not reached them, then awareness-building and trial opportunity are the appropriate instruments. If their lower ratings instead reflect calibrated professional skepticism toward affirmatively framed claims about security capability, as Chapter 4 raises among the candidate explanations, then the same audience calls for demonstrated evidence of benefit in security settings rather than broader exposure. The present data cannot separate the two, and the distinction governs the design of an intervention rather than the standing of the finding.

The instrument does not, however, license capability-by-capability prioritization. Because many items pair a capability with an asserted benefit, and some bundle more than one function, the data speak more securely to which audiences are least reached than to which individual pillar is most or least valued. The instrument refinements below therefore include rewording each item to a single clause so that per-pillar demand reads cleanly.

National Significance

Respondents endorsed, at high rates, the propositions that Digital Engineering could improve compliance with security frameworks and could shorten vulnerability identification and remediation. These are precisely the capabilities that the expanding requirements of SP 800-171 and the Cybersecurity Maturity Model Certification demand (Department of War, 2024; Ross & Pillitteri, 2024). These results bear directly on the national dimension

of significance, which concerns national security and the protection of societal infrastructure. Chapter 1 argued that authoritative traceability of the kind Digital Engineering provides could reduce the burden of compliance verification for operators of federal and national security systems while improving the currency of compliance evidence. Reservations voiced on organization-specific compliance items indicate that demonstrating concrete compliance outcomes will advance the case for Digital Engineering in this setting more effectively than asserting general capability. The future-research agenda below takes up this point directly. These findings address only indirectly the dependency-mapping problem that opened this dissertation: the difficulty of tracing the cascading first-, second-, and third-order effects of a vulnerability across interconnected systems. Survey items measured perceived value for vulnerability identification and remediation and for authoritative traceability in general, not for cascading-impact assessment as a distinct use case. The study therefore establishes professional receptiveness to the class of capability to which dependency mapping belongs without measuring that specific application on its own.

Societal Significance

Compliance documentation consumes scarce capacity in organizations that operate under tight resource constraints: community health centers, tribal organizations, under-resourced school districts, small municipalities, and social-service non-profits. Chapter 1 located the societal significance of this work in exactly these settings. The argument advanced there was distributional. Because the manual labor and specialized expertise that compliance demands fall hardest where capacity is smallest, a capability that lowers that threshold would benefit these organizations most. Such a capability could narrow rather than widen the gap between the security and compliance posture available to well-funded institutions and that available to the smallest and least-resourced organizations.

Sampling did not specifically reach those organizations, so the study cannot speak to adoption within them. The study measured receptiveness in a comparatively

well-resourced, defense-and-aerospace-weighted population, as the limitations detail, so it establishes the precondition for the accessible population rather than demonstrating it for the under-resourced organizations the argument concerns.

What it does establish is the precondition on which that broader benefit depends. Across a diverse professional population, respondents perceived Digital Engineering's capabilities as valuable, at agreement rates above 80%, for precisely the documentation, traceability, and verification work that burdens resource-constrained organizations most heavily. That perceived value held even among the large majority of respondents unfamiliar with the umbrella discipline. By confirming that broad professional receptiveness, the study supplies the empirical warrant for the next step: designing and evaluating accessible, lower-cost Digital Engineering implementations suited to organizations with limited specialized expertise. That work would pursue the equity benefit anticipated in Chapter 1 deliberately rather than leave it to chance.

Nothing about the next step is automatic. Digital Engineering as currently practiced is itself expertise- and tooling-intensive, so extending it to the least-resourced organizations depends on accessible, lower-cost implementations that do not yet exist. This chapter therefore frames the distributional benefit as a precondition and a design-science agenda rather than as a demonstrated outcome.

Limitations

Several limitations bound the conclusions above, some anticipated in Chapter 3 and some surfaced by the results.

Non-probability convenience sampling limits generalizability to the broader population of IT and Information Assurance professionals, so the results should be read as indicative of the accessible population rather than as definitive measures of the entire professional community. Self-selection bias may have over-represented professionals already favorably disposed toward Digital Engineering. The wave analysis did not corroborate this specific concern: early and late respondents did not differ materially on any composite

($|d| \leq 0.04$), which weighs against response-timing self-selection. That analysis cannot, however, exclude a self-selection process operating uniformly across the collection window. Social desirability bias may also have inflated value responses, particularly given the affirmatively framed capability statements, although the anonymous design was intended to mitigate this.

A specific composition concern attends the recruitment strategy. Recruitment relied principally on outreach through the candidate's LinkedIn network, which is itself concentrated in the defense, aerospace, and federal sectors, the very settings in which Digital Engineering originated and is most actively practiced. No industry filter was applied to the invitation, so the frame's composition rather than a selection rule produced the sector weighting. That weighting still tilts the realized sample toward professionals more likely than the broader IT and Information Assurance population to have encountered Digital Engineering. Such weighting plausibly elevates both the awareness and the perceived-value estimates above what a sector-balanced sample would yield. Paid promotion widened circulation beyond that network during the later weeks of collection, and its effect on composition runs in both directions. Reaching professionals outside the candidate's contacts works against the network skew, while delivery through an advertising algorithm the study neither controlled nor recorded introduces a selection mechanism of its own. Wave analysis, however, found that the later respondents it attracted did not differ materially from earlier ones on the value composites. Read conservatively, then, the value findings are upper-leaning estimates for a defense-and-aerospace-weighted accessible population rather than point estimates for the entire professional community.

This limitation is the most consequential the study carries, because the instrument recorded professional field and experience but no measure of employment sector or organization type. No direct adjustment for the frame effect is therefore possible from these data. The proxy adjustments reported below, which hold recent exposure and familiarity-stratum composition constant, narrow the concern without resolving it. Future

replication should carry a sector item so that this confound can be modeled rather than reasoned about. The comparative findings, which contrast groups drawn from the same sample, are more robust to this composition than the absolute levels, because a shared recruitment bias affects the groups being compared in common.

Several limitations follow from the realized data. The instrument produced four professional-field categories rather than the planned dichotomy, and the primary comparison contrasted Information Technology with Information Assurance while retaining Engineering and Other for descriptive context only. A different operationalization of field would therefore change the comparative section. The two investment-willingness items exhibited ceiling effects that rendered them near-invariant and therefore uninformative for subgroup contrasts. Several stems also bundle more than one capability or outcome within a single statement, among them the vulnerability item that pairs identification with remediation, so agreement on those items cannot be attributed to one clause of the stem. Chapter 4 marks the affected items where it reads them individually.

The 16+ year experience cohort, at only eighteen respondents, was too small to characterize, so the experience analysis rests on the three larger cohorts. Composition of that variable also bounds what population the study reached, and it does so on evidence the instrument measured directly. Respondents with more than fifteen years in the field number eighteen, or 1.3% of the sample, while 40.1% report five years or fewer. A cross-section of the IT and Information Assurance professions would carry a substantially larger senior segment than that. Realized this way, the distribution indicates that the recruitment channels reached a predominantly early-career population rather than the professions at large, which is the clearest evidence available here about the accessible frame the study actually obtained. It also reinforces the recruitment-frame caution attached to the field comparison below, since channels that skew this sharply on tenure may skew on other unmeasured attributes. The cross-sectional design captures perceptions at a single point in time and cannot speak to how awareness or perceived value would change

following exposure or education.

A further caution attends the headline field comparison. The Information Technology versus Information Assurance effect sizes ($d = 1.64$ to 1.94, contrasting 501 IT and 501 Information Assurance respondents) are exceptionally large by the standards of survey research, and an effect of that magnitude between two adjacent professional communities itself invites scrutiny.

Beyond the experience confound that the breakout analysis excludes, four alternative explanations remain open. One account is a recruitment-frame difference in how the two groups were reached; another is differential item functioning in how security professionals calibrate affirmatively framed statements. The gap could also reflect a self-categorization effect in the choice of professional label: the combined Information Assurance/Cyber Security response option spanned both governance-oriented and operationally oriented professionals and so mixed two roles the instrument cannot separate. A fourth possibility is that the lower Information Assurance ratings represent calibrated professional judgment, with respondents discounting affirmatively framed claims about security capability because their expertise equips them to appraise those claims critically. That reading gains some support from the observation that Information Assurance respondents rate the security value of Digital Engineering below the rating Information Technology respondents assign it, and it would leave the measured difference intact while changing what the difference means. Present data cannot adjudicate among these, and the magnitude should not be taken at face value as a pure awareness difference.

Three analyses nonetheless bear on the question. Chapter 4's differential-item-functioning screen weakens the measurement-artifact account, and adding a recent-exposure indicator to the regression leaves the field coefficient essentially unchanged. Holding constant the composition of the bimodal familiarity distribution, which divides the sample into a large naive stratum and a small well-acquainted one, likewise retains 87% of the field coefficient.

The anonymous instrument recorded no employment sector and no recruitment channel, so which channel reached which community cannot be reconstructed. An effect this large is therefore as consistent with an unmeasured frame difference as with a true awareness gap, and confirming it is a task for replication rather than for reinterpretation of these data. The appropriate reading is that the field difference is a strong but provisional result. Establishing its size in the population requires a probability or quota sample that records recruitment channel and employment sector, a design the future-research agenda specifies.

A related boundary bears on the adoption argument. The instrument measured awareness, exposure, and perceived value, and it measured no other candidate barrier to adoption. Several plausible barriers went unmeasured. The instrument asked nothing about implementation and licensing cost, about tooling maturity, about the availability of modeling skills, about competing organizational priorities, or about an organization's readiness to absorb a new method. It also left aside the effort expectancy, facilitating conditions, and social influence that the Unified Theory treats as determinants alongside performance expectancy (Venkatesh et al., 2003). Comparing awareness against value perception is therefore the most this design supports. Where the study finds awareness low and value high, the reasonable inference is that awareness and exposure constrain adoption more than doubt about benefit does. Which barrier binds hardest overall remains open, and a barrier inventory administered alongside the awareness and value measures would settle it.

Most fundamentally, the study measured perceived value and anticipated benefit rather than actual experienced benefit. Positive perceptions establish receptiveness, yet they do not demonstrate that Digital Engineering would deliver value if implemented, nor do they establish organizational readiness for adoption. The value finding is further bounded by the instrument's construction. All thirty items are affirmatively keyed, and the instrument carried no reverse-coded or knowledge-verification items, so a respondent

disposed to agree encountered no item on which agreement would register as disconfirmation. Near-universal endorsement is therefore close to the strongest result the instrument could produce, and *receptiveness*, rather than demonstrated conviction, is the ceiling of what the value data support. The negligible straight-lining rate and the priming diagnostic bound crude and uniform acquiescence, but neither can exclude a mild yea-saying tilt that co-varies with familiarity.

Because the design is anonymous, professional field, the basis of the central comparison, is self-reported and cannot be verified; anonymity trades individual verifiability for candor. The response-authenticity screen detects coordinated and duplicated submissions, not individual misrepresentation of role, so the guard against misclassification is indirect. The field groups exhibit coherent, theory-consistent internal structure rather than noise, straight-lining is negligible, and an unpaid anonymous survey offers little evident incentive to misrepresent one's field. The single coordinated attack that reached the instrument was a duplication burst that the screen removed. Individual misrepresentation of role is, by design, undetectable by any content-based screen, so this residual is acknowledged rather than claimed away. A design that records verifiable role or employment, or a screened professional panel, would close this gap and is specified in the future-research agenda.

The supplementary post-hoc analyses reported in Chapter 4 refine two of these cautions and introduce a third. The differential-item-functioning screen found only two of the twenty-three composite items functioning differently across the two professional fields, both by small margins, which weakens the measurement-artifact account of the field gap without excluding it entirely. The construct-validity assessment found the three composites to be reliable but strongly intercorrelated rather than discriminant, so this dissertation treats them as summative indices of related perceptions rather than as independent constructs. Any claim that awareness and perceived value are separable dimensions is correspondingly bounded. Because Research Questions 2 and 3 draw on the same item

pool, their separate answers do not constitute independent findings. Because the study added these analyses after data collection and did not specify them in the approved proposal, they are exploratory and should be read as robustness corroboration of the reported findings rather than as pre-registered confirmatory tests.

Future Research

Five complementary lines of future research follow from these findings.

From Perceived to Measured Value

Case-study and pilot-implementation research in enterprise IT and Information Assurance settings offers the most direct extension, converting perceived value into measured value. Such work would address the measurement deficit that Chapter 2 documented across the broader Digital Engineering literature, where most claimed benefits rest on perceived rather than measured evidence (Henderson & Salado, 2021; E. B. Rogers & Mitchell, 2021). A pilot would test whether the value this population perceives materializes in practice, under what organizational conditions, and at what cost. Coupling such a pilot to an explicit benefits-measurement framework of the kind Henderson et al. developed would convert anecdotal outcomes into quantified evidence suitable for enterprise business cases (Henderson et al., 2023).

Explaining the Field and Experience Patterns

Why Information Technology so consistently outpaces Information Assurance calls for investigation in its own right. Because cross-sectional perception data cannot distinguish a genuine awareness deficit from a sampling artifact, future work should use stratified or probability sampling within the Information Assurance community specifically. Qualitative inquiry should accompany that sampling, probing why these professionals report lower familiarity and value despite operating under the compliance pressures that Digital Engineering is positioned to address.

The breakout analysis also surfaced a pattern that merits dedicated study: the genuine mid-career reservation at 11 to 15 years of experience. The reservation persists

across both focal fields and net of statistical control for field and familiarity, yet it lies outside what the main-effects model explains and remains substantively unaccounted for. Three hypotheses, each grounded in existing theory, are worth testing. The first is a sunk-cost or switching-cost account: mid-career professionals have the deepest personal investment in the incumbent document-centric methods and tooling. The perceived cost of a paradigm shift is therefore highest for them and depresses their appraisal of its value. Earlier-career professionals carry less such investment, and senior professionals evaluate the shift as a portfolio choice rather than a personal retooling. The second is a diffusion-of-innovations account (Call et al., 2024): the 11-to-15-year cohort may occupy a pragmatist position on the adoption curve, past the exploratory openness of newer entrants but short of the strategic vantage of senior staff. Pragmatists are precisely the segment that a preventative innovation with low observability persuades least. The third is a role-structure account: at this tenure professionals often hold hands-on senior-technical or first-line-management roles in which they would absorb the operational burden of adoption without yet holding the authority to direct it or the budget to resource it. That position rationally tempers enthusiasm.

These accounts are not mutually exclusive, and they make different predictions about which interventions would move the cohort, so distinguishing them matters practically. Interviews and focus groups would be well suited to the task, since professional narrative captures a mid-career shift in how professionals appraise emerging methods more naturally than additional survey items would.

Intervention, Longitudinal, and Replication Studies

Because the evidence is consistent with awareness, rather than perceived value, being the limiting factor for adoption, intervention research becomes a natural next step and the direct test of that reading. Studies that measure awareness, perceived value, and adoption intention before and after structured exposure or training would test whether closing the awareness gap shifts adoption intent. They would also connect this work to the

enterprise adoption frameworks synthesized in Chapter 2 (McDermott et al., 2024). Longitudinal designs would address the temporal limitation of the present cross-sectional snapshot by tracking how perceptions evolve as Digital Engineering diffuses into enterprise contexts. Replication against probability frames, across additional sectors, and in other national contexts would establish the external validity that convenience sampling cannot, and it would test whether the field and experience patterns observed here generalize. Any of these designs should carry a barrier inventory, measuring cost, tooling maturity, skills availability, and organizational readiness alongside awareness and perceived value, so that the obstacles to adoption can be ranked against one another rather than assessed one at a time.

Instrument and Methodological Refinement

The instrument itself warrants refinement. Its ceiling effects on the investment-willingness items indicate that future versions should replace binary or ternary willingness questions with graduated or scenario-anchored measures capable of discriminating among the favorably disposed. That low familiarity coexists with near-universal agreement also argues for incorporating reverse-coded items and knowledge-verification items carrying scoreable correct answers. Such items would measure comprehension directly, which the present design could only approach through recognition, and they would separate substantive knowledge from acquiescence. Refining the field item to elicit the four categories deliberately, rather than recovering them after the fact, would support a planned four-group comparison in place of the proposal's dichotomy.

The measurement analyses in Chapter 4 point to two further, more specific refinements. First, the average variance extracted for each composite fell just below the conventional 0.50 benchmark, which reflects item sets that each span a heterogeneous range of capabilities rather than a single tightly defined facet. Tightening each composite toward a more homogeneous set of indicators, or lengthening it with items that load more cleanly on one construct, would raise convergent validity in a future administration.

Second, the heterotrait–monotrait analysis showed the IT-value and security-value dimensions to be empirically inseparable. A revised instrument should therefore measure perceived value as a single scale, which the present data support. If theory instead predicts a substantive IT-versus-security distinction, the revision should redesign the items so that the two facets are worded to diverge rather than to restate a common appraisal in two settings. Several of these refinements—including reverse-coded, knowledge-verification, and per-capability items—are carried forward into the reproducibility artifact described under Dissemination and Reproducibility below. A subsequent study can therefore adopt them without reconstructing the instrument from the proposal.

Technical and Design-Science Directions

Technical barriers identified in Chapter 2 define a final, design-oriented agenda. Developing and validating modeling-language extensions and domain-specific profiles for representing IT services, network configurations, and security controls would close the gap between the general-purpose modeling languages of systems engineering and the specific representational needs of enterprise IT and Information Assurance. Design-science research could build and evaluate a Digital Engineering reference implementation that maintains an authoritative compliance traceability thread across the Risk Management Framework and the Cybersecurity Maturity Model Certification, then measure the reduction in audit-preparation effort against current document-centric practice. Such research would extend the preliminary reference-model work that motivated this study (Bonar & Hastings, 2024; Chick et al., 2023).

A distinct and equity-oriented strand of this agenda would target the small and under-resourced organizations identified earlier. Design-science work could develop and evaluate lightweight, low-cost, and template-driven Digital Engineering implementations that minimize the specialized expertise required for adoption. These implementations would be packaged with pre-built compliance profiles for the regulations these organizations most commonly face, such as HIPAA for community health providers and

FERPA for under-resourced schools. Field studies in such settings, building on the precedent of Model-Based Systems Engineering in rural healthcare (Berg et al., 2023), would then measure whether the documentation and compliance burden actually falls and whether organizations genuinely redirect scarce staff capacity toward direct service provision. This line would convert the distributional promise identified here into measured outcomes for the organizations that stand to benefit most.

Related work on tool interoperability would address the integration challenge of connecting systems-engineering modeling environments to the IT service management, security information, and configuration platforms that enterprises already operate. Finally, Chapter 2 documented that inadequately secured digital twins expand the attack surface they are meant to reduce. Research into securing enterprise digital twins within an integrated Digital Engineering framework would therefore ensure that adoption advances security posture rather than undermining it (Suhail et al., 2024).

Dissemination and Reproducibility

Two components of this work are positioned for peer-reviewed dissemination in their own right. The first is the structured cross-disciplinary review of Chapter 2, which delineates a research gap that no prior synthesis has mapped. The second is the perception survey of Chapters 3 and 4, whose sample of 1,411 professionals is large by the standards of this research area. To support replication and reuse, the survey instrument, the scripted analysis pipeline, and a synthetic dataset that reproduces the reported results are openly available in a public research repository at https://doi.org/10.17605/OSF.IO/VKDTE. This privacy-preserving reproduction dataset is distinct from the separate, purpose-built synthetic ground-truth set used to establish the operating characteristics of the response-authenticity screen; the two serve different roles and are not conflated. Because the instrument was fully anonymous and collected no identifiers, the de-identified response data underlying the study are available from the author on request. Their public deposit is under consideration, subject to the anonymity and exempt-review terms of the study.

Concluding Remarks

This research set out to determine whether IT and Information Assurance professionals recognize the potential of Digital Engineering for their work, and it answers that question with empirical data where previously only proposition existed. Respondents are largely unfamiliar with Digital Engineering as a discipline, yet they endorse its constituent capabilities and perceive substantial value in them for IT operations, security operations, and compliance. That perceived value is strong enough that nearly all respondents would recommend their organizations explore adoption. Of the two barriers this work measured, awareness and exposure are the binding one, and skepticism about value is not. Adoption faces other obstacles the study did not assess, among them implementation cost, tooling maturity, and organizational readiness, so this finding locates one constraint rather than ranking the full set. That deficit appears most pronounced within the Information Assurance community that the expanding compliance landscape most directly affects. This reading remains associational, and the magnitude of the gap warrants the cautions raised in the limitations.

Chapter 2's theoretical lens frames what such receptiveness makes possible. Should the models of Digital Engineering prove able to act as boundary objects, carrying transformed knowledge across the pragmatic divide that separates systems engineering from Information Assurance and IT service management, then the willingness measured here becomes the precondition for that transfer. By establishing that perceived value is high while awareness is low, this dissertation provides the evidentiary foundation for the next stage of inquiry: demonstrating, measuring, and enabling the practical application of Digital Engineering capabilities in enterprise IT and Information Assurance. The need in those domains is documented, and the receptiveness, this study shows, is already present.

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Appendix A: Survey Instrument and Research Question Mapping

This appendix reproduces the survey instrument as administered and provides traceability between survey questions and research questions, following the systems engineering lifecycle approach outlined in the dissertation methodology. The survey structure addresses three core dimensions (**Awareness**, **Applicability**, and **Perceived Value**) as they pertain to Digital Engineering and its four pillars: Model-Based Systems Engineering, the Digital Thread, Digital Twin, and Product Lifecycle Management.

Survey Instrument as Administered

Participants first viewed an informed-consent notice. It stated that the survey was part of a doctoral research project at Dakota State University; that participation was anonymous, voluntary, and limited to respondents aged eighteen or older; that no personally identifiable information was collected; that risks were minimal and no compensation was offered; and that completion and submission constituted informed consent. The instrument used two five-point response scales, presented in Table 29: a familiarity scale for the Section 1 awareness item and an agreement scale for the Likert items in Sections 2 through 5.

Table 29: The two five-point response scales used in the survey instrument.

Point	Familiarity scale (Section 1)	Agreement scale (Sections 2–5)
1	Not at all familiar	Strongly disagree
2	Slightly familiar	Disagree
3	Moderately familiar	Neither agree nor disagree
4	Very familiar	Agree
5	Extremely familiar	Strongly agree

The thirty items, by section, were as follows.

. **Section 1: Awareness and Familiarity.**

Q1.1 Please rate your level of familiarity with Digital Engineering concepts and practices. *(Familiarity scale)*

Q1.2 Have you encountered Digital Engineering methodologies, frameworks, or tools in your professional work within the past two years? *(Yes / No)*

. **Section 2: Understanding of Capabilities.** *(Agreement scale)*

Q2.1 Digital Engineering includes model-based systems engineering approaches that can improve development processes.

Q2.2 Digital Engineering can enable digital twin development and virtual prototyping for Information Technology systems.

Q2.3 Digital Engineering supports continuous integration and data-driven decision-making in technology development.

Q2.4 Digital Engineering enables digital twin technology that can simulate security scenarios and test defensive measures without impacting production systems.

Q2.5 Digital Engineering supports continuous security validation and data-driven threat analysis throughout the development lifecycle.

Q2.6 Digital Engineering can improve security control implementation through automated compliance checking and verification.

. **Section 3: Applicability.** *(Agreement scale)*

Q3.1 Digital Engineering methodologies have relevant applications within the Information Technology sector.

Q3.2 Digital Engineering methodologies have relevant applications for addressing Information Assurance challenges.

Q3.3 The ability to utilize digital twins to test changes against accurate replicas of production environments would provide value to my organization.

Q3.4 The use of digital models to map and document an organization's environment and configurations would provide value to my organization.

Q3.5 The use of digital lifecycle management to meet compliance and service delivery requirements would provide value to my organization.

Q3.6 My organization faces regulatory or compliance requirements that could benefit from Digital Engineering approaches.

. **Section 4: Value for Information Technology.** *(Agreement scale; Q4.5 Yes / No / Unsure)*

Q4.1 Digital Engineering could deliver meaningful value to my organization's Information Technology processes.

Q4.2 Digital Engineering could reduce development cycle time in my organization.

Q4.3 Digital Engineering could improve product quality and reduce defects in my organization.

Q4.4 Digital Engineering could improve collaboration effectiveness across development teams in my organization.

Q4.5 My organization would be willing to invest in Digital Engineering capabilities if clear return on investment could be demonstrated.

. **Section 5: Value for Information Assurance and Cybersecurity.** *(Agreement scale; Q5.7 Yes / No / Unsure)*

Q5.1 Digital Engineering could deliver meaningful value to my organization's Information Assurance and cybersecurity operations.

Q5.2 Digital Engineering could reduce the time required to identify and remediate security vulnerabilities in my organization.

Q5.3 Digital Engineering could improve security posture and reduce successful cyber incidents in my organization.

Q5.4 Digital Engineering could enhance threat modeling and risk assessment capabilities in my organization.

Q5.5 Digital Engineering could improve collaboration between security teams, development teams, and operations teams in my organization.

Q5.6 Digital Engineering could help my organization achieve better compliance with security frameworks and regulatory requirements.

Q5.7 My organization would be willing to invest in Digital Engineering capabilities for cybersecurity purposes if clear return on investment could be demonstrated.

. **Section 6: Interest and Demographics.**

Q6.1 Would you be interested in learning more about Digital Engineering applications for Information Assurance, Cybersecurity, or Information Technology in your industry? *(Yes / No)*

Q6.2 Would you recommend that your organization explore Digital Engineering adoption for improving security operations? *(Yes / No)*

Q6.3 Please indicate your field of practice. *(Information Technology / Information Assurance or Cyber Security / Engineering / Other)*

Q6.4 Please indicate your level of experience in your field of practice. *(1–5 / 6–10 / 11–15 / 16+ years)*

Consistent with the design described in Chapter 3, the Section 2 through Section 5 stems carried only minimal functional context for each capability; respondents did not see a separate definition of Digital Engineering, which bears on the interpretation of the high endorsement reported in Chapter 4.

. **Instrument Limitations.** Because the instrument is reproduced as administered, several design limitations bear on interpretation rather than on revision. Several stems bundle related capabilities in a single item (e.g., Q2.3, Q2.5, Q3.5, Q5.2, Q5.3, Q5.6), so

item-level agreement cannot be attributed to any single clause. All Likert items carry positive keying, which leaves residual acquiescence risk; the straight-lining screen in Appendix B diagnoses that risk. The phrasing of Q3.3 ("accurate replicas") presupposes digital-twin fidelity rather than asking respondents to evaluate it. Q4.5 and Q5.7 pose hypothetical organizational investment questions that individual respondents answered on their organizations' behalf. Finally, the demographic items offer no "prefer not to answer" option, and the experience item (Q6.4) provides no category below one year.

Recruitment Announcement as Posted

Recruitment proceeded primarily through a LinkedIn post, as described in Chapter 3. That announcement was deliberately descriptive rather than persuasive. It named the study topic and the eligible population without asserting that Digital Engineering delivers value, so that its wording would not prime respondents toward agreement. Below, the post appears verbatim as administered.[2]

> **Participants Wanted for Research Study**
>
> We are conducting a research study about Digital Engineering, Information Assurance, and Information Technology Service Management, and are inviting individuals to participate.
>
> **Who can participate?**
>
> - IT, Information Assurance, and Cybersecurity Professionals
>
> **What will you do?**
>
> - Complete a 31 question survey
> - Estimated time: 10 minutes
>
> **No Compensation**

[2] The announcement stated a "31 question survey." One item was subsequently identified as a duplicate and removed before administration, so the instrument as administered contained thirty items, listed in the Survey Instrument as Administered section above. The recruitment text is reproduced here with its original count.

Voluntary Participation

Your participation is completely voluntary.

Interested?

Click here to learn more or participate: `https://lnkd.in/guNhiyky`

For questions, contact: John Bonar, `john.bonar@trojans.dsu.edu`

The paid LinkedIn promotion described in Chapter 3 boosted the visibility of this same post. It altered reach only, and changed neither the wording nor the incentive structure. Because the survey was administered anonymously through Microsoft Forms, the platform did not attribute individual responses to specific referral channels, so channel-level response counts are not available. Chapter 3 characterizes the sector composition of the realized sample, and Chapter 5 addresses its implications for generalizability.

Research Questions

Table 30: Research questions.

RQ	Research Question
RQ1	To what extent are IT and Information Assurance professionals aware of Digital Engineering capabilities, including MBSE, the Digital Thread, digital twin technologies, and PLM principles?
RQ2	Do IT and Information Assurance professionals perceive Digital Engineering capabilities as potentially valuable or important for their work in IA, security compliance, and IT service delivery?
RQ3	Do IT and Information Assurance professionals believe DE practices would help them perform their jobs, meet compliance requirements, or enhance organizational capabilities?

Survey Structure Aligned to Core Dimensions

Table 31: Survey section structure.

Section	Core Dimension	Primary RQ	Description
Section 1	Awareness	RQ1	Baseline familiarity and professional exposure
Section 2	Awareness	RQ1	Understanding of specific DE capabilities/pillars
Section 3	Applicability	RQ2, RQ3	Perceived relevance to IT and IA domains
Section 4	Perceived Value	RQ2, RQ3	Value assessment for IT operations
Section 5	Perceived Value	RQ2, RQ3	Value assessment for IA/Cybersecurity operations
Section 6	Demographics	Supporting	Professional field and experience level

Section 1: Awareness and Familiarity with Digital Engineering

Core Dimension: Awareness

Primary Research Question: RQ1

Table 32: Section 1 question mapping.

Q#	Question Text	Format	DE Pillar	RQ
1.1	Please rate your level of familiarity with Digital Engineering concepts and practices.	5-point Likert (Familiarity)	All/General	RQ1
1.2	Have you encountered Digital Engineering methodologies, frameworks, or tools in your professional work within the past two years?	Binary (Yes/No)	All/General	RQ1

Rationale

These questions establish baseline awareness metrics essential to all subsequent analysis. Question 1.1 measures self-assessed familiarity (theoretical awareness), while Question 1.2 measures practical professional exposure (applied awareness). Together they distinguish between those who have merely heard of Digital Engineering and those who have encountered it in the course of their professional duties.

Section 2: Understanding of Digital Engineering Capabilities

Core Dimension: Awareness

Primary Research Question: RQ1

Table 33: Section 2 question mapping.

Q#	Question Text	Format	DE Pillar	RQ
2.1	DE includes model-based systems engineering approaches that can improve development processes.	5-point Likert (Agreement)	MBSE	RQ1
2.2	DE can enable digital twin development and virtual prototyping for IT systems.	5-point Likert (Agreement)	Digital Twin	RQ1
2.3	DE supports continuous integration and data-driven decision-making in technology development.	5-point Likert (Agreement)	Digital Thread	RQ1
2.4	DE enables digital twin technology that can simulate security scenarios and test defensive measures without impacting production systems.	5-point Likert (Agreement)	Digital Twin (Security)	RQ1
2.5	DE supports continuous security validation and data-driven threat analysis throughout the development lifecycle.	5-point Likert (Agreement)	Digital Thread (Security)	RQ1
2.6	DE can improve security control implementation through automated compliance checking and verification.	5-point Likert (Agreement)	PLM / Traceability	RQ1

Rationale

This section probes understanding of specific Digital Engineering pillars applied to IT and security contexts. Questions address all four pillars: MBSE (Q2.1), Digital Twin (Q2.2, Q2.4), the Digital Thread (Q2.3, Q2.5), and PLM with its authoritative traceability capabilities (Q2.6). Including both general IT applications (Q2.1-Q2.3) and security-specific applications (Q2.4-Q2.6) enables meaningful comparison across professional domains.

Section 3: Applicability of Digital Engineering

Core Dimension: Applicability

Primary Research Questions: RQ2, RQ3

Table 34: Section 3 question mapping.

Q#	Question Text	Format	DE Pillar	RQ
3.1	DE methodologies have relevant applications within the Information Technology sector.	5-point Likert (Agreement)	All/General	RQ2
3.2	DE methodologies have relevant applications for addressing Information Assurance challenges.	5-point Likert (Agreement)	All/General	RQ2
3.3	The ability to utilize digital twins to test changes against accurate replicas of production environments would provide value to my organization.	5-point Likert (Agreement)	Digital Twin	RQ2, RQ3
3.4	The use of digital models to map and document an organization's environment and configurations would provide value to my organization.	5-point Likert (Agreement)	MBSE	RQ2, RQ3
3.5	The use of digital lifecycle management to meet compliance and service delivery requirements would provide value to my organization.	5-point Likert (Agreement)	PLM	RQ2, RQ3
3.6	My organization faces regulatory or compliance requirements that could benefit from Digital Engineering approaches.	5-point Likert (Agreement)	All/General	RQ2, RQ3

Rationale

This section bridges awareness and value by examining whether respondents perceive Digital Engineering as applicable to their professional contexts. Questions 3.1 and 3.2 assess domain-level applicability (IT versus IA), while Questions 3.3 through 3.5 assess pillar-specific organizational value (Digital Twin, MBSE, PLM). Question 3.6 identifies compliance-driven need, which is central to the research focus on Information Assurance and regulatory compliance.

Section 4: Value Assessment for Information Technology

Core Dimension: Perceived Value (IT Domain)

Primary Research Questions: RQ2, RQ3

Table 35: Section 4 question mapping.

Q#	Question Text	Format	Value Category	RQ
4.1	DE could deliver meaningful value to my organization's Information Technology processes.	5-point Likert (Agreement)	Overall IT Value	RQ2, RQ3
4.2	DE could reduce development cycle time in my organization.	5-point Likert (Agreement)	Efficiency Benefit	RQ3
4.3	DE could improve product quality and reduce defects in my organization.	5-point Likert (Agreement)	Quality Benefit	RQ3
4.4	DE could improve collaboration effectiveness across development teams in my organization.	5-point Likert (Agreement)	Collaboration Benefit	RQ3
4.5	My organization would be willing to invest in DE capabilities if clear ROI could be demonstrated.	Ternary (Yes/No/Unsure)	Investment Willingness	RQ2

Rationale

This section measures IT-specific value perceptions. Question 4.1 provides an overall IT value assessment. Questions 4.2 through 4.4 examine specific operational benefits (efficiency, quality, and collaboration) that directly relate to job performance as addressed by RQ3. Question 4.5 measures organizational adoption interest, indicating whether perceived value rises to a level sufficient to warrant investment.

Section 5: Value Assessment for Information Assurance

Core Dimension: Perceived Value (Information Assurance Domain)

Primary Research Questions: RQ2, RQ3

Table 36: Section 5 question mapping.

Q#	Question Text	Format	Value Category	RQ
5.1	DE could deliver meaningful value to my organization's Information Assurance and cybersecurity operations.	5-point Likert (Agreement)	Overall Security Value	RQ2, RQ3
5.2	DE could reduce the time required to identify and remediate security vulnerabilities in my organization.	5-point Likert (Agreement)	Vulnerability Mgmt Benefit	RQ3
5.3	DE could improve security posture and reduce successful cyber incidents in my organization.	5-point Likert (Agreement)	Security Posture Benefit	RQ3
5.4	DE could enhance threat modeling and risk assessment capabilities in my organization.	5-point Likert (Agreement)	Threat Modeling Benefit	RQ3
5.5	DE could improve collaboration between security teams, development teams, and operations teams in my organization.	5-point Likert (Agreement)	Cross-Team Collaboration	RQ3
5.6	DE could help my organization achieve better compliance with security frameworks and regulatory requirements.	5-point Likert (Agreement)	Compliance Benefit	RQ3
5.7	My organization would be willing to invest in DE capabilities for cybersecurity purposes if clear ROI could be demonstrated.	Ternary (Yes/No/Unsure)	Investment Willingness	RQ2

Rationale

This section measures Information Assurance-specific value perceptions. Question 5.1 provides an overall security value assessment. Questions 5.2 through 5.6 examine specific security operational benefits directly related to job performance and compliance requirements as addressed by RQ3. The emphasis on compliance (Q5.6) directly addresses the dissertation's focus on Information Assurance compliance frameworks. Question 5.7 measures security-specific investment willingness.

Section 6: Interest and Demographic Information

Core Dimension: Supporting/Demographics

Purpose

Enable subgroup analysis and assess future research/adoption interest

Table 37: Section 6 question mapping.

Q#	Question Text	Format	Category	RQ
6.1	Would you be interested in learning more about DE applications for Information Assurance, cybersecurity, or Information Technology in your industry?	Binary (Yes/No)	Learning Interest	Supporting
6.2	Would you recommend that your organization explore DE adoption for improving security operations?	Binary (Yes/No)	Recommendation	Supporting
6.3	Please indicate your field of practice.	Categorical (IT / IA or Cyber Security / Engineering / Other)	Professional Field	Demographics
6.4	Please indicate your level of experience in your field of practice.	Categorical (Experience ranges)	Experience Level	Demographics

Rationale

Questions 6.1 and 6.2 measure forward-looking interest that complements value perception, indicating whether positive perceptions translate into desire for learning and willingness to recommend organizational exploration. Questions 6.3 and 6.4 enable comparative analysis between IT and security professionals and across experience levels, addressing whether awareness and perceptions vary systematically by professional background.

Summary: Question Distribution by Research Question

Table 38: Question distribution by research question.

Research Question	Primary Questions	Supporting Questions	Total
RQ1 (Awareness)	Q1.1, Q1.2, Q2.1-Q2.6	—	8
RQ2 (Perceived Value)	Q3.1-Q3.6, Q4.1, Q4.5, Q5.1, Q5.7	Q6.1, Q6.2	12
RQ3 (Job/Compliance Help)	Q3.3-Q3.6, Q4.1-Q4.4, Q5.1-Q5.6	—	14
Demographics	Q6.3, Q6.4	—	2

Note: Several questions map to multiple research questions as they address both perceived value (RQ2) and anticipated benefits for job performance and compliance (RQ3).

Summary: Question Distribution by DE Pillar

Table 39: Question distribution by Digital Engineering pillar.

DE Pillar	Questions
MBSE	Q2.1, Q3.4
Digital Twin	Q2.2, Q2.4, Q3.3
Digital Thread	Q2.3, Q2.5
PLM	Q2.6, Q3.5
General/All Pillars	Q1.1, Q1.2, Q3.1, Q3.2, Q3.6, Q4.1-Q4.5, Q5.1-Q5.7, Q6.1-Q6.4

Analysis Framework

Primary Analysis for Each Research Question

RQ1 Analysis (Awareness):

- Mean familiarity score (Q1.1) with 95% confidence interval
- Percentage with professional exposure (Q1.2) with 95% confidence interval
- Mean agreement scores for capability understanding (Q2.1-Q2.6)
- Percentage indicating agreement ($\geq$4) with each capability statement

- Comparison of awareness levels across professional fields (Q6.3) and experience levels (Q6.4)

RQ2 Analysis (Perceived Value):

- Mean agreement scores for applicability and pillar-specific organizational value (Q3.1-Q3.6)
- Mean agreement scores for overall value (Q4.1, Q5.1)
- Percentage indicating investment willingness (Q4.5, Q5.7)
- Percentage indicating learning interest (Q6.1) and recommendation (Q6.2)
- Comparison of value perceptions across professional fields and experience levels

RQ3 Analysis (Job/Compliance Help):

- Mean agreement scores for specific benefits (Q4.2-Q4.4, Q5.2-Q5.6)
- Percentage agreeing with compliance benefits (Q3.5, Q3.6, Q5.6)
- Percentage agreeing with organizational capability benefits (Q3.3-Q3.5)
- Comparison of benefit perceptions across professional fields and experience levels

Composite Scores

Table 40: Composite score definitions.

Composite	Questions	Items	Interpretation
Awareness Composite	Q1.1, Q2.1-Q2.6	7	Higher = greater awareness/understanding
IT Value Composite	Q3.1, Q3.3-Q3.5, Q4.1-Q4.4	8	Higher = greater perceived IT value
Security Value Composite	Q3.2, Q3.6, Q5.1-Q5.6	8	Higher = greater perceived security value

The Awareness composite averages the familiarity-anchored item Q1.1 with the agreement-anchored items Q2.1–Q2.6; the composite rests on the assumption that both item types, as five-point ordinal scales, are comparable, and Appendix B notes the larger variance of Q1.1 relative to the Section 2 items. Cronbach's alpha assessed internal consistency (threshold $\alpha \geq 0.70$); Appendix B reports the values.

Scale Reference

Familiarity Scale (Q1.1)

Table 41: Familiarity scale.

Score	Label	Interpretation
1	Not at all familiar	No awareness
2	Slightly familiar	Minimal awareness
3	Moderately familiar	Basic awareness
4	Very familiar	Good awareness
5	Extremely familiar	Expert awareness

Agreement Scale (Q2.1–Q4.4 and Q5.1–Q5.6)

Table 42: Agreement scale.

Score	Label	Interpretation
1	Strongly disagree	Strong negative perception
2	Disagree	Negative perception
3	Neither agree nor disagree	Neutral/uncertain
4	Agree	Positive perception
5	Strongly agree	Strong positive perception

Experience Level Categories (Q6.4)

Table 43: Experience level categories.

Category	Label	Career Stage
1–5 Years	Entry Level	Early Career
6–10 Years	Mid-Level (Early)	Mid Career
11–15 Years	Mid-Level (Late)	Mid Career
16+ Years	Senior Level	Late Career

Appendix B: Statistical Analysis and Methods

This appendix documents the complete statistical analysis underlying Chapter 4 and demonstrates each computation functionally. For every procedure it states the method and its originating reference, gives the estimator or test statistic in closed form, and then works a representative case through to its numerical result using the survey data. The detailed tabulations referenced below appear as tables and figures in Chapter 4; the purpose here is to show the arithmetic that produced them, so that every reported value is both traceable to its definition and reproducible by hand.

Reproducibility and Tooling

A fully scripted Python pipeline (NumPy, pandas, and SciPy, with statsmodels for the regression and differential-item-functioning models and semopy for the confirmatory measurement model) produced all statistics, tables, and figures from the response file. The pipeline computes every quantity reported here and writes the results, figures, and LaTeX table fragments deterministically, so that the analysis is reproducible from the raw data. A companion computational notebook reproduces the same results with the mathematics shown inline. Sample standard deviations and variances use the unbiased $(n - 1)$ denominator throughout, and the worked figures below are carried to four decimal places before rounding so that intermediate steps can be checked.

. **Data and materials availability.** A public reproducibility artifact accompanies this dissertation. The survey instrument, the analysis code, and a synthetic dataset that reproduces the structure of the analyses reported in Chapter 4 are openly available at https://doi.org/10.17605/OSF.IO/VKDTE. Because the survey was administered anonymously and received an Institutional Review Board exempt determination, the deposited dataset is synthetic and contains no real response record.

Data Preparation and Sample

The survey platform recorded 1,600 submissions. A pre-registered screen removed 188 rapid or abandoned cases (completion times under three minutes, open longer than five

days, or answering at most the two Section 1 questions). A post-hoc forensic authenticity screen then removed one further submission belonging to a coordinated automated cluster, in which a single identical response vector had been submitted 71 times in a roughly one-minute burst, leaving an analytic sample of $N = 1{,}411$. Decoding mapped Likert items from their anchored string form to the integer scale $\{1, 2, 3, 4, 5\}$. Complete data across all twenty-three Likert items (the twenty-two agreement-scale items together with the familiarity item Q1.1) are present for 1,390 respondents (98.5%); item-level missingness never exceeds six cases, and available-case analysis handled the remaining gaps.

Descriptive Statistics

For each Likert item the sample mean and unbiased sample standard deviation are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{10}$$

. **Worked example (familiarity, Q1.1).** The familiarity item was answered by all $n = 1{,}411$ respondents with the frequency distribution $f_1 = 700$, $f_2 = 404$, $f_3 = 59$, $f_4 = 192$, $f_5 = 56$. Because the data are discrete, the mean is computed from the frequencies,

$$\bar{x} = \frac{\sum_{v=1}^{5} v\, f_v}{n} = \frac{(1)(700) + (2)(404) + (3)(59) + (4)(192) + (5)(56)}{1411} = \frac{2733}{1411} = 1.9369. \tag{11}$$

The corresponding sum of squared deviations gives the variance and standard deviation,

$$s^2 = \frac{1}{n-1}\sum_{v=1}^{5} f_v\,(v - \bar{x})^2 = 1.4364, \qquad s = \sqrt{1.4364} = 1.1985. \tag{12}$$

These match the values reported for Q1.1 in Table 4 ($\bar{x} = 1.94$, $s = 1.20$). Table 4 gives the full set of item-level descriptives.

Agreement Proportions and the Wilson Interval

The agreement proportion is $\hat{p} = k/n$, the share of responses at or above 4. The Wilson score method (Wilson, 1927) formed each 95% confidence interval for a proportion and remains well-behaved near the response ceiling,

$$\frac{\hat{p} + \dfrac{z^2}{2n} \;\pm\; z\sqrt{\dfrac{\hat{p}\,(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \qquad z = z_{0.975} = 1.96. \tag{13}$$

. **Worked example (Q1.1 agreement).** For Q1.1, $k = 248$ respondents answered 4 or 5, so $\hat{p} = 248/1411 = 0.1758$ with $n = 1411$ and $z = 1.96$. The denominator and the two half-terms are

$$1 + \frac{z^2}{n} = 1.0027, \qquad \hat{p} + \frac{z^2}{2n} = 0.1771, \qquad z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}} = 0.0199. \tag{14}$$

Combining gives the interval

$$\frac{0.1771 \pm 0.0199}{1.0027} = [\,0.1568,\ 0.1965\,] = [\,15.7\%,\ 19.6\%\,], \tag{15}$$

which matches the Wilson bounds reported for Q1.1 in Table 4.

Composite Scores and Internal Consistency

Three composites were formed by averaging their constituent items: Awareness (Q1.1, Q2.1–Q2.6), IT Value (Q3.1, Q3.3–Q3.5, Q4.1–Q4.4), and Security Value (Q3.2, Q3.6, Q5.1–Q5.6). Internal-consistency reliability was assessed with Cronbach's α (Cronbach, 1951), where σ_j^2 is the variance of item j among k items and σ_T^2 the variance of the summed score,

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{j=1}^{k}\sigma_j^2}{\sigma_T^2}\right). \tag{16}$$

. **Worked example (Awareness composite, $k = 7$).** The seven item variances are

$$\begin{aligned} &\sigma^2_{Q1.1} = 1.4364, \;\; \sigma^2_{Q2.1} = 0.3295, \;\; \sigma^2_{Q2.2} = 0.3192, \;\; \sigma^2_{Q2.3} = 0.3215, \\ &\sigma^2_{Q2.4} = 0.3352, \;\; \sigma^2_{Q2.5} = 0.5155, \;\; \sigma^2_{Q2.6} = 0.3295, \end{aligned} \tag{17}$$

which sum to $\sum_j \sigma^2_j = 3.5869$. The variance of the seven-item summed score over complete cases is $\sigma^2_T = 11.1691$. Substituting,

$$\alpha = \frac{7}{6}\left(1 - \frac{3.5869}{11.1691}\right) = (1.1667)(1 - 0.3211) = (1.1667)(0.6789) = 0.792. \tag{18}$$

The IT Value ($\alpha = 0.858$) and Security Value ($\alpha = 0.858$) composites are computed identically; all three exceed the conventional $\alpha \geq 0.70$ threshold (Table 6). The 95% confidence interval for each composite mean follows from Student's t distribution, $\bar{x} \pm t_{0.975,\, n-1}\, s/\sqrt{n}$.

Tests of Normality

The Shapiro–Wilk test assessed each composite for normality (Shapiro & Wilk, 1965). The statistic compares the squared correlation between the ordered sample and the expected normal order statistics,

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \tag{19}$$

where $x_{(i)}$ are the order statistics and a_i the tabulated normal weights. Every composite departed from normality (Awareness $W = 0.978$; IT Value $W = 0.957$; Security Value $W = 0.965$; each $p < .001$). Because the test is highly powered at this sample size and the composites are left-skewed, the analysis adopted non-parametric procedures for inference and foregrounded effect sizes over p-values.

Effect Sizes

Cohen's d standardizes the mean difference by the pooled standard deviation (J. Cohen, 1988),

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \qquad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}. \tag{20}$$

. **Worked example (Awareness, IT vs IA)..** The Information Technology group ($n_1 = 501$) has mean $\bar{x}_1 = 4.379$ and $s_1 = 0.38$; the Information Assurance group ($n_2 = 501$) has $\bar{x}_2 = 3.732$ and $s_2 = 0.41$. The pooled standard deviation is

$$s_p = \sqrt{\frac{(500)(0.38^2) + (500)(0.41^2)}{501 + 501 - 2}} = \sqrt{\frac{72.200 + 84.050}{1000}} = \sqrt{0.15625} = 0.3953, \tag{21}$$

so that

$$d = \frac{4.379 - 3.732}{0.3953} = \frac{0.647}{0.3953} = 1.64. \tag{22}$$

This is the value reported for Awareness in Table 8. A 95% confidence interval for d was obtained from its large-sample standard error (Hedges & Olkin, 1985),

$$\mathrm{SE}_d = \sqrt{\frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2\,(n_1 + n_2)}} = \sqrt{\frac{1002}{251001} + \frac{1.64^2}{2(1002)}} = 0.0730, \tag{23}$$

so that $d \pm 1.96\,\mathrm{SE}_d = 1.64 \pm 0.14 = [\,1.50,\ 1.78\,]$, the interval reported in Table 8. The IT Value and Security Value intervals ($[1.79, 2.09]$ and $[1.79, 2.09]$) were computed identically; all three lie well above the conventional "large" threshold of 0.8, so the field difference is both very large and precisely estimated. The rank-biserial correlation, derived from the Mann–Whitney statistic, expresses the same comparison on the ordinal scale (Kerby, 2014). The statistic entering the formula is U_1 for the Information Technology group, the reference group, computed from that group's rank sum; under the convention $r_{rb} = 1 - 2U_1/(n_1 n_2)$, a negative value indicates that the reference group ranks higher. Because Information Technology respondents score above Information Assurance

respondents, $U_1 = 218{,}766$ exceeds $n_1 n_2/2 = 125{,}500.5$ and the coefficient comes out negative,

$$r_{rb} = 1 - \frac{2U_1}{n_1 n_2} = 1 - \frac{2(218766)}{(501)(501)} = 1 - \frac{437532}{251001} = -0.743. \tag{24}$$

The sign therefore encodes direction (Information Technology higher), consistent with the positive Cohen's d above and with the sign convention stated in Chapter 4, while the magnitude 0.743 carries the effect size.

Group Comparison by Professional Field

The Mann–Whitney U test (Mann & Whitney, 1947) compared Information Technology ($n = 501$) and Information Assurance/Cyber Security ($n = 501$) respondents on each composite; the test evaluates whether one group tends to rank above the other without assuming normality. With R_1 the rank sum of the first group in the pooled ranking,

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}, \qquad U_2 = n_1 n_2 - U_1. \tag{25}$$

The significance computation may take the smaller of U_1 and U_2; the rank-biserial effect size worked in the preceding section instead uses U_1 for the Information Technology group directly, so that its sign encodes which group ranks higher. Welch's t was reported alongside for continuity with the parametric literature. Differences are large on all three composites (Table 8): Awareness $d = 1.64$ ($r_{rb} = -0.743$, worked above), IT Value $d = 1.94$, and Security Value $d = 1.94$, with Information Technology exceeding Information Assurance throughout. At the item level within the Awareness family, the Holm–Bonferroni step-down correction controlled the family-wise error rate (Holm, 1979): the m ordered p-values $p_{(1)} \le \cdots \le p_{(m)}$ are compared in turn against $\alpha/(m - i + 1)$, and the procedure stops at the first i for which the inequality fails, retaining all remaining hypotheses. Because the unadjusted p-values are already extreme, every item survives correction (Table 9).

Group Comparison by Experience Level

The four experience cohorts were compared with the Kruskal–Wallis test (Kruskal & Wallis, 1952), the rank-based generalization of one-way analysis of variance,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{g} \frac{R_j^2}{n_j} - 3(N+1), \tag{26}$$

with the effect size $\eta^2 = (H - g + 1)/(N - g)$ interpreted on the small/medium/large scale of J. Cohen (1988).

. **Worked example (Awareness, η^2).** With the omnibus statistic $H = 213.096$ across $g = 4$ groups and $N = 1{,}411$,

$$\eta^2 = \frac{H - g + 1}{N - g} = \frac{213.096 - 4 + 1}{1411 - 4} = \frac{210.096}{1407} = 0.149, \tag{27}$$

a medium-to-large effect (Table 13). Post-hoc pairwise contrasts used Dunn's test with Holm adjustment (Dunn, 1964), comparing standardized differences of mean ranks with a tie correction. The contrasts localize the effect in the 11–15 year cohort, which scores below both the 1–5 and 6–10 year cohorts; on the Awareness composite worked here, the 16+ cohort ($n = 18$) yields no significant pairwise contrast after correction ($p = .118$ against the 11–15 year cohort), though it does separate significantly from that cohort on the two value composites, as reported in Chapter 4.

Association Analysis

Associations between categorical variables were tested with the Pearson chi-square test of independence (Pearson, 1900), comparing observed counts O_{ij} against the counts expected under independence $E_{ij} = R_i C_j / N$,

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad V = \sqrt{\frac{\chi^2}{N\,(\min(r, c) - 1)}}. \tag{28}$$

The second expression is Cramér's V (Cramér, 1946), which rescales χ^2 to a 0–1 effect size.

. **Worked example (professional field × prior exposure).** Restricting to the three fields with exposure data, the observed counts (O) and the counts expected under independence ($E = R_i C_j / N$) are given below; row totals reflect available-case analysis of the items crossed, so the Engineering row totals 337 rather than the group size of 339 because two Engineering respondents did not answer the exposure item Q1.2.

	Exposure: Yes	Exposure: No	Row total
IA/Cyber	59 ($E = 113.4$)	442 ($E = 387.6$)	501
IT	161 ($E = 113.4$)	340 ($E = 387.6$)	501
Engineering	83 ($E = 76.3$)	254 ($E = 260.7$)	337
Column total	303	1036	$N = 1339$

Summing $(O - E)^2/E$ over the six cells, for example $(59 - 113.4)^2/113.4 = 26.1$ and $(161 - 113.4)^2/113.4 = 20.0$, yields $\chi^2 = 60.334$ on $(3 - 1)(2 - 1) = 2$ degrees of freedom. The effect size is therefore

$$V = \sqrt{\frac{\chi^2}{N\,(\min(r, c) - 1)}} = \sqrt{\frac{60.334}{1339\,(2 - 1)}} = \sqrt{0.0451} = 0.212, \tag{29}$$

a moderate association (Table 15). Experience-by-investment associations follow the same computation.

Bias Diagnostics

The analysis computed two planned diagnostics, together with a post-hoc straight-lining check. The wave (non-response) analysis ordered the sample by submission time (the timestamp at which each response was submitted) and compared the early- and late-arriving halves with the Mann–Whitney test and Cohen's d, implementing the early-versus-late split of the non-response-analysis tradition (Armstrong & Overton, 1977). Differences are negligible on every composite (IT Value $d = -0.02$, early $\bar{x} = 4.321$, late $\bar{x} = 4.328$; Awareness $d = -0.04$; Security Value $d = -0.04$; all Mann–Whitney $p > .5$), as Table 22 shows, indicating no detectable early-versus-late self-selection in the recruited

sample. The priming diagnostic compared value composites between high-familiarity (Q1.1 $\in \{4, 5\}$) and low-familiarity (Q1.1 $\in \{1, 2\}$) respondents, again with the Mann–Whitney test and Cohen's d. Both strata diverge sharply on both composites (IT Value $\Delta = 0.539$, $d = 1.30$; Security Value $\Delta = 0.573$, $d = 1.25$; Table 23), which weighs against a pure question-framing explanation for the high endorsement observed. Each Cohen's d here is computed by the pooled-standard-deviation formula worked above.

. **Straight-lining (acquiescence) check.** To gauge the contribution of non-differentiating responding to the high agreement rates, the analysis computed the within-respondent standard deviation across the twenty-two agreement-scale items (Sections 2–5) for each of the 1,390 respondents with the complete battery. Of these, 22 (1.6%) returned an identical response to all twenty-two items (within-respondent SD = 0), and 79 (5.7%, cumulative) had near-flat response strings (SD ≤ 0.25), i.e., fifty-seven additional respondents; the median within-respondent standard deviation is 0.50. Only a small minority therefore straight-lined the agreement battery, so acquiescence is a negligible presence in these data and does not manufacture the high endorsement.

Multivariate Model and Robustness

To separate the contributions of professional field, experience, and familiarity to perceived value, each value composite was regressed on those predictors by ordinary least squares (J. Cohen et al., 2003), with Information Assurance/Cyber Security and the 1–5 year cohort as reference categories,

$$\widehat{\text{Value}} = \beta_0 + \beta_1 \,\text{Familiarity} + \boldsymbol{\beta}_F \,\text{Field} + \boldsymbol{\beta}_E \,\text{Experience}. \tag{30}$$

The models were fit on the full analytic sample of $N = 1{,}411$ and account for a substantial share of variance (IT Value $R^2 = 0.64$; Security Value $R^2 = 0.67$; Table 21). Three results follow: professional field remains a strong predictor after experience and familiarity are held constant (Information Technology vs. Information Assurance $b = 0.46$ for IT Value

and 0.50 for Security Value, both $p < .001$); the 11–15 year cohort retains a substantial negative coefficient once field and familiarity are controlled ($b = -0.40$, $p < .001$ for IT Value; $b = -0.53$, $p < .001$ for Security Value), so the mid-career dip persists as an experience effect net of field composition rather than dissolving into it; and familiarity is a positive predictor of perceived value ($b = 0.13$ and 0.14, both $p < .001$).

. **Heteroscedasticity-robust refit.** Because the value composites are bounded on the 1–5 scale and left-skewed, the models were refit with HC3 heteroscedasticity-consistent standard errors (Long & Ervin, 2000). Every coefficient retained its significance and direction, including the substantial negative 11–15 year coefficient (HC3 $p < .001$ for both value composites), so the conclusions do not depend on the homoscedastic ordinary-least-squares assumption.

. **Standardized familiarity coefficient.** To express the familiarity effect on a scale-free basis, the outcome and the familiarity predictor were standardized to zero mean and unit variance and the models refit. The resulting standardized coefficient, equivalently $\beta = b\,(s_{\text{Fam}}/s_{\text{Value}})$, gives the change in perceived value, in standard-deviation units, per one-standard-deviation increase in familiarity ($\beta = 0.34$ for IT Value, $\beta = 0.33$ for Security Value; both $p < .001$), the values quoted in Chapter 4.

Rank-Order Correlation

The monotonic association among familiarity, the Awareness composite, and the value composites was summarized with Spearman's rank-order correlation (Spearman, 1904), which is the Pearson correlation computed on the ranks of the two variables and therefore assumes neither linearity nor normality,

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n\,(n^2 - 1)}, \tag{31}$$

where d_i is the difference between the ranks of the paired observations for case i (a tie correction is applied when observations share a rank). Familiarity correlates moderately

with perceived value ($\rho = 0.632$ with IT Value, $\rho = 0.616$ with Security Value; both $p < .001$), whereas the Awareness composite correlates more strongly with both ($\rho = 0.734$ and $\rho = 0.731$). The Awareness composite shares no items with either value composite, so these stronger correlations reflect shared endorsement-style variance rather than substance; they are recorded here for completeness only and are not reported as substantive results in Chapter 4, consistent with the construct-validity assessment later in this appendix.

Supplementary Post-Hoc Robustness and Validity Analyses

The three analyses in this section were not specified in the approved proposal; they were added after data collection to test the robustness and measurement properties of the Chapter 4 findings and are reported as corroboration rather than as confirmatory tests. Each is defined here and carried through to its numerical result.

Common-Method Variance: Harman's Single-Factor Test

Because every measure is a single self-report collected on one occasion, common-method variance could inflate the observed correlations (Podsakoff et al., 2003). Harman's test inspects the unrotated eigenstructure of the 23×23 correlation matrix R of the composite items and asks whether one factor explains the majority of the common variance. Writing $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{23}$ for the eigenvalues of R, whose trace is fixed at 23, the variance share of the first unrotated factor is

$$\pi_1 = \frac{\lambda_1}{\sum_{j=1}^{23} \lambda_j} = \frac{\lambda_1}{23}. \tag{32}$$

For the analytic battery ($n = 1{,}390$ complete cases), two eigenvalues exceed unity and $\lambda_1 = 10.12$, so $\pi_1 = 10.12/23 = 0.440$. A first factor accounting for 44.0% of the variance, below the 50% dominance heuristic, indicates that no single method factor dominates the data. The test is a lower bound rather than a decisive control, so the conclusion is correspondingly narrow.

Construct Validity: Composite Reliability and Average Variance Extracted

A confirmatory three-factor model, in which each composite is a latent variable indicated by its own items, yields standardized loadings λ_i from which composite reliability (CR) and average variance extracted (AVE) follow (Fornell & Larcker, 1981),

$$\mathrm{CR} = \frac{\left(\sum_i \lambda_i\right)^2}{\left(\sum_i \lambda_i\right)^2 + \sum_i \left(1 - \lambda_i^2\right)}, \qquad \mathrm{AVE} = \frac{1}{k}\sum_{i=1}^{k} \lambda_i^2. \tag{33}$$

Reliability is high for all three composites (CR = 0.86, 0.87, and 0.87), corroborating the Cronbach's α values of the internal-consistency section. Convergent validity, indexed by AVE, falls just short of the 0.50 benchmark for all three composites (0.47, 0.46, and 0.45), reflecting item sets that each span a heterogeneous range of capabilities. Discriminant validity is assessed by the Fornell–Larcker criterion, which requires $\sqrt{\mathrm{AVE}}$ for each construct to exceed its correlation with the others. It is not met: the inter-composite correlations (0.74 to 0.85) exceed the corresponding $\sqrt{\mathrm{AVE}}$ (0.67 to 0.69). Model fit itself is good (CFI = 0.966, RMSEA = 0.040), so the failure is one of separability, not of fit. The heterotrait–monotrait ratio of correlations (Henseler et al., 2015) localizes it: Awareness is discriminant from both value composites (HTMT = 0.83 against each, below the 0.85 threshold), whereas IT Value and Security Value are not discriminant from each other (HTMT = 0.99). Both value dimensions are therefore best read as one perceived-value construct measured in two domains, and all three composites are used as reliable summative indices rather than as discriminant latent variables; the group comparisons rest on the reliability of the indices, not on their separability.

One further statistic bears on the internal structure of the Awareness composite, which combines a familiarity self-report with six capability-agreement items. A corrected item-total correlation relates item y_j to the mean of the remaining $k - 1$ items. For the Awareness composite it is 0.315 for Q1.1 against a range of 0.53 to 0.67 for the six Section 2 items, and Q1.1 correlates with the mean of those six items at $r = 0.32$.

Removing Q1.1 raises Cronbach's α from 0.792 to 0.863, so the item satisfies both conventional deletion criteria at once, namely a corrected item-total correlation below roughly 0.40 and an α that improves on removal. All seven items are retained nonetheless, because Chapter 3 specified the composite in advance of data collection, and the AVE of 0.47 reported above reflects that retained heterogeneity. Chapter 4 records the interpretive consequence: the composite tracks capability endorsement more closely than familiarity, and Q1.1 is reported separately wherever the familiarity finding is at issue.

Measurement Invariance Across Professional Field

To test whether the large Information Technology versus Information Assurance gap could reflect differential item functioning rather than a true difference, each item was regressed on the respondent's own standardized composite score m, a field indicator g (with $g = 1$ for Information Assurance), and their product, adapting the logistic-regression DIF framework (Zumbo, 1999) to a linear implementation on the five-point item scores. Uniform DIF is the partial effect of g after matching on m, assessed by the improvement in fit of

$$y_i = \beta_0 + \beta_1 m_i + \beta_2 g_i + \varepsilon_i \tag{34}$$

over the matched-only model $y_i = \beta_0 + \beta_1 m_i + \varepsilon_i$, with β_2 giving the Information Assurance versus Information Technology difference in item response, in scale points, at a common trait level. Of the 23 items, 21 show no meaningful differential functioning; two (Questions 1.1 and 2.5) meet the flag criterion ($p < .001$ and $|\beta_2| \geq 0.20$), the larger effect falling on the familiarity item Q1.1 ($\beta_2 = -0.99$ scale points) and a small effect on Q2.5 ($\beta_2 = +0.23$). Both flagged items belong to the Awareness composite, so neither value composite contains a differentially functioning item. The large Information Technology versus Information Assurance gaps on the value composites, which approach two standard deviations, therefore cannot be attributed to item-level measurement differences. The same regression's interaction term $m_i g_i$ captures non-uniform DIF, a field difference in how

strongly an item tracks the trait. No value-composite item reached the flag threshold on it either, the largest value-composite non-uniform effect falling on Q4.4 at 0.17 scale points, so the conclusion holds under both uniform and non-uniform screening. That evidence supports reading the field contrast as substantive.

Familiarity-Stratum Adjustment of the Field Gap

Familiarity is bimodal: the two lowest categories hold 78.2% of respondents and the middle category only 4.2%, while "Very familiar" recovers to 13.6%. That shape suggests two strata rather than one graded continuum, and it raises the possibility that the professional-field gap reflects unequal draw from those strata rather than field membership itself. Both focal groups draw from these strata, and they do so unequally: 206 of 501 Information Technology respondents fall in the high-familiarity stratum (Q1.1 ≥ 4) against 11 of 501 Information Assurance respondents.

A binary stratum indicator s_i therefore enters the field-only model estimated on the two focal groups ($n = 1{,}002$),

$$y_i = \beta_0 + \beta_1 f_i + \beta_2 s_i + \varepsilon_i, \tag{35}$$

where $f_i = 1$ for Information Technology and $s_i = 1$ for the high-familiarity stratum, with HC3-robust standard errors. Field coefficients fall from 0.689 to 0.602 on IT Value and from 0.746 to 0.655 on Security Value, retaining 87% of their magnitude on both composites and remaining significant at $p < .001$. Entering the recent-exposure indicator alongside the stratum indicator retains 86% ($b = 0.589$ and $b = 0.642$ respectively). Stratum composition therefore accounts for roughly one seventh of the field gap.

Observed strata serve here in place of a fitted latent class. A finite-mixture model was considered and set aside: on a five-point ordinal item with a binary covariate, mixture likelihoods admit degenerate components with near-zero variance, and the transparent threshold split reproduces the same partition without that instability. That threshold

matches the high-familiarity group already used for the priming diagnostic in Chapter 4, so the two analyses partition the sample identically. This adjustment narrows the recruitment-frame concern without resolving it, since a channel difference that delivered systematically different respondents within a single familiarity stratum would remain invisible to it.

Response-Authenticity Screen: Operating Characteristics

The response-authenticity screen described in Chapter 4 (Section 5) was validated against synthetic data with known ground truth, following standard classifier-evaluation practice (Fawcett, 2006), and the operating characteristics are reported so as to separate what the validation confirms from what it genuinely measures. Against contamination defined in the screen's own terms, exact-duplicate and near-duplicate clusters, sensitivity is at or near 100 percent, but this is a confirmatory check that the rule matches duplicates, not a measure of real-world reach. The informative test plants coordinated fraud from an independent behavioral model, a ring whose members share an answer key but differ in discipline. Detection is then a decay curve: a disciplined ring that shares a demographic profile is caught in full, sensitivity falls as the ring grows sloppier, and it collapses toward zero for a ring whose members each vary their demographic profile, because the screen requires a repeated whole vector. The screen is therefore effective against disciplined, demographically consistent coordinated duplication and, by construction, blind to loosely coordinated, random-responding, or fluent machine-generated submissions, which a complementary content-only detector would be needed to catch.

Specificity was measured against the case most likely to trip the screen: authentic data with genuine ceiling collisions, where strong endorsers land on the same high-agreement profile. On ordinary synthetic data the screen wrongly flags fewer than one authentic row in seven hundred, but on ceiling-heavy data the mechanical rule over-excludes genuine strong endorsers at roughly one to two percent. That rate is close to the borderline-flag rate seen in this study's own data: the 31 respondents that separate the

$N = 1{,}411$ primary sample from the $N = 1{,}380$ fully screened sample. This is the empirical reason the screen is paired with human adjudication rather than run alone: the automated rule cannot distinguish a genuine ceiling coincidence from a small scripted cluster without the timing and context a reviewer applies. As a further check on real rather than synthetic data, the screen also ran on two large public Likert datasets that bracket this instrument in length. At the twenty-two-item length used here, the false-positive rate on genuinely organic data is negligible, about one row in four thousand, although it rises sharply on much shorter or coarser instruments. Default application is therefore appropriate for instruments of this length and requires recalibration for short ones.

A confirmatory battery of 500 independently generated datasets bounds the false-positive behavior across a wide distribution of dataset sizes and cluster counts. On 100 fully organic datasets, comprising 47,901 authentic rows, the screen produced only 7 false positives, a false-positive rate near 0.01 percent, and returned a completely clean result on 93 of the 100 datasets. The reproducibility artifact carries the complete battery and every validation harness, among them the independent ring model, the ceiling-heavy hard case, and the cross-method complementarity experiment.

Summary

The analysis applied the estimators and tests above exactly as defined, with non-parametric procedures and effect sizes carrying the interpretive weight given the large sample. The worked examples confirm, step by step, that the headline quantities of Chapter 4 follow directly from the raw responses: the familiarity mean of 1.94, the Awareness reliability of $\alpha = 0.792$, the field effect size of $d = 1.64$, the experience effect size of $\eta^2 = 0.149$, and the field-by-exposure association of $V = 0.212$. Every figure in Chapter 4 derives from one of these computations, and each computation is reproducible from the response data through the scripted pipeline described in the Reproducibility and Tooling section.

The Independent Analysis, Reproduced: Code and Output

The methodology above defines each procedure and works a representative case by hand. This section supplies the executable proof, drawn from the independent statistical analysis and its companion computational notebook. The analysis is a fully scripted Python program (NumPy, pandas, SciPy, and statsmodels, with semopy for the confirmatory measurement model used in the post-hoc validity check) that recomputes every value reported in Chapter 4 directly from the raw response file. Each analysis below presents the exact code followed by its verbatim output, so that no reported quantity rests on assertion: each one is reproducible from the response data by running the code shown. The ordering follows the methodology sections above, and the figures these computations produce appear, with alternative-text descriptions, in Chapter 4.

Setup and Data Loading

Loads the response file and maps the raw columns onto the Chapter 3 question numbering. **Code:**

```
import os, json, warnings
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
from matplotlib.patches import Patch
warnings.filterwarnings("ignore")
pd.set_option("display.max_columns", 60)
pd.set_option("display.width", 160)

# The notebook lives in Analysis/; the workbook is one directory up.
ROOT = os.path.dirname(os.getcwd()) if os.path.basename(os.getcwd()) == "Analysis" else os.getcwd()
SRC = os.path.join(ROOT, "Survey Responses Final Export.xlsx")   # formerly "...Final Dump.xlsx"
assert os.path.exists(SRC), f"Cannot find {SRC}"
raw = pd.read_excel(SRC)
cols = list(raw.columns)
print("Rows:", len(raw), "| Columns:", len(cols))
```

Output:

```
Rows: 1411 | Columns: 35
```

Data Cleaning and Sample

Decodes each Likert response to its integer score and confirms the missing-data rule; no retained response crosses the twenty-percent exclusion threshold. The analytic sample follows a documented funnel: 1,600 raw submissions, from which a pre-registered quality screen removed 188 (completion under three minutes, an open interval exceeding five days,

or answering at most Section 1), and a post-hoc response-authenticity screen removed one further submission belonging to a coordinated automated cluster (a single identical response vector submitted 71 times at roughly one-minute completions), giving the genuine $N = 1{,}411$. Complete data on all 23 Likert items are present for 1,390 respondents (98.5%). **Code:**

```
LIKERT = ["Q1.1"]+[f"Q2.{i}" for i in range(1,7)]+[f"Q3.{i}" for i in range(1,7)]+\
         [f"Q4.{i}" for i in range(1,5)]+[f"Q5.{i}" for i in range(1,7)]
BINARY  = ["Q1.2","Q6.1","Q6.2"]
TERNARY = ["Q4.5","Q5.7"]


def to_score(v):
    if pd.isna(v): return np.nan
    s = str(v).strip()
    return int(s[0]) if s and s[0].isdigit() else np.nan


for q in LIKERT:
    df[q] = df[q].map(to_score)
for q in BINARY+TERNARY+["Q6.3","Q6.4"]:
    df[q] = df[q].map(lambda v: np.nan if pd.isna(v) else str(v).strip())


N_TOTAL = len(df)
complete_likert = int(df[LIKERT].notna().all(axis=1).sum())
missing = pd.Series({q:int(df[q].isna().sum()) for q in LIKERT+BINARY+TERNARY+["Q6.3","Q6.4"]})
print(f"N total = {N_TOTAL}")
print(f"Complete across all 23 Likert items = {complete_likert} ({100*complete_likert/N_TOTAL:.1f}%)")
print("Max item-level missing =", int(missing.max()))
missing[missing>0]
```

Output:

```
N total = 1411
Complete across all 23 Likert items = 1390 (98.5%)
Max item-level missing = 6
Q2.2    6
Q2.3    4
Q2.4    4
Q2.5    3
Q2.6    4
Q3.1    1
Q3.2    3
Q3.3    4
Q3.4    4
Q3.5    3
Q3.6    3
Q4.2    4
Q4.3    3
Q4.4    3
Q5.1    2
```

```
Q5.2    3
Q5.3    3
Q5.4    3
Q5.5    3
Q5.6    3
Q1.2    2
Q6.2    6
dtype: int64
```

Item-Level Descriptive Statistics

Computes the mean, standard deviation, median, interquartile range, agreement rate, and Wilson interval for every Likert item (methodology: *Descriptive Statistics*).

Code:

```
def ci_mean(x, conf=0.95):
    x = pd.Series(x).dropna(); n = len(x)
    if n < 2: return (np.nan, np.nan)
    se = x.std(ddof=1)/np.sqrt(n)
    h = se*stats.t.ppf((1+conf)/2, n-1)
    return (x.mean()-h, x.mean()+h)

def ci_wilson(k, n, conf=0.95):
    if n == 0: return (np.nan, np.nan)
    z = stats.norm.ppf((1+conf)/2); p = k/n; den = 1+z**2/n
    centre = (p+z**2/(2*n))/den
    half = z*np.sqrt(p*(1-p)/n + z**2/(4*n**2))/den
    return (centre-half, centre+half)

rows = []
for q in LIKERT:
    x = df[q].dropna(); n = len(x); k = int((x>=4).sum())
    lo,hi = ci_mean(x); plo,phi = ci_wilson(k,n)
    rows.append({"Item":q, "n":n, "Mean":round(x.mean(),2), "SD":round(x.std(ddof=1),2),
                 "Mdn":x.median(), "IQR":f"{x.quantile(.25):.0f}-{x.quantile(.75):.0f}",
                 "Agree%":round(100*k/n,1), "CI_low":round(100*plo,1), "CI_high":round(100*phi,1)})
desc = pd.DataFrame(rows).set_index("Item")
desc
```

Output:

Item	n	Mean	SD	Mdn	IQR	Agree%	CI_low	CI_high
Q1.1	1411	1.94	1.20	2.0	1-2	17.6	15.7	19.6
Q2.1	1411	4.42	0.57	4.0	4-5	95.7	94.6	96.7
Q2.2	1405	4.38	0.56	4.0	4-5	95.8	94.6	96.7
Q2.3	1407	4.39	0.57	4.0	4-5	95.8	94.6	96.7
Q2.4	1407	4.50	0.58	5.0	4-5	95.7	94.5	96.7
Q2.5	1408	4.29	0.72	4.0	4-5	84.6	82.6	86.4
Q2.6	1407	4.41	0.57	4.0	4-5	95.7	94.5	96.7

```
Q3.1  1410  4.27  0.72  4.0  4-5    84.6    82.6     86.4
Q3.2  1408  4.39  0.57  4.0  4-5    95.7    94.6     96.7
Q3.3  1407  4.51  0.58  5.0  4-5    95.8    94.6     96.7
Q3.4  1407  4.42  0.57  4.0  4-5    95.9    94.7     96.8
Q3.5  1408  4.30  0.72  4.0  4-5    84.8    82.8     86.6
Q3.6  1408  3.89  0.81  4.0  3-5    61.1    58.5     63.6
Q4.1  1411  4.39  0.57  4.0  4-5    95.9    94.7     96.8
Q4.2  1407  4.29  0.71  4.0  4-5    85.5    83.6     87.2
Q4.3  1408  4.30  0.72  4.0  4-5    84.8    82.8     86.6
Q4.4  1408  4.12  0.64  4.0  4-5    84.7    82.7     86.4
Q5.1  1409  4.12  0.64  4.0  4-5    85.0    83.0     86.7
Q5.2  1408  4.13  0.80  4.0  3-5    73.8    71.4     76.0
Q5.3  1408  4.31  0.72  4.0  4-5    84.8    82.8     86.6
Q5.4  1408  4.32  0.75  4.0  4-5    83.0    81.0     84.9
Q5.5  1408  4.22  0.71  4.0  4-5    83.3    81.3     85.2
Q5.6  1408  4.23  0.71  4.0  4-5    83.5    81.5     85.4
```

Composites and Cronbach's Alpha

Forms the three composites and computes Cronbach's α for each (methodology: *Composite Scores and Internal Consistency*). **Code:**

```
composites = {
    "Awareness":  ["Q1.1"]+[f"Q2.{i}" for i in range(1,7)],
    "IT_Value":   ["Q3.1","Q3.3","Q3.4","Q3.5","Q4.1","Q4.2","Q4.3","Q4.4"],
    "Sec_Value":  ["Q3.2","Q3.6","Q5.1","Q5.2","Q5.3","Q5.4","Q5.5","Q5.6"],
}
def cronbach_alpha(frame):
    f = frame.dropna(); k = f.shape[1]
    item_var = f.var(axis=0, ddof=1).sum()
    total_var = f.sum(axis=1).var(ddof=1)
    return (k/(k-1))*(1-item_var/total_var), len(f)

rows = []
for name, items in composites.items():
    s = df[items].mean(axis=1); df[f"COMP_{name}"] = s
    a, na = cronbach_alpha(df[items]); lo,hi = ci_mean(s)
    rows.append({"Composite":name, "k":len(items), "Mean":round(s.mean(),3),
                 "SD":round(s.std(ddof=1),3), "Median":round(s.median(),3),
                 "CI":f"[{lo:.3f}, {hi:.3f}]", "alpha":round(a,3)})
pd.DataFrame(rows).set_index("Composite")
```

Output:

```
           k   Mean     SD  Median             CI  alpha
Composite
Awareness  7  4.044  0.478   4.000  [4.019, 4.069]  0.792
IT_Value   8  4.325  0.463   4.375  [4.301, 4.349]  0.858
Sec_Value  8  4.202  0.509   4.250  [4.176, 4.229]  0.858
```

Shapiro–Wilk Normality Tests

Applies the Shapiro–Wilk normality test to each composite (methodology: *Tests of Normality*). **Code:**

```
for name in composites:
    s = df[f"COMP_{name}"].dropna()
    W,p = stats.shapiro(s.sample(min(5000,len(s)), random_state=1))
    print(f"{name:10s}  W={W:.4f}  p={p:.2e}  -> {'non-normal' if p<0.05 else 'normal'}")
```

Output:

```
Awareness   W=0.9780  p=7.49e-14  -> non-normal
IT_Value    W=0.9571  p=5.93e-20  -> non-normal
Sec_Value   W=0.9647  p=4.80e-18  -> non-normal
```

Effect-Size Estimators

Defines the effect-size estimators: Cohen's d, the rank-biserial correlation, and the large-sample confidence interval for d (methodology: *Effect Sizes*). **Code:**

```
def cohen_d(a,b):
    a,b = np.asarray(a), np.asarray(b); na,nb = len(a),len(b)
    sp = np.sqrt(((na-1)*a.var(ddof=1)+(nb-1)*b.var(ddof=1))/(na+nb-2))
    return (a.mean()-b.mean())/sp if sp>0 else np.nan
def rank_biserial(U,n1,n2):
    return 1 - 2*U/(n1*n2)
def d_ci(d, n1, n2):
    """Large-sample 95% CI for Cohen's d (Hedges & Olkin, 1985)."""
    se = np.sqrt((n1+n2)/(n1*n2) + d**2/(2*(n1+n2)))
    return d-1.96*se, d+1.96*se
print("Effect-size helpers defined.")
```

Output:

```
Effect-size helpers defined.
```

Field Comparison: Mann–Whitney, Cohen's d, Rank-Biserial

Compares Information Technology and Information Assurance on each composite with the Mann–Whitney U test, Cohen's d with its interval, and the rank-biserial correlation (methodology: *Group Comparison by Professional Field*). **Code:**

```
it_mask = df["Q6.3"]=="Information Technology"
ia_mask = df["Q6.3"]=="Information Assurance / Cyber Security"
rows = []
for name in composites:
    a = df.loc[it_mask, f"COMP_{name}"].dropna()
    b = df.loc[ia_mask, f"COMP_{name}"].dropna()
    U,p = stats.mannwhitneyu(a,b,alternative="two-sided")
    t,tp = stats.ttest_ind(a,b,equal_var=False)
    d = round(cohen_d(a,b),3); lo,hi = d_ci(d,len(a),len(b))  # round d as in the published table pipeline
    rows.append({"Composite":name,"IT n":len(a),"IA n":len(b),
                 "IT M":round(a.mean(),3),"IA M":round(b.mean(),3),
                 "U":U,"p":f"{p:.2e}","Cohen d":round(d,3),
                 "d 95% CI":f"[{lo:.2f}, {hi:.2f}]",
                 "r_rb":round(rank_biserial(U,len(a),len(b)),3)})
pd.DataFrame(rows).set_index("Composite")
```

Output:

Composite	IT n	IA n	IT M	IA M	U	p	Cohen d	d 95% CI	r_rb
Awareness	501	501	4.379	3.732	218765.5	8.46e-93	1.641	[1.50, 1.78]	-0.743
IT_Value	501	501	4.669	3.980	227984.0	1.29e-111	1.940	[1.79, 2.09]	-0.817
Sec_Value	501	501	4.566	3.820	227813.5	4.60e-111	1.944	[1.79, 2.09]	-0.815

Item-Level Comparison with Holm–Bonferroni

Repeats the comparison item by item and applies the Holm–Bonferroni step-down correction within each item family. **Code:**

```
families = {
    "Awareness (S1-S2)": ["Q1.1"]+[f"Q2.{i}" for i in range(1,7)],
    "Applicability (S3)": [f"Q3.{i}" for i in range(1,7)],
    "IT Value (S4)": [f"Q4.{i}" for i in range(1,5)],
    "IA Value (S5)": [f"Q5.{i}" for i in range(1,7)],
}
def holm(pairs):
    m = len(pairs); order = sorted(pairs, key=lambda kv: kv[1])
    out = {}; prev = 0.0
    for i,(k,p) in enumerate(order):
        adj = min(1.0,(m-i)*p); adj = max(adj,prev); prev = adj
        out[k] = {"p":p,"p_adj":adj,"reject":adj<0.05}
    return out

allrows = []
for fam, items in families.items():
    pairs = []; cache = {}
    for q in items:
        a = df.loc[it_mask,q].dropna(); b = df.loc[ia_mask,q].dropna()
        U,p = stats.mannwhitneyu(a,b,alternative="two-sided")
```

```
        cache[q] = {"IT M":round(a.mean(),2),"IA M":round(b.mean(),2),
                    "r_rb":round(rank_biserial(U,len(a),len(b)),3)}
        pairs.append((q,p))
    h = holm(pairs)
    for q in items:
        allrows.append({"Family":fam,"Item":q,**cache[q],
                        "p_Holm":f"{h[q]['p_adj']:.1e}","Sig":"yes" if h[q]['reject'] else "no"})
pd.DataFrame(allrows).set_index(["Family","Item"])
```

Output:

Family	Item	IT M	IA M	r_rb	p_Holm	Sig
Awareness (S1-S2)	Q1.1	2.88	1.20	-0.775	6.4e-115	yes
	Q2.1	4.65	4.17	-0.415	5.8e-37	yes
	Q2.2	4.60	4.15	-0.395	4.3e-34	yes
	Q2.3	4.61	4.16	-0.398	2.1e-34	yes
	Q2.4	4.73	4.25	-0.416	1.3e-37	yes
	Q2.5	4.53	4.02	-0.365	2.1e-27	yes
	Q2.6	4.65	4.17	-0.428	3.6e-39	yes
Applicability (S3)	Q3.1	4.62	3.93	-0.519	2.9e-53	yes
	Q3.2	4.74	4.05	-0.613	1.0e-79	yes
	Q3.3	4.85	4.13	-0.634	1.0e-85	yes
	Q3.4	4.80	4.05	-0.660	2.4e-91	yes
	Q3.5	4.64	3.97	-0.494	9.1e-49	yes
	Q3.6	4.21	3.54	-0.448	5.5e-39	yes
IT Value (S4)	Q4.1	4.73	4.03	-0.621	1.1e-81	yes
	Q4.2	4.61	3.93	-0.503	2.5e-50	yes
	Q4.3	4.69	3.94	-0.553	7.0e-61	yes
	Q4.4	4.42	3.85	-0.445	1.0e-41	yes
IA Value (S5)	Q5.1	4.47	3.76	-0.564	5.1e-66	yes
	Q5.2	4.43	3.78	-0.444	2.3e-38	yes
	Q5.3	4.70	3.92	-0.573	5.4e-65	yes
	Q5.4	4.69	3.95	-0.525	1.6e-55	yes
	Q5.5	4.64	3.77	-0.637	2.6e-78	yes
	Q5.6	4.64	3.78	-0.639	9.8e-79	yes

Experience: Kruskal–Wallis, Eta-Squared, and Dunn

Compares the four experience cohorts with the Kruskal–Wallis test, computes η^2, and runs Dunn's post-hoc test with Holm adjustment (methodology: *Group Comparison by Experience Level*). **Code:**

```
EXP = ["1-5 Years","6-10 Years","11-15 Years","16+ Years"]
def dunn(groups, names):
    allv = np.concatenate(groups); ranks = stats.rankdata(allv); N=len(allv)
    idx=0; rg=[]
```

```
    for g in groups: rg.append(ranks[idx:idx+len(g)]); idx+=len(g)
    Rbar=[r.mean() for r in rg]; ns=[len(g) for g in groups]
    _,counts = np.unique(allv, return_counts=True)
    ties=(counts**3-counts).sum(); sigma2=(N*(N+1)/12)-ties/(12*(N-1))
    pairs=[]
    for i in range(len(groups)):
        for j in range(i+1,len(groups)):
            se=np.sqrt(sigma2*(1/ns[i]+1/ns[j])); z=(Rbar[i]-Rbar[j])/se
            p=2*(1-stats.norm.cdf(abs(z))); pairs.append((f"{names[i]} vs {names[j]}",p,round(z,2)))
    h=holm([(k,p) for k,p,_ in pairs])
    return pd.DataFrame([{"pair":k,"z":z,"p_adj":f"{h[k]['p_adj']:.2e}","sig":h[k]['reject']} for k,p,z in pairs])

for name in composites:
    groups=[df.loc[df["Q6.4"]==l,f"COMP_{name}"].dropna().values for l in EXP]
    H,p = stats.kruskal(*groups); Ntot=sum(len(g) for g in groups)
    eta2=(H-len(groups)+1)/(Ntot-len(groups))
    print(f"=== {name} ===  H={H:.1f}  df={len(groups)-1}  p={p:.2e}  eta^2={eta2:.3f}")
    print({l:round(g.mean(),3) for l,g in zip(EXP,groups)})
    display(dunn(groups,EXP))
```

Output:

```
=== Awareness ===  H=213.1  df=3  p=6.24e-46  eta^2=0.149
{'1-5 Years': np.float64(4.063), '6-10 Years': np.float64(4.236), '11-15 Years': np.float64(3.735), '16+ Years': np.float64(4.008)}
                       pair      z     p_adj    sig
0    1-5 Years vs 6-10 Years  -6.09  4.58e-09   True
1   1-5 Years vs 11-15 Years   9.51  0.00e+00   True
2     1-5 Years vs 16+ Years   0.82  4.11e-01  False
3  6-10 Years vs 11-15 Years  14.56  0.00e+00   True
4    6-10 Years vs 16+ Years   2.39  5.11e-02  False
5   11-15 Years vs 16+ Years  -1.89  1.18e-01  False=== IT_Value ===  H=229.4  df=3  p=1.88e-49  eta^2=0.161
{'1-5 Years': np.float64(4.402), '6-10 Years': np.float64(4.462), '11-15 Years': np.float64(3.982), '16+ Years': np.float64(4.59)}
                       pair      z     p_adj    sig
0    1-5 Years vs 6-10 Years  -2.20  8.42e-02  False
1   1-5 Years vs 11-15 Years  12.52  0.00e+00   True
2     1-5 Years vs 16+ Years  -1.79  1.46e-01  False
3  6-10 Years vs 11-15 Years  14.09  0.00e+00   True
4    6-10 Years vs 16+ Years  -1.22  2.21e-01  False
5   11-15 Years vs 16+ Years  -5.33  3.88e-07   True=== Sec_Value ===  H=291.2  df=3  p=7.80e-63  eta^2=0.205
{'1-5 Years': np.float64(4.323), '6-10 Years': np.float64(4.352), '11-15 Years': np.float64(3.769), '16+ Years': np.float64(4.431)}
                       pair      z     p_adj    sig
0    1-5 Years vs 6-10 Years  -0.80  1.00e+00  False
1   1-5 Years vs 11-15 Years  15.07  0.00e+00   True
2     1-5 Years vs 16+ Years  -0.84  1.00e+00  False
3  6-10 Years vs 11-15 Years  15.35  0.00e+00   True
4    6-10 Years vs 16+ Years  -0.63  1.00e+00  False
5   11-15 Years vs 16+ Years  -5.12  1.24e-06   True
```

Associations: Chi-Square and Cramér's V

Tests the categorical associations with the Pearson chi-square test and Cramér's V (methodology: *Association Analysis*). **Code:**

```
def chi_assoc(rowvar, colvar, roworder=None, colorder=None):
    sub = df[[rowvar,colvar]].dropna()
```

```
    ct = pd.crosstab(sub[rowvar], sub[colvar])
    if roworder: ct = ct.reindex(roworder).dropna(how="all")
    if colorder: ct = ct.reindex(columns=colorder).dropna(axis=1,how="all")
    chi2,p,dof,_ = stats.chi2_contingency(ct)
    n = ct.values.sum(); k = min(ct.shape)-1
    V = np.sqrt(chi2/(n*k)) if k>0 else np.nan
    return ct, chi2, dof, p, V

field_order=["Information Assurance / Cyber Security","Information Technology","Engineering"]
for lab,(rv,cv,ro,co) in {
    "Field x exposure":("Q6.3","Q1.2",field_order,["Yes","No"]),
    "Experience x IT investment":("Q6.4","Q4.5",EXP,["Yes","No","Unsure"]),
    "Experience x Sec investment":("Q6.4","Q5.7",EXP,["Yes","No","Unsure"]),
}.items():
    ct,chi2,dof,p,V = chi_assoc(rv,cv,ro,co)
    print(f"{lab}: chi2({dof})={chi2:.1f}, p={p:.2e}, Cramer V={V:.3f}, n={ct.values.sum()}")
```

Output:

```
Field x exposure: chi2(2)=60.3, p=7.92e-14, Cramer V=0.212, n=1339
Experience x IT investment: chi2(6)=6.0, p=4.21e-01, Cramer V=0.046, n=1411
Experience x Sec investment: chi2(6)=2.2, p=9.02e-01, Cramer V=0.028, n=1411
```

Bias Diagnostics

Runs the three bias diagnostics: the wave (non-response) analysis, the priming diagnostic, and the straight-lining check (methodology: *Bias Diagnostics*). **Code:**

```
# Wave -- "Completion time" is the submission timestamp, so sorting it orders by arrival
ds = df.sort_values("Completion time"); half=len(ds)//2
early, late = ds.iloc[:half], ds.iloc[half:]
print("WAVE (early vs late):")
for name in composites:
    a=early[f"COMP_{name}"].dropna(); b=late[f"COMP_{name}"].dropna()
    U,p=stats.mannwhitneyu(a,b,alternative="two-sided")
    print(f"  {name:10s} early={a.mean():.3f} late={b.mean():.3f} d={cohen_d(a,b):+.3f} p={p:.2e}")
# Priming
high=df[df["Q1.1"].isin([4,5])]; low=df[df["Q1.1"].isin([1,2])]
print("\nPRIMING (high vs low familiarity):")
for name in ["IT_Value","Sec_Value"]:
    a=high[f"COMP_{name}"].dropna(); b=low[f"COMP_{name}"].dropna()
    U,p=stats.mannwhitneyu(a,b,alternative="two-sided")
    print(f"  {name:10s} high={a.mean():.3f} low={b.mean():.3f} diff={a.mean()-b.mean():+.3f} d={cohen_d(a,b):.3f} p={p:.2e}")

# Straight-lining / acquiescence check across the 22 agreement-scale items
AGREE22 = ([f"Q2.{i}" for i in range(1,7)]+[f"Q3.{i}" for i in range(1,7)]
           +[f"Q4.{i}" for i in range(1,5)]+[f"Q5.{i}" for i in range(1,7)])
batt = df[AGREE22]; complete = batt.notna().all(axis=1)
rowsd = batt[complete].std(axis=1, ddof=0)
```

```
nC = int(complete.sum()); sd0 = int((rowsd==0).sum()); flat = int((rowsd<=0.25).sum())
print(f"\nSTRAIGHT-LINING (22 agreement items, n={nC} with complete battery):")
print(f"  identical response to all 22 (within-respondent SD=0): {sd0} ({100*sd0/nC:.1f}%)")
print(f"  near-flat (SD <= 0.25):                                {flat} ({100*flat/nC:.1f}%)")
print(f"  median within-respondent SD:                           {rowsd.median():.3f}")
```

Output:

```
WAVE (early vs late):
  Awareness  early=4.033 late=4.054 d=-0.044 p=5.23e-01
  IT_Value   early=4.321 late=4.328 d=-0.016 p=8.02e-01
  Sec_Value  early=4.192 late=4.212 d=-0.040 p=5.27e-01

PRIMING (high vs low familiarity):
  IT_Value   high=4.751 low=4.212 diff=+0.539 d=1.302 p=2.71e-67
  Sec_Value  high=4.654 low=4.081 diff=+0.573 d=1.253 p=2.78e-61

STRAIGHT-LINING (22 agreement items, n=1390 with complete battery):
  identical response to all 22 (within-respondent SD=0): 22 (1.6%)
  near-flat (SD <= 0.25):                                79 (5.7%)
  median within-respondent SD:                           0.498
```

Multivariate Regression with HC3 Robustness

Fits the ordinary-least-squares regression of each value composite on familiarity, field, and experience, refits with HC3-robust standard errors, and reports the coefficient table and standardized betas (methodology: *Multivariate Model and Robustness*). **Code:**

```
import statsmodels.formula.api as smf

reg = df.copy()
reg["field4"] = pd.Categorical(
    reg["Q6.3"].map({"Information Technology": "IT",
                     "Information Assurance / Cyber Security": "IA",
                     "Engineering": "Eng", "Other": "Other"}),
    categories=["IA", "IT", "Eng", "Other"])          # IA/Cyber = reference
reg["expc"] = pd.Categorical(
    reg["Q6.4"].map({"1-5 Years": "1-5", "6-10 Years": "6-10",
                     "11-15 Years": "11-15", "16+ Years": "16+"}),
    categories=["1-5", "6-10", "11-15", "16+"])        # 1-5 years = reference
reg["Fam"] = reg["Q1.1"]
reg = reg.dropna(subset=["COMP_IT_Value", "COMP_Sec_Value", "field4", "expc", "Fam"])

LABELS = {
```

```
    "Intercept": "Intercept",
    "Fam": "Familiarity (Q1.1)",
    "C(field4, Treatment('IA'))[T.IT]":    "Field: Information Technology",
    "C(field4, Treatment('IA'))[T.Eng]":   "Field: Engineering",
    "C(field4, Treatment('IA'))[T.Other]": "Field: Other",
    "C(expc, Treatment('1-5'))[T.6-10]":   "Experience: 6-10 years",
    "C(expc, Treatment('1-5'))[T.11-15]":  "Experience: 11-15 years",
    "C(expc, Treatment('1-5'))[T.16+]":    "Experience: 16+ years",
}
stars = lambda p: "***" if p < .001 else "**" if p < .01 else "*" if p < .05 else ""

cols_out, fits = {}, {}
for col, title in [("COMP_IT_Value", "IT Value"), ("COMP_Sec_Value", "Security Value")]:
    m = smf.ols(f"{col} ~ Fam + C(field4, Treatment('IA')) + C(expc, Treatment('1-5'))",
                data=reg).fit()
    fits[title] = m
    cols_out[title] = {LABELS.get(k, k): f"{v:.3f}{stars(m.pvalues[k])} ({m.bse[k]:.3f})"
                       for k, v in m.params.items()}

order = ["Familiarity (Q1.1)", "Field: Information Technology", "Field: Engineering",
         "Field: Other", "Experience: 6-10 years", "Experience: 11-15 years",
         "Experience: 16+ years", "Intercept"]
tab = pd.DataFrame(cols_out).reindex(order)
tab.loc["R-squared"]      = [f"{fits[t].rsquared:.3f}"     for t in tab.columns]
tab.loc["Adj. R-squared"] = [f"{fits[t].rsquared_adj:.3f}" for t in tab.columns]
tab.loc["N"]              = [f"{int(fits[t].nobs):,}"      for t in tab.columns]
print("Reference categories: Information Assurance / Cyber Security; 1-5 years of experience.")
print("Coefficients unstandardized (SE in parentheses); * p<.05, ** p<.01, *** p<.001.")
print("The 11-15 year coefficient is large, negative, and significant in both models: the")
print("mid-career dip persists net of field and familiarity, a genuine experience effect.")
display(tab)

# HC3 heteroscedasticity-robust refit (bounded, left-skewed composites)
hc3_rows = []
for col, title in [("COMP_IT_Value", "IT Value"), ("COMP_Sec_Value", "Security Value")]:
    m = fits[title]; mh = m.get_robustcov_results(cov_type="HC3")
    names = list(m.params.index)
    for key, lab in [("Fam", "Familiarity (Q1.1)"),
                     ("C(field4, Treatment('IA'))[T.IT]", "Field: Information Technology"),
                     ("C(expc, Treatment('1-5'))[T.11-15]", "Experience: 11-15 years")]:
        j = names.index(key)
```

```
        hc3_rows.append({"Model": title, "Predictor": lab,
                         "b": round(float(m.params[key]), 3),
                         "SE (classical)": round(float(m.bse[key]), 3),
                         "p (classical)": round(float(m.pvalues[key]), 3),
                         "SE (HC3)": round(float(mh.bse[j]), 3),
                         "p (HC3)": round(float(mh.pvalues[j]), 3)})
print("\nHC3 robustness check (bounded, left-skewed composites):")
print("Significance and direction are unchanged; the 11-15 coefficient stays significant.")
pd.DataFrame(hc3_rows).set_index(["Model", "Predictor"])
```

Output:

```
Reference categories: Information Assurance / Cyber Security; 1-5 years of experience.
Coefficients unstandardized (SE in parentheses); * p<.05, ** p<.01, *** p<.001.
The 11-15 year coefficient is large, negative, and significant in both models: the
mid-career dip persists net of field and familiarity, a genuine experience effect.
                                    IT Value     Security Value
Familiarity (Q1.1)             0.132*** (0.008)   0.142*** (0.008)
Field: Information Technology   0.458*** (0.022)   0.496*** (0.023)
Field: Engineering              0.301*** (0.020)   0.350*** (0.021)
Field: Other                    0.232*** (0.036)   0.305*** (0.038)
Experience: 6-10 years          0.082*** (0.017)    0.055** (0.018)
Experience: 11-15 years        -0.402*** (0.019)  -0.534*** (0.020)
Experience: 16+ years             0.168* (0.067)      0.092 (0.070)
Intercept                      3.888*** (0.018)   3.759*** (0.019)
R-squared                                 0.635             0.672
Adj. R-squared                            0.633             0.670
N                                         1,411             1,411
HC3 robustness check (bounded, left-skewed composites):
Significance and direction are unchanged; the 11-15 coefficient stays significant.
                                           b  SE (classical)  p (classical)  SE (HC3)  p (HC3)
Model          Predictor
IT Value       Familiarity (Q1.1)             0.132          0.008            0.0     0.007      0.0
               Field: Information Technology  0.458          0.022            0.0     0.021      0.0
               Experience: 11-15 years       -0.402          0.019            0.0     0.020      0.0
Security Value Familiarity (Q1.1)             0.142          0.008            0.0     0.008      0.0
               Field: Information Technology  0.496          0.023            0.0     0.023      0.0
               Experience: 11-15 years       -0.534          0.020            0.0     0.020      0.0
```

Spearman Rank-Order Correlations

Computes the Spearman rank-order correlations among familiarity, the Awareness composite, and the value composites (methodology: *Rank-Order Correlation*). **Code:**

```
from scipy.stats import spearmanr


pairs = [("Q1.1", "COMP_IT_Value", "Familiarity (Q1.1)", "IT Value"),
         ("Q1.1", "COMP_Sec_Value", "Familiarity (Q1.1)", "Security Value"),
         ("COMP_Awareness", "COMP_IT_Value", "Awareness composite", "IT Value"),
         ("COMP_Awareness", "COMP_Sec_Value", "Awareness composite", "Security Value"),
```

```
        ("COMP_IT_Value", "COMP_Sec_Value", "IT Value", "Security Value")]
rows = []
for a, b, la, lb in pairs:
    s = df[[a, b]].dropna()
    rho, p = spearmanr(s[a], s[b])
    rows.append({"X": la, "Y": lb, "Spearman rho": round(rho, 3),
                 "p": f"{p:.2e}", "n": len(s)})
pd.DataFrame(rows)
```

Output:

	X	Y	Spearman rho	p	n
0	Familiarity (Q1.1)	IT Value	0.632	4.53e-158	1411
1	Familiarity (Q1.1)	Security Value	0.616	2.27e-148	1411
2	Awareness composite	IT Value	0.734	3.48e-239	1411
3	Awareness composite	Security Value	0.731	4.90e-236	1411
4	IT Value	Security Value	0.842	0.00e+00	1411

Post-Hoc Common-Method Variance (Harman's Single-Factor Test)

A robustness check added after data collection and not specified in the proposal. It confirms that no single factor dominates the item covariances. **Code:**

```
# Harman's single-factor test on the 23 composite items
Z = (df[LIKERT] - df[LIKERT].mean()) / df[LIKERT].std(ddof=1)
R = np.corrcoef(Z.dropna().values, rowvar=False)
eig = np.sort(np.linalg.eigvalsh(R))[::-1]
print("Items:", len(LIKERT))
print("Factors with eigenvalue > 1:", int((eig > 1).sum()))
print(f"First unrotated factor variance: {100*eig[0]/eig.sum():.1f}%")
```

Output:

```
Items: 23
Factors with eigenvalue > 1: 2
First unrotated factor variance: 44.0%
```

Post-Hoc Construct Validity (CR, AVE, Fornell–Larcker)

A confirmatory three-factor model (semopy) yielding composite reliability, average variance extracted, the discriminant-validity check, and fit indices. **Code:**

```
import semopy
san = lambda s: s.replace(".", "_")
X = df[LIKERT].dropna()
desc = "\n".join(f"{n} =~ " + " + ".join(san(i) for i in items)
                 for n, items in composites.items())
cfa = semopy.Model(desc); cfa.fit(X.rename(columns=san))
ins = cfa.inspect(std_est=True)
load = {r["lval"]: abs(float(r["Est. Std"])) for _, r in ins[ins["op"] == "~"].iterrows()}
```

```
for n, items in composites.items():
    lam = np.array([load[san(i)] for i in items])
    ave = np.mean(lam**2); cr = lam.sum()**2 / (lam.sum()**2 + np.sum(1 - lam**2))
    print(f"{n:10s} CR={cr:.3f} AVE={ave:.3f} sqrtAVE={ave**0.5:.3f}")
comp = pd.DataFrame({n: df[items].mean(axis=1) for n, items in composites.items()})
print(comp.corr().round(3))
stt = semopy.calc_stats(cfa).T
print(f"CFI={float(stt.loc['CFI'].iloc[0]):.3f} RMSEA={float(stt.loc['RMSEA'].iloc[0]):.3f}")
def htmt(a, b):
    C = df[a + b].corr().abs()
    het = np.mean([C.loc[i, j] for i in a for j in b])
    mono = lambda g: np.mean([C.loc[g[i], g[j]] for i in range(len(g)) for j in range(i+1, len(g))])
    return het / np.sqrt(mono(a) * mono(b))
nm = list(composites); it = list(composites.values())
for i in range(3):
    for j in range(i+1, 3):
        print(f"HTMT {nm[i]}-{nm[j]}: {htmt(it[i], it[j]):.3f}")
```

Output:

```
Awareness  CR=0.858 AVE=0.471 sqrtAVE=0.686
IT_Value   CR=0.868 AVE=0.455 sqrtAVE=0.675
Sec_Value  CR=0.867 AVE=0.453 sqrtAVE=0.673
           Awareness  IT_Value  Sec_Value
Awareness      1.000     0.740      0.737
IT_Value       0.740     1.000      0.845
Sec_Value      0.737     0.845      1.000
CFI=0.966 RMSEA=0.040
HTMT Awareness-IT_Value: 0.830
HTMT Awareness-Sec_Value: 0.830
HTMT IT_Value-Sec_Value: 0.987
```

Post-Hoc Measurement Invariance: Linear DIF Screen (Zumbo's framework, linear implementation), IT vs IA

An item-level differential-item-functioning screen across the two focal fields, matching each item on its composite score. **Code:**

```
import statsmodels.api as sm
from scipy.stats import f as _f
foc = df[df["Q6.3"].isin(["Information Technology",
                          "Information Assurance / Cyber Security"])].copy()
foc["grp"] = (foc["Q6.3"] == "Information Assurance / Cyber Security").astype(int)
comp_of = {i: n for n, items in composites.items() for i in items}
compcol = {n: df[items].mean(axis=1) for n, items in composites.items()}
flagged = []
for it in LIKERT:
    sub = pd.DataFrame({"y": foc[it], "m": compcol[comp_of[it]].reindex(foc.index),
                        "g": foc["grp"]}).dropna()
    mm = (sub["m"] - sub["m"].mean()) / sub["m"].std()
    y = sub["y"].values
```

```
    r  = sm.OLS(y, sm.add_constant(np.column_stack([mm, sub["g"].values]))).fit()
    m0 = sm.OLS(y, sm.add_constant(mm)).fit()
    F = (m0.ssr - r.ssr) / (r.ssr / (len(y) - 3))
    p = 1 - _f.cdf(F, 1, len(y) - 3)
    if p < 0.001 and abs(r.params[2]) >= 0.20:
        flagged.append((it, round(float(r.params[2]), 3)))
print("Items tested:", len(LIKERT), "| flagged:", len(flagged))
print(flagged)
```

Output:

```
Items tested: 23 | flagged: 2
[('Q1.1', -0.988), ('Q2.5', 0.23)]
```

Post-Hoc Awareness-Composite Item Diagnostics and Familiarity-Stratum Adjustment

Two checks added in revision 1.9. The first reports the corrected item-total correlations and alpha-if-deleted for the Awareness composite, documenting why the familiarity item is a deletion candidate on the conventional criteria and what the pre-specified retention decision costs. The second holds the composition of the bimodal familiarity distribution constant and re-estimates the professional-field gap. **Code:**

```
AWARE_Q2 = [f"Q2.{i}" for i in range(1, 7)]
AWARE = ["Q1.1"] + AWARE_Q2

def cronbach(frame):
    k = frame.shape[1]
    return k / (k - 1) * (1 - frame.var(ddof=1).sum() / frame.sum(axis=1).var(ddof=1))

X = df[AWARE].dropna()
print(f"alpha, all seven items : {cronbach(X):.4f}")
print(f"alpha, Q1.1 deleted    : {cronbach(X[AWARE_Q2]):.4f}")
for q in AWARE:
    r = X[q].corr(X[[j for j in AWARE if j != q]].mean(axis=1))
    print(f"  {q:6s} {r:+.3f}")

# familiarity-stratum adjustment of the field gap
IA = "Information Assurance / Cyber Security"
work = df.copy()
for _n, _items in composites.items():
    work[_n] = df[_items].mean(axis=1)
foc = work[work["Q6.3"].isin(["Information Technology", IA])].dropna(
    subset=["Q1.1", "IT_Value", "Sec_Value"]).copy()
foc["IT_field"] = (foc["Q6.3"] == "Information Technology").astype(int)
foc["expo"]  = (foc["Q1.2"] == "Yes").astype(float)
foc["famhi"] = (foc["Q1.1"] >= 4).astype(int)
specs = {"field only": ["IT_field"], "+ exposure (Q1.2)": ["IT_field", "expo"],
         "+ familiarity stratum": ["IT_field", "famhi"],
         "+ both": ["IT_field", "expo", "famhi"]}
for dv in ["IT_Value", "Sec_Value"]:
    base = None
    for name, cols in specs.items():
        m = sm.OLS(foc[dv], sm.add_constant(foc[cols])).fit(cov_type="HC3")
```

```
        b = m.params["IT_field"]; base = b if base is None else base
        print(f"{dv} {name:22s} b={b:.3f} retained={100*b/base:5.1f}%")
```

Output:

```
alpha, all seven items : 0.7923
alpha, Q1.1 deleted    : 0.8627
  Q1.1   +0.315
  Q2.1   +0.655
  Q2.2   +0.613
  Q2.3   +0.618
  Q2.4   +0.671
  Q2.5   +0.532
  Q2.6   +0.671
IT_Value  field only             b=0.689 retained=100.0%
IT_Value  + exposure (Q1.2)      b=0.667 retained= 96.8%
IT_Value  + familiarity stratum  b=0.602 retained= 87.3%
IT_Value  + both                 b=0.589 retained= 85.5%
Sec_Value field only             b=0.746 retained=100.0%
Sec_Value + exposure (Q1.2)      b=0.724 retained= 97.0%
Sec_Value + familiarity stratum  b=0.655 retained= 87.7%
Sec_Value + both                 b=0.642 retained= 86.0%
```

Synthesis and Points for Cross-Validation

The independent analysis reproduces the coherent story of Chapter 4 with three important wrinkles. For Research Question 1, self-reported familiarity and recent exposure are low (means near 1.9; 22% exposure), yet endorsement of specific capability descriptions is near-universal: a gap best described as low awareness coexisting with high capability-endorsement rather than as evidence of awareness as such. For Research Questions 2 and 3, perceived value and anticipated benefit are high and precisely estimated across the applicability, IT, and security blocks, with the caveat that the two investment-willingness items are at ceiling and contribute conviction but not discrimination. The comparative analyses are dominated by a very large Information Technology–versus–Information Assurance gap favoring Information Technology, and by a non-monotonic experience pattern in which mid-career respondents are the most reserved.

The following are the points where analytic choices most affect the conclusions, and

therefore where two good-faith analyses of the same data are most likely to diverge; each is addressed in Chapter 4 or its limitations.

1. **Field operationalization.** The instrument yields four field categories, not two. Whether Engineering and Other are excluded, pooled, or analyzed as a four-group Kruskal–Wallis changes the comparative section; the analysis honors the proposal's IT-vs-IA contrast but reports all four groups in the focused breakouts.
2. **Ceiling items.** Questions 4.5 and 5.7 are near-invariant. Treating them as ordinary outcomes versus acknowledging the ceiling materially changes how Research Question 2 receptiveness is reported; the analysis characterizes them as conviction rather than discrimination.
3. **Awareness composite composition.** Averaging Q1.1 with Q2.1–Q2.6 blends two constructs that move in opposite directions; Q1.1 is therefore reported separately alongside the composite so its low central tendency is not obscured.
4. **Experience shape.** The omnibus effect is real but non-linear; a monotonic narrative would misrepresent it. The 11–15 year dip and the instability of the 16+ cell ($n = 18$) are reported explicitly.
5. **Wave (non-response) bias.** The early-versus-late comparison shows no meaningful difference on any composite (all $|d| \leq 0.04$, $p > .5$), so response timing gives no evidence of self-selection; the upper-bound caveat therefore rests on the recruitment frame rather than on a wave signal, and Chapter 5 treats it as such.

A fully scripted pipeline produced every statistic, figure, and table above from the response file, so each value is reproducible and directly checkable against an independent analysis of the same data. Taken together, the methodology, the worked examples, and this executable code and output leave the analysis of Chapter 4 fully traceable from raw responses to reported result.

Appendix C: Acronyms and Terms

This appendix provides two references for the reader. The first is a list of acronyms and abbreviations used throughout the dissertation. The second is a glossary of disciplinary terms, each accompanied by a concise definition intended to make the technical narrative accessible to readers across systems engineering, Information Assurance, computer science, and IT service management. Established artifacts and frameworks retain the names under which they were published; consistent with the convention noted in Chapter 1, the institution itself appears as the Department of War (DoW).

Acronyms

Acronym	Definition
ACM	Association for Computing Machinery
ANOVA	Analysis of Variance
API	Application Programming Interface
AVE	Average Variance Extracted
BoE	Body of Evidence
CAE	Center of Academic Excellence
CFI	Comparative Fit Index
CFR	Code of Federal Regulations
CISA	Cybersecurity and Infrastructure Security Agency
CM	Configuration Management
CMDB	Configuration Management Database
CMMC	Cybersecurity Maturity Model Certification
COBIT	Control Objectives for Information and Related Technologies
CPS	Cyber-Physical System
CR	Composite Reliability
CSA	Cloud Security Alliance
CSF	Cybersecurity Framework (NIST)
CUI	Controlled Unclassified Information
DAU	Defense Acquisition University
DE	Digital Engineering
DevOps	Development and Operations
DevSecOps	Development, Security, and Operations
DIF	Differential Item Functioning
DISA	Defense Information Systems Agency
DMM	Domain Metamodel (UAF)
DoD	Department of Defense

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(Acronyms, continued from previous page)

Acronym	**Definition**
DoDAF	Department of Defense Architecture Framework
DoDI	Department of Defense Instruction
DoW	Department of War
DSU	Dakota State University
EA	Enterprise Architecture
FAR	Federal Acquisition Regulation
FEAF	Federal Enterprise Architecture Framework
FERPA	Family Educational Rights and Privacy Act
GDPR	General Data Protection Regulation
GRC	Governance, Risk, and Compliance
GSN	Goal Structuring Notation
HC3	Heteroscedasticity-Consistent standard errors (type 3)
HIPAA	Health Insurance Portability and Accountability Act
HTMT	Heterotrait-Monotrait ratio (of correlations)
IA	Information Assurance
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
INCOSE	International Council on Systems Engineering
IoT	Internet of Things
IRB	Institutional Review Board
ISACA	Information Systems Audit and Control Association
ISO	International Organization for Standardization
IT	Information Technology
ITIL	Information Technology Infrastructure Library
ITSM	Information Technology Service Management
MBSE	Model-Based Systems Engineering
ML	Modeling Language (UAF usage; not machine learning)
MODAF	Ministry of Defence Architecture Framework (United Kingdom)
NAF	NATO Architecture Framework
NASA	National Aeronautics and Space Administration
NATO	North Atlantic Treaty Organization
NDIA	National Defense Industrial Association
NIST	National Institute of Standards and Technology
NSA	National Security Agency
NSTISSI	National Security Telecommunications and Information Systems Security Instruction
OMG	Object Management Group
OSCAL	Open Security Controls Assessment Language
OSF	Open Science Framework
PCI DSS	Payment Card Industry Data Security Standard

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(Acronyms, continued from previous page)

Acronym	Definition
PLM	Product Lifecycle Management
PLS-SEM	Partial Least Squares Structural Equation Modeling
RMF	Risk Management Framework
RMSEA	Root Mean Square Error of Approximation
ROI	Return on Investment
RQ	Research Question
SACM	Structured Assurance Case Metamodel
SD	Standard Deviation
SE	Systems Engineering
SEBoK	Systems Engineering Body of Knowledge
SEI	Software Engineering Institute
SERC	Systems Engineering Research Center
SoS	System of Systems
SP	Special Publication (NIST)
SSP	System Security Plan
SysML	Systems Modeling Language
TAM	Technology Acceptance Model
TOGAF	The Open Group Architecture Framework
UAF	Unified Architecture Framework
ULMC	Unmeasured Latent Method Construct
UML	Unified Modeling Language
UPDM	Unified Profile for DoDAF and MODAF
UTAUT	Unified Theory of Acceptance and Use of Technology

Terms and Definitions

Term	Definition
Acquiescence bias	The tendency of survey respondents to agree with statements regardless of content, which can inflate agreement on affirmatively framed items.
Assurance case	A structured, evidence-supported argument that a system satisfies a claimed property such as safety or security.
Authoritative source of truth	A single, trusted baseline of system data and models from which all stakeholders operate, ensuring consistency and traceability across the lifecycle.
Authorization boundary	The set of information system components covered by a security authorization, defining the scope of risk-management responsibility under the Risk Management Framework.

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(Terms, continued from previous page)

Term	Definition
Boundary object	An artifact, such as a shared model or specification, flexible enough to serve local needs yet stable enough to coordinate work across different communities of practice.
Common-method variance	The portion of observed covariance among measures attributable to the shared self-report method rather than to the constructs, screened here with Harman's single-factor test.
Composite score	A summary measure formed by averaging a set of related Likert items; this study uses the Awareness, IT Value, and Security Value composites.
Configuration management	The discipline of establishing and maintaining the integrity of a system's components, attributes, and relationships through controlled change.
Continuous monitoring	The ongoing assessment of security-control effectiveness and system state required by the Risk Management Framework.
Convenience sampling	A non-probability sampling method that recruits accessible participants, which constrains the generalizability of results.
Cramér's V	An effect-size measure of association for a chi-square test of independence, ranging from 0 to 1.
Cronbach's alpha	A coefficient of internal-consistency reliability for a set of items, with values of 0.70 or higher conventionally considered acceptable.
Differential item functioning	A condition in which respondents from different groups but at the same underlying trait level answer an item differently, screened here across professional field.
Digital Engineering	An integrated, model-based approach to designing, developing, and sustaining complex systems through authoritative digital models and data.
Digital thread	The authoritative, traceable connections among requirements, design, implementation, and validation artifacts that persist across a system's lifecycle.
Digital twin	A virtual representation of a physical or logical system that maintains synchronization with its real-world counterpart through continuous data exchange.
Effect size	A measure of the magnitude of a difference or association, such as Cohen's d or Cramér's V, that is independent of sample size.
Enterprise architecture	A discipline and set of artifacts describing the structure and relationships among an organization's business processes, information, applications, and technology.

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(Terms, continued from previous page)

Term	Definition
Eta-squared (η^2)	A non-parametric effect-size measure for the Kruskal–Wallis test, expressing the share of rank variance attributable to group membership.
Holm–Bonferroni correction	A sequential procedure that adjusts significance thresholds to control the family-wise error rate across multiple related comparisons.
Information Assurance	The broader discipline of protecting and governing information and information systems, encompassing security policy, risk management, and compliance, within which cybersecurity is the operational component.
Kruskal–Wallis test	A non-parametric test for differences among three or more groups, used here for experience-level comparisons.
Likert scale	An ordered response format, here five points, used to measure attitudes, agreement, or familiarity.
Mann–Whitney U test	A non-parametric test comparing the distributions of two independent groups without assuming normality, used here as the primary field comparison.
Model-Based Systems Engineering	The formalized application of digital models, rather than documents, as the primary means of capturing system requirements, design, analysis, and verification.
Non-response bias	Systematic error arising when survey non-respondents differ from respondents; assessed here through wave analysis.
Priming	In survey methodology, the influence of question wording or context on responses, diagnosed here by comparing value perceptions across familiarity strata.
Product Lifecycle Management	Frameworks and tools for managing a system's information, configuration baselines, and changes from conception through retirement.
Rank-biserial correlation	A non-parametric effect-size measure derived from the Mann–Whitney statistic.
Risk Management Framework	The NIST seven-step process (prepare, categorize, select, implement, assess, authorize, and monitor) for managing security and privacy risk.
Security control	A safeguard or countermeasure, for example from NIST SP 800-53, selected to protect the confidentiality, integrity, and availability of a system.
Straight-lining	A response pattern in which a respondent gives the same answer across many items; used here as an acquiescence diagnostic on the agreement-scale battery.
System of systems	A set of independently useful systems integrated to deliver capabilities that none could provide alone.

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(Terms, continued from previous page)

Term	Definition
Systems engineering	An interdisciplinary engineering approach to designing, integrating, and managing complex systems across their lifecycle.
Unified Architecture Framework	An ISO-standardized enterprise and system-of-systems architecture framework, expressed through UML and SysML profiles, that consolidates earlier military frameworks.
Wave analysis	A non-response diagnostic that compares early and late survey respondents, treating late responders as proxies for non-respondents.
Wilson score interval	A confidence interval for a proportion that remains well-behaved near zero or one, used here for agreement rates.

Appendix D: Institutional Review Board Human Subjects Application

This appendix reproduces the Institutional Review Board (IRB) Human Subjects Application submitted to the Dakota State University IRB for this study. The application records the protocol as reviewed: its purpose, participant tasks, recruitment method, target population, privacy and data-retention safeguards, and consent mechanism. Consistent with the Ethical Considerations discussion in Chapter 3, the study collected no personally identifiable information and received an exempt review determination (DSUIRB-20260421-13EX, April 21, 2026) before any recruitment or data collection began. The responses below are reproduced as submitted. The principal investigator's personal telephone number has been withheld for privacy.

Investigators and Administrative Details

Table 46: IRB application administrative details.

Field	Value
Project title	Transforming Information Assurance and IT Service Management Through Digital Engineering
Date submitted	27 March 2026
Principal Investigator	Patrick Engebretson, Beacom College of Computer and Cyber Sciences (`pat.engebretson@dsu.edu`)
Co-Investigator	John Bonar, Beacom College of Computer and Cyber Sciences (`john.bonar@trojans.dsu.edu`)
CITI training (Co-Investigator)	22 January 2025
Co-Investigator signature date	20 April 2026
Faculty supervisor	Patrick Engebretson

Application Responses

1. **Potential conflict of interest.** No.
2. **External funding.** No.
3. **Required to earn a DSU academic degree.** Yes. The dissertation and its Dissertation Proposal Defense Approval were provided as the required supporting

documentation.

4. **Submitted to another IRB.** No.
5. **Non-IRB approval letters (committees, schools, or external organizations).** Not applicable.
6. **Purpose of the research.** Identify perceptions of applicability and value among Information Technology and Information Assurance professionals with regard to Digital Engineering.
7. **Tasks participants perform.** Complete a six-part survey containing 31 questions in total.[3]
8. **Contacting participants and protecting privacy.** Recruit through LinkedIn posts to a professional social-media network and through email solicitation to industry peers: (1) a LinkedIn post, reproduced in Appendix A, and (2) an email containing the solicitation text. No identifying information is collected, which preserves participant privacy.
9. **Maximum participants enrolled.** A minimum of 385 completed responses are required to achieve a margin of error of 5%.
10. **Vulnerable study populations.** None. The study targeted no children, prisoners, pregnant persons, non-English speakers, students, employees of the investigators, or other listed vulnerable groups.
11. **Targeting a specific ethnic group.** Not applicable.
12. **Age range of subjects.** 18 to 64 years.
13. **Inclusion and exclusion criteria.** Inclusion is specific to invited Information Assurance and Information Technology professionals.

[3] One item was subsequently identified as a duplicate and removed before administration, so the instrument as administered contained thirty items (Appendix A).

14. **Research location.** Online participation within the United States.
15. **Participant compensation.** No.
16. **Direct benefits to participants.** No direct benefit.
17. **Private information collected.** None. The study collected no names, contact information, device or serial identifiers, geographic detail, dates, records, or other private information.
18. **Traceability of a record to a participant.** No.
19. **Agreement to destroy identifying data at study conclusion.** Yes.
20. **Agreement to retain de-identified data for three years.** Yes.
21. **Associated risks (physical, economic, psychological, legal, social, or use of private records).** None of the listed risks apply.
22. **Consent type.** Consent Statement for adults, permissible because no identifiers are collected.
23. **Method of obtaining consent.** Web-based.
24. **HIPAA authorization or waiver.** Not applicable.

The co-investigator signed the application on 20 April 2026. Faculty supervisor Dr. Patrick Engebretson provided the supervisor assurance that the research is scientifically and scholarly sound, that competencies and resources are adequate, and that no conflicts of interest exist.

Appendix E: Artificial Intelligence Assistance Disclosure

In accordance with academic integrity standards and emerging best practices for transparent research documentation, this appendix discloses the use of artificial intelligence tools in the preparation of this dissertation. Such disclosure reflects the author's commitment to scholarly transparency and recognition of evolving norms within the academic community.

AI Tools Utilized

This dissertation utilized Claude Opus 4.8 & Sonnet 4.7, developed by Anthropic, to assist with specific aspects of document preparation. The AI tool was employed under the direct supervision of the author, with all outputs subjected to critical review, verification, and revision as necessary to ensure accuracy, appropriateness, and alignment with the research objectives. The author exercised editorial judgment over all AI-assisted contributions.

Scope of AI Assistance

AI assistance was limited to the following editorial and mechanical activities:

- Grammar and spelling review to identify and correct mechanical errors
- Sentence structure refinement to improve clarity and readability
- Organizational suggestions to enhance document flow and logical progression
- Formatting consistency verification across sections and chapters
- Citation format verification for compliance with APA standards

Scope Limitations

AI tools were explicitly excluded from the following substantive activities:

- Generation of original research ideas, hypotheses, or theoretical frameworks

- Literature search, source identification, or citation selection
- Data analysis, statistical calculations, or interpretation of results
- Writing of substantive original content without subsequent author revision
- Determination of research methodology or survey instrument design

The intellectual contributions of this research, namely the identification of research gaps, the development of the theoretical framework, the design of the survey instrument, and the interpretation of findings, remain solely the work of the author.

Author Responsibility

The author maintains full responsibility for all content, arguments, analyses, and conclusions presented in this dissertation. All AI-assisted text was critically reviewed and revised by the author to ensure it accurately represents the author's original thinking and research contributions. The employment of AI tools for editorial assistance does not diminish the author's intellectual ownership of this work, nor does it transfer scholarly credit for the ideas and arguments herein.

Rationale for Disclosure

This disclosure is provided in the spirit of transparency and in recognition of evolving academic standards regarding AI tool usage. As AI capabilities continue to advance, clear documentation of how these tools are employed in academic work supports research integrity and enables appropriate evaluation of scholarly contributions. The author believes that transparent disclosure serves the academic community better than silent utilization, and that establishing norms for responsible AI assistance in scholarly work represents an important contribution to academic discourse.